EC 246: DECISIONS & GAMES

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Term IV, 2026

Decision Theory is a study of "rational decision making in an exogenous environment" – where the environment's reaction to a decision-maker's choice is insignificant. Game Theory extends Decision Theory to study "interactive decision-making in an endogenous environment" where multiple decision-makers respond to each other's decisions. The course will present a systematic introduction to the core ideas in Game Theory, and will study the applications of these ideas in diverse strategic situations.

We shall focus on the following strategic scenarios: (*a*) competition *vs*. cooperation, (*b*) coordination and anti-coordination, (*c*) intentional and inadvertent information transmission, and (*d*) bargaining. Game-equilibrium analysis will be applied to understand problems in sustaining cooperation, making credible threats and promises, achieving coordination, transmitting signals, and bidding in auctions. We shall conclude with a discussion on the strategic aspects of negotiations, focusing on the interplay between value-creation and value-capture that is inherent in formulating business strategies.

Course Topics

1. DECISIONS, GAMES, & NASH EQUILIBRIA

From Decision Theory to Game Theory; *Game Formulations and Game Structures*; *Nash Equilibrium in Strategic-form Games of Complete Information*

2. EPISTEMIC FOUNDATIONS OF STRATEGIC BEHAVIOUR

Rationalizability and Dominance-solved Nash Equilibria in Simultaneous-move Games; *Backward Induction and Subgame-perfect Nash Equilibria in Sequential-move Games*; *Forward Induction Refinements in Sequential-move Games*

3. COOPERATION DILEMMAS

Prisoners' Dilemmas in Economic and Social Interactions; *Strategic Substitutes and Complements*; *Collective Action Problems*; *Commitment Moves and 'Meta-strategies' to mitigate Cooperation Dilemmas*

4. COORDINATION, ANTI-COORDINATION, AND DIS-COORDINATION

Multiplicity of Nash Equilibria in Games of Coordination and Anti-coordination; *Non-existence of Pure-strategy Nash Equilibria in Dis-Coordination Games*; *Interpreting Mixed-strategy Nash Equilibria*

5. NASH EQUILIBRIA AS SELF-ENFORCING RECOMMENDATIONS

Self-enforcing Recommendations and Self-enforcing Agreements; *Renegotiation-proof Nash Equilibria*; *Coalition-proof Nash Equilibria*; *Correlated Equilibria*

6. Repeated and Dynamic Games

Repeated Games and Sustenance of Tacit Collusion; *Repeated Games and Fair Coordination*; *Dynamic Games of Timing*: *Preemption, Concession, and Wars of Attrition*

7. Strategic Interactions under Asymmetric Information

Bayes-Nash & Perfect-Bayesian Equilibria in Games of Incomplete Information; *Reputation-building in Multi-stage Games*; *Wars of Attrition under Asymmetric Information*; *Signalling Games*: *Belief-updating and Forward Induction Refinements*

Required Reading *Course Lecture Notes*

Basic Text Joel Watson: *Strategy – An Introduction to Game Theory* (2nd edition)

Bedside Books Thomas Schelling: *The Strategy of Conflict*; Avinash Dixit *&* Barry Nalebuff: *The Art of Strategy*; Adam Brandenburger *&* Barry Nalebuff: C*o-opetition*

Course Pedagogy One way to teach a rigorous course in Decision Theory and Game Theory is to take the formal "definitions – theorems – proofs" approach. The current course will, in contrast, take a largely "informal" approach, where decision-theoretic and game-theoretic concepts will be introduced and analyzed through a series of "examples". These examples will be presented in two ways. In some cases, a Discussion Scenario presented in the *Lecture Notes* will be studied in class (it will help immensely if students ponder over the scenario before coming to class). In other cases, a Game Experiment will be held in class, and the experimental outcomes will be discussed in subsequent lectures. Theoretical concepts and formal results will be presented *via* discussions of a series of such examples.

Office Hours For clarification of doubts regarding class lectures, I will hold office-hours from 4.15 pm to 5.15 pm on Mondays and Wednesdays [except for July 1, 8, and 22] in my office - Room 212, Faculty Block B.

Information on in-class Game Experiments There will be ***four*** game experiment sessions held in class over the entire term (each session will be for about 15-20 minutes). They will be pen-and-paper experiments, where each student will play against the "average strategy" of a "rival group of students". Relative performance will determine individual score in an experiment. The first experiment will be held during the **3rd** lecture session.

Course Evaluation Performance in the class experiments will constitute 15% of a student's course grade. Of the four game experiments that will be held over the term, a student's ***three best performances*** will count towards his/her course grade.

A 70-minute mid-term exam will constitute 35% of course grade, and a 100-minute end-term exam will constitute 50% of course grade.

For a student who misses the mid-term exam (for reasons considered valid by the MBA Office), the end-term exam will constitute 85% of the grade (and the student will suffer a grade drop). If a student misses the end-term exam (for reasons considered valid by the MBA Office), (s)he will have to appear for a make-up exam (and will suffer a grade drop).

- "Life is the sum of all your choices" ***Albert Camus***
- Given that, maybe everyone should study a bit about "how to make smart choices / decisions"
- That theory, quite predictably, is called **Choice Theory** and has two parts
  - the interesting part is **Psychological / Behavioural Choice Theory** whose domain is the neuro/psychological analysis of how people ***do*** make choices / decisions [*Thinking Fast and Slow*]
  - the less interesting part is **Rational Choice Theory** whose domain is the mathematical / economic analysis of how people ***ought to*** make decisions
- In contrast, the related discipline of DECISION SCIENCES studies the science of gathering, organizing, and interpreting data in order to permit informed decision-making

- The ambit of **Choice Theory**: Given the relevant data, **how should / do we make decisions**?

The sub-disciplines of "**Choice Theory**"

Rational Choice Theory	*Psychological Choice Theory*
DECISION THEORY	**Behavioural Decision Theory**
GAME THEORY	**Behavioural Game Theory**

***Basic Tenet of Rational Choice Theory*:**

Rational individuals (ought to) make decisions by solving "well specified" ***constrained optimization problems***

***Central questions in Rational Choice Theory*:**

When can we represent a decision-maker's preferences *via* an objective function? And when we can do that, what should be our logical analysis for predicting what constitutes her "rational (sensible/intelligent) decision" ***in non-interactive environments***, ***and in interactive environments*** ?

- When I go with friends to a restaurant, and I order my meal with the understanding that I will pay for my order, mine is a decision theoretic problem
- When I go with friends to a restaurant, and I order my meal with the understanding that the total bill will be shared equally by all, mine is a game theoretic problem

For decision-making in a non-interactive environment, the decision-maker requires comprehensive data about the exogenous environment;
but for decision-making in an interactive environment, over and above the environment data, the decision-maker also needs to formulate sensible ***inferences / conjectures / predictions*** *regarding decisions of others*

… and now for some name-dropping …

Rational Choice Theory	*Psychological Choice Theory*
DECISION THEORY	**Behavioural Decision Theory**
GAME THEORY	**Behavioural Game Theory**
Herbert Simon | **Richard Bellman** **John von Neumann** | **Oscar Morgenstern** **John Nash** | **John Harsanyi** **Reinhart Selten**	**Herbert Simon** **Daniel Kahneman** | **Amos Tversky** **Colin Camerer** | **Vincent Crawford**

- The defense for focusing on **Rational** Choice Theory:

1. Some effectively analytical decisions ***should be*** taken rationally, especially when stakes are very high

   https://www.theguardian.com/science/2021/sep/12/algebra-the-maths-working-to-solve-the-uks-supply-chain-crisis

   discusses how mathematics is working to solve the UK's supply chain crisis

   https://www.theguardian.com/us-news/ng-interactive/2021/nov/12/gerrymander-redistricting-map-republicans-democrats-visual

   discusses how election outcomes can be affected by redistricting constituencies

2. In situations where ***other people*** are making decisions on your behalf, ***you would want them*** to solve constrained optimization problems

3. Rational solutions give benchmarks to compare and contrast the nature and the extent our ***behavioural biases***

Question**: Do management students *really* need to study Game Theory?**

I will defer to Prof. Preston McAfee for an unequivocal answer

Preston McAfee [has been a professor of Game Theory at Cal Tech and the chief economist at Yahoo / Microsoft / Google]:

> "In most scenarios of business decision-making, players are *strategic* – aware of each other's existence and cognizant of how others will behave.
>
> That is precisely a scenario that *Game Theory* is designed to study.
>
> The idea of putting oneself in a rival's position – which is a central strategic notion in *Game Theory* – is an enormously valuable tool in intelligent decision-making in an interactive environment."

Course Logistics

- This will be an analytically rigorous course – more emphasis on logic than on math
- I will post class lecture slides, problem sets, and answer keys on the DropBox Course Site [feel free to use the textbook as a back-up]
- There will be four in-class experiments over the term – the first one will be in the third session (your performance in the best three will matter)
- There will be a 70-minute mid-term in the Mid-term exam week
- There will be a 100-minute end-term exam in the End-term exam week
- Exam questions will involve "solving a set of games"
- For communication send email to *arijitsen@iimcal.ac.in*
  - I will also be available for "recent" doubt-clarifications on Mondays and Wednesdays 4.20 to 5.20 in my office (Room 212, B Block)
- ***Please try not to pollute the class-room environment***
  - ***if you plan on coming to class, please DO NOT come late***
  - ***please DO NOT have laptops open***

First order of business

1. I will begin by discussing Decision Theory under Certainty
2. I will then discuss to Decision Theory under Uncertainty, and present the Expected Utility Theory of "decision-making under uncertainty" (a central model of decision-making for rational actors)

Schematic representation of a *static* decision problem under *certainty*

Choice Set (CS)	Outcomes of Interest	My Preference Ranking	My Utility Function (one representation)	Feasible Choice Set (FCS)
$c1$	$O(c1)$	3	$U_I(O(1)) = 6-3=3$	$c\,1$
$c2$	$O(c2)$	2	$U_I(O(2)) = 6-2=4$	
$c3$	$O(c3)$	5	$U_I(O(3)) = 6-5=1$	$c\,3$
$c4$	$O(c4)$	1	$U_I(O(4)) = 6-1=5$	
$c5$	$O(c5)$	4	$U_I(O(5)) = 6-4=2$	$c\,5$
$c6$	$O(c6)$	6	$U_I(O(6)) = 6-6=0$	

Q: How to represent my decision problem as a constrained optimization problem?

Answer: By constructing an ***ordinal utility function*** $U_I(O(c\,j))$ that has the property that $U_I(O(m)) \geq U_I(O(n))$ if and only if I ***prefer*** $O(m)$ to $O(n)$ and maximizing $U_I(O(j))$ subject to the constraint: $c\,j \in$ **FCS** $= \{c1, c3, c5\}$

I can create "my" $U_I(.)$ function if and only if (***iff***) I can "consistently" rank all outcomes in terms of their desirability to me

which I can do iff my preferences over **outcome pairs** are ***complete & transitive***

Note: ***Any rank-preserving transformation of*** $U_I(.)$ also represents my preferences

I can construct a consistent ranking over all possible outcomes if and only if my preferences over outcome-pairs and triples are ***complete*** and ***transitive***

Axioms of Completeness & Transitivity regarding Preferences

1. Completeness Axiom: One can compare any two outcomes ***A*** and ***B*** in terms of her preferences: ***A*** *preferred to* ***B*** OR ***B*** *preferred to* ***A*** OR *indifferent between* ***A*** and ***B***
2. Transitivity Axiom: If outcome ***A*** is preferred to outcome ***B***, and ***B*** is preferred to ***C***, then ***A*** is preferred to ***C***

[Note that when indifference between ***A*** and ***B*** holds then the 'preference rankings' of ***A*** and ***B*** will be identical, and thus the 'utility numbers' ***U***(***A***) and ***U***(***B***) will be the same.]

Issue to think about: *When can we extend* "formal Choice Theory" *from 'obviously analytical' problems to 'effectively analytical' problems*:

Travelling Salesman Problem I : *minimize travel time*
Travelling Salesman Problem II : *minimize travel distance*

Travelling Salesman Problem III : *minimize discomfort in traveling*
Travelling Salesman Problem IV : *minimize a combination of the three issues*

Summary of analyzing a ***static*** **decision problem under** ***certainty***

1. Identify the **universal choice (options) set CS**
2. Identify the mapping from choices to outcomes: **the outcome map O(C)**
3. **Completely and consistently rank all outcomes according to the individual's preferences**
4. Identify the feasible choices in **CS**: **feasible choice set FCS $\subset$ CS**
5. Choose that option **c*** whose outcome **O(c*)** has the highest rank among all feasible choices **c** $\in$ **FCS**

- If **Step 3** can be executed, then an "ordinal utility (ranking) function" ***U*: O(C) $\to$ R** can be "constructed" such that $U(O(c)) \geq U(O(c'))$ if and only if outcome O(c) is (weakly) more desirable than outcome O(c′)

Then the decision-making problem can be expressed as a constrained optimization problem:

maximize the utility function U(**O(c)**) *over the set of choices* **c** *belonging in the feasible choice set* **FCS $\subset$ CS**

CHOICE UNDER UNCERTAINTY & EXPECTED UTILITY THEORY

Arijit Sen

In many real-world situations, an individual has to take a decision without precisely knowing the specific outcome that will result from that decision – the individual might simply have some idea about the relative probabilities of the different possible outcomes that her decision can lead to. For instance, when a person invests some amount of money in a mutual fund, she will only know the "historical probability distribution" of the possible realized returns generated by the mutual fund in the past.

Polymaths John von Neumann and Oskar Morgenstern developed the ***Expected Utility Theory*** to model "choice under uncertainty". The ***Expected Utility Theory*** presents an analytical framework for studying decision-making under uncertainty, by taking the following two steps:

[Step 1] positing a specific kind of 'expected-utility-maximizing behaviour' in scenarios of uncertainty, and

[Step 2] identifying conditions in which expected-utility-maximizing behaviour will constitute rational decision-making.

Decision-making by Expected Utility Maximization

We begin our discussion by focusing on [Step 1]: What are the specific features of "expected-utility-maximizing behaviour" that the ***Expected Utility Theory*** posits?

The theory considers a scenario where an individual's objects of choice are "random/uncertain prospects" of the form: $\boldsymbol{Z}^j = \{z_1, \ldots, z_i, \ldots, z_N;\ p_1^j, \ldots, p_i^j, \ldots, p_N^j\}$ where $(z_1, \ldots, z_i, \ldots, z_N)$ is the set of possible deterministic outcomes (where each z_i can be a "distinct money amount") and $(p_1^j, .., p_i^j, .., p_N^j)$ is the "associated probability vector of the alternative outcomes" in the prospect $\boldsymbol{Z}^j$ (where p_i^j is the probability of the occurrence of outcome z_i under the prospect $\boldsymbol{Z}^j$).

For concreteness, think of the following bet that you can place against a counterparty: You hold Rs. 100, and you will pay the counterparty Rs.50 if it rains this afternoon, while the counterparty will pay you Rs.50 if it shines this afternoon. The Weather Channel informs you that there is an objective probability of 40% that it will rain this afternoon. Then the bet that you are considering is the following random prospect: $\{50, 150;\ 0.4, 0.6\}$. Note two important features of this random prospect: (1) The deterministic money amounts in the prospect – Rs.50 and Rs.150 – are the two possible final money amounts that you will end up with if you take the bet; this is the only sensible way to specify random prospects – in terms of the possible "deterministic final outcomes". (2) The probabilities associated with the alternative deterministic outcomes are "objective probabilities" on which everybody can agree upon – this feature of objective probabilities make the expected utility theory that much more convincing, but the theory survives (in a more complex form) even when the probabilities are "subjective" (i.e., a set of beliefs held by the decision-maker).

According to the Expected Utility Theory, an individual who makes choices among random prospects is an ***expected utility maximizer*** if and only if the following statements are true for her:

(i) ***Existence of a specific kind of utility function***: The individual possesses a utility function $\boldsymbol{u}(.)$ over the set of all deterministic outcomes $(z_1, \ldots, z_i, \ldots, z_N)$, and the individual's utility from a random prospect $\boldsymbol{Z}^j$ is given by the following "expected utility expression"

$\boldsymbol{EU}(\boldsymbol{Z}^j) = \boldsymbol{p_1^j.u(z_1) + p_2^j.u(z_2) + \ldots + p_N^j.u(z_N)}$.

The individual's utility function $\boldsymbol{u}(.)$ over deterministic outcomes is called her "Bernoulli utility function" (named after the French mathematician Daniel Bernoulli), and the expected utility expression $\boldsymbol{EU}(.)$ is referred to as the individual's "Expected utility function".

(ii) ***Decision-making on the basis of expected utility ranking***: Faced with a set of random prospects $\boldsymbol{Z}^1$, $\boldsymbol{Z}^2$, $\boldsymbol{Z}^3$, ..., the individual "chooses" that prospect $\boldsymbol{Z}^j$ for which the expected utility $\boldsymbol{EU}(\boldsymbol{Z}^j)$ is maximal.

We study three examples of "choice under uncertainty" to clarify how an expected utility maximizer behaves / is supposed to behave.

Example 1: Joy is an expected utility maximizer, who is going out for a long evening walk in the neighbourhood park, and has to decide whether or not to carry an umbrella. Under the Expected Utility Theory, this is how we should model Joy's decision- making:

First, Joy has to ascertain the probability that it might rain during the evening. He can get this information from an authoritative source like the Weather Channel. Let us suppose that the Weather Channel states that there is a probability "$\pi \in (0, 1)$" that it will rain in the evening.

Next, Joy has to identify the possible deterministic outcomes that can occur during his evening walk. The following are the mutually exclusive and exhaustive set of events that might occur:

Event [*CR*] : deterministic outcome that Joy **carries** an umbrella and it **rains**

Event [*CD*] : deterministic outcome that Joy **carries** an umbrella and it remains **dry**

Event [*NR*] : deterministic outcome that Joy does **not carry** an umbrella and it **rains**

Event [*ND*] : deterministic outcome that Joy does **not carry** an umbrella and it remains **dry**

Recognize that when Joy carries an umbrella for his evening walk, he faces the random prospect $\boldsymbol{Z}^C = \{CR, CD;\ \pi, 1-\pi\}$. Alternatively, when Joy does not carry an umbrella for his evening walk, he faces the random prospect $\boldsymbol{Z}^N = \{NR, ND;\ \pi, 1-\pi\}$.

Let Joy's Bernoulli utility function over the above-specified deterministic outcomes be $\boldsymbol{u}(.)$. Given $\boldsymbol{u}(.)$, and given the fact that Joy is an expected utility maximizer, we can state that Joy's expected utility when carrying an umbrella is: $\boldsymbol{EU}(\boldsymbol{Z}^C) = \pi.u(CR) + (1-\pi)u(CD)$, while Joy's expected utility when not carrying an umbrella is: $\boldsymbol{EU}(\boldsymbol{Z}^N) = \pi.u(NR) + (1-\pi)u(ND)$.

We can now state Joy's decision as an expected utility maximizer: He will carry an umbrella for his evening walk if and only if: $\boldsymbol{EU}(\boldsymbol{Z}^C) > \boldsymbol{EU}(\boldsymbol{Z}^N)$.[1]

[1] If $\boldsymbol{u}(CR) > \boldsymbol{u}(NR)$ and $\boldsymbol{u}(ND) > \boldsymbol{u}(CD)$ for Joy, then $\boldsymbol{EU}(\boldsymbol{Z}^C) > \boldsymbol{EU}(\boldsymbol{Z}^N)$ will hold if and only if the rain probability is greater than the *threshold* $\pi^* \equiv [u(ND)-u(CD)]/\{[u(ND)-u(CD)] + [u(CR)-u(NR)]\} \in (0, 1)$.

Example 2: There are four entrepreneurs – ***A***, ***B***, ***C***, ***D*** – all of whom are expected utility maximizers. Each entrepreneur independently considers his/her choice between two mutually exclusive "uncertain projects" X and Y. The two projects have the following characteristics: None of the projects require any up-front monetary investment (each only requires identical amount of "effort" by the entrepreneur), and both of the projects will generate a single "random" monetary return one year later.

Specifically, project X has equal probability of generating returns of Rs.**1** million *or* Rs.**2** million; while project Y has equal probability of generating returns of Rs.**1.32** million *or* Rs.**1.6** million Y. [In terms of statistical means and variances, note that project X has higher mean (expected value) and also higher variance than project Y.]

The above information implies that each entrepreneur, who cares only about final monetary returns from his/her chosen project, faces the problem of choosing one between the following two random prospects: $\boldsymbol{X} = \{1,\ 2;\ \ 0.5,\ 0.5\}$, $\boldsymbol{Y} = \{1.32,\ 1.6;\ \ 0.5,\ 0.5\}$.

While all four entrepreneurs are expected utility maximizers, they value different levels of deterministic money amounts ***differently***. The "differences in their preferences" over different money amounts mean that the four entrepreneurs have ***different*** Bernoulli utility functions. Specifically, letting $z \in [0,\infty)$ denote a typical "sure money return", let us posit that the four entrepreneurs ***A***, ***B***, ***C***, and ***D*** have the following Bernoulli utility functions: $u^{A}(z) = z^{2}$; $u^{B}(z) = z$; $u^{C}(z) = (z)^{1/2}$; and $u^{D}(z) = (z)^{1/3}$.

The distinct preferences over sure money returns for the four entrepreneurs (as captured by their distinct Bernoulli utility functions) imply that their choices over the two random prospects ***X*** and ***Y*** will be as follows:

- ***A*** will choose ***X*** over ***Y*** since $0.5(1)^2 + 0.5(2)^2 > 0.5(1.32)^2 + 0.5(1.6)^2$
- ***B*** will choose ***X*** over ***Y*** since $0.5(1) + 0.5(2) > 0.5(1.32) + 0.5(1.6)$
- ***C*** will be indifferent between ***X*** and ***Y*** since $0.5(1)^{1/2} + 0.5(2)^{1/2} = 0.5(1.32)^{1/2} + 0.5(1.6)^{1/2}$
- ***D*** will be choose ***Y*** over ***X*** since $0.5(1)^{1/3} + 0.5(2)^{1/3} < 0.5(1.32)^{1/3} + 0.5(1.6)^{1/3}$

Example 3: Investor I is an expected utility maximizer. She has funds of Rs.1 million and she wants to invest these funds across <u>two</u> financial assets – ***X*** and ***Y*** – for ***one year***. Asset ***X*** is a bond with a "sure" interest of 10% per annum, while asset ***Y*** is a stock whose return is uncertain. Historical data convinces all market participants that over one year, the value of ***Y*** will either go up by 21% OR will remain unchanged, and both these outcomes are equally likely. [Note that on the basis of the historical data, ***Y*** has *expected annual returns* of10.5%.]

Investor I has a Bernoulli utility function $\boldsymbol{u}^{I}(.)$ over sure returns. Her investment problem is to determine what fraction of her funds she should invest in the stock ***Y*** (and invest the remainder in bond ***X***) if she wants to maximize her expected utility of her "final money holdings".

Recognize that if I invests fraction $\boldsymbol{y}$ of Rs.1 million in stock ***Y*** (and the remainder of her funds in bond ***X***) she will get the prospect: $\boldsymbol{Z}(\boldsymbol{y}) = \{1.21\boldsymbol{y} + 1.10[1-\boldsymbol{y}],\ \boldsymbol{y} + 1.10[1-\boldsymbol{y}];\ \ 0.5,\ 0.5\}$.

Consequently, her expected utility from the choice of $\boldsymbol{y}$ will be: $\boldsymbol{EU^I}(\boldsymbol{Z}(\boldsymbol{y})) = 0.5\ \boldsymbol{u^I}(1.21\boldsymbol{y} + 1.10[1-\boldsymbol{y}]) + 0.5\boldsymbol{u^I}(\boldsymbol{y} + 1.10[1-\boldsymbol{y}])$. The investor's decision problem is to choose the fraction "$\boldsymbol{y} \in [0, 1]$" to maximize her expected utility function $\boldsymbol{EU^I}(\boldsymbol{Z}(\boldsymbol{y}))$. This expected-utility-maximization problem will have a solution $\boldsymbol{y}^* \in [0, 1]$ that will depend not only on the return structures of the financial assets X and Y, but also on the "shape" of the investor's Bernoulli utility function $\boldsymbol{u^I}(.)$. For example, if $\boldsymbol{u^I}(z)$ is "increasing and convex in z" then we will have $\boldsymbol{y}^* = 1$, i.e., the investor will invest all her funds in the stock $\boldsymbol{Y}$ (precisely due to the fact that $\boldsymbol{Y}$ has a higher expected return than $\boldsymbol{X}$**)**. If, on the other hand, $\boldsymbol{u^I}(z)$ is "increasing and strictly concave in z" then the investor might *optimally diversify her portfolio* and invest partly in the stock and partly in the bond. Specifically, if $\boldsymbol{u^I}(z) = \log z$, then the investor's optimal $\boldsymbol{y}^*$ will be "0.5", i.e., with a logarithmic Bernoulli utility function, the investor will divide her total investment funds equally between $\boldsymbol{X}$ and $\boldsymbol{Y}$.[2]

Rationality of Expected-Utility-Maximizing Behaviour

We now address the following theoretical issue (the [Step 2] mentioned in the first page): Let us suppose that a rational individual has "well-defined preferences" over all possible random prospects that she might encounter (just as she has well-defined preferences over all deterministic consumption bundles that are available in the market). Can we then assert that if the individual's preferences over random prospects satisfy a set of "behavioural axioms", then rationality will *require* the individual to make choices as an "expected utility maximizer"?

The ***Expected Utility Theory*** of Von Neumann and Morgenstern answers the above question in the affirmative, and identifies the set of behavioural axioms on individual preferences that lead to expected-utility-maximization.

The Expected Utility Theorem :

Consider an individual who has preferences over the set of random prospects $\{\boldsymbol{Z}^1, \boldsymbol{Z}^2, \boldsymbol{Z}^3, ...\}$. Suppose that the individual's preferences satisfy the ***completeness*** axiom, the ***transitivity*** axiom, the ***continuity*** axiom, the ***monotonicity*** axiom, and the ***independence*** axiom.[3]

Then the individual is an "expected utility maximizer" in the following sense: For the individual, there exists a "continuous and increasing Bernoulli utility function" $\boldsymbol{u}(.)$ defined over the set of all deterministic outcomes $\{z_1, \dots, z_i, \dots, z_N\}$. Further, once we define the individual's "expected utility function" over the set of random prospects $\{ \boldsymbol{Z}^1, \boldsymbol{Z}^2, \boldsymbol{Z}^3, ..., \}$:

$\boldsymbol{EU}(\boldsymbol{Z}^j) = p_1{}^j.\boldsymbol{u}(z_1) + p_2{}^j.\boldsymbol{u}(z_2) + \dots + p_N{}^j.\boldsymbol{u}(z_N)$ for every random prospect $\boldsymbol{Z}^j$,

then the following statement regarding the individual's choice under uncertainty is true: she prefers to choose a prospect $\boldsymbol{Z}^j$ over another prospect $\boldsymbol{Z}^k$ if and only if $\boldsymbol{EU}(\boldsymbol{Z}^j) \geq \boldsymbol{EU}(\boldsymbol{Z}^k)$.

[2] Note that for the logarithmic Bernoulli utility function, the "slope" of the expected utility function (with respect to the choice of $\boldsymbol{y}$) will be: $(1/2)[\{(0.11)/(1.1 + 0.11\boldsymbol{y})\} - \{(0.1)/(1.1 - 0.1\boldsymbol{y})\}]$. Setting this slope value equal to zero and solving for $\boldsymbol{y}$ gives $\boldsymbol{y}^* = 0.5$.

[3] The first four axioms are exactly as defined in the case of deterministic choices; the "independence axiom" will be discussed in details below.

Furthermore, for any expected-utility maximizer $\boldsymbol{I}$ with Bernoulli utility function $\boldsymbol{u^I}(.)$, any "increasing linear transformation" $\boldsymbol{v^I}(z) = \boldsymbol{a} + \boldsymbol{b}.\boldsymbol{u^I}(z)$ with $\boldsymbol{b} > 0$ will also represent his preferences and satisfy the expected utility property, but an increasing *non-linear transformation* [e.g. $\boldsymbol{v^I}(z) = \log.\boldsymbol{u^I}(z)$] ***will not*** represent his preferences.[4]

The Independence Axiom and the Allais Paradox

The important restriction on an individual's preferences over random prospects that makes her an "expected utility maximizer" is the ***independence axiom***. To understand this axiom, consider the following two raffles that have incredibly high odds of winning:

Raffle $\boldsymbol{R1} \equiv \{1 \text{ million, zero}; 99\%, 1\%\}$ and Raffle $\boldsymbol{R2} \equiv \{3 \text{ million, zero}; 80\%, 20\%\}$.

Next, consider the following "compound raffles" that have much more modest chances of winning a large amount of money.

Raffle ***R*3** is constructed as follows: in this raffle, zero payment will be chosen with 95% probability, while the raffle ***R*1** will be chosen with 5% probability. Recognize that this compound Raffle ***R*3** can be represented in one of two equivalent ways:

$$\text{Raffle } \boldsymbol{R3} = \{\boldsymbol{R1}, \underline{\mathbf{0}}; 5\%, 95\%\} \equiv \{1 \text{ million, zero}; 4.95\%, 95.05\%\}$$

Raffle ***R*4** is constructed as follows: In this raffle, zero payment will be chosen with 95% probability, while the raffle ***R*2** will be chosen with 5% probability. Thus the compound Raffle ***R*4** can be represented in one of two equivalent ways:

$$\text{Raffle } \boldsymbol{R4} = \{\boldsymbol{R2}, \underline{\mathbf{0}}; 5\%, 95\%\} \equiv \{3 \text{ million, zero}; 4\%, 96\%\}$$

Consider an expected utility maximizer Alice with Bernoulli utility function $\boldsymbol{u}(.)$ [with $\boldsymbol{u}(0)= 0$]. Note that Alice's expected utility from Raffle ***R*3** is 5% of her expected utility from Raffle ***R*1**:

$$\boldsymbol{EU}(\boldsymbol{R3}) = 0.0495\times\boldsymbol{u}(1000000) \quad \text{while} \quad \boldsymbol{EU}(\boldsymbol{R1}) = 0.99\times\boldsymbol{u}(1000000),$$

and that Alice's expected utility from Raffle ***R*4** is 5% of her expected utility from Raffle ***R*2**:

$$\boldsymbol{EU}(\boldsymbol{R4}) = 0.04\times\boldsymbol{u}(3000000) \quad \text{while} \quad \boldsymbol{EU}(\boldsymbol{R2}) = 0.8\times\boldsymbol{u}(3000000).$$

Thus, if Alice chooses ***R*1** over ***R*2**, then she must choose ***R*3** over ***R*4**. This "consistency in choice" is precisely what the *independence axiom* requires.

The Independence Axiom: An individual's preferences over random prospects satisfies the independence axiom if and only if the following is true: If the individual prefers a prospect $\boldsymbol{X}$ to another prospect $\boldsymbol{Y}$, then for <u>any distinct prospect</u> $\boldsymbol{Z}$ and any probability $\boldsymbol{p} \in (0, 1)$, she must prefer the compound prospect $\{\boldsymbol{X}, \boldsymbol{Z}; \boldsymbol{p}, 1-\boldsymbol{p}\}$ to the compound prospect $\{\boldsymbol{Y}, \boldsymbol{Z}; \boldsymbol{p}, 1-\boldsymbol{p}\}$.

In many laboratory experiments where undergraduate students are told to pick first between Raffles ***R*1** and ***R*2**, and then between Raffles ***R*3** and ***R*4**, a majority of the students choose ***R*1** over ***R*2** but ***R*4** over ***R*3**. French economist (and Nobel laureate) Maurice Allais was the first to document this "experimental refutation" of the Independence Axiom, and his finding has been documented for posterity as the "Allais paradox".

[4] Note that due to the "increasing linear transformation" result, the Bernoulli utility function of every *expected utility maximize* can be "normalized" to set $u(0) = 0$.

Attitudes towards Risk and Risk-Averse Behaviour

We now discuss how the shape of an expected-utility-maximizer's Bernoulli utility function determines her "attitude towards risk-taking". Reconsider the simple bet that we studied earlier: $\boldsymbol{B} = \{50, 150;\ 0.4, 0.6\}$. Note that if a person Bob – who is an expected utility maximize – has the Bernoulli utility function $\boldsymbol{u}(.)$, then Bob's expected utility from $\boldsymbol{B}$ will be: $\boldsymbol{EU}(\boldsymbol{B}) = 0.4\boldsymbol{u}\,(50) + 0.6\boldsymbol{u}(150)$. Next, define the "expected value" of the bet $\boldsymbol{B}$ as $\boldsymbol{EB} = 0.4\times50 + 0.6\times150 = 110$.[5] Note that Bob's Bernoulli utility from the certain amount 110 will be $\boldsymbol{u}(\boldsymbol{EB}) = \boldsymbol{u}(110)$. Then, consider the question: Does Bob prefer the bet $\boldsymbol{B}$ to its expected value $\boldsymbol{EB}$ or not? That is, is "$\boldsymbol{EU}(\boldsymbol{B}) = 0.4\boldsymbol{u}\,(50) + 0.6\boldsymbol{u}(150)$" greater than or less than "$\boldsymbol{u}(\boldsymbol{EB}) = \boldsymbol{u}(110)$". A little reflection will convince the reader of the following result:

If Bob's Bernoulli utility function is a "strictly concave function" then: $\boldsymbol{u}(\boldsymbol{EB}) > \boldsymbol{EU}(\boldsymbol{B})$; if his Bernoulli utility function is a "linear function" then: $\boldsymbol{u}(\boldsymbol{EB}) = \boldsymbol{EU}(\boldsymbol{B})$; and if his Bernoulli utility function is a "strictly convex function" then: $\boldsymbol{u}(\boldsymbol{EB}) < \boldsymbol{EU}(\boldsymbol{B})$.

We thus find the link between the curvature of an individual's Bernoulli utility function and her "attitude towards risk":

- If an expected utility maximize has a ***strictly concave*** Bernoulli utility function, then she is ***strictly risk averse*** since for every random prospect, she strictly prefers to get the expected value of the random prospect for sure than to get the random prospect itself.
- If an expected utility maximize has a ***linear*** Bernoulli utility function, then she is ***risk neutral*** since for every random prospect, she is indifferent between getting the expected value of the random prospect for sure and getting the random prospect itself.[6]
- If an expected utility maximize has a ***strictly convex*** Bernoulli utility function, then she is ***strictly risk loving*** since for every random prospect, she strictly prefers to get the random prospect than to get the expected value of the random prospect for sure.

Numerous experimental studies have demonstrated the following conclusions about individual behavior: (*a*) Almost all individuals are risk averse [See the anecdote about the *St. Petersburg Gamble* presented below.] (*b*) Different individuals differ in their aversion to risk (a "more risk averse" person has a "more concave Bernoulli utility function" than a less risk averse person.) (*c*) In general, a richer person is less risk averse than a poorer person (the extent of concavity of a person's Bernoulli utility function decreases as her wealth increases).

The St. Petersburg Paradox

Consider the following "St. Petersburg gamble": I will keep on tossing an "unbiased coin" repeatedly until a tail first appears, ending the game. The pot starts at 1 dollar and is doubled every time a head appears. You win whatever is in the pot after the game ends. Thus you win

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[5] Treating $\boldsymbol{B}$ to be a binary random variable, this is just the specification of the "mean" of $\boldsymbol{B}$.

[6] Note that if an individual is *risk-neutral*, then by taking a linear transform of her Bernoulli utility function, we can represent her preferences over sure money amounts by the identity function: $\boldsymbol{u}(\boldsymbol{z}) = \boldsymbol{z}$.

1 dollar if a tail appears on the first toss, 2 dollars if on the second, 4 dollars if on the third, 8 dollars if on the fourth, and so on. But there is a catch: You have to pay me some amount of money up-front in order to take part in this gamble. What is the maximum amount of money that you would be willing to pay up-front to play the gamble?

Note that the "expected value" of the gamble is: $[(1/2)\times 1] + [(1/4)\times 2] + [(1/8)\times 4] + \ldots$ This sum diverges to infinity, and so the *expected win* for a participant in this gamble is an infinite amount of money. So, *if* a person is risk-neutral (or risk-loving), then (s)he should be willingly to pay an unboundedly large sum of money to play this lottery. But in reality, nobody is; "few of us would pay even $25 to enter such a game".

This is the famous 'St. Petersburg paradox' that convinced Daniel Bernoulli to claim that almost everybody has a strictly concave and bounded utility function $u(.)$ over 'sure money amounts'. He wrote: "There is no doubt that a gain of one thousand ducats is more significant to the pauper than to a rich man though both gain the same amount."

We now describe a set of "behavior under uncertainty" that is qualitatively true for ***all risk-averse expected utility maximizers***:

1. When a risk-averse individual has the option of investing in a risky asset and a risk-less asset, she will invest in a risky asset if and only if the asset's expected return contains a ***risk premium*** over and above the return of a risk-less asset.
2. When a risk-averse individual has the option of investing in a set of risky assets that are independently distributed and offer an identical risk premium, then she will ***strictly prefer to fully diversify her portfolio*** by holding identical amounts of each risky asset.
3. When facing a risky prospect, a risk-averse individual will be willing to ***pay an insurance premium*** to a counterparty for the latter to *take away* the risk from the prospect.
4. Given an ***actuarially fair insurance premium,*** a risk-averse person will *fully insure herself.*

The first two points noted above can be understood by reviewing our earlier **Example 3**. In that example, investor *I* is an expected utility maximizer with Bernoulli utility function ***u***(.). Given the sure rate of return of 10% per annum on bond ***X***, the following results can easily be established for investor ***I***: if her Bernoulli utility function is "strictly concave" then she will invest "some" of her funds in the (uncertain) stock ***Y*** if and only if the "expected annual return" of stock ***Y*** is strictly greater than 10%. This is a general result: *a risk-averse investor needs to be given a risk premium to hold a risky asset*. Further, if there are multiple stocks – say ***Y*** and ***Z*** – that are not perfectly positively correlated, and if each stock has "expected annual return" greater than 10%, then the ***risk-averse investor will necessarily diversify*** – she will invest some funds in ***Y*** and some funds in ***Z***.

The above results describe how a risk-averse individual will behave when faced with an option of having to "accept some risk". The following discussion – which elaborates on the last two points stated above – pertains to "risk avoidance" by a risk-averse individual.

First, let us define the notion of "maximal insurance premium" as follows. Let us reconsider Bob's situation when he is contemplating the risky prospect $\boldsymbol{B} = \{50, 150;\ 0.4, 0.6\}$. Assume that Bob is risk-averse so that for him: $\boldsymbol{u}(\boldsymbol{EB}) = \boldsymbol{u}(110) > \boldsymbol{EU}(\boldsymbol{B}) = 0.4\boldsymbol{u}(50) + 0.6\boldsymbol{u}(150)$. This means that there exists a "sure money amount" smaller than 110 for which Bob will be indifferent between getting that amount of money for sure and getting the prospect B. That money amount –whatever its value is – is considered to be Bob's "certainty equivalent" of the bet $\boldsymbol{B}$. Formally, for a risk-averse individual with Bernoulli utility function $\boldsymbol{u}(.)$: the ***certainty equivalent*** $\boldsymbol{C}(\boldsymbol{Z})$ of a monetary prospect $\boldsymbol{Z}$ is the "sure money amount" associated with prospect $\boldsymbol{Z}$ such that: $\boldsymbol{u}(\boldsymbol{C}(\boldsymbol{Z})) = \boldsymbol{EU}(\boldsymbol{Z})$. Then, given the certainty equivalent $\boldsymbol{C}(\boldsymbol{Z})$ of a monetary prospect $\boldsymbol{Z}$ for the risk-averse individual, we define his ***maximal insurance premium*** $\rho(\boldsymbol{Z})$ associated with $\boldsymbol{Z}$ to be: $\rho(\boldsymbol{Z}) = \boldsymbol{EZ} - \boldsymbol{C}(\boldsymbol{Z})$. Recognize that for the risk-averse individual, $\rho(\boldsymbol{Z})$ is the maximum amount of money that he is will to pay to a counterparty for the latter to "take away" all the risk inherent in $\boldsymbol{Z}$ and to replace $\boldsymbol{Z}$ by its expected value $\boldsymbol{EZ}$. [To see this, note that: $\boldsymbol{u}(\boldsymbol{EZ} - \rho(\boldsymbol{Z})) = \boldsymbol{u}(\boldsymbol{C}(\boldsymbol{Z})) = \boldsymbol{EU}(\boldsymbol{Z})$.]

Extending the above idea – that risk-averse individuals have positive demand for insurance against risk – the following example discusses the notions of "fair insurance premia" and of "full insurance".

Example 4 Mr. Rich is an expected-utility-maximizer who is strictly risk-averse with Bernoulli utility function $\boldsymbol{u}(z)$. His home (with contents) is valued at Rs.4 million. There is a **15**% probability that his home will be burgled. If it is burgled, the home will lose Rs.1 million in value (as some of the paintings in the home will be stolen).

A risk-neutral insurance company offers Mr. Rich ***theft insurance*** under the following terms: premium of P per rupee of indemnity. This means that if Mr. Rich buys X insurance contracts for which he will have to pay Rs. XP of total premium, the insurance company will pay Mr. Rich a total indemnity of Rs. X if and only if his house is burgled.

Recognize that if Mr. Rich buys no theft insurance, he will be facing the random prospect: $\boldsymbol{Z}(0) = \{4000000, 3000000;\ 0.85,\ 0.15\}$. Alternatively, if he does buy $\boldsymbol{X}$ units of insurance, he in effect gets the random prospect: $\boldsymbol{Z}(\boldsymbol{X}) = \{4000000 - \boldsymbol{X}P,\ 3000000 - \boldsymbol{X}P + \boldsymbol{X};\ 0.85,\ 0.15\}$.

[This specification is of course not quite correct, as the premium is paid up front at a fixed date, and the indemnity is received if and only if burglary occurs at any point in time over the next 365 days. In writing the random prospect in the above form, we are essentially assuming that the within-year interest rate is zero (a relatively minor simplification).]

Thus, Mr. Rich will choose $\boldsymbol{X}$ to maximize his expected utility from the prospect $\boldsymbol{Z}(\boldsymbol{X})$ which equals: $\boldsymbol{EU}(\boldsymbol{Z}(\boldsymbol{X})) = 0.85\times\boldsymbol{u}(4000000 - \boldsymbol{X}P) + 0.15\times\boldsymbol{u}(3000000 - \boldsymbol{X}P + \boldsymbol{X})$.

Note that the "slope of the expected utility function with respect to $\boldsymbol{X}$" is:

$$0.85\times\boldsymbol{u}'(4000000 - \boldsymbol{X}P)(-P) + 0.15\times\boldsymbol{u}'(3000000 - \boldsymbol{X}P + \boldsymbol{X})(1 - P)$$

Recognize from the above expression that if the per-unit premium P equals 0.15, Mr. Rich will set $\boldsymbol{X} = 1$ million because his "expected utility hill" will reach its peak at $\boldsymbol{X} = 1$ (i.e., the slope of his expected utility function will be zero at $\boldsymbol{X} = 1$ when P equals 0.15).

This result indicates the following features of optimal insurance provided by a risk-neutral insurer to a risk-averse individual. First, note that if $P = 0.15$, then the risk-neutral insurance company will make "zero" expected profits per unit of insurance (expected profits being $\{PX - 0.15X\}$).

A per-unit premium that generates zero expected profits for a risk-neutral insurer is termed an "***actuarially fair insurance premium***". Such a premium would obtain if the insurance market was perfectly competitive with all insurers being risk-neutral.

Second, when Mr. Rich buys 1 million units of insurance, he manages to "completely insure himself" as his final wealth (Rs. 3.85 million) becomes independent of whether his home is burgled or not. This is a robust result: *A risk-averse individual facing actuarially fair insurance premium will fully insure himself.* [For a higher insurance premium, a risk-averse person will *partially insure himself*, and to determine the extent of insurance that he will buy, we will need to know the exact functional form of his Bernoulli utility function.]

Dissecting the vN-M Expected Utility Theory

- The ***Expected Utility Theory*** presents an analytical framework for studying decision-making under uncertainty
- The theory can be understood as the following two-step procedure:

[Step 1] Posit a specific kind of '***expected-utility-maximizing behaviour***' in scenarios of uncertainty [*a decision-making algorithm*]

[Step 2] Identify preference structures for which 'expected-utility-maximizing behaviour' will constitute ***rational decision-making***

Shape of an individual's Bernoulli utility function

- In most of our examples (but not all), the component outcomes will be money amounts,
  and if we assume that an individual's preferences over deterministic money amounts are also ***monotonic***,
  then her Bernoulli utility function on money amounts will be increasing

Transformations of Bernoulli Utility Functions

- For any expected-utility maximizer I with Bernoulli utility function $u_I(.)$ any 'increasing linear transformation' $v_I(z) = a + b.u_I(z)$ with $b > 0$ **will also represent his preferences** and satisfy the expected utility property
  but a non-linear transformation [e.g. $v(z) = \log.u_I(z)$] ***will not*** represent his preferences
- As a result, we can use two "normalizations":
  1. We can normalize $u(0) = 0$ for all expected utility maximizers.
  2. For an expected utility maximizer with linear Bernoulli utility function, we can "replace" $v(z) = m + n.z$ by $u(z) = z = -(m/n) + (1/n).[m + n.z]$

The "Expected Utility Maximization" Algorithm

1. Represent every 'uncertain outcome' as a random prospect/lottery of the form $Z = \{z_1, \ldots, z_i, \ldots, z_N;\ p_1, \ldots, p_i, \ldots, p_N\}$
2. View individuals as choosing among a set of such prospects $\{Z^j$ for $j = 1, 2, 3, \ldots\}$
3. Each individual i is assumed to have a Bernoulli utility function $u_i(.)$ over the set of all deterministic outcomes $\{z_1, \ldots, z_i, \ldots, z_N\}$
4. For each individual i, construct **expected utility** of every prospect as follows: $EU_i(Z^j) = p_1^j.\ u_i(z_1) + p_2^j.\ u_i(z_2) + \ldots + p_N^j.\ u_i(z_N)$
5. Posit that individual i will choose a prospect Z^J if and only if it has the highest expected utility for I

- An individual is an "Expected Utility maximizer" if she makes "choices under uncertainty" in the above manner given her Bernoulli utility function $u(.)$

1. If $u(.)$ is the **person**'s Bernoulli utility function, then $v(.) = a + b.u(.)$, for some $b > 0$ **must also be her** Bernoulli utility function
2. If prospect probabilities are ***objective***, **two different** expected utility maximizers ***might*** (respectively, ***will***) choose differently **if** (respectively, **only if**) their Bernoulli utility functions are distinct

The Expected Utility Theorem [von Neumann / Morgenstern]

It will be **rational** for an individual to be an expected-utility maximizer **iff** her preferences over prospects satisfy ***completeness*** and ***transitivity*** axioms, and the ***continuity axiom*** and the ***independence axiom***.

- Suppose **I** am an "expected utility maximizer" choosing between two "compound prospects" A & B constructed from 3 simple prospects X, Y, Z :
  $A = \{X, Z; \pi, 1-\pi\}$, and $B = \{Y, Z; \pi, 1-\pi\}$
- **I** will prefer A to B if and only if:
  I prefer prospect X over prospect Y $\Leftrightarrow$ $EU_I(X) > EU_I(Y)$
- Thus, if I am an "expected utility maximizer", then my choice between A and B is ***required to be independent*** of my like/dislike of the "common" component prospect Z, *given that it is going to be realized with the "same probability" in both the compound prospects*

The Independence Axiom regarding preferences over random prospects

Given any three prospects X, Y, and Z, and any probability p, an individual's preference ordering over the two (compound) prospects $\{X, Z; p, 1-p\}$ and $\{Y, Z; p, 1-p\}$ and her preference ordering over X and Y must 'coincide'

The Expected Utility Theorem [von Neumann / Morgenstern]

It will be **rational** for an individual to be an expected-utility maximizer **if and only if** her preferences over prospects satisfy ***completeness*** and ***transitivity*** properties, and the ***continuity axiom*** and ***independence axiom***.

The Continuity Axiom If I prefer prospect X to prospect Y, and Y to prospect Z, then there must exist a unique probability value $\pi \in (0, 1)$ for which I am indifferent between Y and the **compound prospect** $\{X, Z; \pi, 1-\pi\}$

- Let an **individual**'s universal choice set **CS** be the set of all lotteries over the set of deterministic outcomes $(z_1, \ldots, z_i, \ldots, z_N)$, where z_1 is **her** worst choice and z_N is **her** best choice.
- Then for every intermediate deterministic outcome z_i there will exist a unique probability value $\pi_i \in (0, 1)$ such that for **her**: $z_i \sim \{z_N, z_1; \pi_i, 1-\pi_i\}$
- Then, without loss of generality, set $u(z_1) = 0$ and $u(z_N) = 100$
- And then for every other z_i, set $u(z_i) = \pi_i.100$
- *Voila*! We've identified one Bernoulli utility function for the individual

Implication: If $u(.)$ is a **person**'s Bernoulli utility function, then a non-linear transform of $u(.)$ **cannot also be that person's** Bernoulli utility function

Problem with the Independence Axiom

- Consider two "heavily biased" raffles: $R1$ = {1 million, zero; 99%, 1%} and $R2$ = {3 million, zero; 80%, 20%}
- Now make the raffles a bit more realistic by "compounding them":
  $R3 = \{R1,\ \underline{0};\ 5\%,\ 95\%\} \equiv$ {1 million, zero; 4.95%, 95.05%}
  $R4 = \{R2,\ \underline{0};\ 5\%,\ 95\%\} \equiv$ {3 million, zero; 4%, 96%}
- The ***independence axiom*** requires that an individual's preference ordering over the two (compound) prospects $R3 = \{R1, \underline{0}; 5\%, 95\%\}$ and $R4 = \{R2, \underline{0}; 5\%, 95\%\}$ and her preference ordering over $R1$ and $R2$ must 'coincide'
- ***Allais' paradox***: Many people choose $R1$ over $R2$ BUT $R4$ over $R3$
- Note that there are great advantages in working with Expected Utility Functions defined over probabilistic outcomes, where **overall utility is linear in probabilities**

Another Example of "Expected Utility Maximizing Behaviour"

Projects ***X*** and ***Y*** with uncertain returns
- they require no up-front monetary investment (only identical effort)
- ***X*** will give returns of **1** million *or* **2** million with equal probability
  Y will give returns of **1.32** million *or* **1.6** million with equal probability

• Choice between $X = \{1, 2; 0.5, 0.5\}$ and $Y = \{1.32, 1.6; 0.5, 0.5\}$ for 4 entrepreneurs ***A***, ***B***, ***C***, ***D*** with different Bernoulli utility functions:
$u^A(z) = z^2$; $u^B(z) = z$; $u^C(z) = (z)^{1/2}$; $u^D(z) = (z)^{1/3}$

A compares $0.5(1)^2+0.5(2)^2$ to $0.5(1.32)^2+0.5(1.6)^2$
B compares $0.5(1)+0.5(2)$ to $0.5(1.32)+0.5(1.6)$
C compares $0.5(1)^{1/2}+0.5(2)^{1/2}$ to $0.5(1.32)^{1/2}+0.5(1.6)^{1/2}$
D compares $0.5(1)^{1/3}+0.5(2)^{1/3}$ to $0.5(1.32)^{1/3}+0.5(1.6)^{1/3}$

• ***A*** and ***B*** prefer ***X*** to ***Y***, ***C*** indifferent between ***X*** & ***Y***, ***D*** prefers ***Y*** to ***X***
⇒ Curvature of a person's Bernoulli utility function determines her attitude towards risk, and thus her choice over random prospects

Some consistent behaviours of risk-averse individuals

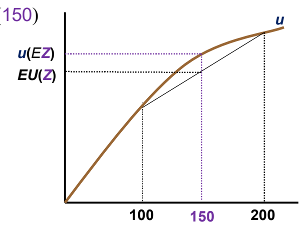
1. A risk-averse individual will invest in a risky asset if and only if the asset's expected return contains a ***risk premium*** over and above the return of a risk-less asset.
2. If there are multiple risky assets that are not perfectly correlated, and if each such asset offers a similar risk premium, then a risk-averse individual will ***strictly prefer to diversify her portfolio*** of investments over these risky assets.
3. When facing a risky prospect, a risk-averse individual will be willing to ***pay an insurance premium*** to a counterparty for the latter to take away (some of) the risk from the prospect.

Curvature of Bernoulli Utility Function & Risk Preference

• Given the prospect ***Z***, the expected utility of ***Z*** is $EU(Z)$ and the Bernoulli utility of the expected value of ***Z*** is $u(EZ) = u(150)$

• Note that whether $u(EZ)$ is bigger, equal, or smaller than $EU(Z)$ depends on whether the Bernoulli utility function $u(.)$ is concave, linear, or convex

Z = {100, 200; 0.5, 0.5},



• when $u(EZ) > EU(Z)$ 'sure mean' preferred to prospect [***risk aversion***]
• when $u(EZ) < EU(Z)$ prospect preferred to 'sure mean' [***risk loving***]
• when $u(EZ) = EU(Z)$ there is indifference [***risk neutrality***]

Defining Attitudes toward Risk

• A person is ***globally risk averse*** if it is always true that she prefers getting the 'mean' of a random prospect for sure rather than getting the prospect
⇔ A person is *globally risk averse* if and only if her ***u*(.)** is *strictly concave*

• A person is ***globally risk neutral*** if it is always true that she is indifferent between a getting a random prospect and getting is 'mean' for sure
⇔ A person is *globally risk neutral* if and only if her ***u*(.)** is *linear*
[a linear ***u*(.)** can be represented by identity function $u(z)=z$ for all z]

• A person is ***globally risk loving*** if it is always true that she prefers getting a random prospect rather than getting its 'mean' for sure
⇔ A person is *globally risk loving* if and only if her ***u*(.)** is *strictly convex*

• Most individuals are risk averse [read about the *St. Petersburg paradox*]
- individuals differ in their aversion to risk (a 'more risk averse' person has a 'more concave ***u*** function' than a less risk averse person)
- in general, a richer person is less risk averse than a poorer person (the extent of concavity of ***u*** function decreases as wealth increases)

Some consistent behaviours of risk-averse individuals

1. A risk-averse individual will invest in a risky asset if and only if the asset's expected return contains a ***risk premium*** over and above the return of a risk-less asset.
2. If there are multiple risky assets that are not perfectly correlated, and if each such asset offers a similar risk premium, then a risk-averse individual will ***strictly prefer to diversify her portfolio*** of investments over these risky assets.
3. When facing a risky prospect, a risk-averse individual will be willing to ***pay an insurance premium*** to a counterparty for the latter to take away (some of) the risk from the prospect.

- The most general decision theory problems are of

Finite-horizon / Infinite-horizon Decision-making under uncertainty

Examples:

1. *Sequentially managing a finite-horizon stochastic investment project*

I am a risk-averse expected-utility-maximizer with $u(z) = \log z$. I have 1 million rupees to invest in two investment projects A and B over two years. Project A is safe giving a 10% return per year, while project B is risky: per year, it gives a high return of 30% and a low return of 0% with equal probability. How much should I invest in each of the two projects in each of the two years, given that I have decided to re-invest my total first year project returns in the same two projects in the second year?

2. *Sequentially managing an infinite-horizon stochastic investment project*

Optimization strategies involve:

(i) Decision making by "Backward Induction over a Value Function"

(ii) Engaging in "Value Function Optimization" by following ***Bellman's principle of optimality***

A Portfolio Choice Example

- You have Rs. 1 million to invest in *two* financial securities for a year
  Asset X is a bond with a certain interest of 10% p.a.
  Asset Y is a stock that will go up by 21% over the year
  or stay unchanged with equal probability [*expected return* = 10.5%]

Q: Given your u, what is your expected utility if invest y (part of 1 mill) in Y?

- You get a random prospect: $\{1.1[1-y] + 1.21y,\ 1.1[1-y] + y;\ 0.5,\ 0.5\}$
  So, your *expected utility* is: $0.5u(1.1[1-y] + 1.21y) + 0.5u(1.1[1-y] + y)$
- Your optimal choice is that y^* that maximizes your expected utility
- If $u(z) = \log z$, then $y^* = 0.5$ million [50-50 split]
- If I knew you were risk averse, but I did not know your u function, I would still be able to predict the following:

(i) If X is a 'safe asset' with return r_X% p.a., and Y is 'risky asset' with 'expected return' r_Y% p.a., you will hold "some" Y if & only if $r_Y > r_X$

(ii) If X safe with r_X, and Y & Z are risky and not perfectly correlated, with expected $r_Y \approx r_Z > r_X$, you will "diversify" between Y and Z

Decision Scenario I: *Siblings' Day Out* (pp. 5 in Notes)

On a weekend, a parent takes siblings Alice & Bob to the city mall

Problem A: The parent has to decide whether or not to carry umbrellas.

Problem B: The parent has to decide which public transport to take, in order to minimize travel time and maximize chance of getting seats.

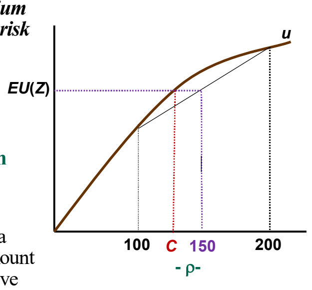
Problem C: In the mall, the parent aims to find a way to "encourage" the siblings to consume healthy food (like soup) rather than unhealthy food (like French Fries) – without engaging in *command and control*

Certainty Equivalents and Insurance Premia

- A risk averse person is always willing to pay a ***premium to an insurer*** who will 'reduce the risk / provide insurance'
- There's a ***maximum insurance premium*** that he will pay for the ***elimination of risk***
- For a risk averse u, the **certainty equivalent** $C(Z)$ is the sure amount associated with prospect Z such that: $u(C(Z)) = EU(Z)$

  and the **maximal insurance premium** $\rho(Z)$ associated with Z is: $\rho(Z) = EZ - C(Z)$
- The ***maximal insurance premium*** of a prospect Z for a person is the max amount of money that she would pay to receive the 'mean' of Z rather than Z : $u(EZ - \rho(Z)) = EU(Z)$

Z = {100, 200; 0.5, 0.5},



Decision-making under Uncertainty: Schematic representation of Problem A

Choice options	Random events	Random outcome/Prospect
C [*carry umbrellas*]	***R*** [*rain*] \w prob π ***D*** [*dry*] \w prob $(1-\pi)$	$\{CR, CD; \pi, 1-\pi\}$
N [*not carry*]	***R*** [*rain*] \w prob π ***D*** [*dry*] \w prob $(1-\pi)$	$\{NR, ND; \pi, 1-\pi\}$

The von Neumann-Morgenstern algorithm of expected utility maximization

1. Determine ***the*** probabilities of the random event of rain: $\{\pi, 1-\pi\}$
2. Determine ***parent's*** Bernoulli utilities of the 'component deterministic outcomes' $\{u(NR),\ u(CR),\ u(CD),\ u(ND)\}$
3. Construct ***parent's*** 'expected utilities' of prospects associated with decision options $EU(C) = \pi.u(CR) + (1-\pi)u(CD)$; $EU(N) = \pi.u(NR) + (1-\pi)u(ND)$
4. Compare the expected utilities $EU(C)$ and $EU(N)$ and choose the ***maximum***

Decision Scenario I: *Siblings' Day Out*

On a weekend, a parent takes siblings Alice & Bob to the city mall

Problem A: The parent has to decide whether or not to carry umbrellas.

Problem B: The parent has to decide on what public transport to take, in order to minimize travel time and maximize chance of getting seats.

Problem C: In the mall, the parent aims to find a way to "encourage" the siblings to consume healthy food (like soup) rather than unhealthy food (like French Fries) – without engaging in *command and control*

Problem B is a problem in parametric equilibrium theory, where each decision-maker makes her best decisions given the values of a "vector of aggregate parameters" and the aggregation of individual decisions precisely generate those values of the "vector of aggregate parameters"

Problem C: a game designed by the parent to be played by siblings

- In the mall, soup sells for Rs. 50 per bowl and Fries sell for Rs. 10 per ounce; soup orders are indivisible; Fries can be ordered by weight.
- Consider the parent announcing the following scheme:

  Each sibling is given a budget of Rs. 150. Each has to go to a distinct Soup counter and simultaneously order the no. of bowls of Soup that (s)he wants, with a limit of three.

  If $(S_A + S_B)$ bowls of soup are ordered by the two, each sibling has to pay $0.5[50(S_A + S_B)]$ out of the budget of Rs. 150, and can spend the remaining money on as many ounces of French Fries as (s)he wishes.

  If sibling ***i*** consumes S_i bowls of Soup & F_i ounces of Fries (s)he has ***Bernoulli utility*** $= [5S_i + 2\sqrt{(10F_i)}]$ (and this is commonly known)

- The parent's ***mechanism*** creates a ***game*** between the siblings, where each child's soup order determines the amount of Fries that the other can have
- If instead, parent had given each child Rs.150 and the child was free to choose no. of bowls of soup and oz. of fries, each siblings would maximize $\{5S_i + 2\sqrt{(10F_i)}\}$ subject to constraints: $S_i \in \{0, 1, 2, 3\}$ & $50S_i + 10F_i \leq 150$, and would consequently have ordered one bowl of soup each.

The "payoff-matrix" of the Sibling Rivalry Game

		Bob			
		$S_B = 0$	$S_B = 1$	$S_B = 2$	$S_B = 3$
Alice	$S_A = 0$	24.5, 24.5	22.36, 27.36	20, 30	17.33, 32.33
	$S_A = 1$	27.36, 22.36	25, 25	22.33, 27.33	19.14, 29.14
	$S_A = 2$	30, 20	27.33, 22.33	24.14, 24.14	20, 25
	$S_A = 3$	32.33, 17.33	29.14, 19.14	25, 20	15, 15

The payoff matrix explicitly describes:
the players, their deterministic strategy sets, and their payoffs (Bernoulli utilities) from alternative strategy combinations,
and implicitly describes outcomes, and the following game features:
number of stages, order of moves, and
information structure which includes structure of strategic uncertainty

EC 246: Decisions & Games

PROBLEM SET 1 – ON CHOICE UNDER UNCERTAINTY

1. Alice and Bob are sitting in a bar at 8 pm, each with Rs.100. They are both expected utility maximizers; Alice's Bernoulli utility function is $u(z) = +\sqrt{z}$, and Bob's Bernoulli utility function is $u(z) = z^2$. They both agree that there is a 28% probability that it will rain within the next one hour. Consider the following bet: If it rains within the next one hour, then Alice will give Bob her Rs.100; while if it does not rain within the next one hour, then Bob will give Alice his Rs.100. Verify that each person will prefer to take this bet rather than sit with his/her Rs.100.

2. Ms. A tells Mr. B and Mr. C that she believes that India will win the next Cricket World Cup with a 70% probability. Then Mr. B offers her the bet: If India wins the Cup Mr. B will give her Rs. 1000, but if India loses she will have to pay Rs. 2000 to Mr. B. And Mr. C offers her the bet: If India wins the Cup Mr. C will give her Rs. 5000, but if India loses she will have to pay Rs. 8000 to Mr. C. Ms. A is a risk-averse expected-utility-maximizer, with Bernoulli utility function $u(z) = +\sqrt{z}$. If she is holding Rs. 10,000, to begin with, whose bet will she accept?

3. I am sitting in a bar at the beginning of the 2020 Cricket World Cup Final between England and Australia. I have Rs.1000 in my pocket, and my drinks for the evening have been paid for. The bar-tender offers me the choice of picking one of the following two bets (for free).

BET A: If England wins I will be paid Rs.600 and if Australia wins I will have to pay Rs.100.

BET B: If Australia wins I will be paid Rs.1500 and if England wins I will have to pay Rs.1000.

I am an expected utility maximizer with Bernoulli utility function $u(x) = +\sqrt{x}$. After a quick mental calculation, I pick BET A. My choice reveals that I must consider the probability of England winning to be at least $\boldsymbol{p}$. What is the value of $\boldsymbol{p}$?

4. I have to travel from point A to point B, and there are two bus routes – I and II – that make this journey. Bus route I takes 20 minutes to travel from A to B, and has 30% probability of having at least one seat empty (i.e., available for me to sit); bus route II takes 30 minutes to travel from A to B, and has 40% probability of having at least one seat empty. If I travel sitting for t minutes then my Bernoulli utility will be $(100 - 2t)$, while if I travel standing for t minutes then my Bernoulli utility will be $(70 - 2t)$. If I am an expected utility-maximizer, which bus route should I choose?

5. A farmer has 100 acres of land, and has to allocate his land between the production of two crops A and B. The technology is simple ‑ for each crop, one acre of land is used to produce one unit of output. The market price of crop B is certain, Rs. 100 per unit. But the market price of crop A is random − it will be either Rs.50 or Rs.200 with equal probability. The farmer is only interested in the revenue generated from selling the crops in the market, and has to decide on the amount of land, X acres, in which to cultivate A (with B being cultivated in the remaining land, 100-X acres). The farmer is an "expected utility maximizer" with Bernoulli utility function $u(.)$.

If the farmer is "risk neutral", i.e., $u(r) = r$ for all positive real numbers r, what is his optimal X? Alternatively, if he is "risk averse", with $u(r) = \log r$ for all positive real numbers r, can you determine what value of X will maximize his expected utility?

A game is completely specified by the following details:

(1) the ***identity*** of the players;
(2) players' ***pure strategy sets*** (sets of their deterministic strategy choices),
(3) the ***outcome function*** – mapping from strategy vectors to outcomes
(4) the players' ***payoff functions*** – mapping from outcomes to utilities

- Specification of (2), (3), and (4) also incorporate specification of the (single/multiple) ***stages*** of the game, of the ***order of moves*** in the different stages, and the ***information structure*** of the game [describing who knows what at each stage of the game]
- We classify games according to
  number of stages: ***single-stage*** vs. ***multi-stage*** games
  order of moves: ***simultaneous-move*** vs. ***sequential-move*** games
  info structure: ***complete-information*** vs. ***incomplete-information*** games

Rock-Scissors-Paper: single-stage, simultaneous-move game of complete info
Noughts-and-Crosses: multi-stage, sequential-move game of complete info
Tender Auction: single-stage, simultaneous-move game of incomplete info
Bridge: multi-stage, sequential-move game of incomplete info

Thinking about "strategic equilibria" in the 19th century

- Augustine Cournot was the first to think about this issue in 1838
- He studied a duopoly game with the following kind of payoff structure:

	Low volume	Medium volume	High volume
Low vol	1800, 1800	1500, 2000	900, 1800
Medium vol	2000, 1500	1600, 1600	800, 1200
High vol	1800, 900	1200, 800	0, 0

each cell records the profits of the two firms

- Note that the above game has the following property:
  The ***sequence of response-adjustments to 'planned rival play'*** converges to a unique strategy pair "Medium vol, Medium vol" irrespective of which 'planned rival play' we start with
- Cournot concluded that such a "point of convergence" must constitute a "strategic equilibrium of the game"

The Sibling Rivalry Game

Bob / Alice

	$S_B = 0$	$S_B = 1$	$S_B = 2$	$S_B = 3$
$S_A = 0$	24.5, 24.5	22.36, 27.36	20, 30	17.33, 32.33
$S_A = 1$	27.36, 22.36	25, 25	22.33, 27.33	19.14, 29.14
$S_A = 2$	30, 20	27.33, 22.33	24.14, 24.14	20, 25
$S_A = 3$	32.33, 17.33	29.14, 19.14	25, 20	15, 15

- It's a **2-player 1-stage *simultaneous-move complete-information game,*** and the payoff matrix presented above is its ***normal-form representation***
- Each player's ***pure strategy set*** is {0, 1, 2, 3}
- As each player can (in principle) also play a ***mixed strategy*** (randomization over any subset of pure strategies),
  so, each player's ***extended strategy set*** is the 3-dimensional unit simplex
  (i.e., all vectors $\{p_0, p_1, p_2, p_3\}$ such that each $p_i \in [0, 1]$ and they add up to 1)
- As it is a simultaneous-move game (i.e., *a game of complete but imperfect information*), there can be ***strategic uncertainty -***
  If Alice 'conjectures' that Bob is equally likely to order 2 or 3 Soup bowls, then her expected utility from ordering 3 bowls will be: {0.5(25) + 0.5(15)}

... and in the 20th century

	Low volume	Medium volume	High volume
Low vol	1800, 1800	1500, 2000	900, 1800
Medium vol	2000, 1500	1600, 1600	800, 1200
High vol	1800, 900	1200, 800	0, 0

- In the 1950s, John Nash realized (and proved) three things:

(1) the '***sequence of response-adjustments to planned rival play***' does not necessarily converge to a unique strategy vector in every game

(2) if a sequence does converge, then following property [*] holds at that point
[*] ≡ "***each payer behaves optimally given the behaviour of others***"

(3) one (or more) strategy vector(s) which has (have) the property [*]: "***each payer behaves optimally given the behaviour of others***" exist(s) in almost all games, especially when we admit the possibility of players playing mixed (i.e., randomized) strategies

… such a strategy vector seems to be an appropriate candidate for equilibrium

How will the siblings play the game?

Bob / Alice

	$S_B = 0$	$S_B = 1$	$S_B = 2$	$S_B = 3$
$S_A = 0$	24.5, 24.5	22.36, 27.36	20, 30	17.33, 32.33
$S_A = 1$	27.36, 22.36	25, 25	22.33, 27.33	19.14, 29.14
$S_A = 2$	30, 20	27.33, 22.33	24.14, 24.14	20, 25
$S_A = 3$	32.33, 17.33	29.14, 19.14	25, 20	15, 15

Alice's possible thought process:

Playing [$S_A = 0$] or [$S_A = 1$] makes no sense — ignore dominated strategies

Whether I play [$S_A = 2$] or [$S_A = 3$] depends on what I think Bob will play — best-respond to conjectured rival behaviour

If I am unsure about what Bob will do, maybe I should play [$S_A = 2$] & limit my downside risk — contemplate playing your maximin strategy

But if I think that Bob is thinking the same way, then he'll be thinking of playing [S_B=2] and in that case I should play [S_A=3] .... — put yourself in rival's shoes to determine the most sensible conjecture about rival actions

Nash Equilibrium: General Definition

In a complete-information game (a [*C-I*] game): a ***Nash equilibrium*** is a collection of '(complete contingent) strategies' – one for each player – such that each strategy is a ***best response*** to the collection of rival strategies in the stated collection of strategies.

If each '(complete contingent) strategy' in the Nash equilibrium vector is a *pure-strategy*, then the strategy vector is a ***pure-strategy Nash equilibrium***; if that is not the case then it is a ***mixed-strategy Nash equilibrium***.

Elon Kohlberg [*a venerable game-theorist at HBS*] **on Nash's theory:**

"Given a game, can we predict what rational players will actually do?
Except for when the game is unusually simple, our answer must be 'No!'
The reason is that in determining his own choice each player must take into account what the others will do. This leads to an endless ***chain of deductions*** about the other players, about other players' deductions about all players, etc.
The main idea of John Nash is to make a bold simplification and, instead of asking how the process of deductions might unfold, ask where its **rest points** might be.
The great advantage of this simplification is that it leads to a well-defined algorithm to find the set of equilibria, the set being almost always non-empty."

NON-COOPERATIVE GAMES

A DISSERTATION

Presented to the Faculty of Princeton University in Candidacy for the Degree of Doctor of Philosophy

Recommended for Acceptance by the Department of Mathematics

May, 1960

Abstract

This paper introduces the concept of a non-cooperative game and develops methods for the mathematical analysis of such games. The games considered are n-person games represented by means of pure strategies and pay-off functions defined for the combinations of pure strategies.

The distinction between cooperative and non-cooperative games is unrelated to the mathematical description by means of pure strategies and pay-off functions of a game. Rather, it depends on the possibility or impossibility of coalitions, communication, and side-payments.

The concepts of an equilibrium point, a solution, a strong solution, a sub-solution, and values are introduced by mathematical definitions. And in later sections the interpretation of these concepts in non-cooperative games is discussed.

The main mathematical result is the proof of the existence in any game of at least one equilibrium point. Other results concern the geometrical structure of the set of equilibrium points of a game with a solution, the geometry of sub-solutions, and the existence of a symmetrical equilibrium point in a symmetrical game.

As an illustration of the possibilities for application a treatment of a simple three-man poker model is included.

Table of Contents

Introduction

Von Neumann and Morgenstern have developed a very fruitful theory of two-person zero-sum games in their book Theory of Games and Economic Behavior. This book also contains a theory of n-person games of a type which we would call cooperative. This theory is based on an analysis of the interrelationships of the various coalitions which can be formed by the players of the game.

Our theory, in contradistinction, is based on the absence of coalitions in that it is assumed that each participant acts independently, without collaboration or communication with any of the others.

The notion of an equilibrium point is the basic ingredient in our theory. This notion yields a generalization of the concept of the solution of a two-person zero-sum game. It turns out that the set of equilibrium points of a two-person zero-sum game is simply the set of all pairs of opposing "good strategies."

In the immediately following sections we shall define equilibrium points and prove that a finite non-cooperative game always has at least one equilibrium point. We shall also introduce the notions of solvability and strong solvability of a non-cooperative game and prove a theorem on the geometrical structure of the set of equilibrium points of a solvable game.

As an example of the application of our theory we include a solution of a simplified three person poker game.

The motivation and interpretation of the mathematical concepts employed in the theory are reserved for discussion in a special section of this paper.

Formal Definitions and Terminology

In this section we define the basic concepts of this paper and set up standard terminology and notation. Important definitions will be preceeded by a sub-title indicating the concept defined. The non-cooperative idea will be implicit, rather than explicit, below.

Finite Game:

For us an n-person game will be a set of n players, or positions, each with an associated finite set of pure strategies; and corresponding to each player, i, a pay-off function, p_i, which maps the set of all n-tuples of pure strategies into the real numbers. When we use the term n-tuple we shall always mean a set of n items, with each item associated with a different player.

Mixed Strategy, s_i:

A mixed strategy of player i will be a collection of non-negative numbers which have unit sum and are in one to one correspondence with his pure strategies.

We write $s_i = \sum_{\alpha} c_{i\alpha} \pi_{i\alpha}$ with $\sum_{\alpha} c_{i\alpha} = 1$ and $c_{i\alpha} \geq 0$ to represent such a mixed strategy, where the $\pi_{i\alpha}$'s are the pure strategies of player i. We regard the s_i's as points in a simplex whose vertices are the $\pi_{i\alpha}$'s. This simplex may be regarded as a convex subset of a real vector space, giving us a natural process of linear combination for the mixed strategies.

We shall use the suffixes i, j, k for players and α, β, γ to indicate various pure strategies of a player. The symbols s_i, t_i and r_i, etc. will indicate mixed strategies; $\pi_{i\alpha}$ will indicate the i th player's α th pure strategy, etc.

Pay-off Function, p_i :

The pay-off function, p_i , used in the definition of a finite game above, has a unique extension to the n-tuples of mixed strategies which is linear in the mixed strategy of each player [n-linear]. This extension we shall also denote by p_i , writing $p_i(s_1, s_2, \cdots s_n)$.

We shall write $\mathscr{s}$ or $\mathscr{t}$ to denote an n-tuple of mixed strategies and if $\mathscr{s} = (s_1, \cdots s_n)$ then $p_i(\mathscr{s})$ shall mean $p_i(s_1, s_2, \cdots s_n)$. Such an n-tuple, $\mathscr{s}$, will also be regarded as a point in a vector space, which space could be obtained by multiplying together the vector spaces containing the mixed strategies. And the set of all such n-tuples forms, of course, a convex polytope, the product of the simplices representing the mixed strategies.

For convenience we introduce the substitution notation

$(\mathscr{s}; t_i)$ to stand for $(s_1, s_2, \cdots s_{i-1}, t_i, s_{i+1}, \cdots s_n)$

where $\mathscr{s} = (s_1, s_2, \cdots s_n)$. The effect of successive substitutions $((\mathscr{s}; t_i); r_j)$ we indicate by $(\mathscr{s}; t_i; r_j)$, etc.

Equilibrium Point:

An n-tuple $\mathscr{s}$ is an <u>equilibrium point</u> if and only if for every i

$$(1) \qquad p_i(\mathscr{s}) = \max_{\text{all } r_i\text{'s}} \left[p_i(\mathscr{s}; r_i) \right] .$$

Thus an equilibrium point is an n-tuple $\mathscr{s}$ such that each player's mixed strategy maximizes his pay-off if the strategies of the others are held fixed. Thus each player's strategy is optimal against those of the others. We shall occasionally abbreviate equilibrium point by eq. pt.

We say that a mixed strategy s_i uses a pure strategy $\pi_{i\beta}$ if $s_i = \sum_\alpha c_{i\alpha} \pi_{i\alpha}$ and $c_{i\beta} > 0$. If $\mathscr{s} = (s_1, s_2, \cdots s_n)$ and s_i uses $\pi_{i\alpha}$ we also say that $\mathscr{s}$ uses $\pi_{i\alpha}$.

From the linearity of $p_i(s_1, \cdots s_n)$ in s_i,

$$(2) \qquad \max_{\text{all } t_i\text{'s}} \left[p_i(\mathscr{s}; t_i) \right] = \max_\alpha \left[p_i(\mathscr{s}; \pi_{i\alpha}) \right] .$$

We define $p_{i\alpha}(\mathscr{s}) = p_i(\mathscr{s}; \pi_{i\alpha})$. Then we obtain the following trivial necessary and sufficient condition for $\mathscr{s}$ to be an equilibrium point:

$$(3) \qquad p_i(\mathscr{s}) = \max_\alpha p_{i\alpha}(\mathscr{s}) .$$

If $\mathscr{s} = (s_1, s_2, \cdots s_n)$ and $s_i = \sum_\alpha c_{i\alpha} \pi_{i\alpha}$ then $p_i(\mathscr{s}) = \sum_\alpha c_{i\alpha} p_{i\alpha}(\mathscr{s})$, consequently for (3) to hold we must have $c_{i\alpha} = 0$ whenever $p_{i\alpha}(\mathscr{s}) < \max_\beta p_{i\beta}(\mathscr{s})$, which is to say that $\mathscr{s}$ does not use $\pi_{i\alpha}$ unless it is an optimal pure strategy for player i. So we write

$$(4) \qquad \text{if } \pi_{i\alpha} \text{ is used in } \mathscr{s} \text{ then } p_{i\alpha}(\mathscr{s}) = \max_\beta p_{i\beta}(\mathscr{s})$$

as another necessary and sufficient condition for an equilibrium point.

Since a criterion (3) for an eq. pt. can be expressed as the equating of two continuous functions on the space of n-tuples $\mathscr{s}$ the eq. pts. obviously form a closed subset of this space. Actually, this subset is formed from a ~~lot~~ number of pieces of algebraic varieties, cut out by other algebraic varieties.

Existence of Equilibrium Points

I have previously published [Proc. N. A. S. 36 (1950) 48–49] a proof of the result below based on Kakutani's generalized fixed point theorem. The proof given here uses the Brouwer theorem.

The method is to set up a sequence of continuous mappings:

$\mathscr{A} \rightarrow \mathscr{A}'(\mathscr{A},1)$; $\mathscr{A} \rightarrow \mathscr{A}'(\mathscr{A},2)$; $\cdots$ whose fixed points have an equilibrium point as limit point. A limit mapping exists, but is discontinuous, and need not have any fixed points.

THEO. 1: Every finite game has an equilibrium point.

Proof: Using our standard notation, let $\mathscr{A}$ be an n-tuple of mixed strategies, and $p_{i\alpha}(\mathscr{A})$ the pay-off to player i if he uses his pure strategy $\pi_{i\alpha}$ and the others use their respective mixed strategies in $\mathscr{A}$. For each integer λ we define the following continuous functions of $\mathscr{A}$:

$$q_i(\mathscr{A}) = \max_{\alpha} p_{i\alpha}(\mathscr{A}) \, ,$$

$$\phi_{i\alpha}(\mathscr{A},\lambda) = p_{i\alpha}(\mathscr{A}) - q_i(\mathscr{A}) + 1/\lambda \, , \text{ and}$$

$$\phi^{+}_{i\alpha}(\mathscr{A},\lambda) = \max\left[0, \phi_{i\alpha}(\mathscr{A},\lambda)\right] \, .$$

Now $\sum_{\alpha} \phi^{+}_{i\alpha}(\mathscr{A},\lambda) \geq \max_{\alpha} \phi^{+}_{i\alpha}(\mathscr{A},\lambda) = 1/\lambda > 0$ so that

$$C'_{i\alpha}(\mathscr{A},\lambda) = \frac{\phi^{+}_{i\alpha}(\mathscr{A},\lambda)}{\sum_{\beta} \phi^{+}_{i\beta}(\mathscr{A},\lambda)}$$ is continuous.

Define $S'_i(\mathscr{A},\lambda) = \sum_{\alpha} \pi_{i\alpha} C'_{i\alpha}(\mathscr{A},\lambda)$ and $\mathscr{A}'(\mathscr{A},\lambda) = (S'_1, S'_2, \ldots S'_n)$. Since all the operations have preserved continuity, the mapping $\mathscr{A} \rightarrow \mathscr{A}'(\mathscr{A},\lambda)$ is continuous; and since the space of n-tuples, $\mathscr{S}$, is a cell, there must be a fixed point for each λ. Hence there will be a subsequence $\mathscr{S}_\mu$, converging to $\mathscr{S}^*$, where $\mathscr{S}_\mu$ is fixed under the mapping $\mathscr{S} \to \mathscr{S}'(\mathscr{S}, \lambda(\mu))$.

Now suppose $\mathscr{S}^*$ were not an equilibrium point. Then if $\mathscr{S}^* = (s_1^*, \dots s_n^*)$ some component s_i^* must be non-optimal against the others, which means s_i^* uses some pure strategy $\pi_{i\alpha}$ which is non-optimal. [see (4), pg. 4] This means that

$$p_{i\alpha}(\mathscr{S}^*) < q_i(\mathscr{S}^*) \qquad \text{which justifies}$$

writing

$$p_{i\alpha}(\mathscr{S}^*) - q_i(\mathscr{S}^*) < -\epsilon \quad .$$

From continuity, if μ is large enough,

$$\left| \left[p_{i\alpha}(\mathscr{S}_\mu) - q_i(\mathscr{S}_\mu) \right] - \left[p_{i\alpha}(\mathscr{S}^*) - q_i(\mathscr{S}^*) \right] \right| < \epsilon/2 \quad \text{and} \quad 1/\lambda(\mu) < \epsilon/2 .$$

Adding,

$$p_{i\alpha}(\mathscr{S}_\mu) - q_i(\mathscr{S}_\mu) + 1/\lambda(\mu) < 0 \qquad \text{which is}$$

simply

$$\phi_{i\alpha}(\mathscr{S}_\mu, \lambda(\mu)) < 0, \text{ whence } \phi^+_{i\alpha}(\mathscr{S}_\mu, \lambda(\mu)) = 0, \text{ whence}$$

$$C'_{i\alpha}(\mathscr{S}_\mu, \lambda(\mu)) = 0$$

. From this last equation we know that $\pi_{i\alpha}$ is not used in $\mathscr{S}_\mu$ since

$$\mathscr{S}_\mu = \sum_\alpha \pi_{i\alpha} \, C'_{i\alpha}(\mathscr{S}_\mu, \lambda(\mu))$$

, because $\mathscr{S}_\mu$ is a fixed point.

And since $\mathscr{S}_\mu \to \mathscr{S}^*$, $\pi_{i\alpha}$ is not used in $\mathscr{S}^*$, which contradicts our assumption.

Hence $\mathscr{S}^*$ is indeed an equilibrium point.

Symmetries of Games

An *automorphism*, or *symmetry*, of a game will be a permutation of its pure strategies which satisfies certain conditions, given below.

If two strategies belong to a single player they must go into two strategies belonging to a single player. Thus if ϕ is the permutation of the pure strategies it induces a permutation ψ of the players.

Each n-tuple of pure strategies is therefore permuted into another n-tuple of pure strategies. We may call χ the induced permutation of these n-tuples. Let ξ denote an n-tuple of pure strategies and $p_i(\xi)$ the pay-off to player i when the n-tuple ξ is employed. We require that if

$$j = i^{\psi} \quad \text{then} \quad p_j(\xi^{\chi}) = p_i(\xi)$$

which completes the definition of a symmetry.

The permutation ϕ has a unique linear extension to the mixed strategies. If

$$s_i = \sum_{\alpha} c_{i\alpha} \pi_{i\alpha} \quad \text{we define} \quad (s_i)^{\phi} = \sum_{\alpha} c_{i\alpha} (\pi_{i\alpha})^{\phi}.$$

The extension of ϕ to the mixed strategies clearly generates an extension of χ to the n-tuples of mixed strategies. We shall also denote this by χ.

We define a *symmetric n-tuple* $\mathscr{L}$ of a game by

$$\mathscr{L}^{\chi} = \mathscr{L} \quad \text{for all } \chi\text{'s}$$

it being understood that χ means a permutation derived from a symmetry ϕ.

THEO. 4: Any finite game has a symmetric equilibrium point.

Proof: First we note that

$S_{i\alpha} = \frac{\sum_{\alpha} \pi_{i\alpha}}{\sum_{\alpha} 1}$ has the property $(S_{i\alpha})^{\phi} = S_{j\alpha}$ where $j = i^{\psi}$, so that the n-tuple $\mathscr{s}_o = (S_{1\alpha}, S_{2\alpha}, \dots S_{n\alpha})$ is fixed under any χ; hence any game has at least one symmetric n-tuple.

If $\mathscr{s} = (s_1, \dots s_n)$ and $\mathscr{t} = (t_1, \dots t_n)$ are symmetric then $\frac{\mathscr{s}+\mathscr{t}}{2} = \left(\frac{s_1+t_1}{2}, \dots \frac{s_n+t_n}{2}\right)$ is so too because $\mathscr{s}^{\chi} = \mathscr{s} \Leftrightarrow s_j = (s_i)^{\phi}$ where $j = i^{\psi}$, hence

$$\frac{s_j+t_j}{2} = \frac{(s_i)^{\phi}+(t_i)^{\phi}}{2} = \left(\frac{s_i+t_i}{2}\right)^{\phi}, \text{ hence } \left(\frac{\mathscr{s}+\mathscr{t}}{2}\right)^{\chi} = \frac{\mathscr{s}+\mathscr{t}}{2}.$$

This shows that the set of symmetric n-tuples is a convex subset of the space of n-tuples since it is obviously closed.

Now observe that for each λ the mapping $\mathscr{s} \to \mathscr{s}'(\mathscr{s}, \lambda)$ used in the proof of existence theorem was intrinsically defined. Therefore, if $\mathscr{s}_2 = \mathscr{s}'(\mathscr{s}_1, \lambda)$ and χ is an auto morphism of the game we will have $\mathscr{s}_2^{\chi} = \mathscr{s}'(\mathscr{s}_1^{\chi}, \lambda)$. If $\mathscr{s}_1$ is symmetric $\mathscr{s}_1^{\chi} = \mathscr{s}_1$ and therefore $\mathscr{s}_2^{\chi} = \mathscr{s}'(\mathscr{s}_1, \lambda) = \mathscr{s}_2$. Consequently this mapping maps the set of symmetric n-tuples into itself.

Since this set is a cell there must be a symmetric fixed point $\mathscr{s}_{\lambda}$. And, as in the proof of the existence theorem we could obtain a limit point $\mathscr{s}^{*}$ which would have to be symmetric.

Solutions

We define here solutions, strong solutions, and sub-solutions. A non-cooperative game does not always have a solution, but when it does the solution is unique. Strong solutions are solutions with special properties. Sub-solutions always exist and have many of the properties of solutions, but lack uniqueness.

S_i will denote a set of mixed strategies of player i and $\mathcal{S}$ a set of n-tuples of mixed strategies.

Solvability:

A game is solvable if its set, $\mathcal{S}$, of equilibrium points satisfies the condition

$$(1) \qquad (\mathfrak{t}; r_i) \in \mathcal{S} \quad \text{and} \quad \mathfrak{s} \in \mathcal{S} \Rightarrow (\mathfrak{s}; r_i) \in \mathcal{S} \quad \text{for all } i\text{'s}.$$

This is called the _interchangeability_ condition. The _solution_ of a solvable game is its set, $\mathcal{S}$, of equilibrium points.

Strong Solvability:

A game is _strongly solvable_ if it has a solution, $\mathcal{S}$, such that for all i's

$$\mathfrak{s} \in \mathcal{S} \quad \text{and} \quad p_i(\mathfrak{s}; r_i) = p_i(\mathfrak{s}) \Rightarrow (\mathfrak{s}; r_i) \in \mathcal{S}$$

and then $\mathcal{S}$ is called a _strong solution_.

Equilibrium Strategies:

In a solvable game let S_i be the set of all mixed strategies s_i

such that for some t the n-tuple $(t; s_i)$ is an equilibrium point. [s_i is the i th component of some equilibrium point.] We call S_i the set of __equilibrium strategies__ of player i.

Sub-solutions:

If $\mathcal{L}$ is a subset of the set of equilibrium points of a game and satisfies condition (1); and if $\mathcal{L}$ is maximal relative to this property then we call $\mathcal{L}$ a __sub-solution__.

For any sub-solution $\mathcal{L}$ we define the i th __factor set__, S_i, as the set of all s_i's such that $\mathcal{L}$ contains $(t; s_i)$ for some t.

Note that a sub-solution, when unique, is a solution; and its factor sets are the sets of equilibrium strategies.

THEO. 2: A sub-solution, $\mathcal{L}$, is the set of all n-tuples $(s_1, s_2, \cdots s_n)$ such that each $s_i \in S_i$ where S_i is the i th factor set of $\mathcal{L}$. Geometrically, $\mathcal{L}$ is the product of its factor sets.

Proof: Consider such an n-tuple $(s_1, \cdots s_n)$. By definition $\exists\ t_1, t_2, \cdots, t_n$ such that for each i $(t_i; s_i) \in \mathcal{L}$. Using the condition (1) n-1 times we obtain successively $(t_1; s_1; s_2) \in \mathcal{L}$, $\cdots$, $(t_1; s_1; s_2; s_3; \cdots; s_n) \in \mathcal{L}$ and the last is simply $(s_1, s_2, \cdots s_n) \in \mathcal{L}$, which we needed to show.

THEO. 3: The factor sets $S_1, S_2, \cdots S_n$ of a sub-solution are closed and convex as subsets of the mixed strategy spaces.

Proof: It suffices to show two things: (a) if s_i and $s_i' \in S_i$

then $s_i^* = (s_i + s_i')/2 \in S_i$; (b) if $s_i^\#$ is a limit point of S_i then $s_i^\# \in S_i$.

Let $\not t \in \mathscr{L}$. Then we have

$$p_j(\not t; s_i) \geq p_j(\not t; s_i; t_j) \quad \text{and} \quad p_j(\not t; s_i') \geq p_j(\not t; s_i'; t_j)$$

for any t_j , by using the criterion of (1) , pg. 3 for an eq. pt. Adding these inequalities, using the linearity of $p_j(s_1, \ldots s_n)$ in s_i , and dividing by 2, we get $p_j(\not t; s_i^*) \geq p_j(\not t; s_i^*; t_j)$ since $s_i^* = (s_i + s_i')/2$. From this we know that $(\not t; s_i^*)$ is an eq. pt. for any $\not t \in \mathscr{L}$. If the set of all such eq. pts. $(\not t; s_i^*)$ is added to $\mathscr{L}$ the augmented set clearly satisfies condition (1), and since $\mathscr{L}$ was to be maximal it follows that $s_i^* \in S_i$.

To attack (b) note that the n-tuple $(\not t; s_i^\#)$, where $\not t \in \mathscr{L}$ will be a limit point of the set of n-tuples of the form $(\not t; s_i)$ where $s_i \in S_i$, since $s_i^\#$ is a limit point of S_i . But this set is a set of eq. pts. and hence any point in its closure is an eq. pt., since the set of all eq. pts. is closed [see pg. 4] . Therefore $(\not t; s_i^\#)$ is an eq. pt. and hence $s_i^\# \in S_i$ from the same argument as for s_i^* .

Values:

Let $\mathscr{L}$ be the set of equilibrium points of a game. We define

$$v_i^+ = \max_{\mathcal{A} \in \mathscr{L}} [p_i(\mathcal{A})] \quad , \quad v_i^- = \min_{\mathcal{A} \in \mathscr{L}} [p_i(\mathcal{A})] \quad .$$

If $v_i^+ = v_i^-$ we write $v_i = v_i^+ = v_i^-$. v_i^+ is the <u>upper value</u> to player i of the game; v_i^- the <u>lower value</u>; and

V: the <u>value</u>, if it exists.

Values will obviously have to exist if there is but one equilibrium point.

One can define <u>associated</u> <u>values</u> for a sub-solution by restricting $\mathcal{L}$ to the eq. pts. in the sub-solution and then using the same defining equations as above.

A two-person zero-sum game is always solvable in the sense defined above. The sets of equilibrium strategies S_1 and S_2 are simply the sets of "good" strategies. Such a game is not generally strongly solvable; strong solutions exist only when there is a "saddle point" in <u>pure</u> strategies.

Geometrical Form of Solutions

In the two-person zero-sum case it has been shown that the set of "good" strategies of a player is a convex polyhedral subset of his strategy space. We shall obtain the same result for a player's set of equilibrium strategies in any solvable game.

THEO. 5: The sets $S_1, S_2, \ldots S_n$ of equilibrium strategies in a solvable game are polyhedral convex subsets of the respective mixed strategy spaces.

Proof: An n-tuple $\mathcal{S}$ will be an equilibrium point if and only if for every i

$$(1) \qquad p_i(\mathcal{S}) = \max_{\alpha} p_{i\alpha}(\mathcal{S})$$

which is condition (3) on page 4. An equivalent condition is for every i and α

$$(2) \qquad p_i(\mathcal{S}) - p_{i\alpha}(\mathcal{S}) \geq 0 \ .$$

Let us now consider the form of the set S_j of equilibrium strategies, s_j, of player j. Let $\mathcal{t}$ be any equilibrium point, then $(\mathcal{t}; s_j)$ will be an equilibrium point if and only if $s_j \in S_j$, from Theo. 2. We now apply conditions (2) to $(\mathcal{t}; s_j)$, obtaining

$$(3) \quad s_j \in S_j \iff \text{for all } i, \alpha \quad p_i(\mathcal{t}; s_j) - p_{i\alpha}(\mathcal{t}; s_j) \geq 0 \ .$$

Since p_i is n-linear and $\mathcal{t}$ is constant these are a set of linear inequalities of the form $F_{i\alpha}(s_j) \geq 0$. Each such inequality is either satisfied for all s_j or for those lying on and to one side of some hyperplane passing through the strategy simplex. Therefore, the

Simple Examples

These are intended to illustrate the concepts defined in the paper and display special phenomena which occur in these games.

The first player has the roman letter strategies and the pay-off to the left, etc.

Ex. 1 $5\ a\alpha\ -3$ $-4\ a\beta\ 4$ $-5\ b\alpha\ 5$ $3\ b\beta\ -4$	Weak Solution: $\left(\frac{9}{16}a + \frac{7}{16}b,\ \frac{7}{17}\alpha + \frac{10}{17}\beta\right)$ $v_1 = \frac{-5}{17}$, $v_2 = +\frac{1}{2}$
Ex. 2 $1\ a\alpha\ 1$ $-10\ a\beta\ 10$ $10\ b\alpha\ -10$ $-1\ b\beta\ -1$	Strong Solution: (b, β) $v_1 = v_2 = -1$
Ex. 3 $1\ a\alpha\ 1$ $-10\ a\beta\ -10$ $-10\ b\alpha\ -10$ $1\ b\beta\ 1$	Unsolvable; equilibrium points (a,α), (b,β), and $(a/2 + b/2,\ \alpha/2 + \beta/2)$. The strategies in the last case have maxi-min and mini-max properties.
Ex. 4 $1\ a\alpha\ 1$ $0\ a\beta\ 1$ $1\ b\alpha\ 0$ $0\ b\beta\ 0$	Strong Solution: all pairs of mixed strategies. $v_1^+ = v_2^+ = 1$, $v_1^- = v_2^- = 0$
Ex. 5 $1\ a\alpha\ 2$ $-1\ a\beta\ -4$ $-4\ b\alpha\ -1$ $2\ b\beta\ 1$	Unsolvable; eq. pts. (a,α), (b,β) and $(\frac{1}{4}a + \frac{3}{4}b,\ \frac{3}{8}\alpha + \frac{5}{8}\beta)$. However, empirical tests show a tendency toward (a,α).
Ex. 6 $1\ a\alpha\ 1$ $0\ a\beta\ 0$ $0\ b\alpha\ 0$ $0\ b\beta\ 0$	Eq. pts.: (a,α) and (b,β), with (b,β) an example of instability.

complete set [which is finite] of conditions will all be satisfied simultaneously on some convex polyhedral subset of player j's strategy simplex. [Intersection of half-spaces.]

As a corollary we may conclude that S_k is the convex closure of a finite set of mixed strategies [vertices].

Dominance and Contradiction Methods

We say that s_i' dominates s_i if $p_i(t; s_i') > p_i(t; s_i)$ for every t.

This amounts to saying that s_i' gives player i a higher pay-off than s_i no matter what the strategies of the other players are. To see whether a strategy s_i' dominates s_i it suffices to consider only pure strategies for the other players because of the n-linearity of p_i.

It is obvious from the definitions that no equilibrium point can involve a dominated strategy s_i.

The domination of one mixed strategy by another will always entail other dominations. For suppose s_i' dominates s_i and t_i uses all of the pure strategies which have a higher coefficient in s_i than in s_i'. Then for a small enough $\rho > 0$

$$t_i' = t_i + \rho(s_i' - s_i)$$

is a mixed strategy; and t_i' dominates t_i by linearity.

One can prove a few properties of the set of undominated strategies. It is simply connected and is formed by the union of some collection of faces of the strategy simplex.

The information obtained by discovering dominances for one player may be of relevance to the others, insofar as the elimination of classes of mixed strategies as possible components of an equilibrium point is concerned. For the t's whose components are all undominated are all that need be considered and this eliminating some of the strategies of one player may make possible the elimination of a new class of strategies for another player.

Another procedure which may be used in locating equilibrium points is the contradiction-type analysis. Here one assumes that an equilibrium point exists having component strategies lying within certain regions of the strategy spaces and proceeds to deduce further conditions which must be satisfied if the hypothesis is true. This sort of reasoning may be carried through several stages to eventually obtain a contradiction indicating that there is no equilibrium point satisfying the initial hypothesis.

A Three-Man Poker Game

As an example of the application of our theory to a more or less realistic case we include the simplified poker game given below. The rules are as follows:

(1) The deck is large, with equally many <u>high</u> and <u>low</u> cards, and a hand consists of one card.

(2) Two chips are used to ante, open, or call.

(3) The players play in rotation and the game ends after all have passed or after one player has opened and the others have had a chance to call.

(4) If no one bets the antes are retrieved.

(5) Otherwise the pot is divided equally among the highest hands which have bet.

We find it more satisfactory to treat the game in terms of quantities we call "behavior parameters" than in the normal form of "Theory of Games and Economic Behavior." In the normal form representation two mixed strategies of a player may be equivalent in the sense that each makes the individual choose each available course of action in each particular situation requiring action on his part with the same frequency. That is, they represent the same behavior pattern on the part of the individual.

Behavior parameters give the probabilities of taking each of the various possible actions in each of the various possible situations which may arise. Thus they describe behavior patterns.

In terms of behavior parameters the strategies of the players may be represented as follows, assuming that since there is no point in passing with a <u>high</u> card at one's last opportunity to bet that this will not be done. The greek letters are the probabilities of the various acts.

	First Moves	Second Moves
I	α Open on <u>high</u> β Open on <u>low</u>	κ Call III on <u>low</u> λ Call II on <u>low</u> μ Call II and III on <u>low</u>
II	γ Call I on <u>low</u> δ Open on <u>high</u> ϵ Open on <u>low</u>	ν Call III on <u>low</u> ξ Call III and I on <u>low</u>
III	ζ Call I and II on <u>low</u> η Open on <u>low</u> θ Call I on <u>low</u> ι Call II on <u>low</u>	Player III never gets a second move.

We locate all possible equilibrium points by first showing that most of the greek parameters must vanish. By dominance mainly with a little contradiction-type analysis β is eliminated and with it go γ, ζ, and θ by dominance. Then contradictions eliminate $\mu, \xi, \iota, \lambda, \kappa$, and ν in that order. This leaves us with α, δ, ϵ and η. Contradiction analysis shows that none of these can be zero or one and thus we obtain a system of simultaneous algebraic equations. The equations happen to have but one solution with the variables in the range $(0,1)$. We get

$\alpha = \dfrac{21-\sqrt{321}}{10}$, $\eta = \dfrac{5\alpha+1}{4}$, $\delta = \dfrac{5-2\alpha}{5+\alpha}$, and

$\epsilon = \dfrac{4\alpha-1}{\alpha+5}$. These yield $\alpha = .308$, $\eta = .635$, $\delta = .826$, and

$\epsilon = .044$.

Since there is only one equilibrium point the game has values; these are

$$V_1 = -.147 = \frac{-(1+17\alpha)}{9(5+\alpha)}, \qquad V_2 = -.096 = -\frac{1-2\alpha}{4}, \text{ and}$$

$$V_3 = .243 = \frac{79}{40}\left(\frac{1-\alpha}{5+\alpha}\right).$$

Investigation of the coalition powers yields the following "good strategies" and values for the various coalitions. Parameters not mentioned are zero.

I and II	versus	III
$\alpha = 3/4$		$\iota = 1/4$, $0 \le \eta \le 2/3$
$\delta = \epsilon = 1$		value to III: $.03125 = 1/32$

II and III	versus	I
$\delta = 1$, $\epsilon = 0$		$\alpha = 2/3$
$\eta = 2/3$		value to I: $-.1667 = -1/6$

I and III versus II

I high	I low	III	
bet	pass	$\eta = 0$	$3/11$
pass	pass	$\eta = 13/16$	$8/11$

II: $\delta = 7/11$, $\epsilon = 3/11$

value to II: $-.1136 = -5/44$

The coalition members have the power to agree upon a pattern of play before the game is played. This advantage becomes significant only in the case of coalition I III where III may open after two passes when I had planned to pass on both <u>high</u> and <u>low</u> but will not open if

I had planned to bet if he got high. The values given are, of course, what the single player assures himself with his "safe" strategy.

A more detailed treatment of this game is being prepared for publication elsewhere. This will consider different relative sizes of ante and bet.

Motivation and Interpretation

In this section we shall try to explain the significance of the concepts introduced in this paper. That is, we shall try to show how equilibrium points and solutions can be connected with observable phenomena.

The basic requirements for a non-cooperative game is that there should be no pre-play communication among the players [unless it has no bearing on the game]. Thus, by implication, there are no coalitions and no side-payments. Because there is no extra-game utility [pay-off] transfer, the pay-offs of different players are effectively incomparable; if we transform the pay-off functions linearly: $p_i' = a_i p_i + b_i$, where $a_i > 0$ the game will be essentially the same. Note that equilibrium points are preserved under such transformations.

We shall now take up the "mass-action" interpretation of equilibrium points. In this interpretation solutions have no great significance. It is unnecessary to assume that the participants have full knowledge of the total structure of the game, or the ability and inclination to go through any complex reasoning processes. But the participants are supposed to accumulate empirical information on the relative advantages of the various pure strategies at their disposal.

To be more detailed, we assume that there is a population [in the sense of statistics] of participants for each position of the game. Let us also assume that the "average playing" of the game involves n participants selected at random from the n populations, and that there is a stable average frequency with which each pure strategy is employed by the "average member" of the appropriate population.

Since there is to be no collaboration between individuals playing in different positions of the game, the probability that a particular n-tuple of pure strategies will be employed in a playing of the game should be the product of the probabilities indicating the chance of each of the n pure strategies to be employed in a random playing.

Let the probability that $\pi_{i\alpha}$ will be employed in a random playing of the game be $c_{i\alpha}$, and let $s_i = \sum_{\alpha} c_{i\alpha} \pi_{i\alpha}$ and $\mathcal{J} = (s_1, s_2, \dots s_n)$. Then the expected pay-off to an individual playing in the i th position of the game and employing the pure strategy $\pi_{i\alpha}$ is $p_i(\mathcal{J}; \pi_{i\alpha}) = p_{i\alpha}(\mathcal{J})$.

Now let us consider what effects the experience of the participants will produce. To assume, as we did, that they accumulated empirical evidence on the pure strategies at their disposal is to assume that those playing in position i learn the numbers $p_{i\alpha}(\mathcal{J})$. But if they know these they will employ only optimal pure strategies, i.e., those pure strategies $\pi_{i\alpha}$ such that

$$p_{i\alpha}(\mathcal{J}) = \max_{\beta} p_{i\beta}(\mathcal{J}) .$$

Consequently since s_i expresses their behavior s_i attaches positive coefficients only to optimal pure strategies, so that

$$\pi_{i\alpha} \text{ is used in } s_i \Rightarrow p_{i\alpha}(\mathcal{J}) = \max_{\beta} p_{i\beta}(\mathcal{J}) .$$

But this is simply a condition for $\mathcal{J}$ to be an equilibrium point. [see (4), pg 3]

Thus the assumptions we made in this "mass-action" interpretation lead to the conclusion that the mixed strategies representing the average behavior in each of the populations form an equilibrium point.

The populations need not be large if the assumptions still hold.

... hold. There are situations in economics or international politics in which, effectively, a group of interests are involved in a non-cooperative game without being aware of it; the non-awareness helping to make the situation truly non-cooperative.

Actually, of course, we can only expect some sort of approximate equilibrium, since the information, its utilization, and the stability of the average frequencies will be imperfect.

We now sketch another interpretation, one in which solutions play a major role, and which is applicable to a game played but once.

We proceed by investigating the question: what would be a "rational" prediction of the behavior to be expected of rational playing the game in question? By using the principles that a rational prediction should be unique, that the players should be able to deduce and make use of it, and that such knowledge on the part of each player of what to expect the others to do should not lead him to act out of conformity with the prediction, one is led to the concept of a solution defined before.

If $S_1, S_2, \ldots S_n$ were the sets of equilibrium strategies of a solvable game, the "rational" prediction should be: "The average behavior of rational men playing in position i would define a mixed strategy s_i in S_i if an experiment were carried out."

In this interpretation we need to assume the players know the full structure of the game in order to be able to deduce the prediction for themselves. It is quite strongly a rationalistic and idealizing interpretation.

In an unsolvable game it sometimes happens that good heuristic reasons can be found for narrowing down the set of equilibrium points to those in a single sub-solution, which then plays the role of a solution.

In general a sub-solution may be looked at as a set of mutually compatible equilibrium points, forming a coherent whole. The sub-solutions appear to give a natural subdivision of the set of equilibrium points of a game.

Applications

The study of n-person games for which the accepted ethics of fair play imply non-cooperative playing is, of course, an obvious direction in which to apply this theory. And poker is the most obvious target. The analysis of a more realistic poker game than our very simple model should be quite an interesting affair.

The complexity of the mathematical work needed for a complete investigation increases rather rapidly, however, with increasing complexity of the game; so that it seems that analysis of a game much more complex than the example given here would only be feasible using approximate computational methods.

A less obvious type of application is to the study of cooperative games. By a cooperative game we mean a situation involving a set of players, pure strategies, and pay-offs as usual; but with the assumption that the players can and will collaborate as they do in the von Neumann and Morgenstern theory. This means the players may communicate and form coalitions which will be enforced by an umpire. It is unnecessarily restrictive, however, to assume any transferability, or even comparability of the pay-offs [which should be in utility units] to different players. Any desired transferability can be put into the game itself instead of assuming it possible in the extra-game collaboration.

The writer has developed a "dynamical" approach to the study of cooperative games based upon reduction to non-cooperative form. One proceeds by constructing a model of the pre-play negotiation so that the steps of negotiation become moves in a larger non-cooperative game [which will have an infinity of pure strategies] describing the total situation.

This larger game is then treated in terms of the theory of this paper [extended to infinite games] and if values are obtained they are taken as the values of the cooperative game. Thus the problem analysing a cooperative game becomes the problem of obtaining a suitable, and convincing, non-cooperative model for the negotiation.

The writer has, by such a treatment, obtained values for all finite two person cooperative games, and some special n-person games.

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Acknowledgements

Drs. Tucker, Gale, and Kuhn gave valuable criticism and suggestions for improving the exposition of the material in this paper. David Gale suggested the investigation of symmetric games. The solution of the Poker model was a joint project undertaken by Lloyd S. Shapley and the author. Finally, the author was sustained financially by the Atomic Energy Commission in the period 1949-50 during which this work was done.

Nash Equilibrium: In a complete-information game (a [*C-I*] game), a ***Nash equilibrium*** is a collection of '(complete contingent) strategies' – one for each player – such that each strategy is a ***best response*** to the collection of rival strategies in the stated collection of strategies.

If each '(complete contingent) strategy' in the Nash equilibrium vector is a *pure-strategy*, then the strategy vector is a ***pure-strategy Nash equilibrium***;
if that is not the case then it is a ***mixed-strategy Nash equilibrium***.

Formally, in an N-player game of complete information, a strategy vector $\{\mathbf{x}_1^*, \mathbf{x}_2^*, \ldots, \mathbf{x}_i^*, \ldots, \mathbf{x}_N^*\}$ such that each $\mathbf{x}_i^*$ maximizes each player i's expected utility given the specified strategies of the other $(N-1)$ players

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- **Given Nash's thesis, we have two issues to discuss:**

1. How to identify all Nash Equilibria of a *CI* game?

2a. For each identified Nash equilibrium in a *CI* game, how sensible will it be to predict that the players will play according to that Nash equilibrium?

2b. For each strategy combination that is <u>not</u> a Nash equilibrium in a *CI* game, how sensible will it be to predict that the players will <u>not</u> play that strategy combination?

At this time, we will address the above questions by focusing only on "single-stage simultaneous-move games of complete information" - i.e., to [*ss-Sim-CI*] games

even though the concept of Nash Equilibrium applies for all games of complete information - i.e., to all [*CI*] games

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Nash Equilibrium: Definition & Algorithm

Definition:

In a complete-information game (a [*C-I*] game):

a ***Nash equilibrium*** is a stated vector of strategies – one strategy for each player – such that each strategy is a ***best response*** to the remaining strategies in the stated vector of strategies.

If each of the strategies in the Nash equilibrium vector is a *pure-strategy*, then the vector is a ***pure-strategy Nash equilibrium***;
if that is not the case then it is a ***mixed-strategy Nash equilibrium***.

Algorithm:

1. Determine the ***best response correspondences*** of all the players (i.e., responses to all possible deterministic and stochastic conjectures)
2. Find all the ***points of intersection*** of the ***best response correspondences*** of all players

[In principle, algorithm identifies all *pure-strategy and mixed-strategy Nash equilibria,* but computations to find only pure-strategy equilibria much easier]

Conjectures, Best-responses, and Nash equilibria in *Sim-CI* games

	$S_B = 0$	$S_B = 1$	$S_B = 2$	$S_B = 3$
$S_A = 0$	24.5, 24.5	22.36, 27.36	20, 30	17.33, 32.33
$S_A = 1$	27.36, 22.36	25, 25	22.33, 27.33	19.14, 29.14
$S_A = 2$	30, 20	27.33, 22.33	24.14, 24.14	20, 25
$S_A = 3$	32.33, 17.33	29.14, 19.14	25, 20	15, 15

Alice's pure-strategy best-response correspondence to deterministic **conjectures**:
$[S_A^*(S_B = 0) = S_A^*(S_B = 1) = S_A^*(S_B = 2) = 3]$, $[S_A^*(S_B = 3) = 2]$

Bob's pure-strategy best-response correspondence to deterministic **conjectures**:
$[S_B^*(S_A = 0) = S_B^*(S_A = 1) = S_B^*(S_A = 2) = 3]$, $[S_B^*(S_A = 3) = 2]$

- A player's ***pure-strategy best-response*** to a ***deterministic conjecture about rival behaviour*** is the set of pure strategies (with one or more elements) which maximizes the player's utility when rivals play the conjectured deterministic strategies

➢ *In a [**Sim-CI**] game, all intersections of all* **p-s b-r correspondences to d-c** *of all players constitute the set of pure-strategy Nash equilibria*

Conjectures, Best-responses, and Nash equilibria in *Sim-CI* games

	$S_B = 0$	$S_B = 1$	$S_B = 2$	$S_B = 3$
$S_A = 0$	24.5, 24.5	22.36, 27.36	20, 30	17.33, 32.33
$S_A = 1$	27.36, 22.36	25, 25	22.33, 27.33	19.14, 29.14
$S_A = 2$	30, 20	27.33, 22.33	24.14, 24.14	20, 25
$S_A = 3$	32.33, 17.33	29.14, 19.14	25, 20	15, 15

- A player's "***<u>pure</u> best-response correspondence***" is the mapping showing her ***pure-strategy best-responses*** for <u>all</u> ***conjectures*** – ***deterministic*** and ***probabilistic*** – that she can hold about rival behaviour
- If **Alice** conjectures that **Bob** will play $\{(S_B = 2), (S_B = 3); 0.5, 0.5\}$ then **Alice**'s unique best response is to play $(S_A = 2)$
- But if **Alice** conjectures **Bob** will play $\{(S_B = 2), (S_B = 3); 0.853, 0.147\}$ then **Alice** has two pure-strategy best responses: $(S_A = 2)$ and $(S_A = 3)$
- Alice's (***relevant – i.e., ignoring*** $S_B = 0, 1$) pure best-response correspondence:
  $[S_A^*(S_B = 2) = \mathbf{3}]$, $[S_A^*(S_B = 3) = \mathbf{2}]$
  $S_A^*(\{(S_B = 2), (S_B = 3); p, 1-p\}) = \mathbf{2}$ for $p < 0.853$
  $S_A^*(\{(S_B = 2), (S_B = 3); p, 1-p\}) = \mathbf{3}$ for $p > 0.853$
  $S_A^*(\{(S_B = 2), (S_B = 3); p, 1-p\}) = \{\mathbf{2, 3}\}$ for $p = 0.853$

Conjectures, Best-responses, and Nash equilibria in *Sim-CI* games

	$S_B = 0$	$S_B = 1$	$S_B = 2$	$S_B = 3$
$S_A = 0$	24.5, 24.5	22.36, 27.36	20, 30	17.33, 32.33
$S_A = 1$	27.36, 22.36	25, 25	22.33, 27.33	19.14, 29.14
$S_A = 2$	30, 20	27.33, 22.33	24.14, 24.14	20, 25
$S_A = 3$	32.33, 17.33	29.14, 19.14	25, 20	15, 15

- A player's "***best-response correspondence***" is the mapping showing the set of **all her *best-responses*** – *pure* and *mixed* – for all ***conjectures*** – *deterministic* and *probabilistic* – about rival behaviour

Theorem: In a ***decision*** environment, and in a ***Sim-CI*** game, a rational person will "randomize" over a set of deterministic decisions **if and only if** she is "indifferent" between the outcomes that the different decisions generate

- Alice's (***relevant – i.e., ignoring*** $S_B = 0, 1$) **best-response correspondence**:
  $S_A^*(S_B = 2) = \{3\}$, $S_A^*(S_B = 3) = \{2\}$
  $S_A^*(\{(S_B = 2), (S_B = 3); p, 1-p\}) = \{2\}$ for $p < 0.853$
  $S_A^*(\{(S_B = 2), (S_B = 3); p, 1-p\}) = \{3\}$ for $p > 0.853$
  $S_A^*(\{(S_B = 2), (S_B = 3); p, 1-p\}) = \{\mathbf{2, 3}\}$ & $\{[(S_A=2), (S_A=3); q, 1-q]$ for all $q \in (0,1)\}$ for $p = 0.853$

All Nash Equilibria in the ***Sibling Rivalry Game***

	$S_B = 0$	$S_B = 1$	$S_B = 2$	$S_B = 3$
$S_A = 0$	24.5, 24.5	22.36, 27.36	20, 30	17.33, 32.33
$S_A = 1$	27.36, 22.36	25, 25	22.33, 27.33	19.14, 29.14
$S_A = 2$	30, 20	27.33, 22.33	24.14, 24.14	20, 25
$S_A = 3$	32.33, 17.33	29.14, 19.14	25, 20	15, 15

Two pure-strategy Nash equilibria: $[S_A = 3, S_B = 2]$, $[S_A = 2, S_B = 3]$

- ***Equilibrium outcome*** associated with $[S_A = 3, S_B = 2]$: Alice has 3 bowls of soup / 2.5 oz of fries, Bob has 2 bowls of soup / 2.5 oz of fries
- ***Equilibrium payoffs*** associated with $[S_A = 3, S_B = 2]$: Bernoulli utility of 25 to Alice and Bernoulli utility of 20 to Bob

One mixed-strategy Nash equilibrium :

$[\{(S_A = 2), (S_A = 3); 0.853, 0.147\}, \{(S_B = 2), (S_B = 3); 0.853, 0.147\}]$

[Each sibling randomizes in a way that the other is indifferent between playing '2' & '3']

- *what are the outcomes and the payoffs*? [23.53]

Question: Has the parent designed a good game?

Claim Game 1: each gets {rival claim + 0.1(rival claim – own claim)}

	10K	11K	12K	13K
10K	10 , 10	11.1, 9.9	12.2 , 9.8	13.3 , 9.7
11K	9.9, 11.1	11 , 11	12.1, 10.9	13.2, 10.8
12K	9.8 , 12.2	10.9, 12.1	12 , 12	13.1, 11.9
13K	9.7, 13.3	10.8 , 13.2	11.9 , 13.1	13 , 13

- The unique Nash equilibrium: {10K, 10K}
- Claim Game 1 is in effect a "prisoner's dilemma" game

Experiment:

10K	11K	12K	13K
42 (1st)	1 (2nd)	2 (3rd)	1 (4th)
86%	2%	3%	9%

Results about Best-responses and Nash Equilibria

Result 1: A game will have a pure-strategy Nash equilibrium if and only if the players' pure best-response correspondences intersect.
If that is not the case, then the game will necessarily have at least one mixed-strategy Nash equilibrium in either of the following cases:
(1) each player has a finite number of pure strategies, or
(2) each player has a compact strategy space, and all players' payoff functions are "sufficiently continuous"

Result 2: A mixed-strategy is a player's best-response to her conjectured rival strategy *if and only if* all pure strategies played with positive probabilities in that mixed-strategy are also best responses individually.
This fact is used to locate mixed-strategy Nash equilibria in a game.

Claim Game 2: lower/higher claim gets 120% / 80% of avg. claim

	10K	11K	12K	13K
10K	10 , 10	12.6, 8.4	13.2 , 8.8	13.8 , 9.2
11K	8.4, 12.6	11 , 11	13.8, 9.2	14.4, 9.6
12K	8.8 , 13.2	9.2, 13.8	12 , 12	15, 10
13K	9.2, 13.8	9.6 , 14.4	10 , 15	13 , 13

- The unique Nash equilibrium: {10K, 10K}

Experiment:

10K	11K	12K	13K
42 (1st)	3 (4th)	1 (3rd)	0 (2nd)
74%	6%	6%	14%

The ***Matching Pennies*** game

	heads	*tails*
heads	10, 0	0, 10
tails	0, 10	10, 0

→

	heads	*tails*
heads	10, 0	0, 10
tails	0, 10	10, 0

- There is no pure-strategy Nash equilibrium in this game
- A player will randomize between ***heads*** and ***tails*** if and only if she expects to receive the same (expected) utility when she plays heads deterministically or tails deterministically
- Thus, each player will randomize between ***heads*** and ***tails*** if and only if she expects her opponent to play {***heads***, ***tails***; **0.5**, **0.5**}
- Thus, the unique Nash equilibrium of this game is: each player plays{***heads***, ***tails***; **0.5**, **0.5**}

Claim Game 4: lower / higher bid gets 105% / 95% of avg. claim

	10K	11K	12K	13K
10K	10 , 10	11.025, 9.975	11.55 , 10.45	12.075 , 10.925
11K	9.975, 11.025	11 , 11	12.075, 10.925	12.6, 11.4
12K	10.45 , 11.55	10.925, 12.075	12 , 12	13.125, 11.875
13K	10.925, 12.075	11.4 , 12.6	11.875, 13.125	13 , 13

- A Nash equilibrium: each player plays{11K, 12K, 13K; 5/24, 2/3, 1/8}

Experiment:

10K	11K	12K	13K
3 (4th)	3 (3rd)	5 (2nd)	35 (1st)
7%	5%	16%	72%

***Claim Game* 3**:

	10K	11K	12K	13K
10K	10 , 10	9.9, 9.9	10.8 , 10.8	11.7, 11.7
11K	9, 9	11 , 11	10.8, 10.8	11.7, 11.7
12K	9, 9	9.9, 9.9	12 , 12	11.7, 11.7
13K	9, 13.5	9.9, 9.9	10.8, 10.8	13 , 13

- [12K, 12K] and [{10K,13K; 5/22, 17/22}, {10K,13K; 13/23, 10/23}] are two Nash equilibria

Experiment:

	10K	11K	12K	13K
Row	5 (3rd)	1 (4th)	16 (2nd)	4 (1st)
	15%	11%	53%	21%
Column	0 (4th)	0 (3rd)	4 (2nd)	6 (1st)
	18%	12%	44%	26%

Answer Key to Problem Set 1

Problem 1. Note that for Alice: $0.72\times(\sqrt{200}) > (\sqrt{100})$, while for Bob: $0.28\times(200)^2 > (100)^2$.

Problem 2. A's expected utility if she takes B's bet is: $0.7\times\sqrt{11{,}000} + 0.3\times\sqrt{8{,}000} = 100.25$. A's expected utility if she takes C's bet is : $0.7\times\sqrt{15{,}000} + 0.3\times\sqrt{2{,}000} = 99.15$. So, A will accept B's bet.

Problem 3. Let $\boldsymbol{p}$ denote my subjective probability belief that England will win. Then my expected utility from Bet A = $\boldsymbol{p}\sqrt{(1000 + 600)} + (1-\boldsymbol{p})\sqrt{(1000 - 100)} = 30 + 10\boldsymbol{p}$, and my expected utility from Bet B = $\boldsymbol{p}\sqrt{(1000 - 1000)} + (1-\boldsymbol{p})\sqrt{(1000 + 1500)} = 50 - 50\boldsymbol{p}$. So I will pick Bet A if and only if $\boldsymbol{p} \geq 1/3$.

Problem 4. If I choose bus route I, my expected utility will be: $0.3(100 - 40) + 0.7(70 - 40)$. If I choose bus route II, my expected utility will be: $0.4(100 - 60) + 0.6(70 - 60)$. Thus, I should choose bus route I.

[In this context, the "parametric equilibrium issue" is the following: Where are the probabilities of 'having at least one seat empty' arising from? From the choices made by all other city residents, with different people having different Bernoulli utility functions (defined over travel time and sitting comfort). So, suppose there are a 1000 distinct people choosing over the two bus routes at a given time. The probability vector {30% for I, and 40% for II} will be a parametric equilibrium if and only if the following is true: when the 1000 people choose bus routes on the basis of these conjectured probabilities, ex post it turns out that bus route I actually has 30% probability of having at least one seat empty and bus route II actually has 40% probability of having at least one seat empty.]

Problem 5. If the farmer plants crop A in X acres of land, he gets a random prospect of: $\{200X + 100(100 - X),\ 50X + 100(100 - X);\ 0.5,\ 0.5\}$; and so his *expected utility* is: $0.5u(10000 + 100X) + 0.5u(10000 - 50X)$.
His optimal crop allocation choice is that X^* that maximizes his expected utility. If the farmer is "risk neutral" with $u(r) = r$, his expected utility is linearly rising in X, and thus he will set $X^* = 100$; this makes perfect sense as crop A has a higher "expected price" than crop B, and a risk-neutral individual cares only about expected values of prospects.
If the farmer is "risk averse" with $u(z) = \log z$, then $X^* = 50$, i.e., he splits his "portfolio" half-and-half. To see this note that in this case the expected utility is: $0.5\log(10000 + 100X) + 0.5\log(10000 - 50X)$, which is a strictly concave function of X. So, taking the derivative of the function with respect to X and setting is equal to zero gives the "first-order condition": $[100/(10000 + 100X^*)] = [50/(10000 - 50X^*)]$. The solution $X^* = 50$ follows immediately.

Problem Set 2

Problem 1: Consider the following "Extended Matching Pennies Game – 1" which is a complete-information simultaneous-move game. In this game, each of two players can play one of three pure strategies: 'Heads' (i.e., place her coin with heads up), 'Tails' (i.e., place her coin with tails up), and 'Covered' (place her coin with any face up but immediately cover the coin with her hands so that nobody can see which face is up). The commonly-known payoff matrix of the game is as follows:

Column Player

Row Player		***Heads***	***Tails***	***Covered***
	Heads	10, 0	0, 10	6, 6
	Tails	0, 10	10, 0	6, 6
	Covered	6, 6	6, 6	6, 6

.

Determine the set of Nash equilibria of the game in which "exactly one player plays a pure strategy". Do all these Nash equilibria have *distinct* payoffs for the two players?

Problem 2: Consider the following "Extended Matching Pennies Game – 2" which is similar to the game above except that only one player – in this case the Row player – can player 'Heads' or 'Tails' or 'Covered'.

Column Player

Row Player		***Heads***	***Tails***
	Heads	10, 0	0, 10
	Tails	0, 10	10, 0
	Covered	7, 5	7, 7

.

There exists a unique Nash equilibrium of the game. Find it *via* the algorithm of "intersection of best response correspondences".

On Nash Equilibria as predictions of behaviour

Question: In simultaneous-move complete-information [***Sim-CI***] games, do Nash equilibria constitute sensible predictions of actual strategic behaviour?

Fundamental cause of concern

- In a simultaneous-move complete-information game, a player has to (*i*) form conjectures about rival play & (*ii*) respond to conjectures

  For such strategic behaviour by all the players of a game to coincide with a Nash equilibrium, it must be the case that each player holds "correct" conjectures about rival behavior, and then best-responds.

- ***While it might be sensible to argue that players will best-respond to their "conjectures about others' behaviours", what guarantees that these conjectures will turn out to be correct?***

  … maybe something! … maybe nothing!

A Nash equilibrium as a "*self-enforcing pre-play recommendation*"

- Consider an impartial strategy consultant. Suppose that before a simultaneous-move game is played, the players jointly approach the consultant, and ask him to publicly make a set of strategy recommendations (one strategy recommendation for each player), with the understanding that once all players hear the public strategy recommendations, each one of them will go back to her "private space" and then commit to her strategy of choice.

  If the consultant wants to ensure that his "***recommendation is self enforcing***" in the sense that that each player will want to follow his recommendation if she believes that all other players will follow his recommendation, what strategy combinations can he recommend?

- In a ***large class of Sim-CI games*** (that includes all games with a unique Nash eqm) – but certainly **not** in **all** ***Sim-CI games*** – the following is true: a strategy vector is a ***self-enforcing recommendation*** if and only if it is a Nash equilibrium of the game

Skepticism regarding Nash Equilibria as predictions of behaviour

Elon Kohlberg writes: **"**What is the relevance of Nash equilibrium for predicting how the players will play? Possibly none!

The one connection between the Nash equilibrium concept and an actual outcome is this: **if** the choices of rivals were correctly announced to each player in advance, **then** each must best-respond to the announcements made to her.

Thus, while the Nash equilibria of a game need not delineate its possible outcomes; they do delineate a set of ***self-enforcing norms of behavior***"

We'll seek justification for Nash equilibria from two distinct perspectives:

- Nash equilibria as a set of ***self-enforcing pre-play recommendations***
- Nash equilibria as consequences of players' ***rational introspection***

Nash equilibria as consequences of players' *rational introspection*

	10K	11K	12K	13K
10K	10 , 10	11.1, 9.9	12.2 , 9.8	13.3 , 9.7
~~11K~~	~~9.9, 11.1~~	~~11 , 11~~	~~12.1, 10.9~~	~~13.2, 10.8~~
~~12K~~	~~9.8 , 12.2~~	~~10.9, 12.1~~	~~12 , 12~~	~~13.1 , 11.9~~
~~13K~~	~~9.7, 13.3~~	~~10.8 , 13.2~~	~~11.9 , 13.1~~	~~13 , 13~~

***Claim Game* 1**: each gets {rival claim + 0.1(rival claim – own claim)}

- For each player, the pure strategies 11K, 12K, and 13K are ***strictly dominated*** by strategy 10K (for any conjecture regarding rival play)

$\Rightarrow$ A rational player must play [10K] (must never play a strictly dominated strategy)

- The unique Nash equilibrium {10K, 10K} is ***dominant strategy equilibrium***
- Claim Game 1 is in effect a "prisoner's dilemma" game

The Sibling Rivalry Game

	$S_B = 0$	$S_B = 1$	$S_B = 2$	$S_B = 3$
$S_A = 0$	24.5, 24.5	22.36, 27.36	20, 30	17.33, 32.33
$S_A = 1$	27.36, 22.36	25, 25	22.33, 27.33	19.14, 29.14
$S_A = 2$	30, 20	27.33, 22.33	24.14, 24.14	20, 25
$S_A = 3$	32.33, 17.33	29.14, 19.14	25, 20	15, 15

Two pure-strategy Nash equilibria: [$S_A = 3$, $S_B = 2$], [$S_A = 2$, $S_B = 3$]

One mixed-strategy Nash equilibrium :

[{($S_A = 2$), ($S_A = 3$); 0.853, 0.147}, {($S_B = 2$), ($S_B = 3$); 0.853, 0.147}]

Consider the parent ***publicly*** giving ***pre-play strategy recommendation*** to the two siblings as to how many bowls of soup each should order under the recognition that the siblings, when ordering soup, will not be formally bound to follow the recommendations.

What's the minimal condition that must hold for siblings to follow recommendations?

… that the recommendations be self-enforcing (i.e., if one believes that other will follow recommendation, it is in his/her interest to follow recommendation)

	10K	11K	12K	13K
10K	10 , 10	12.6 8.4	13.2 8.8	13.8 , 9.2
~~11K~~	~~8.4, 12.6~~	~~11 , 11~~	~~13.8, 9.2~~	~~14.4, 9.6~~
~~12K~~	~~8.8 , 13.2~~	~~9.2, 13.8~~	~~12 , 12~~	~~15, 10~~
~~13K~~	~~9.2, 13.8~~	~~9.6 , 14.4~~	~~10 , 15~~	~~13 , 13~~

***Claim Game* 2**: lower/higher claim gets 120% / 80% of avg. claim

- For each player, strategy 13K strictly dominated by strategy 10K
  - if 13K is deleted, strategy 12K strictly dominated by strategy 10K
  - if 13K & 12K are deleted, strategy 11K strictly dominated by 10K

$\Rightarrow$ A rational player, sufficiently convinced of rival rationality, must play 10K

- The unique Nash equilibrium {10K, 10K} is a ***dominance-solved equilibrium***

Nash Equilibrium & Rational Introspection: good news / bad news

- In some games, the Nash equilibrium is unique, and is more than just a "pre-play self enforcing recommendation":

  *in such games, **rational introspection** by each player about what strategies each player has no reason to play will **inexorably** lead a player to play her Nash equilibrium strategy, especially when it is sufficiently well known among players that all players are rational*

  ... ***but that is not the case in all games***

  [These issues are in the domain of ***Epistemic Game Theory***]
- *When will **rational introspection** work?*
- Answer depends on two factors:

(i) a game's payoff structure, which determines extent to which **algorithm of "iterated deletions of strictly dominated strategy"** can progress; and

(ii) the players' "degree of knowledge about rival rationality"

The Strict-Dominance Algorithm & Nash Equilibria

- In **Sim-CI** games in which the strict deletion algorithm converges to a unique pure strategy vector for all players, that unique pure strategy vector is necessarily the unique Nash equilibrium of the game – we refer to such Nash equilibria as ***dominance-solved equilibria***
  - This is the case for Nash equilibria in Schemes 1 & 2 of Claim Game
- Recognize that a "dominance-solved Nash equilibrium" can be reached *via* **one or more** steps of iterated deletions of strictly dominated strategies. If convergence is reached in "one step", then the Nash equilibrium is a "***dominant-strategy equilibrium***"
  - This is the case for Nash equilibrium in Scheme 1 of Claim Game
- In games in which the strict deletion algorithm does not converge, all Nash equilibria of the game are "contained" in the set of pure strategies of all players that "survive" the process
  - This is the case for Nash equilibria in Schemes 3 & 4 of Claim Game

Algorithm of '*iterated deletions of strictly dominated strategies*'

	10K	11K	12K	13K
10K	10 , 10	11.1, 9.9	12.2 9.8	13.3 9.7
11K	9.9, 11.1	11 , 11	12.1, 10.9	13.2, 10.8
12K	9.8 , 12.2	10.9 , 12.1	12 , 12	13.1 , 11.9
13K	9.7, 13.3	10.8 , 13.2	11.9 , 13.1	13 , 13

	Low volume	Medium volume	High volume
Low vol	1800 , 1800	1500 , 2000	900 , 1800
Medium vol	2000 , 1500	1600 , 1600	800 , 1200
High vol	1800 , 900	1200 , 800	0 , 0

	heads	*tails*
heads	10 , 0	0 , 10
tails	0 , 10	10 , 0
covered	4 , 6	4 , 6

	heads	*tails*
heads	10 , 0	0 , 10
tails	0 , 10	10 , 0
cover	6 , 6	6 , 6

***Claim Game* 4**: lower / higher bid gets 105% / 95% of avg. claim

	10K	11K	12K	13K
10K	10 , 10	11.025, 9.975	11.55 , 10.45	12.075 , 10.925
11K	9.975, 11.025	11 , 11	12.075, 10.925	12.6, 11.4
12K	10.45 , 11.55	10.925, 12.075	12 , 12	13.125, 11.875
13K	10.925, 12.075	11.4 , 12.6	11.875, 13.125	13 , 13

- 10K is strictly dominated for each player (by 13K)
- All other pure strategies survive the "strict deletion algorithm"
- In this game, there is no pure-strategy Nash equilibrium
- In this game 11K,12K, & 13K are Nash eqm strategies for each player
- A Nash equilibrium: each player plays {11K, 12K, 13K; 5/24, 2/3, 1/8}

Algorithm of '*iterated deletions of strictly dominated strategies*'

- In a **Sim-CI** game, a *pure strategy* s_P^i of player P is a ***strictly dominated strategy*** if and only if there exists another strategy for P – either *pure or mixed* – which gives strictly higher payoff to P than s_P^i for every (deterministic) conjecture that P can hold about rival play.
- Consider a **Sim-CI** game where at least one player has a strictly dominated pure strategy in the (full) game.

(1) Delete all strictly dominated pure strategies of each player in the full game, and so create the "***first-order pruned game***".

(2) In the ***first-order pruned game***, if no player has a strictly dominated pure strategy, then stop. If at least one player has a strictly dominated pure strategy, delete all strictly dominated pure strategies of each player to create the ***second-order pruned game***.

(3) Continue as above in the ***second-order pruned game*** ... then in the ***third-order pruned game*** ... and so on, stopping when a pruned game is reached in which no player has a strictly dominated pure strategy.

***Claim Game* 3**:

	10K	11K	12K	13K
10K	10 , 10	9.9, 9.9	10.8 , 10.8	11.7, 11.7
11K	9, 9	11 , 11	10.8, 10.8	11.7, 11.7
12K	9, 9	9.9, 9.9	12 , 12	11.7, 11.7
13K	9, 13.5	9.9, 9.9	10.8, 10.8	13 , 13

- 11K is strictly dominated for Column by 13K

⇒ 11K is strictly dominated for Row by {10K, 12K, 13K; 1/3, 1/3 1/3}

- All other pure strategies survive the "strict deletion algorithm"
- In this game, there is a unique pure-strategy Nash equilibrium
- There is also a mixed-strategy Nash equilibrium:

  [{10K,13K; 5/22, 17/22}, {10K,13K; 13/23, 10/23}]

IDSDS and the requirements of players' knowledge of rival rationality

	w	x	y	z
~~a~~	~~10, 09~~	~~09, 10~~	~~08, 09~~	~~07, 08~~
~~b~~	~~09, 07~~	~~10, 08~~	~~09, 09~~	~~08, 10~~
~~c~~	~~08, 05~~	~~09, 06~~	~~10, 07~~	~~09, 08~~
d	07, 03	08, 04	09, 05	10, 06

- Start from the assumption that each player knows that "both are rational"
- In the full game, **w** is strictly dominated by **x**

  [to cancel **w** from the game, "Red must know that Blue knows that Red is rational"]
- In the 1st order pruned game, **a** is strictly dominated by **b**

  [to cancel **a** from the game, "Blue must know that Red knows that Blue knows that Red is rational"]
- In the 2nd order pruned game, **x** and **y** are strictly dominated by **z**
- In the 3rd order pruned game, **b** and **c** are strictly dominated by **d**

Rationalizability in *Claim Game* 1:

	10K	11K	12K	13K
10K	10 , 10	11.1, 9.9	12.2 , 9.8	13.3 , 9.7
11K	9.9, 11.1	11 , 11	12.1, 10.9	13.2, 10.8
12K	9.8 , 12.2	10.9, 12.1	12 , 12	13.1, 11.9
13K	9.7, 13.3	10.8 , 13.2	11.9 , 13.1	13 , 13

- One can rationalize playing 10K in a single statement: "I am rational, therefore I will play 10K"
- One cannot rationalize playing any other strategy, because for every other strategy the following statement is true: "the strategy is not a best response to any conjecture about rival behavior"
- Here 10K is uniquely rationalizable without requiring any knowledge of rival rationality

Orders of Knowledge

- A group of individuals have 1st-order knowledge about fact ***f*** if all of them know ***f*** [*f* could be the fact that "all players are rational"]
- They have 2nd-order knowledge about fact ***f*** if each of them know that all of them have 1st-order knowledge about fact ***f***
- They have 3rd-order knowledge about fact ***f*** if each of them know that all of them have 2nd-order knowledge about fact *f*.
- ….
- A fact ***f*** is ***common knowledge*** among the group if all individuals know *f*, they all know that each knows *f*, they all know that they all know that each knows *f*, and so on *ad infinitum*.

[See *Friends*, S 5, Ep 14 for illuminating implications of lack of common knowledge]

- In a **game of complete information** in ***rational choice theory***, it is maintained that "all the details of the game" (its constituent strategy sets, and outcome and payoff functions) is common knowledge among the players, as is the fact that "all players are rational"

Rationalizability in *Claim Game* 2:

	10K	11K	12K	13K
10K	10 , 10	12.6, 8.4	13.2 , 8.8	13.8 , 9.2
11K	8.4, 12.6	11 , 11	13.8, 9.2	14.4, 9.6
12K	8.8 , 13.2	9.2, 13.8	12 , 12	15, 10
13K	9.2, 13.8	9.6 , 14.4	10 , 15	13 , 13

- When there is common knowledge (*or sufficiently high order of knowledge*) of rationality among the players, 10K is "uniquely rationalizable" for each player, for the following reasons:

1. 10K is a best response to 10K
2. For every other pure strategy of each player,

   the attempt to rationalize playing that strategy in terms of higher-order conjectures eventually breaks down [***verify this***]

Bernheim (1984) and Pearce (1984) presented an epistemic interpretation of the algorithm of IDSDS

and called it the logic of "Rationalizability"

- A basic rationality requirement for a player should be that if (s)he has played a strategy then she ought to be able to ***rationalize*** (i.e., rationally justify) the play
- In order to ***rationalize*** playing a pure strategy, a player must be able to argue that the strategy is a best response to a conjecture which is "itself rationalizable in terms of higher-order conjectures"

Rationalizability in *Claim Game* 3:

	10K	11K	12K	13K
10K	10 , 10	9.9, 9.9	10.8 , 10.8	11.7, 11.7
11K	9, 9	11 , 11	10.8, 10.8	11.7, 11.7
12K	9, 9	9.9, 9.9	12 , 12	11.7, 11.7
13K	9, 13.5	9.9, 9.9	10.8, 10.8	13 , 13

- Strict deletion algorithm first prunes 11K and then prunes 11K and stops

Row cannot ***rationalize*** 11K given any order knowledge of rival rationality:

To do so Row *must* say "I claim 11K b'cuz I conjecture that Col will claim 11K"

But then *cannot rationalize* why (s)he conjectures that Col will play a strategy that is strictly dominated in full game

Row can ***rationalize*** 13K when players' rationality is common knowledge:

"I, Row, claim 13K b'cuz I conjecture that Col will claim 13K

… b'cuz I conjecture that Col conjectures that I'll claim 10K

… b'cuz I conjecture that Col conjectures that I conjecture that Col will claim10K

… b'cuz I conjecture that Col conjectures that I conjecture that Col conjectures that I'll claim 13K

… *and then repeat this "cycle of rationalizations" unendingly*

Results regarding *Rationalizability*

- Bernheim (1984) / Pearce (1984)

1. If player rationality is ***common knowledge*** among players, then a player can rationalize a strategy <u>if and only if</u> it survives the ***strict deletion algorithm***
2. Every "Nash strategy of each player" (i.e., a strategy that she plays with positive probability in some Nash equilibrium of the game) survives the ***strict deletion algorithm*** and therefore is ***rationalizable***
3. If a Nash Equilibrium is dominance-solved, the Nash Equilibrium strategy for each player is her ***uniquely rationalizable*** strategy
4. In a game, there may exist a rationalizable strategy for a player that is not a "Nash strategy for the player"

Do 'dominance-solved equilibria' predict human behaviour accurately?

Even a dominance-solved equilibrium might not constitute a good prediction about real-world strategic behaviour because of two reasons:

(*a*) players might not be confident that rational rivals will not make mistakes in executing strategy, and/or

(*b*) players might not be confident about holding 'high-orders of knowledge about rival rationality'

Fear of mistakes by rival, and the logic of playing maximin strategies

	left	right
up	20, 10	15, 9.5
down	50, 09	-500, 8.5

- Here {*down*, *left*} is the dominance-solved Nash equilibrium
- But if ***Row*** thinks that there is a small chance (say 0.06) that ***Column*** will make a mistake and play *right* (a mistake that doesn't cost ***Column*** too much), ***Row*** will play her "maximin strategy" *up*
- A player's ***maximin strategy*** is that pure strategy which maximizes the minimum payoff that (s)he can receive when playing a pure strategy

Predicting strategic behaviour in simultaneous-move games

- If a player has a strictly dominant strategy that should be the prediction
- If that is not the case, find the Nash equilibria of the game
- If the Nash equilibrium is dominance-solved (and thus unique), then that's the best prediction of behaviour, subject to two caveats:

(*a*) payoff-profiles may be such that players have strong reasons to play their maximin strategies, and/or

(*b*) dominance-solution might require many iterations (and thus high orders of knowledge about rival rationality)

In these cases, consider maximin strategies and/or best responses to "naïve play" and/or best responses to {best responses to naïve play}

- If there are multiple rationalizable strategies for a player, then choose among these strategies on the basis of : Nash equilibrium strategies, Maximin strategies, Best responses to "naïve play", and {Best responses to best responses to naïve play}

The <u>Guessing Game</u> [with a dominance-solved Nash eqm after many iterations]

Each of ten players simultaneously writes a bid between 0 and 100. Game conductor determines 70% of the average. The player whose bid is closest to that number wins Rs.100, others get nothing. If there is a tie among the closest bids, the tied players share Rs.100 equally.

- ***The strict-dominance algorithm at work:***

I am rational ⇒ "bid > **70**" is strictly dominated by "bid **70**"

I'm rational and I know all players are rational

⇒ "bid > **49**" is strictly dominated by "bid **49**"

I'm rational and I know all players are rational, and I know that all players know that all players are rational ⇒

⇒ "bid > **34.3**" is strictly dominated by "bid **34.3**"

…

⇒ "bid > **24.01**" is strictly dominated by "bid **24.01**"

…

⇒ "bid > **16.807**" is strictly dominated by "bid **16.807**"

…

⇒ "bid > **0**" is strictly dominated by "bid **0**"

Multi-stage Games of Complete-Information

- A ***multi-stage game of complete-information*** is one where at least some actions have to be <u>implemented</u> over multiple stages
- ***Examples***: Noughts-and-Crosses, Chess, Nim, ...

A modified Nim game: There are 7 marbles in a jar. Two players alternate in collecting marbles. Each must collect 1 or 2 or 3 in each turn. Last collector gets Rs. [500 + 10×no. of marbles in possession], other gets Rs. [50 + 10×no. of marbles in possession].

- Such a game <u>can be played</u> either ***simultaneously*** or ***sequentially***
- When the game is played ***simultaneously***, each player has to commit to a "***complete contingent strategy***" up front ***that cannot be revised later***
- When the game is played ***sequentially***, each player has to specify/execute her <u>stage-specific</u> "***behavior strategy***" when every feasible stage is reached (no public pre-commitment to future behavior strategies are made or required)

<u>A Behavioural Analysis of the Guessing Game</u>

- In numerous experiments of the Guessing Game, bids are clustered between 35 and 15; people bidding between 20 and 16 typically win
- Understanding real-world play in the Guessing Game through a ***"cognitive hierarchy" model of behaviour***

1. A small fraction of players are **naïve**: they bid uniformly randomly
2. A somewhat larger fraction of players possess **1st-degree cognition:** they best-respond to **naïve** players
3. A significant fraction of players possess **2nd-degree cognition:** they best-respond to a mix of **naïve** players and players of **1st-degree cognition**
4. A small fraction of players possess **3rd-degree cognition:** they best-respond to a mix of **naïve** players, players of **1st-degree cognition**, and players of **2nd-degree cognition**

See: https://www.youtube.com/watch?v=szaXR2_Smmk

Answer Key to Problem Set 2

Problem 1: In the "Extended Matching Pennies Game – 1":

Column Player

Row Player

	Heads	*Tails*	*Covered*
Heads	$\underline{10}$, 0	0, $\underline{10}$	$\underline{6}$, 6
Tails	0, $\underline{10}$	$\underline{10}$, 0	$\underline{6}$, 6
Covered	6, $\underline{6}$	6, $\underline{6}$	$\underline{6}$, $\underline{6}$

.

First note that **if**, in a Nash equilibrium, exactly one player plays a pure strategy, that pure strategy must be *Covered*.

Next, note that *Covered* will be a best response for a player when rival is playing a mixed strategy only under the following circumstances:

(i) rival plays $\{$*Heads, Tails*; m, $(1–m)$ for some $m \in [0.4, 0.6]$)$\}$, or

(ii) rival plays $\{$*Heads, Tails*, *Covered*; p, q, $(1–p–q)$ for some non-negative $\{p, q\}$ such that $(p + q) < 1$ and $(2/3) \leq (p/q) \leq (3/2)\}$.

It is then easy to deduce all Nash equilibria in which exactly one player plays a pure strategy, and to conclude that generate identical payoffs of 6 for each player.

The set of all Nash equilibria in which exactly one player plays a pure strategy are the following:

- [Row plays *Covered*, Column plays $\{$*Heads*, *Tails*; m, $(1–m)$ for some $m \in [0.4, 0.6]$)$\}$],
- [Row plays $\{$*Heads*, *Tails*; n, $(1–n)$ for some $n \in [0.4, 0.6]\}$, Column plays *Covered*].
- [Row plays *Covered*, Column plays $\{$*Heads*, *Tails*, *Covered*; p, q, $(1–p–q)$ for some non-negative $\{p, q\}$ such that $(p + q) < 1$ and $(2/3) \leq (p/q) \leq (3/2)\}$].
- [Row plays $\{$*Heads*, *Tails*, *Covered*; x, y, $(1–x–y)$ for some non-negative $\{x, y\}$ such that $(x+y) < 1$ and $(2/3) \leq (x/y) \leq (3/2)\}$, Column plays *Covered*].

Problem 2: In "Extended Matching Pennies Game – 2":

Column Player

Row Player

	Heads	*Tails*
Heads	$\underline{10}$, 0	0, $\underline{10}$
Tails	0, $\underline{10}$	$\underline{10}$, 0
Covered	7, 5	7, $\underline{7}$

.

- Row's pure best-response set to conjecture *Heads* is $\{$*Heads*$\}$
- Row's pure best-response set to conjecture *Tails* is $\{$*Tails*$\}$
- Row's pure best-response set to conjecture [*Heads*, *Tails*; p, $1–p$] is $\{$*Heads*$\}$ for $p \geq 0.7$, $\{$*Tails*$\}$ for $p \leq 0.3$, $\{$*Covered*$\}$ for $p \in [0.3, 0.7]$, $\{$*Heads*, *Covered*$\}$ for $p = 0.7$, and $\{$*Tails*, *Covered*$\}$ for $p = 0.3$.
- Column's pure best-response set to conjecture *Heads* is $\{$*Tails*$\}$

- Column's best-response set to conjecture *Tails* is {*Heads*}
- Column's pure best-response set to conjecture {*Covered*} is {*Tails*}
- Column's pure best-response set to conjecture [*Heads*, *Covered*; p, $1-p$] is {*Tails*} for all $p \in (0, 1)$
- Column's pure best-response set to conjecture [*Tails*, *Covered*; q, $1-q$] is {*Heads*} for $q \geq 1/6$, {*Tails*} for $q \leq 1/6$, and {*Heads*, *Tails*} for $q = 1/6$.

Thus, the unique Nash equilibrium of the game is:

Row plays {*Tails*, *Covered*; 1/6, 5/6}, Column plays {*Heads*, *Tails*; 0.3, 0.7}.

Note: [1] Row's equilibrium payoff is 7, while Column's equilibrium payoff is (35/6). In the "Extended Matching Pennies Game – 2", the additional strategy at Row's disposal enables her to secure a higher payoff. This is not a robust result. In some games, the equilibrium payoff of a player can *increase* if she can "credibly contract" her strategy set.

[2] The "Extended Matching Pennies Game – 2" demonstrates that in a simultaneous-move complete-information game, a player ***can rationalize*** playing a pure strategy that is not a part of any Nash equilibrium. Here, the Row Player can rationalize playing *Heads* even though that strategy is not a part of any Nash equilibrium. [Please take the trouble to figure out the "chain of rationalization" that will allow Row Player to rationalize playing *Heads*.]

Problem Set 3

Problem 1: Consider the complete-information simultaneous-move game described by the following payoff matrix:

		Column			
		a	***b***	***c***	***d***
Row	***x***	3, 11	6, 10	9, 5	12, 8
	y	6, 6	7, 5	8, 20	14, 25

Is the Nash equilibrium of this game also a dominance-solved equilibrium? If so, specify the sequence of "iterated deletion of strictly dominated strategies".

Problem 2: Alice and Barbara are playing a one stage guessing game. Each player must simultaneously write down **an integer** between 1 and 4 (inclusive). If Alice writes down the integer $\boldsymbol{a} \in \{1, 2, 3, 4\}$ and Barbara writes down the integer $\boldsymbol{b} \in \{1, 2, 3, 4\}$, the Alice will receive $\{10 - |\boldsymbol{a} - \boldsymbol{b}|\}$ gold coins and Barbara will receive $\{10 - |2\boldsymbol{a} - \boldsymbol{b}|\}$ gold coins. Each player's objective is to maximize the number of gold coins she gets.

Write down the above game in the "payoff matrix" form.

There is a unique Nash equilibrium in this game. Find it. This Nash equilibrium is a dominance-solved equilibrium. Specify the precise sequence of "iterated deletion of strictly dominated strategies" that will converge to the unique Nash equilibrium strategy vector.

Problem 3: Adam and Eve have to simultaneously select a 'location' to reside in – Fun-city, or Joy-city, or Sin-city. Individual payoffs, which depend on the location choice of both, are given by the following payoff matrix:

		Eve		
		Fun	*Joy*	*Sin*
Adam	*Fun*	20, 8	10, 8	5, 10
	Joy	12, 6	22, 10	12, 6
	Sin	5, 10	10, 8	20, 8

What is the pure-strategy Nash equilibrium of this location choice game?

If rationality is common knowledge between Adam and Eve, identify their "rationalizable" pure strategies.

Problem 4: (*The "Travelers' Dilemma" game developed by Kaushik Basu*)

An airline loses two suitcases belonging to two different travelers. Both suitcases happen to be identical and contain identical antiques. An airline manager tasked to settle the claims of both travelers explains that the airline is liable for a maximum of \$100 per suitcase. In order to determine an honest appraised value of the antiques, the manager separates both travelers so they can't confer, and asks them to write down the amount of their value at no less than \$2 and no larger than \$100. He tells them that if both 'demand' the same number, he will treat that number as the true value of both suitcases and reimburse both travelers that amount. However, if one demands a smaller number than the other, this smaller number will be taken as the true value, and both travelers will receive that amount along with a bonus/malus: \$2 extra will be paid to the traveler who demanded the lower value and a \$2 deduction will be taken from the person who demanded the higher amount.

If you were one of the travelers, how much would you demand? How would you "rationalize" your demand?

Problem 5: *Algorithm to find all Nash equilibria in a single-stage complete-information game*

Step 1. Do the "underlining", i.e., determine each player's (set of) pure best-responses for every deterministic conjecture about rival play.

Step 2. Using the underling as a guide, carry out the algorithm of "iterated deletion of strictly dominated strategies (IDSDS)" in the game. Remember that at each stage of iteration, a strictly-dominated strategy must be "never underlined", and in addition there must exist another strategy for the player (pure or mixed) that strictly vector-dominates it.

Step 3. If the IDSDS algorithm converges to a unique strategy vector of all the players, you have uncovered the dominance-solved Nash equilibrium of the game. If the IDSDS algorithm does not converge, focus only on the "pruned game" in which no player has a strictly dominated strategy. In that pruned game,

(i) determine each player's set of pure best-responses for every stochastic conjecture about rival play given all (surviving) rival pure strategies, and

(ii) then determine her entire best response set for every conjecture about rival play by recognizing that whenever she has multiple pure best responses for a specific conjecture, all probability distributions over those pure best responses will be her mixed best responses;

(iii) finally, determine "all intersections of all best response correspondences" of the players.

Use the above algorithm to determine all Nash equilibria in the following game.

		Column		
		x	***y***	***z***
Row	***a***	10, 11	9, 9	11, 10
	b	9, 9	11, 11	10, 12
	c	9, 9	9, 9	12, 12

Journal of Economic Perspectives—Volume 6, Number 4—Fall 1992—Pages 83–101

Knowledge and Equilibrium in Games

Adam Brandenburger

In recent years, economists have used game-theoretic reasoning in a variety of areas: the economics of business strategy, the economics of incentives, the theory of bargaining, the theory of bidding, trade policy, monetary policy, and others. The part of game theory most frequently applied falls under the heading of "non-cooperative" game theory. A non-cooperative game is one in which each player is faced with a specified set (or series of sets) of choices. In making choices in actual strategic situations, people must grapple with questions such as: What will the other people involved do? What are their motivations? What do they think I will do? And so on. This paper describes an approach to non-cooperative game theory that aims to capture considerations that exercise the minds of real-world strategists.

The most commonly used tool of non-cooperative game theory is the Nash equilibrium (Nash, 1951). This raises the question: Are there assumptions on what the players in a game think—including what they think other players think, and so on—that lead to consideration of Nash equilibrium? The paper provides answers to this, and related, questions.

The approach of this paper involves analyzing the decision problem facing each player in a strategic ("interactive") situation. It might be described as doing "interactive decision theory." In addition to grounding game theory in considerations that are of the essence in actual strategic situations, the approach has a number of other objectives. As conventionally formulated, non-cooperative game theory seems somewhat divorced from single-person decision theory, with its special apparatus of strategy randomization, equilibrium points,

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and the like. By contrast, the approach of this paper aims to make game theory more immediately accessible to people who are trained in decision theory but who are not "game theorists." A related objective is to make game theory easier to teach to students. Finally, the approach suggests new directions for research into the nature of strategic situations. Some further thoughts along these lines are offered in the closing section of the paper.

Common Knowledge of Rationality

A finite non-cooperative game in "strategic" (or "normal") form consists of a finite set of players and, for each player, a finite set of choices and a payoff function associating a payoff to that player with every specification of choices by all the players.[1] This formulation of a game might seem rather restrictive; for example, it appears not to allow for any temporal structure to the game, for any information that one player might have about previous choices made by other players and the like. Nevertheless, von Neumann and Morgenstern (1944) argued that the formulation is in fact entirely general. Their key postulate is that while the rules of a particular game may involve a player making a series of choices, all these choices can be collapsed into the single choice of a *strategy*, where a strategy is a specification of what choice the player would make in any contingency that could arise during the play of the game. Accordingly, this paper will focus on the strategic form of a game.

In the strategic form, each player must decide on a particular strategy in ignorance of the strategies chosen by the other players. This is because a player cannot condition plans on those of other players. Conditioning choices on previously observed choices of other players is, of course, allowed for in the definition of a strategy. A player's choice of strategy, then, must in general depend on a subjective assessment of the relative likelihood of different strategy choices by the other players.

Cast in this form, the problem of choosing a strategy in a game is a decision problem under uncertainty. This perspective on non-cooperative game theory is not the standard one. Traditionally, decision theory has been perceived as appropriate only when the uncertainty pertains to "exogenous" events (or to the "state of nature" as it is sometimes termed), not when the uncertainty is over the strategy choices of players in a game. For the latter, "endogenous" variety of uncertainty, it has been customary to argue that an equilibrium, or other game-theoretic, construct must be employed.[2] By contrast, the view espoused in this paper is that there is no distinction between the two types of

[1] In applications, it is often assumed that the players' choice sets are infinite—say, closed intervals of the real line. However, to avoid technical complications, this paper will treat only finite games.
[2] For example, Harsanyi (1967–68, p. 167) has written: "[E]very player i ... will assign a *subjective* joint probability distribution P_i to all variables unknown to him—or at least to all unknown *independent* variables, i.e., to all variables not depending on the players' own strategy choices."

Figure 1
A Simple Two Person Game

		Player 2		
		L	*C*	*R*
Player 1	*U*	4, 10	3, 0	1, 3
	D	0, 0	2, 10	10, 3

uncertainty and that a player in a game should assign subjective probabilities to *all* uncertainty, including the strategy choices of other players.

Consider, for example, the game depicted in Figure 1, which shows the possible strategy choices for each player beside the rows and above the columns, and the payoff to each player inside the cells, with player 1's payoff coming first. We assume that player 1 chooses a strategy that maximizes expected payoff given some subjective assessment over player 2's choice. Does this assumption allow us to reach any conclusions concerning player 1's choice? The answer is evidently no: either *U* or *D* might be optimal for player 1, depending on that player's precise assessment. What about player 2? Simply knowing that player 2 maximizes, given some assessment over player 1's choice, is sufficient to rule out player 2 choosing *R*. If player 2 assesses probability greater than 1/2 to player 1 choosing *U*, then player 2's best choice is *L*; if player 2 assesses probability less than 1/2 to player 1 choosing *U*, then player 2's best choice is *C*; finally, either *L* or *C* is best if player 2 assesses exactly probability 1/2 to player 1 choosing *U*.

Now comes an important assumption: suppose that the conclusion just reached concerning player 2 is available to player 1. What exactly is entailed by this assumption will be discussed in a moment. First, however, let us see where this line of reasoning leads. If player 1 can indeed eliminate the possibility of player 2 choosing *R* then player 1's subjective assessment should assign probability 0 to this event, in which case *U* is a better choice for player 1 than is *D*. If this latter conclusion is in turn available to player 2 then player 2 will choose *L*. Our reasoning has led to the conclusion that player 1 will choose *U* and player 2 will choose *L*.

What informational assumptions are implicit in this line of reasoning? First, player 1 is assumed to know that player 2 is *rational*, which means that the player chooses a strategy that maximizes expected payoff, given some probabilistic assessment over the other players'—in this case, player 1's—strategy choices. Second, player 1 is also assumed to know player 2's payoffs. These two assumptions together allow player 1 to conclude that player 2 will not choose *R*. Similarly, player 2 is assumed to know that player 1 is rational and to know player 1's payoffs, and to know that player 1 knows player 2 is rational and that player 1 knows player 2's payoffs. These assumptions allow player 2 to infer that player 1 will choose U.

The general version of these assumptions is the supposition that both the structure of the game (the set of players, the sets of strategies, and the payoff functions) and the rationality of the players are "common knowledge." To say that a fact is *common knowledge* means that everyone knows it, everyone knows that everyone knows it, and so on ad infinitum. The term "common knowledge" was first used in this connection by Lewis (1969), who attributes the basic idea to Schelling (1960). Aumann (1976) offered a precise mathematical formulation of the concept. The consequence of this supposition is that for each player any strategy that does not maximize expected payoff for some probabilistic assessment over the other players' choices should be deleted from the game. In the reduced game that results, again for each player any strategy that does not maximize expected payoff for some probabilistic assessment over opponents' choices in the reduced game should be deleted. This procedure continues until no further deletion is possible.

This iterated deletion procedure can be characterized in another useful way. Recall the definition of a strongly dominated strategy: a strategy of a player is strongly dominated if that player has some other (possibly mixed) strategy that, for every specification of choices by the other players, yields a strictly higher expected payoff. A well-known result states that a strategy is strongly dominated if and only if there is no probability distribution over the choices of the other players for which the strategy in question maximizes the player's expected payoff.[3] Thus in the game depicted in Figure 1, the strategy R of player 2 is strongly dominated by a mixed strategy that gives weight 1/2 to each of L and C. A strategy will be called *iteratively undominated* if it remains after all strongly dominated strategies have been deleted from the game, all strategies that become strongly dominated in the reduced game are then eliminated, and so on. The conclusion of the preceding paragraph can now be restated as the following proposition, versions of which are proved, with differing degrees of formality, in Bernheim (1984, 1986), Pearce (1984), Brandenburger and Dekel (1987), and Tan and Werlang (1988).

Proposition 1 *Suppose that the structure of the game and the rationality of the players are common knowledge. Then each player chooses an iteratively undominated strategy.*

A question arises as to whether the full force of common knowledge is really needed here. For example, in the game depicted in Figure 1, player 1's knowledge about player 2, player 2's knowledge about player 1, and player 2's knowledge about player 1's knowledge about player 2 sufficed. In other words, knowledge to (at most) two levels was needed. More generally, for a given finite game there is an obvious upper bound on the number of levels of

[3] The necessity direction of this result is straightforward; sufficiency is a deeper result that relies on the supporting hyperplane theorem. For a proof see, for example, Pearce (1984, Appendix B).

knowledge needed to conclude that players choose iteratively undominated strategies. However, if one is to state a sufficient condition that applies to all finite games, common knowledge is needed.

A point worth emphasizing is that the development so far assumes that each player makes a definite choice of strategy, with no attempt to randomize. In terms of the usual parlance, players choose "pure" and not "mixed" strategies.[4] There is, of course, no difficulty if a player should happen to want to choose a mixed strategy. A commitment to roll a die, for example, can simply be viewed as an extra pure strategy. However, at no point in what follows do we require players to choose mixed strategies.

At this juncture, the reader familiar with this sort of analysis may be asking what is really new here. After all, arguments based on dominance and iterated dominance go back a long way in game theory (for example, Luce and Raiffa, 1957). Nevertheless, it is only rather recently that the decision-theoretic foundations of iterated dominance have been spelled out in detail, as in the papers cited immediately before Proposition 1. Nor is the delay a mystery: a proper understanding could not develop until the crucial notion of common knowledge was in place.

In fact, Bernheim and Pearce are led to consider "rationalizable" strategies, which differ somewhat from iteratively undominated strategies, because they make a further common knowledge assumption: they require it to be common knowledge that each player's probability distribution on the strategy choices of the other players exhibit no dependence between those choices. In the language of probability theory, the distributions are required to be "stochastically independent."

In two-person games rationalizable strategies are clearly the same as iteratively undominated strategies, since a player faces only one other player and hence the stochastic independence requirement has no bite.

For games with more than two players, any strategy which is rationalizable is iteratively undominated. The three-person game depicted in Figure 2 demonstrates that the converse is false. Player 1 chooses the row, player 2 the column, and player 3 the matrix. No strategy of any player is strongly dominated, hence all strategies are iteratively undominated. In particular, notice that B is an optimal strategy for player 3 if that player assesses probability 1/2 to player 1 choosing U and player 2 choosing L, and probability 1/2 to player 1 choosing D and player 2 choosing R. This distribution is stochastically dependent: player 3 believes that if player 1 chooses U, player 2 chooses L; if player 1 chooses D, player 2 chooses R.

On the other hand, if player 3 is restricted to an independent distribution over player 1's and player 2's choices, a simple calculation shows that B can never be optimal. Following the deletion of B, next D and R, and then finally

[4] The reader should not be misled into thinking that play of mixed strategies is somehow involved by virtue of the mention of mixed strategies in the definition of a strongly dominated (pure) strategy. Mixed strategies appear in this context as a formal device only; they are used to test whether a given (pure) strategy should be deleted from the game.

Figure 2
A Simple Three Person Game

	L	R
U	1, 1, 1	1, 0, 1
D	0, 1, 0	0, 0, 0

A

	L	R
U	2, 2, 2	0, 0, 0
D	0, 0, 0	2, 2, 2

B

	L	R
U	1, 1, 0	1, 0, 0
D	0, 1, 1	0, 0, 1

C

C can be deleted leaving only *U*, *L*, and *A* as rationalizable strategies in the sense of Bernheim and Pearce.

Non-cooperative game theory has traditionally taken the route of assuming stochastic independence, on the grounds that this captures the idea that strategies are chosen "independently." Thus in the example just described, player 3's dependent probability distribution would be ruled out for failing to reflect the "independence" of player 1's and player 2's choices. But this argument is far from watertight, as Aumann (1987, p. 16) explains:

> In games with more than two players, correlation may express the fact that what 3, say, thinks that 1 will do may depend on what he thinks 2 will do. This has no connection with any overt or even covert collusion between 1 and 2; they may be acting entirely independently. Thus it may be common knowledge that both 1 and 2 went to business school, or perhaps to the same business school, but 3 may not know what is taught there. In that case 3 would think it quite likely that they would take similar actions, without being able to guess what those actions might be.

An analogy (for which I am grateful to Bob Aumann) may be helpful here. In elementary probability theory a sequence of coin tosses is often represented by an independently and identically distributed (i.i.d.) sequence of random variables. But this is appropriate only when the parameter of the coin is known with certainty before the first toss. In any other case one would surely not assess the same probability to heads on the first toss, and to heads on the 101st toss following 90 heads in the first 100 tosses, yet that is what independence demands.[7] Likewise, in the game theoretic context there would in general be information about a second player's choice of strategy to be gleaned from observation of a first player's choice: dependence in probability assessments

[7]The answer in probability theory is to assume instead that the sequence of random variables is "exchangeable" (in the sense of de Finetti). This captures the "physical independence" of successive tosses, yet allows that successive outcomes contain information that can be used to modify probability estimates.

should therefore be permitted. The assumption of independence, which is normally accepted without discussion in non-cooperative game theory, in fact becomes quite questionable once the decision-theoretic viewpoint is adopted.

Pure-Strategy Nash Equilibrium

In the game depicted in Figure 1, the assumption of common knowledge of the structure of the game and the rationality of the players was sufficient to pin down a unique choice for each player. In many games, however, the assumption has little bite by itself, and places few or no restrictions on the strategies the players can choose. In the game depicted in Figure 3 (based on a game in Bernheim, 1984), no strategy is strongly dominated, and so the common knowledge assumption does not exclude any choice of either player from consideration.

However, consider the case where player 1 decides to play T after assessing probability one to player 2 choosing L, and player 2 chooses L after assessing probability one to 1 choosing B. While this situation is not ruled out by anything in our discussion so far, there is nevertheless an "inconsistency:" player 2's choice is based on an incorrect assessment of player 1's choice. Of course, this "inconsistency" may simply be an accurate reflection of the uncertainty facing player 2 but, while admitting this as a realistic possibility, we are nevertheless going to pursue the consequences of ruling out such situations.

Call a fact *mutual knowledge* if everyone knows it. This is to be distinguished from common knowledge, which also requires higher levels of knowledge—everyone knows that everyone knows, and so on. The "inconsistency" just discussed is eliminated by supposing the players' strategy choices to be mutual knowledge. To see how this works, refer again to the game in Figure 3. Can the pair of strategies (T, L) be mutual knowledge? The answer is "no," provided

Figure 3
Illustrating Mutual Knowledge

		Player 2		
		L	C	R
	T	7, 0	0, 5	0, 7
Player 1	M	5, 0	2, 2	5, 0
	B	0, 7	0, 5	7, 0

player 2 is rational, for then player 2 will choose R rather than L. In fact, the only pair of strategies that can be mutual knowledge, given that both players are rational, is (M, C). This strategy pair also constitutes a *pure-strategy Nash equilibrium*: a profile of (pure) strategy choices such that each player's choice is optimal given the choices of the other players. The general connection is summarized in the following proposition (Aumann and Brandenburger, 1991).

Proposition 2: Suppose that each player is rational and that the strategy choices of the players are mutual knowledge. Then the choices constitute a pure-strategy Nash equilibrium.

The proof is immediate: Since a player is assumed to know the strategy choices of the other players, rationality of the player means that the player's own choice must be optimal against the other choices. But this is exactly the definition of a Nash equilibrium.

While the proposition may seem transparent, it does contain a surprise: common knowledge is not needed. This is despite the folklore that has grown up connecting the notions of Nash equilibrium and common knowledge. The basis of the folklore goes something like this: player 1 plays his part of a Nash equilibrium because he thinks that player 2 is playing her part, a belief which is held because he thinks she thinks he is playing his part, and so on. The circularity of Nash equilibrium does indeed seem related to the infinite hierarchy of beliefs inherent in common knowledge, yet Proposition 2 suggests that the relationship is illusory. This is perhaps the more unexpected given that common knowledge plays a role in Proposition 1. In fact, we shall see later that common knowledge does crop up in the context of Nash equilibrium, but in a manner more subtle than that suggested by conventional wisdom.

The hypothesis of Proposition 2 leads, in the case of the game depicted in Figure 3, to a definite specification of the players' choices (for player 1 the choice M and for player 2 the choice C), but in general such a sharp conclusion is not possible. If a game possesses several pure-strategy Nash equilibria, then play of any of the equilibria is consistent with the hypothesis of Proposition 2; which equilibrium will obtain depends on the precise mutual knowledge that the players possess.

Mixed-Strategy Nash Equilibrium

While some games possess multiple Nash equilibria, others possess none at all, at least if players choose only pure strategies. A simple example is depicted in Figure 4. This game possesses no pure-strategy Nash equilibria, from which it follows (by Proposition 2) that no pure-strategy pair can be mutual knowledge, given that both players are rational.

The traditional way to "solve" a game such as that depicted in Figure 4 is to augment the players' choices to include mixed strategies, that is, randomized choices over the sets of pure strategies. In the example of Figure 4, there is a

Figure 4
A Game with No Pure Stategy Equilibria

		Player 2	
		L	*R*
Player 1	*U*	2, 0	0, 1
	D	0, 1	1, 0

(unique) Nash equilibrium in mixed strategies, in which player 1 chooses *U* with probability 1/2 and *D* with probability 1/2, while player 2 chooses *L* with probability 1/3 and *R* with probability 2/3. Each player's mixed strategy maximizes expected payoff given the other player's mixed strategy.

A fundamental result of non-cooperative game theory (Nash, 1951) states that if mixed strategies are introduced, then every finite game possesses an equilibrium. But the use of mixed strategies, while mathematically convenient, has always been conceptually troubling. To begin with, there is never a strict incentive for a player to randomize. Take player 1 in the example: given player 2's mixed strategy, player 1 is indifferent between carrying out the 50:50 randomization described above and any other randomization between *U* and *D* (including choosing either *U* or *D* for sure). This would not matter if no such "deviation" by player 1 upset the equilibrium in question, but this is not the case. It is only when player 1 performs the 50:50 randomization that player 2 is prepared to play her part of the equilibrium. Further clouding the picture is the feeling that people simply do not randomize when making decisions. Given these difficulties, it is perhaps not surprising that many economists have tended to reject use of equilibria that involve mixed strategies.

Does the decision-theoretic approach to non-cooperative game theory shed some light on the matter? In particular, can a different interpretation be placed on mixed strategies? And, if so, what assumptions on the players' state of knowledge lead to mixed-strategy equilibrium?

In answering, it will be helpful to begin with a definition: the *conjecture* of a player is the player's probability assessment over the strategy choices of the other players. Consider now a mixed-strategy equilibrium. Allow that players do not actually randomize; rather, each player chooses some definite pure strategy. Allow also that the other players need not know which one, or, in other words, that choices need not be mutual knowledge. Each player's mixed strategy can then be thought of as representing not a conscious randomization by that player, but rather the (common) conjecture held by the other players about that player's choice.

This interpretation of mixed-strategy equilibrium, which originated in Harsanyi (1973) and was further developed by Armbruster and Böge (1979), Böge and Eisele (1979), Aumann (1987), Tan and Werlang (1988), and

Brandenburger and Dekel (1989), among others, offers an appealing solution to the interpretational difficulties described above. Wholesale rejection of mixed-strategy equilibria is unnecessary; instead, a mixed-strategy equilibrium can be viewed as an "equilibrium in conjectures."

To take an example, the mixed-strategy equilibrium of the game in Figure 4 can be interpreted as an "equilibrium in conjectures," in which player 1 assesses probability 1/3 to player 2 choosing L and probability 2/3 to player 2 choosing R, and player 2 assesses probability 1/2 to player 1 choosing U and probability 1/2 to player 1 choosing D.

The question, however, remains: how does such an "equilibrium in conjectures" come about? In particular, what assumptions on the players' state of knowledge lead to their conjectures being in equilibrium? It turns out that the answer is significantly different in the two-player and many-player situations. The following result provides the answer for two-player games (Aumann and Brandenburger, 1991).

Proposition 3: Consider a two-person game. Suppose that the structure of the game, the rationality of the players, and their conjectures are mutual knowledge. Then the conjectures constitute a mixed-strategy Nash equilibrium.

To see why Proposition 3 is true, it will help to have in mind the following characterization of Nash equilibrium. The mixed-strategy pair (σ_1, σ_2), where σ_1 is a mixed strategy of player 1 and σ_2 is a mixed strategy of player 2, is a Nash equilibrium if and only if every pure strategy of player 1 assigned positive weight by σ_1 maximizes player 1's expected payoff given that player 2 is playing σ_2 and every pure strategy of player 2 assigned positive weight by σ_2 maximizes player 2's expected payoff given that player 1 is playing σ_1. Turning to the proof of Proposition 3, let us, in a suggestive notation, write σ_1 for the conjecture that player 1 knows player 2 to hold concerning player 1's choice of strategy. Similarly, let σ_2 denote the conjecture that player 2 knows player 1 to hold concerning player 2's choice of strategy. Now, since player 1 knows player 2 to be rational and also knows player 2's payoff function, any strategy of player 2 to which player 1 assigns positive probability, that is, any strategy of player 2 assigned positive weight by σ_2, must maximize player 2's expected payoff given the conjecture σ_1, that player 1 knows player 2 to hold. Similarly, since player 2 knows player 1 to be rational and also knows player 1's payoff function, any strategy of player 1 to which player 2 assigns positive probability, that is, any strategy of player 1 assigned positive weight by σ_1, must maximize player 1's expected payoff given the conjecture σ_2, that player 2 knows player 1 to hold. The preceding two sentences establish that (σ_1, σ_2) satisfies the condition characterizing Nash equilibrium.

As is evident from the proof, a corollary of Proposition 3 is that the players' expected payoffs are also determined: they are equal to the expected payoffs in the mixed-strategy Nash equilibrium to which the players' conjectures correspond. Notice that it is *not* claimed that the players' (pure) strategy choices are

Figure 5
An Interactive Belief System

	L_1	L_2	L_3
U_1	A_1 (2/5)	B_1 (1/5)	
U_2		A_2 (1/5)	B_2 (1/10)
U_3			A_3 (1/10)

in equilibrium. Indeed this may be impossible, as in the case of the game of Figure 4.

The conditions of Proposition 3 are insufficient in games with more than two players. To see this, consider a three-person game in which players 1 and 2 each have just one choice—say U for player 1 and L for player 2—and player 3 has two choices, A and B.[6] We are going to describe a situation in which there is mutual knowledge of conjectures, yet player 1's and player 2's conjectures about player 3 disagree, so that no well-defined mixed-strategy profile emerges. The payoffs are unimportant for our purposes and so will be left unspecified, but they could easily be chosen so as to make both the structure of the game and the players' rationality mutual knowledge. Hence the example will suffice to establish the claim made at the beginning of this paragraph.

To construct the situation, we introduce a little more formal apparatus than has been employed so far. Accordingly, consider the "interactive belief system" depicted in Figure 5. Here it is assumed that there are three different possible "types" of player 1, corresponding to the three rows and labelled U_1, U_2, and U_3, three different possible "types" of player 2, corresponding to the three columns and labelled L_1, L_2, and L_3, and five different possible "types" of player 3, corresponding to the five boxes and labelled A_1, B_1, A_2, B_2, and A_3, where a *type* of a player describes the player's strategy choice and the probabilities the player assesses to the various types of the other players.[7]

The labelling of types indicates each type's strategy choice: U for all types of player 1, L for all types of player 2, and alternately A and B for the various types of player 3.

[6] The following example is taken from Aumann and Brandenburger (1991), and is based on an idea of John Geanakoplos.

[7] More generally, a type of a player also describes the player's payoff function, but we avoid this extra complexity here. See Aumann and Brandenburger (1991) for a full treatment.

As for probabilities, it is assumed in Figure 5 that the types' assessments are consistent with the existence of a "common prior." To understand this idea, it will help to have the notion of a "state of the world," which is simply a specification of a type for each player. Thus in the example there are $3 \times 3 \times 5 = 45$ different possible states of the world. A *common prior* is then a probability distribution on the states of the world such that for each player, the probability assessment of a given type of that player is equal to the conditional distribution obtained from the common prior by conditioning on the event that the player's type is in fact the given type.

The numbers in parentheses in Figure 5 give the common prior probabilities of the five states indicated. (All other states are assumed to have zero prior probability.) What, then, is the probability assessment of type U_1 of player 1? The answer is found by conditioning on the row in question, namely the first one. Thus type U_1 assesses probability 2/3 to player 2 being of type L_1 and player 3 being of type A_1, and probability 1/3 to player 2 being of type L_2 and player 3 being of type B_1. To take another example, the probability assessment of type L_2 of player 2 is found by conditioning on the second column: type L_2 assesses probability 1/2 to player 1 being of type U_1 and player 3 being of type B_1, and probability 1/2 to player 1 being of type U_2 and player 3 being of type A_2. Finally, player 3 always assesses probability one to the true state of the world: if, for example, the true state is (U_2, L_2, A_2), player 3 assesses probability one to player 1 being of type U_2 and player 2 being of type L_2.

An interactive belief system is a formal description of the players' choices, of their conjectures about other players' choices, of their beliefs about other players' conjectures, and so on. It should be clear how a particular state of the world determines the players' choices and also their conjectures. Take the state (U_1, L_1, A_1). The players' choices are evidently U, L, and A, respectively. Player 1's conjecture derives immediately from type U_1's assessment: player 1 assesses probability 2/3 to player 2 choosing L and player 3 choosing A, and probability 1/3 to player 2 choosing L and player 3 choosing B. The other players' conjectures are found in similar fashion.

But what is player 1's belief concerning player 2's conjecture, or player 1's belief concerning player 2's belief concerning player 3's conjecture, and so on? Are these higher-order beliefs already determined in some fashion, or must they too be specified? The beauty of the interactive belief system model is that, appearances to the contrary, all these higher-order beliefs are already determined. To see this, simply note that, at the state (U_1, L_1, A_1) say, player 1 assesses probability 2/3 to player 2 being of type L_1 and probability 1/3 to player 2 being of type L_2. Hence player 1 believes that player 2 either: a) assesses probability one to player 1 choosing U and player 3 choosing A; or b) assesses probability 1/2 to player 1 choosing U and player 3 choosing B, and probability 1/2 to player 1 choosing U and player 3 choosing A. Indeed, player 1 assesses probability 2/3 to the first situation and probability 1/3 to the second. It should be apparent that Figure 5 can be used in like fashion to calculate all higher-order beliefs.

Notice an implicit assumption here: the interactive belief system is in some sense "transparent" to the players themselves. This is not, in fact, a formal assumption; rather, it is a tautology, reflecting the fact that the interactive belief system already describes any ignorance on the part of the players.[8]

Before proceeding with using the interactive belief system of Figure 5 to describe a situation of mutual knowledge of conjectures, let us delay a little longer and explain why it was important to examine the interactive belief system model in such detail. This paper began by promising to be much more explicit than is customary in non-cooperative game theory about the decision problem facing each player in a game. The interactive belief system model gives rigorous shape to this approach, and allows one to state and prove all the results described in this paper as formal theorems. This applies not only to Propositions 4 and 5 below, which rely on the interactive belief system model in an obvious way, but also to Propositions 1–3. Although the argument for the latter proceeded at an intuitive level, the reader should now be in a position to see that they can also be given formal content: they are not simply "stories."

Consider again Figure 5. We assert that if the true state is (U_2, L_2, A_2), then the players' conjectures are mutual knowledge. At this state, a) player 1 assesses probability 2/3 to player 2 choosing L and player 3 choosing A, and probability 1/3 to player 2 choosing L and player 3 choosing B; b) player 2 assesses probability 1/2 to player 1 choosing U and player 3 choosing B, and probability 1/2 to player 1 choosing U and player 3 choosing A; and, finally, c) player 3 assesses probability one to player 1 choosing U and player 2 choosing L. Player 1 knows player 2's and player 3's conjectures, as they are the same at the only other state, (U_2, L_1, B_2), that player 1 considers possible. Similarly, player 2 knows player 1's and player 3's conjectures, as they are the same at the only other state, (U_1, L_2, B_1), that player 2 considers possible. Finally, player 3 knows player 1's and player 2's conjectures, since player 3 knows the true state. So the conjectures are mutual knowledge. Yet player 1's and player 2's conjectures about player 3 disagree: player 1 thinks it twice as likely that player 3 will choose A than B, while player 2 considers the two choices equally likely. Hence no well-defined mixed-strategy profile emerges. The conclusion is that the conditions of Proposition 3 do not suffice once there are more than two players.

Notice that at (U_2, L_2, A_2) player 2 does not know that player 1 knows player 2's conjecture. The reason is that player 2 thinks it possible that player 1 is of type U_1, in which case, as we saw above, player 1 entertains two different

[8] Further discussion of this and other aspects of the interactive belief system model can be found in Aumann (1987) and Aumann and Brandenburger (1991). Relevant foundational papers include Armbruster and Böge (1979), Mertens and Zamir (1985), Tan and Werlang (1988), and Brandenburger and Dekel (1993).

possibilities for the conjecture that player 2 holds. Certainly the players' conjectures are not common knowledge at (U_2, L_2, A_2). It turns out that exactly this extra condition is sufficient for mixed-strategy Nash equilibrium in many-person games, as the following result indicates (Aumann and Brandenburger, 1991).

Proposition 4: Suppose that there is a common prior on the set of states of the world. Suppose further that at some state the structure of the game and the rationality of the players are mutual knowledge, and that the conjectures of the players are common knowledge. Then for each player, all the other players hold the same conjecture about that player, and the resulting profile of conjectures constitutes a mixed-strategy Nash equilibrium.

We discussed earlier the folklore connecting Nash equilibrium and common knowledge and saw that the precise nature of the connection, if indeed any existed, was not obvious. With Proposition 4, common knowledge finally enters the picture. But the connection that emerges is unexpected: it is common knowledge of conjectures, not of the structure of the game nor of the players' rationality, that matters for Nash equilibrium—and then only in games with more than two players.[9] Once on the scene, however, common knowledge is needed in full force. It can be shown, using a modification of the interactive belief system of Figure 5, that Proposition 4 is false if common knowledge of conjectures is replaced by iterated knowledge of conjectures to any finite number of levels. Aumann and Brandenburger (1991) offer details.

Proposition 4 bears a relation to Aumann's (1976) well-known theorem on the impossibility of "agreeing to disagree:" if two individuals start with a common prior and their beliefs concerning some event are common knowledge, then their beliefs must be equal.[10] Likewise, in the game-theoretic context, if two players' conjectures are common knowledge, then their conjectures about a third player's choice of strategy must agree. Hence common knowledge of conjectures rules out the situation earlier in which player 1's and

[9]Ben Polak has pointed out the following interesting result. Suppose that at some state the structure of the game and the players' conjectures are common knowledge. Then if rationality is mutual knowledge it is also common knowledge. (For a proof, see Aumann and Brandenburger, 1991.) In this sense, common knowledge of rationality does play a part in equilibrium, but it should be viewed not as a condition for equilibrium but rather as an "epiphenomenon" of (somewhat tightened) conditions for equilibrium.

[10]To gain some intuition for this result, consider two individuals 1 and 2 whose beliefs concerning some event E are common knowledge. Since 1's belief is common knowledge, it must be that the conditional probability of E is the same, equal to $\frac{1}{3}$ say, whether conditioned on: a) the information 1 actually possesses, or b) any piece of information that 2 thinks 1 might possess (otherwise 2 would not know 1's belief); or c) any piece of information that 1 thinks 2 thinks 1 might possess (otherwise 1 would not know that 2 knows 1's belief); and so on. It follows that the conditional probability of E is again $\frac{1}{3}$ if conditioned only on the union of all these pieces of information. But exactly the same argument can be made for individual 2 and, since the unions are the same for both people (this needs some thought!), the conditional probability of E calculated in this latter fashion must also be $\frac{1}{3}$. This will be so only if the conditional probability of E, conditioned on 2's information, is $\frac{1}{3}$.

player 2's conjectures concerning player 3's choice differed. (Recall that player 1's and player 2's conjectures were mutual knowledge, but not common knowledge.) Proposition 4 goes further than this, however, in that common knowledge of conjectures also implies that each player's conjecture concerning the choices of the other players is stochastically independent. It is this, together with agreement between the players' conjectures, that leads to Nash equilibrium.

We end this section with a brief remark on the assumption of a common prior. This assumption, which is discussed at length in Harsanyi (1967–68) and Aumann (1987), always generates much controversy. Here we content ourselves with observing that the assumption of a common prior is implicit throughout almost all of economic theory. It serves to focus attention on the role of informational differences in interactive situations; without it, analysis is often inconclusive. No doubt, the last word remains to be written on this subject.

Correlated Equilibrium

The conditions of Propositions 1–4 refer to the state of knowledge of the players themselves. This "inside" perspective is natural, given the decision-theoretic approach of the paper.

However, this section alters perspective and asks the following question from the point of view of an outside observer. Suppose we are given an interactive belief system with a common prior, in which each player is rational at every state of the world. What probability distribution will we, as outside observers, assign to the players' strategy choices? At first sight it would seem that very little can be said about this distribution. After all, the answer would appear to depend on all the details of the interactive belief system, which could be very complicated. Nevertheless, a definite answer exists, as the following result indicates (Aumann, 1987).

Proposition 5: Suppose that there is a common prior on the set of states of the world and that each player is rational at every state. Then the distribution of the players' strategy choices constitutes a correlated equilibrium distribution.

To see what a correlated equilibrium distribution is, and to understand why Proposition 5 is true, consider the game, sometimes called "chicken," depicted in Figure 6a.[13] This game has three Nash equilibria: a) one in which player 1 chooses *D* and player 2 chooses *L*; b) one in which player 1 chooses *U* and player 2 chooses *R*; and c) one in which (in the usual terminology) player 1 chooses *U* with probability 2/3 and *D* with probability 1/3 while player 2 chooses *L* with probability 2/3 and *R* with probability 1/3.

[13]Although this is a two-person game, there is no loss of generality since the theory of correlated equilibrium is the same for two-person and many-person games.

Figure 6
A Game of Chicken

Player 2

Player 1	L	R
U	6, 6	2, 7
D	7, 2	0, 0

(a)

	L	R
U	p	q
D	r	s

(b)

	L	R
U	$\frac{1}{3}$	$\frac{1}{3}$
D	$\frac{1}{3}$	0

(c)

A correlated equilibrium distribution, on the other hand, is a probability distribution (p, q, r, s) on the four possible pairs of choices in Figure 6*a*, as illustrated in Figure 6*b*, which satisfies the following system of linear inequalities:

$$6p + 2q \geq 7p,$$
$$7r \geq 6r + 2s,$$
$$6p + 2r \geq 7p,$$
$$7q \geq 6q + 2s.$$

What conditions do these inequalities embody? The first says that if the probability of player 1 choosing U is positive, then U is optimal for player 1 given the distribution on player 2's choices calculated by conditioning on player 1 playing U. (To see this, divide both sides by $(p + q)$.) The remaining inequalities refer to D, L, and R, respectively.

The distribution of strategy choices that arises in a Nash equilibrium satisfies the defining inequalities for a correlated equilibrium distribution, with the added property that the distribution is stochastically independent. Hence, all Nash equilibrium distributions are correlated equilibrium distributions. However, Figure 6*c* displays a correlated equilibrium distribution of "chicken" that is not a Nash equilibrium distribution.

Why should the distribution of the players' strategy choices, under the conditions of Proposition 5, constitute a correlated equilibrium distribution? The argument goes as follows. Fix an interactive belief system and suppose that (p, q, r, s) is the resulting distribution of the players' strategy choices. Player 1 is rational at every state of the world, hence U must be optimal for player 1 whenever he chooses it. Consider the set of states in which player 1 chooses U. If player 1's only information was that the true state was contained in this set, U would still be the optimal choice for player 1. (Notice the similarity between the argument here and in note 10.) But the conditional probabilities that player 1 would, in this situation, assign to player 2 choosing L versus R are exactly $p/(p + q)$ versus $q/(p + q)$. This gives the first inequality above. The argument for the remaining inequalities in similar.

Concluding Remarks

This paper has discussed sufficient conditions for various non-cooperative solution concepts: iterated dominance, rationalizability, pure-strategy Nash equilibrium, mixed-strategy Nash equilibrium in two-person games, mixed-strategy Nash equilibrium in many-person games, and correlated equilibrium. Table 1 summarizes conditions on the players' knowledge that lead to each of these concepts.

Table 1
A Summary of Solution Concepts

	Conditions		
Solution Concept	*Structure of the Game*	*Rationality*	*Choices or Conjectures*
Iterated dominance	Common knowledge	Common knowledge	
Rationalizability	Common knowledge	Common knowledge	Common knowledge of stochastic independence of conjectures
Pure-strategy Nash equilibrium	—	Fact of rationality	Mutual knowledge of choices
Mixed-strategy Nash equilibrium in two-person games	Mutual knowledge	Mutual knowledge	Mutual knowledge of conjectures
Mixed-strategy Nash equilibrium in many-person games	Mutual knowledge	Mutual knowledge	Common prior plus common knowledge of conjectures
Correlated equilibrium		Rationality at every state	Common prior

There are also converse results of the following kind: for a given solution, say a profile of iteratively undominated strategies, there is an interactive belief system in which the appropriate conditions are satisfied.[12] But typically there are also interactive belief systems in which the conditions do *not* hold. Thus the conditions described in this paper are sufficient but not necessary. This is very natural. For example, there is no reason why players should not blunder into a Nash equilibrium "by accident," so to speak, without anybody knowing much of anything.

Conspicuous by its absence has been any discussion in this paper of the large number of so-called "refinements" of Nash equilibrium that have been

[12]For proofs, see Aumann (1987), Brandenburger and Dekel (1987), and Aumann and Brandenburger (1991).

developed in recent years. Nor has this paper discussed the related issue of elimination of weakly, rather than strongly, dominated strategies. This omission was deliberate. Work on providing decision-theoretic foundations for refinements of Nash equilibrium has started, but it would be premature to attempt a survey at this point.

The results summarized in Table 1 suggest many further questions. What do other combinations of assumptions give? What happens if, instead of common knowledge, there is iterated knowledge to a large, but finite, number of levels? What if everyone is rational, but assigns some small positive probability to the event that everyone else is not? And so on.[14] Questions such as these, which reflect very nicely the concerns of the real-world strategist, can be explored in a systematic fashion using the decision-theoretic approach to non-cooperative game theory.

■ *I would like to acknowledge the considerable intellectual debt that I owe my friend, Eddie Dekel. Many hours spent working together have done much to shape the views expressed here. I am deeply indebted to Bob Aumann for correspondence and conversations over a long period of time concerning game theory. Ben Polak, John Pratt, and Gus Stuart provided detailed and insightful comments on an earlier draft. I also wish to thank Ken Binmore, Elon Kohlberg, and Vijay Krishna for much help. Financial support from the Harvard Business School Division of Research is gratefully acknowledged.*

[14] For some answers in the context of specific games, see, for example, Stuart (1990) and Aumann (1992).

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Colin Camerer's Discourse on the *Keynesian Guessing Game* & Experiments on it

In Keynes's famous book *General Theory of Employment, Interest, and Money*, he draws an analogy between the stock market and a newspaper contest in which people guess what faces others will guess are most beautiful: "It is not a case of choosing those which, to the best of one's judgment, are really the prettiest, nor even those which average opinion genuinely thinks the prettiest. We have reached the third degree, where we devote our intelligences to anticipating what average opinion expects the average opinion to be. And there are some, I believe, who practise the fourth, fifth, and higher degrees" (1936, p. 156). This quote is perhaps no more apt than in the year 2001 (when I first wrote this), just after prices of American internet stocks soared to unbelievable heights in the largest speculative bubble in history. (At one point, the market valuation of the e-tailer bookseller Amazon, which had never reported a profit, was worth more than all other American booksellers combined.)

A simple game that captures the reasoning Keynes had in mind is called the "beauty contest" game (see Nagel, 1995, and Ho, Camerer, and Weigelt, 1998). In a typical beauty contest game, each of N players simultaneously chooses a number x_i in the interval [0,100]. Take an average of the numbers and multiply by a multiple $p < 1$ (say $p = 0.7$). The player whose number is closest to this target (70 percent of the average) wins a fixed prize. Before proceeding, think about what number you would pick.

The beauty contest game can be used to distinguish whether people "practise the fourth, fifth, and higher degrees" of reasoning as Keynes wondered. Here's how. Most players start by thinking, "Suppose the average is 50". Then you should choose 35, to be closest to the target of 70 percent of the average and win. But if you think all players will think this way the average will be 35, so a shrewd player such as yourself (thinking one step ahead) should choose 70 percent of 35, around 25. But if you think all players think that way you should choose 70 percent of 25, or 18.

In analytical game theory, players do not stop this iterated reasoning until they reach a best-response point. But, since all players want to choose 70 percent of the average, if they all choose the same number it must be zero. (That is, if you solve the equation $x^* = 0.7x^*$, you've found the unique Nash equilibrium.)

The beauty contest game provides a rough measure of the number of steps of strategic thinking that subjects are doing. It is called a "dominance-solvable game" because it can be "solved"—i.e., an equilibrium can be computed—by iterated application of dominance. A dominated strategy is one that yields a lower payoff than another (dominant) strategy, regardless of what other players do. Choosing a number above 70 is a dominated strategy because the highest possible value of the target number is 70, so you can always do better by choosing a number lower than 70. But if nobody violates dominance by choosing above 70, then the highest the target can be is 70 percent of 70, or 49, so choosing 49–70 is dominated if you think others obey one step of dominance. Deleting dominated strategies iteratively leads you to zero.

Many interesting games are dominance solvable. A familiar example in economics is Cournot duopoly. Two firms each choose quantities of similar products to make. Since their products are the same, the market price is determined by the total quantity they make (and by consumer demand). It is easy to show that there are quantities so high that firms will lose money because flooding the market with so much supply will drive prices too low to cover fixed costs. If you assume your rivals won't produce that much, then somewhat lower quantities are bad (dominated) choices for you. Applying this logic iteratively leads to a precise solution.

In practice, it is unlikely that people perform more than a couple of steps of iterated thinking because it strains the limits of working memory (i.e., the amount of information people can keep active in their mind at one time). Consider embedded sentences such as "Kevin's dog bit David's mailman whose sister's boyfriend gave the dog to him." Who's the "him" referred to at the end of the sentence? By the time you get to the end, many people have forgotten who owned the dog because working memory has only so much space.[11] Embedded sentences are difficult to understand. Dominance-solvable games are similar in mental complexity.

Iterated reasoning also requires you to believe that others are thinking hard, and are thinking that *you* are thinking hard. When I played this game at a Caltech board of trustees meeting, a very clever board member (a well-known Ph.D. in finance) chose 18.1. Later he explained his choice: He knew the Nash equilibrium was 0, but figured the average Caltech board member was clever enough to do two steps of reasoning and pick 25. Then why not pick 17.5 (which is 70 percent of 25)? He added 0.6 so he wouldn't tie with people who picked 17.5 or 18, and because he guessed that a few people would pick high numbers, which would push the average up. Now that's good behavioral game theory! (He didn't win, but was close.)

Instructor's comment: *Colin Camerer considers the clever board member's actions to constitute* "*good behavioural game theory*". *I beg to differ. I consider the board member's behaviour to be a* "*rational best-response*" *to his sensible conjecture that most of his opponents will play behaviourally* (*and not rationally*).

Multi-stage games of complete-information : Consider the following games:

A Nim Game:
- There are 7 marbles in a jar. Two players alternate in collecting marbles.
- Each must collect 1 or 2 or 3 in each turn.
- Last collector gets Rs. [500 + 10 × no. of marbles in possession], other gets Rs. [50 + 10 × no. of marbles in possession].

An Ultimatum Game
- A "**proposer**" is given 100 rupees. She must offer to give a "**responder**" either 50 rupees, or 25 rupees
- The **responder** has to accept or reject the offer
- If the **responder** accepts the offer made, then the 100 rupees will be shared between the two according to the **proposer's** offer.
- If the **responder** rejects, then **responder** will get nothing, while **proposer** will get: {what she would have gotten if her offer was accepted, *less* Rs. 50}

***Nim Game** has a unique Nash eqm outcome when played simultaneously, which is also the unique "backward-inducted eqm outcome" when played sequentially*

***Ultimatum Game** has two Nash eqm outcomes when played simultaneously, but has a unique "backward-inducted eqm outcome" when played sequentially*

Sequential-Move Complete-Information Games [*SeqM-CI* games]

- A ***sequential-move complete-information game*** is a ***multi-stage*** game (where at least some actions/decisions are to be implemented over multiple stages) that is ***necessarily played in a sequential manner***, i.e.: the players are asked to "play" their ***stage-specific moves*** (**behaviour strategies**) as and when the relevant stages arise, and are not required or permitted to commit to "future strategic decisions"
- A sequential-move game is represented by a **game-tree** (***extensive form representation***) through the specification of a specific sequence of "decision nodes", "information sets" and "terminal points"

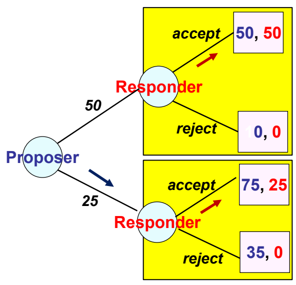
- ***decision node***: a stage where only one player has to ***move***
- ***information set***: a stage where many players have to ***move*** simultaneously
- ***terminal point*** : a specific way the game can end with associated payoffs

- A player's **complete contingent strategy** in a ***SeqM-CI*** game is a "sequenced collection of all her behaviour strategies" where every behavior strategy refers to the action-choice at every specific decision-node / information set

Extensive-form (game tree) representation of the Ultimatum game

- Consider the three pure-strategy Nash equilibria in the game:

[1] **P**: *offer* **25** ; **R**: *accept* **50**, *accept* **25**
[2] **P**: *offer* **50** ; **R**: *accept* **50**, *reject* **25**
[3] **P**: *offer* **25** ; **R**: *reject* **50**, *accept* **25**



- [3] is a ***silly*** Nash eqm, w/ same outcome as [1]
- [2] contains an "***incredible threat***" by **R**, which will not be carried out *ex post* if he is rational and can revise strategy
- [1] is a ***credible*** Nash eqm as it specifies strategies that are 'best response / Nash Eqm' in every subgame / contingency
- ***The backward-induction algorithm picks the outcome of the credible Nash eqm***

	accept if 50, accept if 25	accept if 50, reject if 25	reject if 50, accept if 25	reject if 50, reject if 25
offer 50	50, 50	50, 50	0, 0	0, 0
offer 25	75, 25	25, 0	75, 25	25, 0

A Modified Sequential Claim Game

Two consumers – *X* and *Y* – have won lawsuits against Sleaze Company for selling them defective products. The Judge decides that the monetary compensation to be paid to each consumer will be determined by the following **sequential** process:

In Stage 1 : Consumer *Y* will publicly make one of two statements: "*Statement A*: Let the Judge decide" or "*Statement B*: I'll put in a written claim". If *Y* makes *Statement A*, the process will conclude with the Judge giving compensation of **Rs.22,500** to each consumer.

If *Y* makes *Statement B*, the process will continue to **Stage 2**.

In Stage 2 : Consumer *X* will publicly make one of two statements: "*Statement A*: Let the Judge decide" or "*Statement B*: I'll put in a written claim". If *X* makes *Statement A*, the process will conclude with the Judge giving compensation of **Rs. 22,000** to each consumer.

If *X* makes *Statement B*, the process will continue to **Stage 3**.

In Stage 3 : the two consumers will ***simultaneously and privately*** present written claims to the judge. The claim amount for each consumer is permitted to be either **Rs 20,000** or **Rs 24,000**. Their compensations as functions of their claims are given by following "compensation matrix":

	20K	24K
20K	20, 20	18, 18
24K	18, 18	23, 23

Version v.I

	20K	24K
20K	19, 19	20, 20
24K	20, 20	23, 23

Version v.II

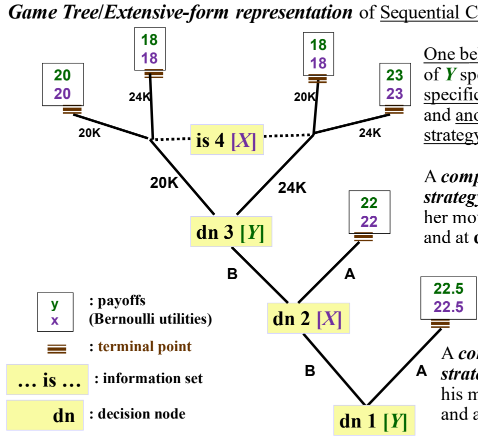
	20K	24K
20K	23, 23	18, 18
24K	18, 18	23, 23

Version v.III

- **In a simultaneously-played muti-stage game of complete information with multiple Nash equilibria:**

  - it is not necessarily the case that "some Nash equilibrium outcomes ***remain*** "sensible predictions of behaviour" when the game is actually played sequentially
  - because when the game is played sequentially, players might want to ***revise*** their complete contingent strategies (which contained incredible threats and/or promises) as game-play unfolds
  - in contrast, the "backward induction algorithm" picks out the ***credible*** Nash equilibria

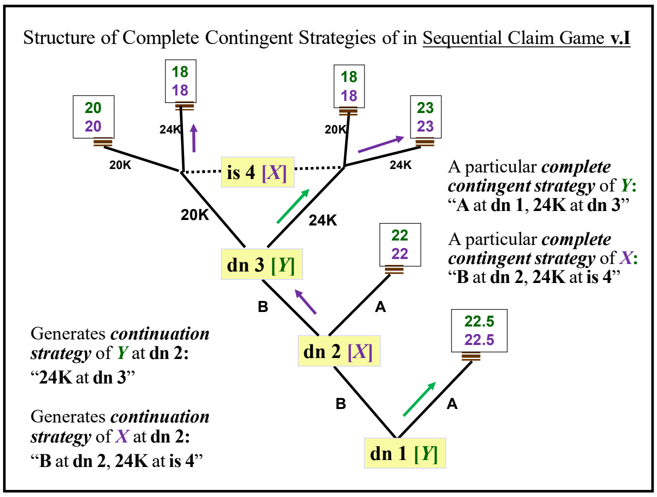
Game Tree/Extensive-form representation of Sequential Claim Game v.I

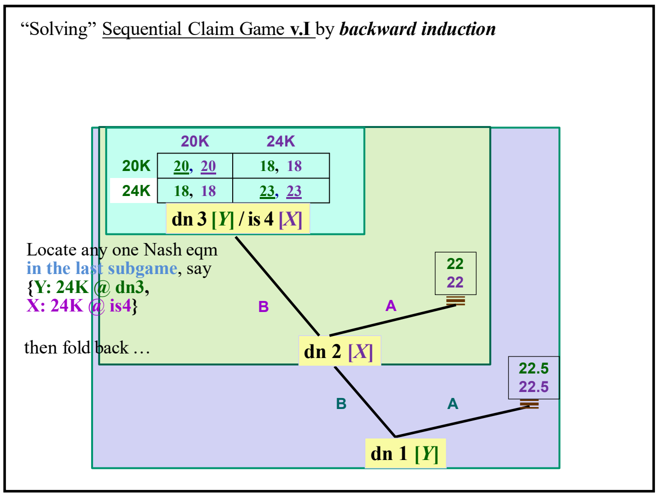


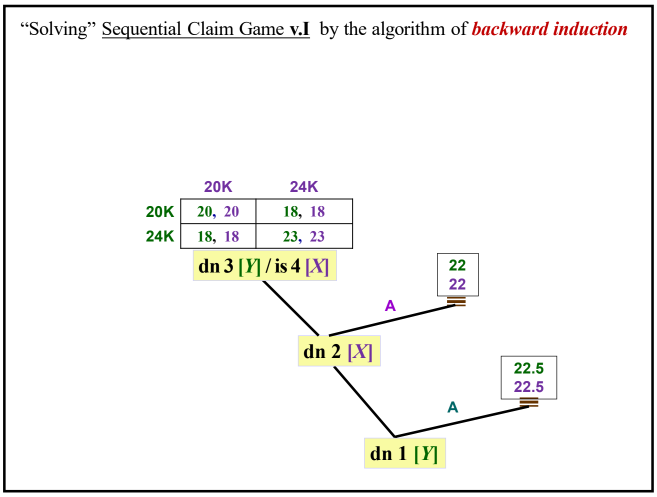
One behaviour strategy of *Y* specifies her stage-specific move at **dn 1** and another behaviour strategy at **dn 3**

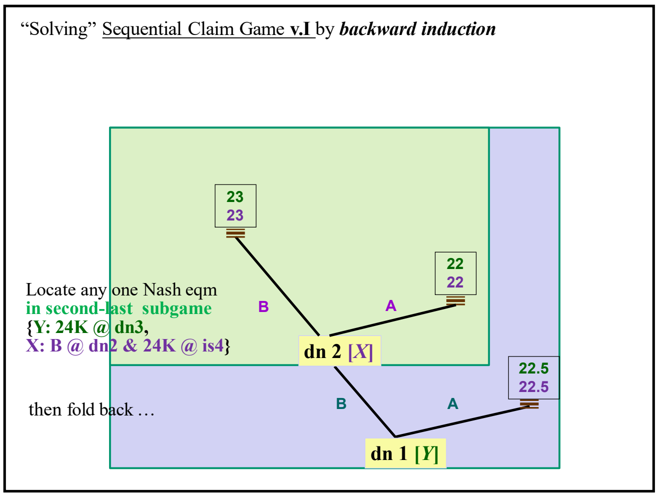
A ***complete contingent strategy*** of *Y* specifies her moves at **dn 1** and at **dn 3**

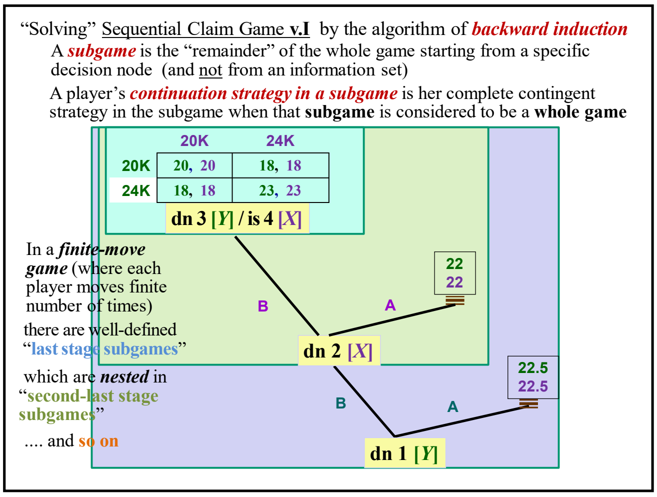
A ***complete contingent strategy*** of *X* specifies his moves at **dn 2** and at **is 4**

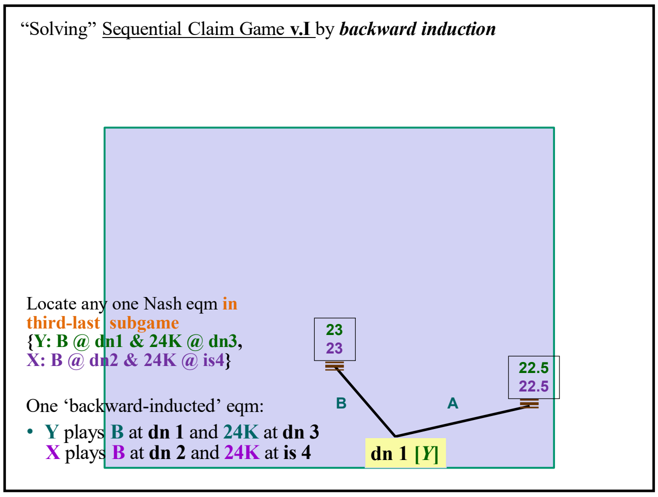

Structure of Complete Contingent Strategies of in Sequential Claim Game v.I
20
20
18
18
18
18
23
23
20K
24K
20K
24K
20K
24K
is 4 [X]
20K
24K
dn 3 [Y]
22
22
B
A
dn 2 [X]
22.5
22.5
B
A
dn 1 [Y]
A particular complete contingent strategy of Y: "A at dn 1, 24K at dn 3"
A particular complete contingent strategy of X: "B at dn 2, 24K at is 4"
Generates continuation strategy of Y at dn 2: "24K at dn 3"
Generates continuation strategy of X at dn 2: "B at dn 2, 24K at is 4"


"Solving" Sequential Claim Game v.I by backward induction
20K
24K
20K
20, 20
18, 18
24K
18, 18
23, 23
dn 3 [Y] / is 4 [X]
Locate any one Nash eqm in the last subgame, say {Y: 24K @ dn3, X: 24K @ is4}
then fold back …
22
22
B
A
dn 2 [X]
22.5
22.5
B
A
dn 1 [Y]

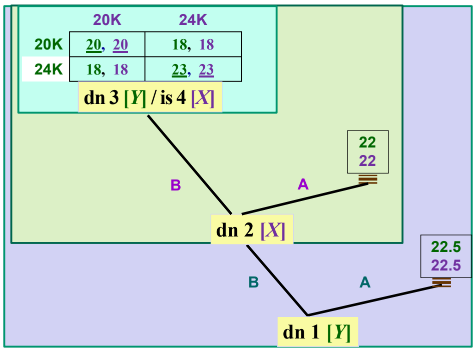

"Solving" Sequential Claim Game v.I by the algorithm of backward induction
20K
24K
20K
20, 20
18, 18
24K
18, 18
23, 23
dn 3 [Y] / is 4 [X]
22
22
A
dn 2 [X]
22.5
22.5
A
dn 1 [Y]


"Solving" Sequential Claim Game v.I by backward induction
23
23
22
22
Locate any one Nash eqm in second-last subgame {Y: 24K @ dn3, X: B @ dn2 & 24K @ is4}
B
A
dn 2 [X]
then fold back …
22.5
22.5
B
A
dn 1 [Y]


"Solving" Sequential Claim Game v.I by the algorithm of backward induction
A subgame is the "remainder" of the whole game starting from a specific decision node (and not from an information set)
A player's continuation strategy in a subgame is her complete contingent strategy in the subgame when that subgame is considered to be a whole game
20K
24K
20K
20, 20
18, 18
24K
18, 18
23, 23
dn 3 [Y] / is 4 [X]
In a finite-move game (where each player moves finite number of times)
there are well-defined "last stage subgames"
which are nested in "second-last stage subgames"
.... and so on
22
22
B
A
dn 2 [X]
22.5
22.5
B
A
dn 1 [Y]


"Solving" Sequential Claim Game v.I by backward induction
Locate any one Nash eqm in third-last subgame {Y: B @ dn1 & 24K @ dn3, X: B @ dn2 & 24K @ is4}
23
23
22.5
22.5
One 'backward-inducted' eqm:
• Y plays B at dn 1 and 24K at dn 3
X plays B at dn 2 and 24K at is 4
B
A
dn 1 [Y]

Question: How many backward inducted equilibria will this game have?



The Backward Induction Algorithm

(1) Determine all Nash equilibria in each of the "last subgames"

(2) Choose one Nash eqm in each of the last subgames (if there are more than one Nash eqm), and ***fold-back their outcomes*** to compute all Nash equilibria in each of the "second-last subgames"

(3) Choose one Nash eqm in each of the second-last subgames ...... and proceed till Nash equilibria of the "first subgame" is determined

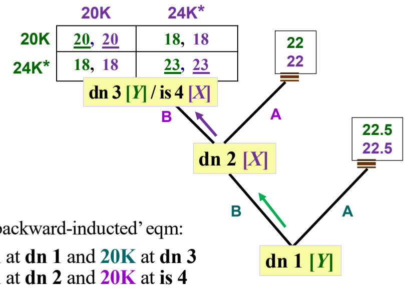
- By specifying the selected Nash equilibrium strategy vectors as the players' behaviour strategies at every decision node /information set of the game, we find out <u>one</u> particular ***sequentially optimal*** (and therefore ***credible***) ***Nash equilibrium*** of the game

  We can find out <u>all</u> the ***credible Nash equilibria*** by considering all possible selections of Nash equilibria in the sequence of subgames
- Backward induction involves "sequentially piecing together" (starting from the end) behaviour strategy vectors that are Nash equilibria in every subgame and ultimately allows us to identify a vector of complete contingent strategies that constitutes a credible/revision-proof Nash equilibrium

<u>Sequential Claim Game **v.I**</u>

One 'backward-inducted' eqm:

Y plays **B** at **dn 1** and **24K** at **dn 3**
X plays **B** at **dn 2** and **24K** at **is 4**



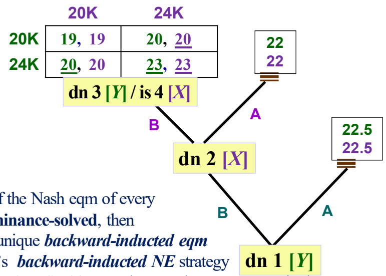
Another 'backward-inducted' eqm:

Y plays **A** at **dn 1** and **20K** at **dn 3**
X plays **A** at **dn 2** and **20K** at **is 4**

<u>Sequential Claim Game **v.II**</u>

Unique ***dominance-solved*** "backward-inducted equilibrium":

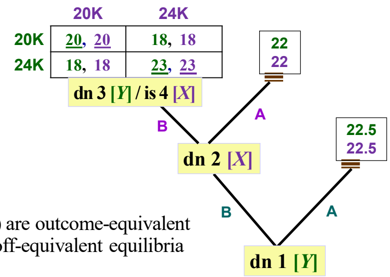
Y plays **B** at **dn 1** and **24K** at **dn 3**
X plays **B** at **dn 2** and **24K** at **is 4**



In a C-I game, if the Nash eqm of every subgame is **dominance-solved**, then the game has a unique ***backward-inducted eqm*** and each player's ***backward-inducted NE*** strategy is his ***uniquely rationalizable*** complete contingent strategy in the game

<u>Sequential Claim Game **v.I**</u> : all 'backward-inducted' equilibria

- (1) **Y** plays **B** at **dn 1** and **24K** at **dn 3**
  X plays **B** at **dn 2** and **24K** at **is 4**
- (2) **Y** plays **A** at **dn 1** and **20K** at **dn 3**
  X plays **A** at **dn 2** and **20K** at **is 4**
- (3) **Y** plays **A** at **dn 1** and {**20K, 24K; 5/7, 2/7**} at **dn 3**
  X plays **A** at **dn 2** and {**20K, 24K; 5/7, 2/7**} at **is 4**



- (2) & (3) are outcome-equivalent and payoff-equivalent equilibria

Implications of Backward Induction

- In a ***finite*** sequential-move game (where each player moves finite no. of times), ***backward induction algorithm*** uncovers exactly those Nash equilibria of the game that are

  sequentially optimal – that is, involves contingent strategies for which every "component behaviour strategy at a decision node / information set" is a best response in that situation

  equivalently, ***credible*** – that is, involves complete contingent strategies that do not contain any non-credible threats or promises

  [equivalently, ***revision-proof*** – that is, involves complete contingent strategies that no player will want to revise *ex post*]

Major issue: What do we do when the ***Seq-CI*** game is not finite?

We can build on the fact that "backward induction involves <u>sequentially piecing together</u> (starting from the end) behaviour strategy vectors that are **Nash equilibria in every subgame**"

Subgame-Perfect Nash Equilibrium (Reinhard Selten, 1965)

- In sequential-move complete information games, "**subgame-perfect Nash equilibrium**" (***SPNE***) is the appropriate equilibrium concept

Definition: An ***SPNE*** in a sequential-move complete-info game is a vector of complete contingent strategies (one for each player) such that ***in every subgame*** of the game, the vector of "continuation strategies for that subgame" (that are contained in the specified complete strategies) is a ***vector of mutual best responses*** (i.e., **Nash eqm**) in that subgame

Existence Result: An ***SPNE*** exists in every finite-horizon ***Seq-CI*** game, and in a large class of infinite-horizon ***Seq-CI*** games, including "infinitely repeated games"

Bilateral Sequential-move "win-lose-draw" games of perfect information

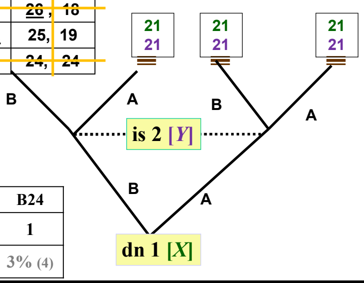
- In such games between two players ***A*** and ***B***, there are no information sets, and all terminal nodes have one of three outcomes: ***A*** wins / ***B*** wins / draw
- In such a game, a player is said to have a "***forcing strategy to win***" if there exists a complete contingent strategy for her such that it ensures that she wins irrespective of the strategy played by her opponent
  - and a player is said to have a "***forcing strategy to draw***" if there exists a complete contingent strategy for her such that it ensures that the game ends in a draw irrespective of her opponent's strategy
- If a player has a forcing strategy S^W to win then S^W is certainly a dominant revision-proof strategy (and so an ***SPNE*** strategy) for the player;
  if there is no ***SPNE*** in which player ***I*** wins with positive probability, and player ***I*** has a forcing strategy S^D to draw, then S^D is certainly a dominant revision-proof strategy for the player
- It may be easier to prove that a player has a forcing strategy than to find it
- ***Zermelo's theorem***: "Chess is determinate. Either White can force a win, or Black can force a win, or either player can force a draw."

Sequential Claim Game 1.1 ***Game Tree***

Unique dominance-solved Nash eqm in claim subgame {22K, 22K}

Two b-i-N-E: [B & 22K, B & 22K] and [A & 22K, A & 22K]

	~~20K~~	22K	~~24K~~
~~20K~~	~~20, 20~~	~~20, 21~~	~~26, 18~~
22K	~~21, 20~~	22, 22	~~25, 19~~
~~24K~~	~~18, 26~~	~~19, 25~~	~~24, 24~~



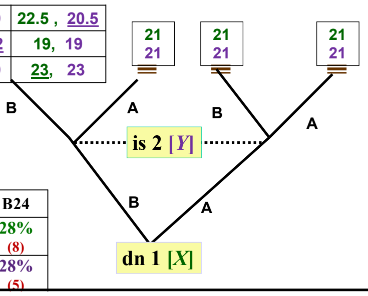
Experiment:

A	B20	B22	B24
0	0	48	1
1% (2)	1% (3)	95% (1)	3% (4)

Sequential Claim Game 1.2 ***Game Tree***

- "Claim subgame" has multiple Nash equilibria
  e.g. [22K, 22K] & [{20K, 24K; 0.5, 0.5}, {20K, 24K; 0.5, 0.5}]
  and the players face a "coordination problem" in the claim subgame

	20K	22K	24K
20K	20, 20	19, 19	22.5 , 20.5
22K	19, 19	22, 22	19, 19
24K	19.5 , 23.5	19, 19	23, 23



Experiment:

	A	B20	B22	B24
X	35% (2)	0% (0)	37% (12)	28% (8)
Y	36% (5)	4% (0)	32% (17)	28% (5)

EC 246: Decisions & Games

Answer Key for Problem Set 3

Problem 1: The unique Nash equilibrium of the game is: {***y***, ***d***}, and this Nash equilibrium is "dominance-solved". The fact that Column "is rational" will preclude her from playing either '***b***' or '***c***' as they are strictly dominated strategies. That fact, in and of itself, does not permit us to "prune/eliminate" ***b*** and ***c***. Rather, the following to things – the fact that Row "is rational" ***and*** the first-order knowledge of Row about Column – "I know that Column is rational" – will convince Row that Column will not play '***c***' and thus lead Row to play '***y***'. Then, second-order knowledge of Column about Row – "I know that Row is rational, and that Row knows that I am rational" – will convince Column that Row will play '***y***'; this knowledge will lead Column to play '***d***'.

Problem 2: Given that each player has to guess an "integer among **1**, **2**, **3**, and **4**", the payoff matrix of the simultaneous-move game with the "best responses underlined" is given below.

It is clear from the best responses that there is a unique pure-strategy Nash equilibrium: {***Alice*** guesses **4**, ***Barbara*** guesses **4**}.

Next, we commence on the algorithm of "iterated deletion of strictly dominated strategies" to determine the following results:

(*i*) In the full game, **1** by ***Barbara*** is strictly dominated (by **2**).

(*ii*) In the resultant "pruned" game, **1** by ***Alice*** is strictly dominated (by **2**).

(*iii*) In the resultant "pruned" game, **2** and **3** by ***Barbara*** is strictly dominated (by **4**).

(*iv*) In the resultant "pruned" game, **2** and **3** by ***Alice*** is strictly dominated (by **4**).

Thus the algorithm converges to {***Alice*** guesses **4**, ***Barbara*** guesses **4**}.

This convergence gives us the following conclusions: {***Alice*** guesses **4**, ***Barbara*** guesses **4**} is the unique Nash equilibrium of the game, and the Nash equilibrium is "dominance-solved".

Barbara

		1	**2**	**3**	**4**
	1	<u>10</u>, 09	09, <u>10</u>	08, 09	07, 08
Alice	**2**	09, 07	<u>10</u>, 08	09, 09	08, <u>10</u>
	3	08, 05	09, 06	<u>10</u>, 07	09, <u>08</u>
	4	07, 03	08, 04	09, 05	<u>10</u>, <u>06</u>

Problem 3: Unique pure-strategy Nash equilibrium: {*Joy*, *Joy*}. All pure strategies of each player are *rationalizable* because no pure strategy of either player is strictly dominated.

Problem 4: The following payoff-matrix applies to a version of the Travelers' Dilemma where each traveler can specify own value between 97 and 100.

	100	**99**	**98**	**97**
100	**100, 100**	**97, <u>101</u>**	**96, 100**	**95, 99**
99	**<u>101</u>, 97**	**99, 99**	**96, <u>100</u>**	**95, 99**
98	**100, 96**	**<u>100</u>, 96**	**98, 98**	**95, <u>99</u>**
97	**99, 95**	**99, 95**	**<u>99</u>, 95**	**<u>97</u>, <u>97</u>**

Note that in the above symmetric game, {97, 97} is the *dominance-solved* Nash equilibrium.

Recognize further that in the full 4×4 game, the pure strategy {100} for each player is strictly dominated by the mixed-strategy {99, 97; 0.9, 0.1}. Then, in the first-order pruned game, the pure strategy {99} for each player is strictly dominated by the mixed-strategy {98, 97; 0.5, 0.5}. Finally, in the third-order pruned game, the pure strategy {98} for each player is strictly dominated by the pure strategy {97}.

Thus, for each player, the pure strategy {97} is the only rationalizable strategy. Thus, the theoretical claim that "if a complete-information simultaneous-move game has a dominance-solved Nash equilibrium, then the pure Nash strategies of the players are the only rationalizable strategies" is upheld in this game.

Similar logic applies to the "full game" where {2, 2} is the *dominance-solved* Nash equilibrium.

Problem 5: In the full game, $\boldsymbol{y}$ is strictly dominated by $\boldsymbol{z}$ – so delete $\boldsymbol{y}$. In the first-order pruned game, $\boldsymbol{b}$ is strictly dominated by $\boldsymbol{a}$ – so delete $\boldsymbol{b}$. At this point, the IDSDS algorithm stops, so do the underling in the second-ordered pruned game, and get the following matrix:

	x	***z***
a	<u>10</u>, <u>11</u>	11, 10
c	9, 9	<u>12</u>, <u>12</u>

Recognize that Row's pure best-responses to conjecture {$\boldsymbol{x}$, $\boldsymbol{z}$; p, $1–p$} is: $\boldsymbol{a}$ when $p > 0.5$, $\boldsymbol{c}$ when $p < 0.5$, and {$\boldsymbol{a}$, $\boldsymbol{c}$} when $p = 0.5$. So, Row's "best-response correspondence" is: $\boldsymbol{a}$ when $p > 0.5$, $\boldsymbol{c}$ when $p < 0.5$, and {$\boldsymbol{a}$, $\boldsymbol{c}$; r, $1–r$} for every $r \in [0, 1]$ when $p = 0.5$.

Column's best-response to conjecture {$\boldsymbol{a}$, $\boldsymbol{c}$; q, $1–q$} is: $\boldsymbol{x}$ when $q > 0.75$, $\boldsymbol{z}$ when $q < 0.75$, and {$\boldsymbol{x}$, $\boldsymbol{z}$} when $q = 0.75$. So, Column's "best-response correspondence" is: $\boldsymbol{x}$ when $q > 0.75$, $\boldsymbol{z}$ when $q < 0.75$, and {$\boldsymbol{x}$, $\boldsymbol{z}$; s, $1–s$} for every $s \in [0, 1]$ when $q = 0.75$.

Thus the set of all Nash equilibria of the game is: [Row: $\boldsymbol{a}$, Column: $\boldsymbol{x}$]; [Row: $\boldsymbol{c}$, Column: $\boldsymbol{z}$]; and [Row: {$\boldsymbol{a}$, $\boldsymbol{c}$; 0.75, 0.25}, Column: {$\boldsymbol{x}$, $\boldsymbol{z}$; 0.5, 0.5}].

Problem Set 4

Problem 1: In the Sequential Claim Game Version **v.1**, we identified all the three Subgame Perfect Nash Equilibria of the game when the game is played ***sequentially***. **IF** the judge forced the two claimants to play the game ***simultaneously*** (where each claimant is required to write down a complete contingent strategy on a piece of paper, and hand it over to the judge, who then would compute and implement the Nash equilibrium outcome from the claimants' specified strategies), there will exist at least "one more distinct" pure-strategy Nash equilibrium of that Claim Game. State the players' complete contingent strategies in one such pure-strategy Nash equilibrium.

Problem 2: *In a strategic scenario* (*as opposed to a decision-theoretic one*) *there might be **benefits of limiting one's options*** https://medium.com/the-ascent/burning-the-ships-aae93a7ad1a5

The Ruritanian government has recently realized that due to decades of protectionist policies the quality of software available in the country is very poor. So, they have decided to permit two foreign companies, Vipro and Sinfosys, to market *one* software product *each* in Ruritania. Each company is required to submit the name of a single product to the Ruritanian government on the "license approval date" which is one month from today. At that time, each company will be granted a license to market its submitted product in Ruritania.

While Vipro has two products – products A and B – ready for marketing in Ruritania, currently Sinfosys has only one product – product C – ready for marketing. However, Sinfosys can easily develop another product – product D – within the next month at virtually no cost (by utilizing open-source codes). Whether it does this product development or not will be observable to all.

The four products – A, B, C, and D – differ in their quality levels, and are imperfect substitutes of each other. When Vipro and Sinfosys compete in the Ruritanian market with their respective "single" products, their present value profits (in million Ruritanian Roubles) will be as follows:

If Vipro sells A and Sinfosys sells C, Vipro profits will be 4 and Sinfosys profits will be 8; if Vipro sells B and Sinfosys sells C, Vipro profits will be 2 and Sinfosys profits will be 2; if Vipro sells A and Sinfosys sells D, Vipro profits will be 16 and Sinfosys profits will be 10; and if Vipro sells B and Sinfosys sells D, Vipro profits will be 20 and Sinfosys profits will be 6.

Consider the following two-stage sequential-move game: In the first stage, Sinfosys decides whether it will develop product D or not. After this decision is implemented (and observed by all), then on the license approval date (Stage 2), each of the two companies simultaneously submits the name of a single product for the marketing license.

In this sequential-move game of complete information, determine the unique "subgame perfect Nash equilibrium" strategy vector.

Problem 3: *Uncovering "forcing strategies" in a "win-lose-draw" sequential-move game*

David Gale invented the following two-player competitive game "Chomp": It is played on a $N{\times}M$ board of squares. Each square is denoted by the coordinates $\boldsymbol{i}{\times}\boldsymbol{j}$ for $1 \leq \boldsymbol{i} \leq N$ and $1 \leq \boldsymbol{j} \leq \boldsymbol{M}$ ($\boldsymbol{i}$ is the horizontal coordinate and $\boldsymbol{j}$ is the vertical coordinate).

Each of the two players – ***A*** and ***B*** – takes (alternate) turns in "marking" squares, staring with player ***A***. When his/her turn comes, a player is required to "mark" a previously unmarked square. When a player marks square (I, J) all squares to the "north and east of (I, J)" (i.e., all squares with horizontal coordinates greater than or equal to I and vertical coordinates greater than or equal to J) are removed from the board – they are "chomped up". The player who (is forced to) mark the square (1, 1) loses the game.

(i) Consider the following 2×3 Chomp board:

(1, 3)	(2, 3)
(1, 2)	(2, 2)
(1, 1)	(2, 1)

Specify player ***A***'s "forcing strategy" to win the game.

(ii) Consider any $P \times P$ 'square' Chomp board, where P is a finite positive integer greater than 1. Specify player ***A***'s "forcing strategy" to win the game.

Three kinds of "introspection" in games

- In any stage of a general multi-stage game (played sequentially), a player's introspection about rival behaviour involves one or more of the following:

(i) making ***predictions*** about future play
(ii) forming ***conjectures*** about contemporaneous play
(iii) drawing ***inferences*** from past play

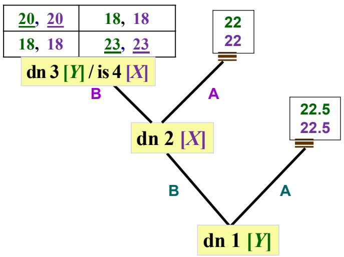
- ***Backward induction*** relates to making "rational predictions"
- ***Strict dominance algorithm*** relates to forming "rational conjectures"
- ***Forward induction*** relates to drawing "rational inferences"

1. In a muti-stage game, the non-empty set of Nash equilibria contains the non-empty set of Subgame-perfect Nash equilibria
2. ... and the latter might contain a subset of "SPNE that satisfy a specific Forward Induction refinement"

Unique SPNE in Sequential Claim Game v.I that satisfies a specific Forward Induction refinement

Y plays **B** at **dn 1** and **24K** at **dn 3**
X plays **B** at **dn 2** and **24K** at **is 4**

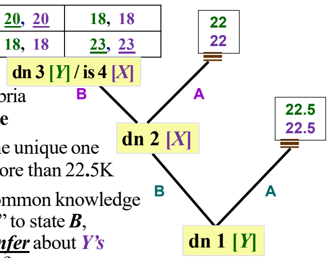
	20K	24K
20K	20, 20	18, 18
24K	18, 18	23, 23



Revisiting multiplicity of SPNE in Sequential Claim Game v.I

- (1) **Y** plays **B** at **dn 1** and **24K** at **dn 3**
  X plays **B** at **dn 2** and **24K** at **is 4**
- (2) **Y** plays **A** at **dn 1** and **20K** at **dn 3**
  X plays **A** at **dn 2** and **20K** at **is 4**
- (3) **Y** plays **A** at **dn 1** and {**20K, 24K; 5/7, 2/7**} at **dn 3**
  X plays **A** at **dn 2** and {**20K, 24K; 5/7, 2/7**} at **is 4**

	20K	24K
20K	20, 20	18, 18
24K	18, 18	23, 23



Multiple Nash equilibria in the **claim subgame**

But {24K, 24K} is the unique one in which each gets more than 22.5K

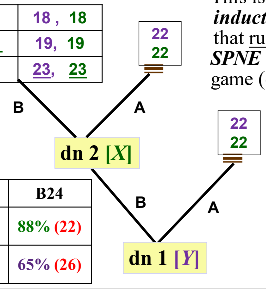
So, if rationality is common knowledge and if *Y* is "observed" to state ***B***, then what should *X* ***infer*** about *Y's* subsequent intentions?

Sequential Claim Game 2.2 (in class experiment)

Multiple Nash equilibria in "claim subgame"
But {24K, 24K} is the unique eqm in which each player gets more than 22K
⇒ If rationality is common knowledge & if *Y* is "observed" to state ***B***, then *X* should ***infer*** that *Y* will claim **24K** in "claim subgame"
- So, *X* should make Statement ***B*** in Stage **2** and then claim **24K**
- If *Y* ***deduces*** this, then *Y* should indeed make Statement ***B*** in Stage **1**

	20K	22K	24K
20K	20, 20	19, 19	18 , 18
22K	19, 19	21, 21	19, 19
24K	18 , 18	19, 19	23, 23

- This is a ***forward induction refinement*** that rules out all other ***SPNE*** outcomes of the game (e.g., {A, A})



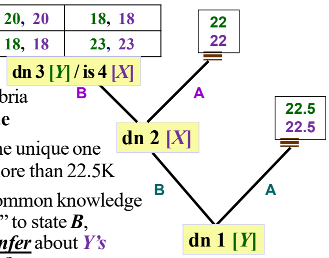
Experiment:

	A	B20	B22	B24
X	12% (0)	0% (0)	0% (0)	88% (22)
Y	29% (0)	0 (0)	6% (1)	65% (26)

Revisiting multiplicity of SPNE in Sequential Claim Game v.I

X should ***infer*** that *Y* will claim **24K** in "claim subgame"
- So, *X* should make Statement ***B*** in Stage **2** and then claim **24K**
- If *Y* can deduce this, then *Y* should indeed make Statement ***B*** in Stage **1**
- This is a ***forward induction logic*** that rules out all other ***SPNE*** outcomes of this game except for **Y** plays **B** at **dn 1** and **24K** at **dn 3**
  X plays **B** at **dn 2** and **24K** at **is 4**

	20K	24K
20K	20, 20	18, 18
24K	18, 18	23, 23



Multiple Nash equilibria in the **claim subgame**

But {24K, 24K} is the unique one in which each gets more than 22.5K

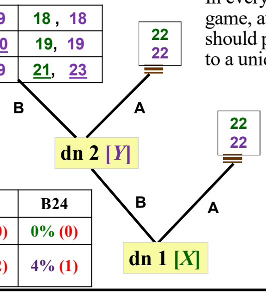
So, if rationality is common knowledge and if *Y* is "observed" to state ***B***, then what should *X* ***infer*** about *Y's* subsequent intentions?

Sequential Claim Game 2.1 *Game Tree*

{20, 20K} is the unique eqm in claim subgame where *X* gets more than 22K
⇒ If rationality is common knowledge & if *X* is "observed" to state ***B***, then *Y* should ***infer*** that *X* will claim **20K** in claim subgame
- So, *Y* should make Statement ***A*** in Stage **2**
- On deducing this, *X* indifferent between Statements ***A*** and ***B*** in Stage **1**

	20K	22K	24K
20K	23, 21	19, 19	18 , 18
22K	19, 19	20, 20	19, 19
24K	18 , 18	19, 19	21, 23

- In every ***SPNE*** in this game, at least one player should play ***A***, leading to a unique outcome

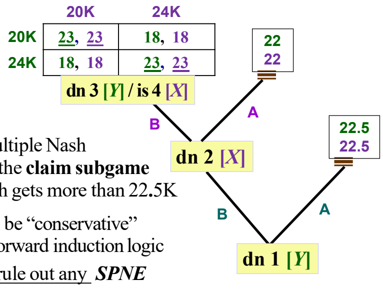


Experiment:

	A	B20	B22	B24
X	93% (21)	7% (1)	0% (0)	0% (0)
Y	94% (24)	0% (0)	2% (2)	4% (1)

***SPNE* in Sequential Claim Game v.III**

- (1) Y plays B at **dn 1** and 24K at **dn 3**
  X plays B at **dn 2** and 24K at **is 4**
- (2) Y plays B at **dn 1** and 20K at **dn 3**
  X plays B at **dn 2** and 20K at **is 4**
- (3) Y plays A at **dn 1** and {20K, 24K; 0.5, 0.5} at **dn 3**
  X plays A at **dn 2** and {20K, 24K; 0.5, 0.5} at **is 4**

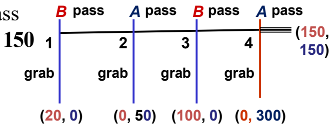


There are multiple Nash equilibria in the **claim subgame** in which each gets more than 22.5K

Here, we will be "conservative" in using the forward induction logic and will not rule out any *SPNE*

The "Grab or Pass" Game [a 4-stage version of the ***Centipede Game*** by Robert Rosenthal]

Stage 1: **Bob** can either grab **20** or pass
Stage 2: **Alice** can either grab **50** or pass
Stage 3: **Bob** can either grab **100** or pass
Stage 4: **Alice** can grab **300** or pass
If **Alice** passes at Stage 4, each gets **150**



Backward induction ⇒ unique ***SPNE*** strategy for each player:
"I will grab at every decision node in which I have to make a move"

- If Bob knows "Alice is rational" then Bob will grab in stage 3; if Alice knows that "(i) Bob is rational, and (ii) Bob knows that Alice is rational" then Alice will grab in stage 2, ...
- Every step of backward induction requires a higher-order knowledge about rival's rationality ... the ***SPNE*** strategy is the only rationalizable strategy for each player when rationality is common knowledge

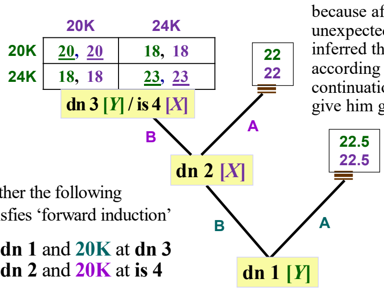
A Specific Forward Induction Refinement under Complete Information

Suppose a player ***P*** [*Y*] will get equilibrium payoff U^P [22.5] under some ***SPNE*** [♠]

Suppose that for some behaviour-strategy at some stage that ***P*** is not supposed to play according to the ***SPNE*** [♠] , following statement is true:

Following ***P***'s "unexpected play", there exist a unique "**continuation *SPNE***" [♣] in the "continuation subgame" under which ***P*** will get a payoff greater than U^P

Then the particular ***SPNE*** [♠] "fails" the (*conservative*) ***forward induction refinement***



because after seeing ***P***'s unexpected play, it must be inferred that ***P*** will then play according to the unique continuation ***SPNE*** [♣] that'll give him greater than U^P

Checking whether the following ***SPNE*** [♠] satisfies 'forward induction'

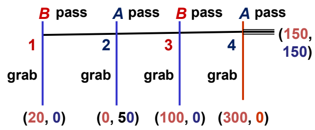
Y plays A at **dn 1** and 20K at **dn 3**
X plays A at **dn 2** and 20K at **is 4**

It's a game where unexpected plays "must" be interpreted as mistakes

Grab or pass game has unique ***SPNE***: *grab when you get a chance*

So if **Bob** is seen to **pass** at Stage **1**, the *forward induction logic* says nothing about what inferences are to be drawn after that move

If **Alice** does infer that **Bob** will **pass** again in Stage **3**, then **Alice** should **pass** in Stage **2**



In which case, **Bob**'s passing in Stage **1** becomes a "profitable deviation" from a strategy that is supposed to be the unique ***SPNE*** strategy

Thus the only inference that **Alice** is permitted to make is that **Bob**'s pass in Stage **1** was a mistake that will not be repeated in Stage **3**

- In many experiments of such a Centipede Game, players typically pass in the initial rounds before someone "grabs"

The Logic of Forward Induction

- The logic of "forward induction" pertains to making "rational inferences" from past play
- In a sequential-move game where it is common knowledge that players are rational, there are two scenarios in which rational inferences from past play can have a role to play in strategic decision-making:

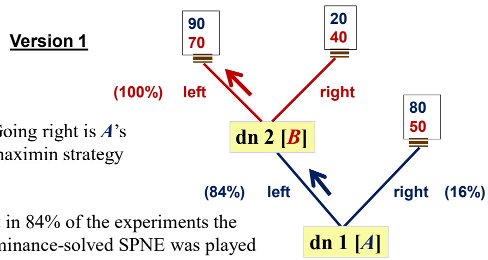
(1) In a complete-information game, if there are multiple ***SPNE***, then past play might indicate "which" ***SPNE*** is actually being played

(2) In an incomplete-information game (where players do not have full information about the preferences of their rival players), past play might allow a player to "deduce" her rivals' preferences

- *At this point, we are concentrating on* **(1)**, *we will discuss* **(2)** *later when we study incomplete-information games*
- ***When a seq-move game has a unique SPNE, no forward induction argument can be made, and sometimes we are faced with a puzzle***

- If a seq-move game has a dominance-solved Nash equilibrium in every subgame, will players necessarily play their unique ***SPNE*** strategies?

A two-stage game experiment by Goeree and Holt (2001)



Going right is *A*'s maximin strategy

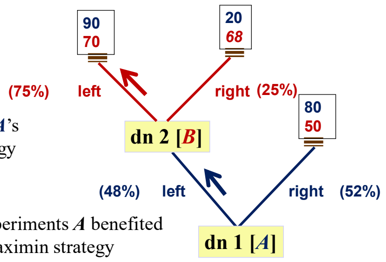
But in 84% of the experiments the dominance-solved SPNE was played

- If a seq-move game has a dominance-solved Nash equilibrium in every subgame, will players necessarily play their unique ***SPNE*** strategies?

Answer: Maybe not, especially when
(*a*) players not confident that rational rivals will not make <u>mistakes</u>,
or (*b*) players do not have very high orders of knowledge about <u>rival rationality</u>,

A two-stage game experiment by Goeree and Holt (2001)

<u>Version 2</u>



Going right is *A*'s maximin strategy

In 25% of the experiments ***A*** benefited by playing her maximin strategy

Game Classifications & Equilibrium Concepts

	Single-stage Games	*Multi-stage Games*	
	Simultaneous move games	**Simultaneous move games**	**Sequential move games**
Complete Info Games	**Nash equilibrium****		**Subgame-perfect Nash equilibrium***
Incomplete Info Games	**Bayesian Nash equilibrium**		**Perfect Bayesian Nash equilibrium**

** Dominance-solved Strategies;
Maximin Strategies;
Best Responses to a Cognitive Hierarchy of beliefs

* Dominance-solved SPNE;
SPNE satisfying Forward Induction

- There is some experimental evidence that when “stakes are raised” in games, strategic behaviour “come closer” to theoretical predictions

Answer Key to Problem Set 4

Problem 1: If the Sequential Claim Game Version v.1 is played in a "simultaneous move manner", then the "normal form representation" of the game, with the "underlined pure best responses" will look as follows:

		X A/20K	A/24K	B/20K	B/24K
Y	**A/20K**	$\underline{22.5}$, $\underline{22.5}$	$\underline{22.5}$, $\underline{22.5}$	$\underline{22.5}$, $\underline{22.5}$	22.5, $\underline{22.5}$
	A/20K	$\underline{22.5}$, $\underline{22.5}$	$\underline{22.5}$, $\underline{22.5}$	$\underline{22.5}$, $\underline{22.5}$	22.5, $\underline{22.5}$
	B/24K	22, $\underline{22}$	22, $\underline{22}$	20, 20	18, 18
	B/24K	22, 22	22, 22	18, 18	$\underline{23}$, $\underline{23}$

You can then figure out all the pure-strategy Nash equilibria of the game, when the game is played in a "simultaneous move manner".

Problem 2: If Sinfosys develops *D*, it will be its dominant strategy to offer *D* on the license approval date, and then it will earn present value profits of 6 million Roubles. On the other hand, if Sinfosys does not develop *D*, it will be constrained to offer *C*, and then it will earn present value profits of 8 million Roubles. So, in the unique dominance-solved Subgame-perfect Nash equilibrium, Sinfosys will not develop *D* in Stage 1, and in Stage 2 Vipro will offer *A* and Sinfosys will offer *C*. *This example demonstrates that in a strategic environment, it might be individually profitable to limit one's future choices.*

Problem 3: (i) The following is ***A***'s forcing strategy to win the game: "Mark (2,3) in the first move. If ***B*** marks (1,2), then mark (2,1). If ***B*** marks (2,1), then mark (1,2). If ***B*** marks (1,3), then mark (2,2); and subsequently mark either (1,2) or (2,2) depending on which is unmarked. If ***B*** marks (2,2), then mark (1,3); and subsequently mark either (1,2) or (2,2) depending on which is unmarked."

(ii) The following is ***A***'s forcing strategy to win in any *P*× square Chomp game: "Mark (2,2) in the first move. From then on, "mark symmetrically" to ***B***"s markings, i.e., if ***B*** marks (*I*, *J*) in one round, mark (*J*, *I*) in the following round."

Comments: The following statements can be proved for any *N*×*M* "rectangular" Chomp board: (1) The game cannot end in a draw. (2) The second-mover cannot have a forcing strategy to win the game. (*Try proving these statements*, *they are not very hard to prove.*) Given these statements, *Zermelo's Theorem* implies that the first-mover must have a forcing strategy to win the game. Search the net to convince yourself that no one has yet been successful in discovering this forcing strategy for the first-mover when *N* and *M* are large.

Problem Set 5

Problem 1: *A different version of the Sequential Claim Game*

Two consumers – X and Y – have won lawsuits against Sleaze Company for selling them defective products. The Judge decides that the monetary compensation to be paid to each consumer will be determined by the following sequential process:

In Stage 1 : Consumer Y will publicly make one of two statements: "Statement A: Let the Judge decide" or "Statement B: I'll put in a written claim". If Y makes Statement A, the process will conclude with the Judge giving compensation of Rs.45,000 to each consumer.

If Y makes Statement B, this act will be observed, and the game will continue to Stage 2.

In Stage 2 : Consumer X will publicly make one of two statements: "Statement A: Let the Judge decide" or "Statement B: I'll put in a written claim". If X makes Statement A, the process will conclude with the Judge giving compensation of Rs.44,000 to each consumer.

If X makes Statement B, this act will be observed, and the game will continue to Stage 3.

In Stage 3 : the two consumers will simultaneously and privately present written claims to the judge. The claim amount for each consumer is permitted to be either Rs. 60,000 or Rs. 40,000. Their compensations as functions of their claims are given by the following "compensation matrix":

Consumer X

Consumer Y	claim 40K	claim 60K
claim 40K	Z, Z	30000, 30000
claim 60K	30000, 30000	60000, 60000

(i) When Z = 40000, find out all the Subgame-perfect Nash equilibria (SPNE) of this game. Then determine which of these SPNE "survive" the Forward Induction Refinement.

(ii) When Z = 50000, find out all the Subgame-perfect Nash equilibria (SPNE) of this game. Then determine which of these SPNE "survive" the Forward Induction Refinement.

Problem 2: Make sure to review the "Trust Game" in the Course lecture Notes (pages 37-38).

Ten Little Treasures of Game Theory and Ten Intuitive Contradictions

By Jacob K. Goeree and Charles A. Holt*

This paper reports laboratory data for games that are played only once. These games span the standard categories: static and dynamic games with complete and incomplete information. For each game, the treasure is a treatment in which behavior conforms nicely to predictions of the Nash equilibrium or relevant refinement. In each case, however, a change in the payoff structure produces a large inconsistency between theoretical predictions and observed behavior. These contradictions are generally consistent with simple intuition based on the interaction of payoff asymmetries and noisy introspection about others' decisions. (JEL C72, C92)

The Nash equilibrium has been the centerpiece of game theory since its introduction about 50 years ago. Along with supply and demand, the Nash equilibrium is one of the most commonly used theoretical constructs in economics, and it is increasingly being applied in other social sciences. Indeed, game theory has finally gained the central role envisioned by John von Neumann and Oscar Morgenstern, and in some areas of economics (e.g., industrial organization) virtually all recent theoretical developments are applications of game theory. The impression one gets from recent surveys and game theory textbooks is that the field has reached a comfortable maturity, with neat classifications of games and successively stronger (more "refined") versions of the basic approach being appropriate for more complex categories of games: Nash equilibrium for static games with complete information, Bayesian Nash for static games with incomplete information, subgame perfectness for dynamic games with complete information, and some refinement of the sequential Nash equilibrium for dynamic games with incomplete information (e.g., Robert Gibbons, 1997). The rationality assumptions that underlie this analysis are often preceded by persuasive adjectives like "perfect," "intuitive," and "divine." If any noise in decision-making is admitted, it is eliminated in the limit in a process of "purification." It is hard not to notice parallels with theology, and the highly mathematical nature of the developments makes this work about as inaccessible to mainstream economists as medieval treatises on theology would have been to the general public.

The discordant note in this view of game theory has come primarily from laboratory experiments, but the prevailing opinion among game theorists seems to be that behavior will eventually converge to Nash predictions under the right conditions.[1] This paper presents a much more unsettled perspective of the current state of game theory. In each of the major types of games, we present one or more examples for which the relevant version of the Nash equilibrium predicts remarkably well. These "treasures" are observed in games played only once by financially motivated subjects who have had prior experience in other, similar, strategic situations. In each of these games, however, we show that a change in the payoff structure can produce a large inconsistency between theoret-

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[1] For example, George J. Mailath's (1998) survey of evolutionary models cites the failure of backward induction as the main cause of behavioral deviations from Nash predictions.

ical predictions and human behavior. For example, a payoff change that does not alter the unique Nash equilibrium may move the data to the opposite side of the range of feasible decisions. Alternatively, a payoff change may cause a major shift in the game-theoretic predictions and have no noticeable effect on actual behavior. The observed contradictions are typically somewhat intuitive, even though they are not explained by standard game theory. In a simultaneous effort-choice coordination game, for example, an increase in the cost of players' "effort" decisions is shown to cause a dramatic decrease in effort, despite the fact that any common effort is a Nash equilibrium for a range of effort costs. In some of these games, it seems like the Nash equilibrium works only by coincidence, e.g., in symmetric cases where the costs of errors in each direction are balanced. In other cases, the Nash equilibrium has considerable drawing power, but economically significant deviations remain to be explained.

The idea that game theory should be tested with laboratory experiments is as old as the notion of a Nash equilibrium, and indeed, the classic prisoner's dilemma paradigm was inspired by an experiment conducted at the RAND Corporation in 1950. Some of the strategic analysts at RAND were dissatisfied with the received theory of cooperative and zero-sum games in von Neumann and Morgenstern's (1944) *Theory of Games and Economic Behavior*. In particular, nuclear conflict was not thought of as a zero-sum game because both parties may lose. Sylvia Nasar (1998) describes the interest at RAND when word spread that a Princeton graduate student, John Nash, had generalized von Neumann's existence proof for zero-sum games to the class of all games with finite numbers of strategies. Two mathematicians, Melvin Dresher and Merrill Flood, had been running some game experiments with their colleagues, and they were skeptical that human behavior would be consistent with Nash's notion of equilibrium. In fact, they designed an experiment that was run on the same day they heard about Nash's proof. Each player in this game had a dominant strategy to defect, but both would earn more if they both used the cooperative strategy. The game was repeated 100 times with the same two players, and a fair amount of cooperation was observed. One of Nash's professors, Al W. Tucker, saw the payoffs for this game written on a blackboard, and he invented the prisoner's dilemma story that was later used in a lecture on game theory that he gave in the Psychology Department at Stanford (Tucker, 1950).

Interestingly, Nash's response to Dresher and Flood's repeated prisoner's dilemma experiment is contained in a note to the authors that was published as a footnote to their paper:

> The flaw in the experiment as a test of equilibrium point theory is that the experiment really amounts to having the players play one large multi-move game. One cannot just as well think of the thing as a sequence of independent games as one can in zero-sum cases. There is just too much interaction ... (Nasar, 1998 p. 119)

In contrast, the experiments that we report in this paper involved games that were played *only once*, although related results for repeated games with random matching will be cited where appropriate. As Nash noted, the advantage of one-shot games is that they insulate behavior from the incentives for cooperation and reciprocity that are present in repeated games. One potential disadvantage of one-shot games is that, without an opportunity to learn and adapt, subjects may be especially prone to the effects of confusion. The games used in this paper, however, are simple enough in structure to ensure that Nash-like behavior can be observed in the "treasure" treatment. In addition, the study of games played only once is of independent interest given the widespread applications of game theory to model one-shot interactions in economics and other social sciences, e.g., the FCC license auctions, elections, military campaigns, and legal disputes.

The categories of games to be considered are based on the usual distinctions: static versus dynamic and complete versus incomplete information. Section I describes the experiments based on static games with complete information: social dilemmas, matching pennies, and coordination games. Section II contains results from dynamic games with complete information: bargaining games and games with threats that are not credible. The games reported in Sections III and IV have incomplete information about other players' payoffs: in static settings (auctions) and two-stage settings (signaling games).

It is well known that decisions can be affected by psychological factors such as framing, aspiration levels, social distance, and heuristics (e.g., Daniel Kahneman et al., 1982; Catherine Eckel and Rick Wilson, 1999). In this paper we try to hold psychological factors constant and focus on payoff changes that are primarily *economic* in nature. As noted below, economic theories can and are being devised to explain the resulting anomalies. For example, the rational-choice assumption underlying the notion of a Nash equilibrium eliminates all errors, but if the costs of "overshooting" an optimal decision are much lower than the costs of "undershooting," one might expect an upward bias in decisions. In a game, the endogenous effects of such biases may be reinforcing in a way that creates a "snowball" effect that moves decisions well away from a Nash prediction. Models that introduce (possibly small) amounts of noise into the decision-making process can produce predictions that are quite far from any Nash equilibrium (Richard D. McKelvey and Thomas R. Palfrey, 1995, 1998; Goeree and Holt, 1999). Equilibrium models of noisy behavior have been used to explain behavior in a variety of contexts, including jury decision-making, bargaining, public goods games, imperfect price competition, and coordination (Simon P. Anderson et al., 1998a, b, 2001a; C. Monica Capra et al., 1999, 2002, Stanley S. Reynolds, 1999; Serena Guarnaschelli et al., 2001).

A second type of rationality assumption that is built into the Nash equilibrium is that beliefs are consistent with actual decisions. Beliefs are not likely to be confirmed out of equilibrium, and learning will presumably occur in such cases. There is a large recent literature on incorporating learning into models of adjustment in games that are played repeatedly with different partners.[2] These models include adaptive learning (e.g., Vincent P. Crawford, 1995; David J. Cooper et al., 1997), naive Bayesian learning (e.g., Jordi Brandts and Holt, 1996; Dilip Mookherjee and Barry Sopher, 1997), reinforcement or stimulus response learning (e.g., Ido Erev and Alvin E. Roth, 1998), and hybrid models with elements of both belief and reinforcement learning (Colin Camerer and Teck-Hua Ho, 1999). Learning from experience is not possible in games that are only played once, and beliefs must be formed from introspective thought processes, which may be subject to considerable noise. Without noise, iterated best responses will converge to a Nash equilibrium, if they converge at all. Some promising approaches to explaining deviations from Nash predictions are based on models that limit players' capacities for introspection, either by limiting the number of iterations (e.g., Dale O. Stahl and Paul W. Wilson, 1995; Rosemarie Nagel, 1995) or by injecting increasing amounts of noise into higher levels of iterated beliefs (Goeree and Holt, 1999; Dorothea Kübler and Georg Weizsäcker, 2000). The predictions derived from these approaches, discussed in Section V, generally conform to Nash predictions in the treasure treatments and to the systematic, intuitive deviations in the contradiction treatments. Some conclusions are offered in Section VI.

I. Static Games with Complete Information

In this section we consider a series of two-player, simultaneous-move games, for which the Nash equilibria show an increasing degree of complexity. The first game is a "social dilemma" in which the pure-strategy Nash equilibrium coincides with the unique rationalizable outcome. Next, we consider a matching pennies game with a unique Nash equilibrium in mixed strategies. Finally, we discuss coordination games that have multiple Nash equilibria, some of which are better for all players.

In all of the games reported here and in subsequent sections, we used cohorts of student subjects recruited from undergraduate economics classes at the University of Virginia. Each cohort consisted of ten students who were paid $6 for arriving on time, plus all cash they earned in the games played. These one-shot games followed an initial "part A" in which the subjects played the same two-person game for ten periods with new pairings made randomly in each period.[3] Earnings for the two-hour ses-

[2] See, for instance, Drew Fudenberg and David K. Levine (1998) for a survey.

[3] We only had time to run about six one-shot games in each session, so the data are obtained from a large number

sions ranged from $15 to $60, with an average of about $35. Each one-shot game began with the distribution and reading of the instructions for that game.[4] These instructions contained assurances that all money earned would be paid and that the game would be followed by "another, quite different, decision-making experiment." Since the one-shot treatments were paired, we switched the order of the treasure and contradiction conditions in each subsequent session. Finally, the paired treatments were always separated by other one-shot games of a different type.

A. *The One-Shot Traveler's Dilemma Game*

The Nash equilibrium concept is based on the twin assumptions of perfect error-free decision making and the consistency of actions and beliefs. The latter requirement may seem especially strong in the presence of multiple equilibria when there is no obvious way for players to coordinate. More compelling arguments can be given for the Nash equilibrium when it predicts the play of the unique justifiable, or *rationalizable*, action (B. Douglas Bernheim, 1984; David G. Pearce, 1984). Rationalizability is based on the idea that players should eliminate those strategies that are never a best response for any possible beliefs, and realize that other (rational) players will do the same.[5]

To illustrate this procedure, consider the game in which two players independently and simultaneously choose integer numbers between (and including) 180 and 300. Both players are paid the *lower* of the two numbers, and, in addition, an amount $R > 1$ is transferred from the player with the higher number to the player with the lower number. For instance, if one person chooses 210 and the other chooses 250, they receive payoffs of $210 + R$ and $210 - R$ respectively. Since $R > 1$, the best response is to undercut the other's decision by 1 (if that decision were known), and therefore, the upper bound 300 is never a best response to any possible beliefs that one could have. Consequently, a rational person must assign a probability of zero to a choice of 300, and hence 299 cannot be a best response to any possible beliefs that rule out choices of 300, etc. Only the lower bound 180 survives this iterated deletion process and is thus the unique rationalizable action, and hence the unique Nash equilibrium.[6] This game was introduced by Kaushik Basu (1994) who coined it the "traveler's dilemma" game.[7]

Although the Nash equilibrium for this game can be motivated by successively dropping those strategies that are never a best response (to any beliefs about strategies that have not yet been eliminated from consideration), this deletion process may be too lengthy for human subjects with limited cognitive abilities. When the cost of having the higher number is small, i.e., for small values of R, one might expect more errors in the direction of high claims, well away from the unique equilibrium at 180, and indeed this is the intuition behind the dilemma. In contrast, with a large penalty for having the higher of the two claims, players are likely to end up with claims that are near the unique Nash prediction of 180.

To test these hypotheses we asked 50 subjects (25 pairs) to make choices in a treatment with $R = 180$, and again in a matched treatment with $R = 5$. All subjects made decisions in each treatment, and the two games were separated by a number of other one-shot games. The

of sessions where part A involved a wide range of repeated games, including public goods, coordination, price competition, and auction games that are reported in other papers. The one-shot games never followed a repeated game of the same type.

[4] These instructions can be downloaded from http://www.people.virginia.edu/~cah2k/datagame.html.

[5] A well-known example for which this iterated deletion process results in a unique outcome is a Cournot duopoly game with linear demand (Drew Fudenberg and Jean Tirole, 1993 pp. 47–48).

[6] In other games, rationalizability may allow outcomes that are not Nash equilibria, so it is a weaker concept than that of a Nash equilibrium, allowing a wider range of possible behavior. It is in this sense that Nash is more persuasive when it corresponds to the unique rationalizable outcome.

[7] The associated story is that two travelers purchase identical antiques while on a tropical vacation. Their luggage is lost on the return trip, and the airline asks them to make independent claims for compensation. In anticipation of excessive claims, the airline representative announces: "We know that the bags have identical contents, and we will entertain any claim between $180 and $300, but you will each be reimbursed at an amount that equals the minimum of the two claims submitted. If the two claims differ, we will also pay a reward R to the person making the smaller claim and we will deduct a penalty R from the reimbursement to the person making the larger claim."

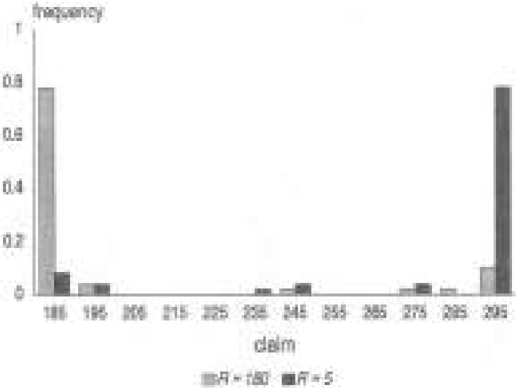

FIGURE 1. CLAIM FREQUENCIES IN A TRAVELER'S DILEMMA FOR $R = 180$ (LIGHT BARS) AND $R = 5$ (DARK BARS)

ordering of the two treatments was alternated. The instructions asked the participants to devise their own numerical examples to be sure that they understood the payoff structure.

Figure 1 shows the frequencies for each 10-cent category centered around the claim label on the horizontal axis. The lighter bars pertain to the high-R "treasure" treatment, where close to 80 percent of all the subjects chose the Nash equilibrium strategy, with an average claim of 201. However, roughly the same fraction chose the *highest* possible claim in the low-R treatment, for which the average was 280, as shown by the darker bars. Notice that the data in the contradiction treatment are clustered at the opposite end of the set of feasible decisions from the unique (rationalizable) Nash equilibrium.[3] Moreover, the "anomalous" result for the low-R treatment does not disappear or even diminish over time when subjects play the game repeatedly and have the opportunity to learn.[4] Since the treatment change does not alter the unique Nash (and rationalizable) prediction, standard game theory simply cannot explain the most salient feature of the data, i.e., the effect of the penalty/reward parameter on average claims.

TABLE 1—THREE ONE-SHOT MATCHING PENNIES GAMES (WITH CHOICE PERCENTAGES)

		Left (48)	*Right* (52)
Symmetric matching pennies	*Top* (48)	80, 40	40, 80
	Bottom (52)	40, 80	80, 40
		Left (16)	*Right* (84)
Asymmetric matching pennies	*Top* (96)	320, 40	40, 80
	Bottom (4)	40, 80	80, 40
		Left (80)	*Right* (20)
Reversed asymmetry	*Top* (8)	44, 40	40, 80
	Bottom (92)	40, 80	80, 40

B. A Matching Pennies Game

Consider a symmetric matching pennies game in which the row player chooses between *Top* and *Bottom* and the column player simultaneously chooses between *Left* and *Right*, as shown in top part of Table 1. The payoff for the row player is \$0.80 when the outcome is (*Top*, *Left*) or (*Bottom*, *Right*) and \$0.40 otherwise. The motivations for the two players are exactly opposite: column earns \$0.80 when row earns \$0.40, and vice versa. Since the players have opposite interests there is no equilibrium in pure strategies. Moreover, in order not to be exploited by the opponent, neither player should favor one of their strategies, and the mixed-strategy Nash equilibrium involves randomizing over both alternatives with equal probabilities. As before, we obtained decisions from 50 subjects in a one-shot version of this game (five cohorts of ten subjects, who were randomly matched and assigned row or column

[3] This result is statistically significant at all conventional levels, given the strong treatment effect and the relatively large number of independent observations (two paired observations for each of the 50 subjects). We will not report specific nonparametric tests for cases that are so clearly significant. The individual choice data are provided in the Data Appendix for this paper on http://www.people.virginia.edu/~cah2k/datapage.html.

[4] In Capra et al. (1999), we report results of a repeated traveler's dilemma game (with random matching). When subjects chose numbers in the range [80, 200] with $R = 5$, the average claim rose from approximately 180 in the first period to 196 in period 5, and the average remained above 190 in later periods. Different cohorts played this game with different values of R, and successive increases in R resulted in successive reductions in average claims. With a penalty/reward parameter of 5, 10, 20, 25, 50, and 80 the average claims in the final three periods were 195, 186, 119, 138, 85, and 81, respectively. Even though there is one treatment reversal, the effect of the penalty/reward parameter on average claims is significant at the 1-percent level. The patterns of adjustment are well explained by a naive Bayesian learning model with decision error, and the claim distributions for the final five periods are close to those predicted by a logit equilibrium (McKelvey and Palfrey, 1995).

roles). The choice percentages are shown in parentheses next to the decision labels in the top part of Table 1. Note that the choice percentages are essentially "50-50," or as close as possible given that there was an odd number of players in each role.

Now consider what happens if the row player's payoff of $0.80 in the (*Top, Left*) box is increased to $3.20, as shown in the asymmetric matching pennies game in the middle part of Table 1. In a mixed-strategy equilibrium, a player's *own* decision probabilities should be such that the *other* player is made indifferent between the two alternatives. Since the column player's payoffs are unchanged, the mixed-strategy Nash equilibrium predicts that *row's* decision probabilities do not change either. In other words, the row player should ignore the unusually high payoff of $3.20 and still choose *Top* or *Bottom* with probabilities of one-half. (Since column's payoffs are either 40 or 80 for playing *Left* and either 80 or 40 for playing *Right*, row's decision probabilities must equal one-half to keep column indifferent between *Left* and *Right*, and hence willing to randomize.)[10] This counterintuitive prediction is dramatically rejected by the data, with 96 percent of the row players choosing the *Top* decision that gives a chance of the high $3.20 payoff. Interestingly, the column players seemed to have anticipated this, and they played *Right* 84 percent of the time, which is quite close to their equilibrium mixed strategy of 7/8. Next, we lowered the row player's (*Top, Left*) payoff to $0.44, which again should leave the row player's own choice probabilities unaffected in a mixed-strategy Nash equilibrium. Again the effect is dramatic, with 92 percent of the choices being *Down*, as shown in the bottom part of Table 1. As before, the column players seemed to have anticipated this reaction, playing *Left* 80 percent of the time. To summarize, the unique Nash prediction is for the bolded row-choice percentages to be unchanged at 50 percent for all three treatments. This prediction is violated

TABLE 2—AN EXTENDED COORDINATION GAME

	L	*H*	*S*
L	90, 90	0, 0	*x*, 40
H	0, 0	180, 180	0, 40

in an intuitive manner, with row players' choices responding to their own payoffs.[11] In this context, *the Nash mixed strategy prediction seems to work only by coincidence*, when the payoffs are symmetric.

C. A Coordination Game with a Secure Outside Option

Games with multiple Nash equilibria pose interesting new problems for predicting behavior, especially when some equilibria produce higher payoffs for all players. The problem of coordinating on the high-payoff equilibrium may be complicated by the possible gains and losses associated with payoffs that are not part of any equilibrium outcome. Consider a coordination game in which players receive $1.80 if they coordinate on the high-payoff equilibrium (*H, H*), $0.90 if they coordinate on the low-payoff equilibrium (*L, L*), and they receive nothing if they fail to coordinate (i.e., when one player chooses *H* and the other *L*). Suppose that, in addition, the column player has a secure option *S* that yields $0.40 for column and results in a zero payoff for the row player. This game is given in Table 2 when $x = 0$. To analyze the Nash equilibria of this game, notice that for the column player a 50-50 combination of *L* and *H* dominates *S*, and a rational column player should therefore avoid the secure option. Eliminating *S* turns the game into a standard 2×2 coordination game that has three Nash equilibria:

[10] The predicted equilibrium probabilities for the row player are not affected if we relax the assumption of risk neutrality. There are only two possible payoff levels for column so, without loss of generality, columns' utilities for payoffs of 40 and 80 can be normalized to 0 and 1. Hence even a risk-averse column player will only be indifferent when row uses choice probabilities of one-half.

[11] This anomaly is persistent when subjects play the game repeatedly. Jack Ochs (1995b) investigates a matching pennies game with an asymmetry similar to that of the middle game in Table 1, and reports that the row players continue to select *Top* considerably more than one-half of the time, even after as many as 50 rounds. These results are replicated in McKelvey et al. (2000). Similarly, Goeree et al. (2000) report results for ten-period repeated matching pennies games that exactly match those in Table 1. The results are qualitatively similar but less dramatic than those in Table 1, with row's choice probabilities showing strong "own-payoff" effects that are not predicted by the Nash equilibrium.

both players choosing L, both choosing H, and a mixed-strategy equilibrium in which both players choose L with probability ½.

The Nash equilibria are independent of x, which is the payoff to the row player when (L, S) is the outcome, since the argument for eliminating S is based solely on column's payoffs. However, the magnitude of x may affect the coordination process: for $x = 0$, row is indifferent between L and H when column selects S, and row is likely to prefer H when column does not select S (since then L and H have the same number of zero payoffs for row, but H has a higher potential payoff). Row is thus more likely to choose H, which is then also the optimal action for the column player. However, when x is large, say 400, the column player may anticipate that row will select L in which case column should avoid H.

This intuition is borne out by the experimental data: in the treasure treatment with $x = 0$, 96 percent of the row players and 84 percent of the column players chose the high-payoff action H, while in the contradiction treatment with $x = 400$ only 64 percent of the row players and 76 percent of the column players chose H. The percentages of outcomes that were coordinated on the high-payoff equilibrium were 80 for the treasure treatment versus 32 for the contradiction treatment. In the latter treatment, an additional 16 percent of the outcomes were coordinated on the low-payoff equilibrium, but more than half of all the outcomes were uncoordinated, non-Nash outcomes.

D. A Minimum-Effort Coordination Game

The next game we consider is also a coordination game with multiple equilibria, but in this case the focus is on the effect of payoff asymmetries that determine the risks of deviating in the upward and downward directions. The two players in this game choose "effort" levels simultaneously, and the cost of effort determines the risk of deviation. The joint product is of the fixed-coefficients variety, so that each person's payoff is the minimum of the two efforts, minus the product of the player's own effort and a constant cost factor, c. In the experiment, we let efforts be any integer in the range from 110 to 170. If $c < 1$, any common effort in this range is a Nash equilibrium, because a unilateral one-unit increase in effort above a

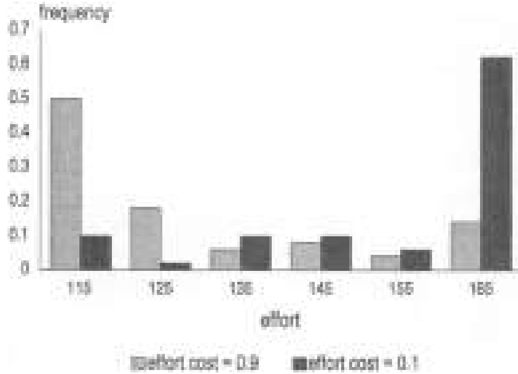

FIGURE 2. EFFORT CHOICE FREQUENCIES FOR A MINIMUM-EFFORT COORDINATION GAME WITH HIGH EFFORT COST (LIGHT BARS) AND LOW EFFORT COST (DARK BARS)

common starting point will not change the minimum but will reduce one's payoff by c. Similarly, a one-unit decrease in effort will reduce one's payoff by $1 - c$, i.e., the reduction in the minimum is more than the savings in effort costs when $c < 1$. Obviously, a higher effort cost increases the risk of raising effort and reduces the risk of lowering effort. Thus simple intuition suggests that effort levels will be inversely related to effort costs, despite the fact that any common effort level is a Nash equilibrium.

We ran one treatment with a low effort cost of 0.1, and the data for 50 randomly matched subjects in this treatment are shown by the dark bars in Figure 2. Notice that behavior is quite concentrated at the highest effort level of 170; subjects coordinate on the Pareto-dominant outcome. The high effort cost treatment ($c = 0.9$), however, produced a preponderance of efforts at the lowest possible level, as can be seen by the lighter bars in the figure. Clearly, the extent of this "coordination failure" is affected by the key economic variable in this model, even though Nash theory is silent.[12]

[12] The standard analysis of equilibrium selection in coordination is based on the John C. Harsanyi and Reinhard Selten's (1988) notion of risk dominance, which allows a formal analysis of the trade-off between risk and payoff dominance. Paul G. Straub (1995) reports experimental evidence for risk dominance as a selection criterion. There is no agreement on how to generalize risk dominance beyond 2 × 2 games, but see Anderson et al. (2001b) for a proposed generalization based on the "stochastic potential." Experiments with repeated plays of coordination games

TABLE 3—TWO VERSIONS OF THE KREPS GAME (WITH CHOICE PERCENTAGES)

		Left (26)	*Middle* (8)	*Non-Nash* (68)	*Right* (0)
Basic game	*Top* (68)	200, 50	0, 45	10, 30	20, −250
	Bottom (32)	0, −250	10, −100	30, 30	50, 40
		Left (24)	*Middle* (12)	*Non-Nash* (64)	*Right* (0)
Positive payoff frame	*Top* (84)	500, 350	300, 345	310, 330	320, 300
	Bottom (16)	300, 50	310, 200	330, 330	350, 340

E. *The Kreps Game*

The previous examples demonstrate how the cold logic of game theory can be at odds with intuitive notions about human behavior. This tension has not gone unnoticed by some game theorists. For instance, David M. Kreps (1995) discusses a variant of the game in the top part of Table 3 (where we have scaled back the payoffs to levels that are appropriate for the laboratory). The pure-strategy equilibrium outcomes of this game are (*Top, Left*) and (*Bottom, Right*). In addition, there is a mixed-strategy equilibrium in which row randomizes between *Top* and *Bottom* and column randomizes between *Left* and *Middle*. The only column strategy that is not part of any Nash equilibrium is labeled *Non-Nash*. Kreps argues that column players will tend to choose *Non-Nash* because the other options yield at best a slightly higher payoff (i.e., 10, 15, or 20 cents higher) but could lead to substantial losses of $1 or $2.50. Notice that this intuition is based on payoff magnitudes out of equilibrium, in contrast to Nash calculations based only on signs of payoff differences.

Kreps did try the high-hypothetical payoff version of this game on several graduate students, but let us consider what happens with financially motivated subjects in an anonymous laboratory situation. As before, we randomly paired 50 subjects and let them make a single choice. Subjects were told that losses would be subtracted from prior earnings, which were quite substantial by that point. As seen from the percentages in parentheses in the top part of the table, the *Non-Nash* decision was selected by approximately two-thirds of the column players. Of course, it is possible that this result is simply a consequence of "loss-aversion," i.e., the disutility of losing some amount of money is greater than the utility associated with winning the same amount (Daniel Kahneman et al., 1991). Since all the other columns contain negative payoffs, loss-averse subjects would thus be naturally inclined to choose *Non-Nash*. Therefore, we ran another 50 subjects through the same game, but with 300 cents added to payoffs to avoid losses, as shown in the bottom part of Table 3. The choice percentages shown in parentheses indicate very little change, with close to two-thirds of column players choosing *Non-Nash* as before. Thus "loss aversion" biases are not apparent in the data, and do not seem to be the source of the prevalence of *Non-Nash* decisions. Finally, we ran 50 new subjects through the original version in the top part of the table, with the (*Bottom, Right*) payoffs of (50, 40) being replaced by (350, 400), which (again) does not alter the equilibrium structure of the game. With this admittedly heavy-handed enhancement of the equilibrium in that cell, we observed 96 percent *Bottom* choices and 84 percent *Right* choices, with 16 percent *Non-Nash* persisting in this, the "treasure," treatment.

II. Dynamic Games with Complete Information

As game theory became more widely used in fields like industrial organization, the complexity of the applications increased to accommodate

have shown that behavior may begin near the Pareto-dominant equilibrium, but later converge to the equilibrium that is worst for all concerned (John B. Van Huyck et al., 1990). Moreover, the equilibrium that is selected may be affected by the payoff structure for dominated strategies (Russell Cooper et al., 1992). See Goeree and Holt (1998) for analysis of a repeated coordination game with random matching. They show that the dynamic patterns of effort choices are well explained by a simple evolutionary model of noisy adjustment toward higher payoffs, and that final-period effort decisions can be explained by the maximization of stochastic potential function.

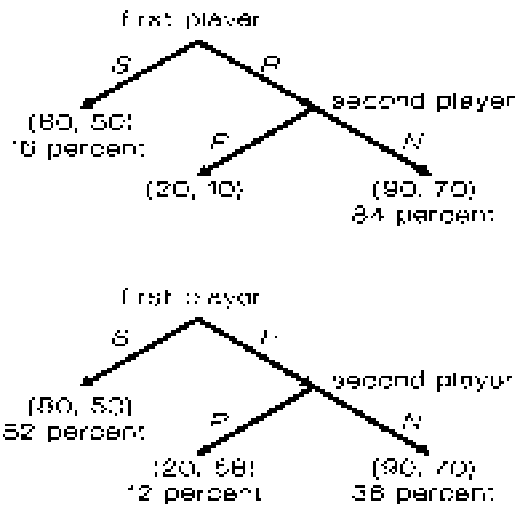

FIGURE 3. SHOULD YOU TRUST OTHERS TO BE RATIONAL?

dynamics and asymmetric information. One of the major developments coming out of these applications was the use of backward induction to eliminate equilibria with threats that are not "credible" (Selten, 1975). Backward induction was also used to develop solutions to alternating-offer bargaining games (Ariel Rubinstein, 1982), which was the first major advance on this historically perplexing topic since Nash's (1950) axiomatic approach. However, there have been persistent doubts that people are able to figure out complicated, multistage backward induction arguments. Robert W. Rosenthal (1981) quickly proposed a game, later dubbed the "centipede game," in which backward induction over a large number of stages (e.g., 100 stages) was thought to be particularly problematic (e.g., McKelvey and Palfrey, 1992). Many of the games in this section are inspired by Rosenthal's (1981) doubts and Randolph T. Beard and Beil's (1994) experimental results. Indeed, the anomalies in this section are better known than those in other sections, but we focus on very simple games with two or three stages, using parallel procedures and subjects who have previously made a number of strategic decisions in different one-shot games.

A. Should You Trust Others to Be Rational?

The power of backward induction is illustrated in the top game in Figure 3. The first player begins by choosing between a safe decision, *S*, and a risky decision, *R*. If *R* is chosen, the second player must choose between a decision *P* that punishes both of them and a decision *N* that leads to a Nash equilibrium that is also a joint payoff maximum. There is, however, a second Nash equilibrium where the first player chooses *S* and the second chooses *P*. The second player has no incentive to deviate from this equilibrium because the self-inflicted punishment occurs off of the equilibrium path. Subgame perfectness rules out this equilibrium by requiring equilibrium behavior in each subgame, i.e., that the second player behave optimally in the event that the second-stage subgame is reached.

Again, we used 50 randomly paired subjects who played this game only once. The data for this treasure treatment are quite consistent with the subgame-perfect equilibrium: a preponderance of first players trust the other's rationality enough to choose *R*, and there are no irrational *P* decisions that follow. The game shown in the bottom part of Figure 3 is identical, except that the second player only forgoes 2 cents by choosing *P*. This change does not alter the fact that there are two Nash equilibria, one of which is ruled out by subgame perfectness. The choice percentages for 50 subjects indicate that a majority of the first players did not trust others to be perfectly rational when the cost of irrationality is so small. Only about a third of the outcomes matched the subgame-perfect equilibrium in this game.[12] We did a third treatment (not shown) in which we multiplied all payoffs by a factor of 5, except that the *P* decision led to (100, 348) instead of (100, 340). This large increase in payoffs produced an even more dramatic result: only 16 percent of the outcomes were subgame perfect, and 80 percent of the outcomes were at the Nash equilibrium that is not subgame perfect.

B. Should You Believe a Threat That Is Not Credible?

The game just considered is a little unusual in that, in the absence of relative payoff effects, the second player has no reason to punish, since

[12] See Beard and Beil (1994) for similar results in a two-stage game played only once.

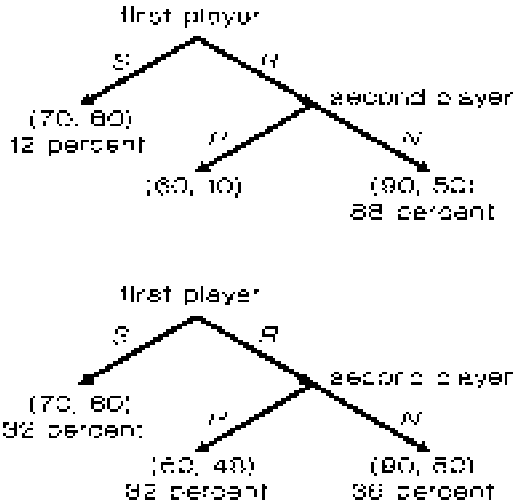

FIGURE 4. SHOULD YOU BELIEVE A THREAT THAT IS NOT CREDIBLE?

the first player's *R* decision also benefits the second player. This is not the case for the game in Figure 4, where an *R* decision by the first player will lower the second player's payoff. As before, there are two Nash equilibria, with the (*R*, *P*) equilibrium ruled out by subgame perfectness. In addition to not being credible, the threat to play *P* is a relatively costly punishment for the second player to administer (40 cents).

The threat to play *P* in the top part of Figure 4 is evidently not believed, and 88 percent of the first players choose the *R* strategy, with impunity. The threat is cheap (2 cents) for the game in the bottom part of the figure, and outcomes for 25 subject pairs are evenly divided between the subgame imperfect outcome, the incredible threat outcome, and the subgame perfect outcome. Cheap threats often are (and apparently should be) believed. Again we see that payoff magnitudes and off-the-equilibrium-path risks matter.

Since the *P* decisions in the bottom games of Figures 3 and 4 only reduce the second player's payoff by 2 cents, behavior may be affected by small variations in payoff preferences or emotions, e.g., spite or rivalry. As suggested by Ernst Fehr and Klaus Schmidt (1999) and Gary E. Bolton and Axel Ockenfels (2000), players may be willing to sacrifice own earnings in order to reduce payoff inequities which would explain the *P* choices in the contradiction treatments. Alternatively, the occurrence of the high fraction of *P* decisions in the bottom game of Figure 4 may be due to negative emotions that follow the first player's *R* decision, which reduces the second player's earnings (Matthew Rabin, 1993). Notice that this earnings reduction does not occur when the first player chooses *R* for the game in the bottom part of Figure 3, which could explain the lower rate of punishments in that game.

The anomalous results of the contradiction treatments may not come as any surprise to Selten, the originator of the notion of subgame perfectness. His attitude toward game theory has been that there is a sharp contrast between standard theory and behavior. For a long time he essentially wore different hats when he did theory and ran experiments, although his 1994 Nobel prize was clearly for his contributions in theory. This schizophrenic stance may seem inconsistent, but it may prevent unnecessary anxiety, and some of Selten's recent theoretical work is based on models of boundedly rational (directional) learning (Selten and Joachim Buchta, 1998). In contrast, John Nash was reportedly discouraged by the predictive failures of game theory and gave up on both experimentation and game theory (Nasar, 1998 p. 150).

C. Two-Stage Bargaining Games

Bargaining has long been considered a central part of economic analysis, and at the same time, one of the most difficult problems for economic theory. One promising approach is to model unstructured bargaining situations "as if" the parties take turns making offers, with the costs of delayed agreement reflected in a shrinking size of the pie to be divided. This problem is particularly easy to analyze when the number of alternating offers is fixed and small.

Consider a bargaining game in which each player gets to make a single proposal for how to split a pie, but the amount of money to be divided falls from \$5 in the first stage to \$2 in the second. The first player proposes a split of \$5 that is either accepted (and implemented) or rejected, in which case the second player proposes a split of \$2 that is either accepted or rejected by the first player. This final rejection results in payoffs of zero for both players, so the second player can (in theory) successfully demand \$1.99 in the second stage if the first player prefers a penny to nothing. Knowing this, the first player should demand $3 and offer $2 to the other in the first stage. In a subgame perfect equilibrium, the first player receives the amount by which the pie shrinks, so a larger cost of delay confers a greater advantage to the player making the initial demand, which seems reasonable. For example, a similar argument shows that if the pie shrinks by $4.50, from $5 to $0.50, then the first player should make an initial demand of $4.50.

We used 60 subjects (six cohorts of ten subjects each), who were randomly paired for each of the two treatments described above (alternating in order and separated by other one-shot games). The average demand for the first player was $2.83 for the $5/$2 treatment, with a standard deviation of $0.29. This is quite close to the predicted $3.00 demand, and 14 of the 30 initial demands were exactly equal to $3.00 in this treasure treatment. But the average demand only increased to $3.38 for the other treatment with a $4.50 prediction, and 28 of the 30 demands were below the prediction of $4.50. Rejections were quite common in this contradiction treatment with higher demands and correspondingly lower offers to the second player, which is not surprising given the smaller costs of rejecting "stingy" offers.

These results fit into a larger pattern surveyed in Douglas D. Davis and Holt (1993 Chapter 5) and Roth (1995); initial demands in two-stage bargaining games tend to be "too low" relative to theoretical predictions when the equilibrium demand is high, say more than 80 percent of the pie as in our $5.00/$0.50 treatment, and initial demands tend to be close to predictions when the equilibrium demand is 50–75 percent of the pie (as in our $5.00/$2.00 treatment). Interestingly, initial demands are "too high" when the equilibrium demand is less than half of the pie. Here is an example of why theoretical explanations of behavior should not be based on experiments in only one part of the parameter space, and why theorists should have more than just a casual, secondhand knowledge of the experimental economics literature.[11] Many of the diverse theoretical explanations for anomalous behavior in bargaining games hinge on models of preferences in which a person's utility depends on the payoffs of both players, i.e., distribution matters (Bolton, 1998; Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000; Miguel Costa-Gomes and Klaus G. Zauner, 2001). The role of fairness is illustrated dramatically in the experiment reported in Goeree and Holt (2000a), who obtained even larger deviations from subgame-perfect Nash predictions than those reported here by giving subjects asymmetric money endowments that were paid independently of the bargaining outcome. These endowments were selected to accentuate the payoff inequities that result in the subgame-perfect Nash equilibrium, and hence their effect was to exaggerate fairness issues without altering the equilibrium prediction. The result (for seven different one-shot bargaining games) was for demands to be *inversely* related to the subgame-perfect Nash predictions.

III. Static Games with Incomplete Information

William Vickrey's (1961) models of auctions with incomplete information constitute one of the most widely used applications of game theory. If private values are drawn from a uniform distribution, the Bayesian Nash equilibrium predicts that bids will be proportional to value, which is generally consistent with laboratory evidence. The main deviation from theoretical predictions is the tendency of human subjects to "overbid" (relative to Nash), which is commonly rationalized in terms of risk aversion, an explanation that has lead to some controversy. Glenn W. Harrison (1989), for instance, argues that deviations from the Nash equilibrium may well be caused by a lack of monetary incentives since the costs of such deviations are rather small: the "flat maximum critique." Our approach here is to specify two auction games with identical Nash equilibria, but with differing incentives not to overbid.

First, consider a game in which each of two bidders receives a private value for a prize to be auctioned in a first-price, sealed-bid auction. In other words, the prize goes to the highest bidder for a price equal to that bidder's own bid. Each bidder's value for the prize is equally likely to be $0, $2, or $5. Bids are constrained to be integer dollar amounts, with ties decided by the flip of a coin.

The relevant Nash equilibrium in this game

[11] Another example is the development of theories of generalized expected utility to explain "fanning out" preferences in Allais paradox situations, when later experiments in other parts of the probability triangle found "fanning in."

TABLE 4—BIDDERS' EXPECTED PAYOFFS FOR THE (0,2,5) TREATMENT (EQUILIBRIUM BIDS MARKED WITH AN ASTERISK)

	Bid = 0	Bid = 1	Bid = 2	Bid = 3	Bid = 4	Bid = 5
Value = $0	0*	−0.5	−1.66	−3	−4	−5
Value = $2	0.33	0.5*	0	−1	−2	−3
Value = $5	0.83	2	2.5*	2	1	0

TABLE 5—BIDDERS' EXPECTED PAYOFFS FOR THE (0,3,6) TREATMENT (EQUILIBRIUM BIDS MARKED WITH AN ASTERISK)

	Bid = 0	Bid = 1	Bid = 2	Bid = 3	Bid = 4	Bid = 5
Value = $0	0*	−0.5	−1.66	−3	−4	−5
Value = $3	0.5	1*	0.83	0	−1	−2
Value = $6	[illegible]	2.5	3.33*	3	2	1

with incomplete information about others' preferences is the Bayesian Nash equilibrium, which specifies an equilibrium bid for each possible realization of a bidder's value. It is straightforward but tedious to verify that the Nash equilibrium bids are $0, $1, and $2 for a value of $0, $2, and $5 respectively, as can be seen from the equilibrium expected payoffs in Table 4. For example, consider a bidder with a private value of $5 (in the bottom row) who faces a rival that bids according to the proposed Nash solution. A bid of 0 has a one-half chance of winning (decided by a coin flip) if the rival's value, and hence the rival's bid, is zero, which happens with probability one-third. Therefore, the expected payoff of a zero bid with a value of $5 equals $1/2 \cdot 1/3 \cdot (\$5 - \$0) = \$5/6 = 0.83$. If the bid is raised to $1, the probability of winning becomes 1/2 (1/3 when the rival's value is $0 plus 1/6 when the rival's value is $2). Hence, the expected payoff of a $1 bid is $1/2 \cdot (\$5 - \$1) = \$2$. The other numbers in Table 4 are derived in a similar way. The maximum expected payoff in each row coincides with the equilibrium bid, as indicated by an asterisk (*). Note that the equilibrium involves bidding about one-half of the value.[14]

Table 5 shows the analogous calculations for the second treatment, with equally likely private values of $0, $3, or $6. Interestingly, this increase in values does not alter the equilibrium bids in the unique Bayesian Nash equilibrium, as indicated by the location of optimal bids for each value. Even though the equilibria are the same, we expected more of an upward bias in bids in the second (0, 3, 6) treatment. The intuition can be seen by looking at payoff losses associated with deviations from the Nash equilibrium. Consider, for instance, the middle-value bidder with expected payoffs shown in the second rows of Tables 4 and 5. In the (0, 3, 6) treatment, the cost of bidding $1 above the equilibrium bid is $1 − $0.83 = $0.17, which is less than the cost of bidding $1 below the equilibrium bid: $1 − $0.50 = $0.50. In the (0, 2, 5) treatment, the cost of an upward deviation from the equilibrium bid is greater than the cost of a downward deviation, see the middle row of Table 4. A similar argument applies to the high-value bidders, while deviation costs are the same in both treatments for the low-value bidder. Hence we expected more overbidding for the (0, 3, 6) treatment.

This intuition is borne out by bid data for the 50 subjects who participated in a single auction under each condition (again, alternating the order of the two treatments and separating the two auctions with other one-shot games). Eighty percent of the bids in the (0, 2, 5) treatment matched the equilibrium: the average bids for low-, medium-, and high-value bidders were $0, $1.06, and $2.64, respectively. In contrast, the average bids for the (0, 3, 6) treatment were $0, $1.82, and $3.40 for the three value levels, and only 50 percent of all bids were Nash bids. The bid frequencies for each value are shown in Table 6. As in previous games, deviations from Nash behavior in these private-value auctions seem to be sensitive to the costs of deviation. Of

[14] The bids would be exactly one half of the value if the highest value were $4 instead of $5, but we had to raise the highest value to eliminate multiple Nash equilibria.

TABLE 6—BID FREQUENCIES
(EQUILIBRIUM BIDS MARKED WITH AN ASTERISK*)

(0, 2, 5) Treatment			(0, 3, 6) Treatment		
	Bid	Frequency		Bid	Frequency
Value = 0	0*	20	Value = 0	0*	17
Value = 2	1*	15	Value = 3	1*	5
	2	1		2	11
	3	0		3	2
Value = 5	1	1	Value = 6	1	0
	2*	5		2*	3
	3	6		3	4
	4	2		4	0
	5	1		5	1
	6	0		6	1

course, this does not rule out the possibility that risk aversion or some other factor may also have some role in explaining the overbidding observed here, especially the slight overbidding for the high value in the (0, 2, 5) treatment.[16]

IV. Dynamic Games with Incomplete Information: Signaling

Signaling games are complex and interesting because the two-stage structure allows an opportunity for players to make inferences and change others' inferences about private information. This complexity often generates multiple equilibria that, in turn, have stimulated a sequence of increasingly complex refinements of the Nash equilibrium condition. Although it is unlikely that introspective thinking about the game will produce equilibrium behavior in a single play of a game this complex (except by coincidence), the one-shot play reveals useful information about subjects' cognitive processes.

In the experiment, half of the subjects were designated as "senders" and half as "responders." After reading the instructions, we began by throwing a die for each sender to determine whether the sender was of type *A* or *B*. Everybody knew that the *ex ante* probability of a type A sender was one half. The sender, knowing his/her own type would choose a signal, *Left* or *Right*. This signal determined whether the payoffs on the right or left side of Table 7 would be used. (The instructions used letters to identify the signals, but we will use words here to facilitate the explanations.) This signal would be communicated to the responder that was matched with that sender. The responder would see the sender's signal, *Left* or *Right*, but not the sender's type, and then choose a response, *C*, *D*, or *E*. The payoffs were determined by Table 7, with the sender's payoff to the left in each cell.

First, consider the problem facing a type A sender, for whom the possible payoffs from sending a *Left* signal (300, 0, 500) seem, in some loose sense, less attractive than those for sending a *Right* signal (450, 150, 1,000). For example, if each response is thought to be equally likely (the "principle of insufficient reason"), then the *Right* signal has a higher expected payoff. Consequently, type A's payoffs have been made bold for the *Right* row in the top right part of Table 7. Applying the principle of insufficient reason again, a type B sender looking at the payoffs in the bottom row of the table might be more attracted by the *Left* signal, with payoffs of (500, 300, 300) as compared with (450, 0, 0).[17] Therefore, sender B's payoffs are in bold for the *Left* signal. In fact, all of the type B subjects did send the *Left* signal, and seven of the ten type A subjects sent the *Right* signal. All responses in this game were *C*, so all but three of the outcomes were in one of the two cells marked by an asterisk. Notice that this is an equilibrium, since neither type of sender would benefit from sending the other signal, and the respondent cannot do any better than the maximum payoff received in the marked cells. This is a separating Nash equilibrium; the signal reveals the sender's type.

The payoff structure for this game becomes a little clearer if you think of the responses as one of three answers to a request: Concede, Deny, or

[16] Goeree et al. (2000) report a first-price auction experiment with six possible values, under repeated random matching for ten periods. A two-parameter econometric model that includes both decision error and risk aversion provides a good fit of 67 value/bid frequencies and shows that both the error parameter and risk aversion parameter are significantly different from zero. David Lucking-Reiley (1999) mentions risk aversion as a possible explanation for overbidding in a variety of auction experiments.

[17] These are not dominance arguments, since the responder can respond differently to each signal, and the lowest payoff from sending one signal is not higher than the highest payoff that can be obtained from sending the other signal.

TABLE 7—SIGNALING WITH A SEPARATING EQUILIBRIUM (MARKED BY ASTERISKS) (SENDER'S PAYOFF, RESPONDER'S PAYOFF)

	Response to *Left* signal				Response to *Right* signal		
	C	*D*	*E*		*C*	*D*	*E*
Type A sends *Left*	300, 300	0, 0	500, 300	Type A sends *Right*	**450**, 900 (*)	**150**, 150	**1,000**, 300
Type B sends *Left*	**500**, 500 (*)	**300**, 450	**300**, 0	Type B sends *Right*	450, 0	0, 300	0, 150

TABLE 8—SIGNALING WITHOUT A SEPARATING EQUILIBRIUM (SENDER'S PAYOFF, RESPONDER'S PAYOFF)

	Response to *Left* signal				Response to *Right* signal		
	C	*D*	*E*		*C*	*D*	*E*
Type A sends *Left*	300, 300	0, 0	500, 300	Type A sends *Right*	**450**, 900	**150**, 150	**1,000**, 300
Type B sends *Left*	**300**, 300	**300**, 450	**300**, 0	Type B sends *Right*	450, 0	0, 300	0, 150

Evade. With some uncertainty about the sender's type, Evade is sufficiently unattractive to respondents that it is never selected. Consider the other two responses and note that a sender always prefers that the responder choose Concede instead of Deny. In the separating equilibrium, the signals reveal the senders' types, the responder always Concedes, and all players are satisfied. There is, however, a second equilibrium for the game in Table 7 in which the responder Concedes to *Left* and Denies *Right*, and therefore both sender types send *Left* to avoid being Denied.[18] Backward induction rationality (of the sequential Nash equilibrium) does not rule out these beliefs, since a deviation does not occur in equilibrium, and the respondent is making a best response to the beliefs. What is unintuitive about these beliefs (that a deviant *Right* signal comes from a type B) is that the type B sender is earning 500 in this (*Left*, *Concede*) equilibrium outcome, and no deviation could conceivably increase this payoff. In contrast, the type A sender is earning 300 in the *Left* side pooling equilibrium, and this type could possibly earn more (450 or even 1,000), depending on the response to a deviation. The In-Koo Cho and Kreps (1987) *intuitive criterion* rules out these beliefs, and selects the separating equilibrium observed in the treasure treatment.[19]

The game in Table 8 is a minor variation on the previous game, with the only change being that the (500, 500) in the bottom left part of Table 7 is replaced by a (300, 300) payoff.[20] As before, consider the sender's expected payoffs when each response is presumed to be equally likely, which leads one to expect that type A senders will choose *Right* and that type B senders will choose *Left*, as indicated by the bold payoff numbers. In the experiment, 10 of the 13 type A senders did choose *Right*, and 9 of the 11 type B senders did choose *Left*. But the separation observed in this contradiction treatment is *not* a Nash equilibrium.[21] All equilibria for this

[18] To check that the responder has no incentive to deviate, note that Concede is a best response to a *Left* regardless of the sender's type, and that Deny is a best response to a deviant *Right* signal if the responder believes that it was sent by a type B.

[19] Brandts and Holt (1992, 1993) report experimental data that contradict the predictions of the intuitive criterion, i.e., the decisions converged to an equilibrium ruled out by that criterion.

[20] Unlike the paired treatments considered previously, the payoff change for these signaling games does alter the set of Nash equilibria.

[21] The respondents would prefer to Concede to a *Right* signal and Deny a *Left* signal. Type B senders would therefore have an incentive to deviate from the proposed separating equilibrium and send a *Right* signal. In the experiment, half of the *Left* signals were Denied, whereas only 2 of the 12 *Right* signals were Denied.

contradiction treatment involve "pooling," with both types sending the same signal.[22]

V. Explaining Anomalous Behavior in One-Shot Games

Although the results for the contradiction treatments seem to preclude a game-theoretic explanation, many of the anomalous data patterns are related to the nature of the incentives. This suggests that it may be possible to develop formal models that explain both treasures and contradictions. Below we discuss several recent approaches that relax the common assumptions of perfect selfishness, perfect decision-making (no error), and perfect foresight (no surprises).

As noted in Section II, the anomalies observed for the dynamic games in Figures 3 and 4 are consistent with models of inequity aversion (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000), which assumes that people like higher payoffs for themselves and dislike earning less than the other person ("envy") or earning more ("guilt"). Inequity aversion also seems to play a role when players bargain over the division of a fixed amount of money (Goeree and Holt, 2000a). However, it cannot explain observed behavior in the contradiction matching pennies treatments. Consider, for example, the "320" version of the matching pennies game in Table 1. Since the column player is averse to the (320, 40) outcome, the column player would only be willing to randomize between *Left* and *Right* if the attractiveness of *Right* is increased by having the row player play *Bottom* more often than the 0.5 probability that would make a purely selfish column player indifferent. This prediction, that the row player should play *Bottom* more often, is sharply contradicted by the data in the middle part of Table 1.[23,24]

Another possibility is that behavior in one-shot games conforms to a simple heuristic. Indeed, some experimental economists have suggested that subjects in the initial period of a repeated game choose the decision that maximizes their security level, i.e., the "maximin" decision. For example, in the Kreps game of Table 5, the frequently observed *Non-Nash* decision maximizes column's security. The strong treatment effects in the matching pennies games cannot be explained in this way, however, since in all three treatments each player's minimum payoff is the same for both decisions. A similar argument applies to the coordination game in Table 2. Moreover, the security-maximizing choices in the traveler's dilemma and the minimum-effort coordination game are the lowest possible decision, which is contradicted by the high claim and effort choices in the contradiction treatments. Subjects may be risk averse in unfamiliar situations, but the extreme risk aversion implied by maximin security is generally not observed. Furthermore, heuristics based on reciprocity or a status quo bias do not apply to single-stage, one-shot games where there is neither a precedent nor an opportunity to reciprocate. Nor can loss aversion be the primary cause, since losses are impossible in most of the games reported here, and the possibility of a loss had no effect in the Kreps game.

As an alternative to simple heuristics, one could try to model players' introspective thought processes. Previous models have typically specified some process of belief formation, assuming that players *best* respond to the

[22] For example, it is an equilibrium for both types to send *Right* if a *Left* signal triggers a *C* or a *D* response. The *D* response to *Left* is appropriate if the respondent thinks the deviant signal comes from a type B sender, and the *C* response is appropriate if the deviant is thought to be of type A. Beliefs that the deviant is of type A are intuitive, since type A earns 450 in equilibrium and could possibly earn more (500) by switching to *Left* (if an *E* response follows). A second pooling equilibrium involves both types sending a *Left* signal, to which the respondents Concede. A deviant *Right* signal is Denied, which is appropriate if the respondent thinks the deviant signal comes from a type B sender. Again these beliefs are intuitive since the type B sender could possibly gain by deviating.

[23] Goeree et al. (2000) report formal econometric tests that reject the predictions of inequity aversion models in the context of a group of repeated asymmetric matching pennies games.

[24] Payoff inequity aversion also has no effect in the minimum-effort coordination game: any common effort level is still a Nash equilibrium. To see this, note that a unilateral effort increase from a common level reduces one's own payoff and creates an disadvantageous inequity. Similarly, a unilateral decrease from a common effort level reduces one's payoff and creates an inequity where one earns more than the other, since their costly extra effort is wasted. Thus inequity aversion cannot explain the strong effect of an increase in effort costs.

resulting beliefs.[19] The experiments reported above indicate that *magnitudes* (not just *signs*) of payoff differences matter, and it is thus natural to consider a decision rule for which choice probabilities are positively but imperfectly related to payoffs. The logit rule, for example, specifies that choice probabilities, p_i, for options $i = 1, \ldots, m$, are proportional to exponential functions of the associated expected payoffs, π_i^e:

$$(1) \quad p_i = \frac{\exp(\pi_i^e/\mu)}{\sum_{j=1,\ldots,m} \exp(\pi_j^e/\mu)}, \quad i = 1, \ldots, m,$$

where the sum in the denominator ensures that the probabilities sum to one, and the "error parameter," μ, determines how sensitive choice probabilities are to payoff differences.[20]

In order to use the "logit best response" in (1), we need to model the process of belief formation, since belief probabilities are used to calculate the expected payoffs on the right side of (1). By the principle of insufficient reason one might postulate that each of the others' actions are equally likely. This corresponds to the Stahl and Wilson (1995) notion of "level one" rationality, which captures many of the first-period decisions in the "guessing game" reported by Nagel (1995).[21] It is easy to verify that level one rationality also provides good predictions for both treasure and contradiction treatments in the traveler's dilemma, the minimum-effort coordination game, and the Kreps game. There is evidence, however, that at least some subjects form more precise beliefs about others' actions, possibly through higher levels of introspection.[22] In the matching pennies games in Table 1, for example, a flat prior makes column indifferent between *Left* and *Right*, and yet most column players seem to anticipate that row will choose *Top* in the 320 version and *Bottom* in the 44 version of this game.

Of course, what the other player does depends on what they think you will do, so the next logical step is to assume that *others* make responses to a flat prior, and then you respond to that anticipated response (Selten, 1991). This is Stahl and Wilson's (1995) "level two" rationality. There is, however, no obvious reason to truncate the levels of iterated thinking. The notion of rationalizability discussed above, for example, involves infinitely many levels of iterated thinking, with "never best" responses eliminated in succession. But rationalizability seems to imply too much rationality, since it predicts that all claims in the traveler's dilemma will be equal to the minimum claim, independent of the penalty/reward parameter. One way to limit the precision of the thought process, without making an arbitrary assumption about the number of iterations, is to inject increasing amounts of noise into higher levels of iterated thinking (Goeree and Holt, 1999; Kübler and Weizsäcker, 2000). Let ϕ_μ denote the logit best-response map (for error rate μ) on the right side of (1). Just as a single logit response to beliefs, p_0, can be represented as $p = \phi_\mu(p_0)$, a series of such responses can be represented as:[23]

[19] Perhaps the best-known model of introspection is Harsanyi and Selten's (1988) "tracing procedure." This procedure involves an axiomatic determination of players' common priors (the "preliminary theory") and the construction of a modified game with payoffs for each decision that are weighted averages of those in the original game and of the expected payoffs determined by the prior distribution. By varying the weight on the original game, a sequence of best responses for the modified game are generated. This process is used to select one of the Nash equilibria of the original game. Gonzalo Olcina and Amparo Urbano (1994) also use an axiomatic approach to select a prior distribution, which is then revised by a simulated learning process that is essentially a partial adjustment from current beliefs to best responses to current beliefs. Since neither the Harsanyi-Selten model nor the Olcina/Urbano model incorporates any noise, they predict that behavior will converge to the Nash equilibrium in games with a unique equilibrium, which is an undesirable feature in light of the contradictions data reported above.

[20] As μ goes to zero, payoff differences are "blown up," and the probability of the optimal decision converges to 1. In the other extreme, as μ goes to ∞, the choice probabilities converge to $1/m$ independently of expected payoffs. See R. Duncan Luce (1959) for an axiomatic derivation of the logit choice rule in (1).

[21] In our own work, we have used a noisy response to a flat prior as a way of starting computer simulations of simulations of behavior in repeated games (Brandts and Holt, 1996; Capra et al., 1999, 2002; Goeree and Holt, 1999).

[22] Costa-Gomes et al. (2001), for example, infer some heterogeneity in the amount of introspection by observing the types of information that subjects acquire before making a decision.

[23] Goeree and Holt (2000b) use continuity arguments to show that the limit in (2) exists for all (nonnegative) error parameters and general spaces.

$$p = \lim_{n \to \infty} \phi_{\mu_1}(\phi_{\mu_2}(\cdots \phi_{\mu_n}(p_0))), \tag{2}$$

where $\mu_1 \leq \mu_2 \leq \ldots$ with μ_n converging to infinity.[30] This assumption captures the idea that it becomes increasingly complex to do more and more iterations.[31] Since ϕ_μ for $\mu = \infty$ maps the whole probability simplex to a single point, the right side of (2) is independent of the initial belief vector p_0. Moreover, the introspection process in (2) yields a unique outcome even in games with multiple Nash equilibria. Note that the choice probabilities on the left side of (2) generally do not match the beliefs at any stage of the iterative process on the right. In other words, the introspective process allows for surprises, which are likely to occur in one-shot games.

For games with very different levels of complexity such as the ones reported here, the error parameters that provide the best fit are likely to be different. In this case, the estimates indicate the degree of complexity, i.e., they serve as a measurement device. For games of similar complexity, the model in (2) could be applied to predict behavior across games. We have used it to explain data patterns in a series of 37 single matrix games, assuming a simple two-parameter model for which $\mu_n = \mu t^n$, where t determines the rate at which noise increases with higher iterations (Goeree and Holt, 2000b). The estimated value ($t = 4.1$) implies that there is more noise for higher levels of introspection, a result that is roughly consistent with estimates obtained by Kübler and Weizsäcker (2000) for data from information-cascade experiments.

The analysis of introspection is a relatively understudied topic in game theory, as compared with equilibrium refinements and learning, for example. Several of the models discussed above do a fairly good job of organizing the qualitative patterns of conformity and deviation from the predictions of standard theory, but there are obvious discrepancies. We hope that this paper will stimulate further theoretical work on models of behavior in one-shot games. One potentially useful approach may be to elicit beliefs directly as the games are played (Theo Offerman, 1997; Andrew Schotter and Yaw Nyarko, 1998).

VI. Conclusion

One-shot game experiments are interesting because many games are in fact only played once; single play is especially relevant in applications of game theory in other fields, e.g., international conflicts, election campaigns, and legal disputes. The decision makers in these contexts, like the subjects in our experiments, typically have experience in similar games with other people. One-shot games are also appealing because they allow us to abstract away from issues of learning and attempts to manipulate others' beliefs, behavior, or preferences (e.g., reciprocity, cooperativeness). This paper reports the results of ten pairs of games that are played only once by subjects who have experience with other one-shot and repeated games. The Nash equilibrium (or relevant refinement) provides accurate predictions for standard versions of these games. In each case, however, there is a matched game for which the Nash prediction clearly fails, although it fails in a way that is consistent with simple (non-game-theoretic) intuition. The results for these experienced subjects show:

(1) Behavior may diverge sharply from the unique rationalizable (Nash) equilibrium in a social (traveler's) dilemma. In these games, the Nash equilibrium is located on one side of the range of feasible decisions, and data for the contradiction treatment have a mode on the *opposite* side of this range. The most salient feature of the data is the extreme sensitivity to a parameter that has no effect on the Nash outcome.

(2) Students suffering through game theory classes may have good reasons when they have trouble understanding why a change in one player's payoffs only affects the

[30] The case of a constant parameter ($\mu_1 = \mu_2 = \ldots = \mu$) is of special interest. In this case, the process may not converge for some games (e.g., matching pennies), but when it does, the limit probabilities, p^*, must be invariant under the logit map: $\phi_\mu(p^*) = p^*$. A fixed point of this type constitutes a "logit equilibrium," which is a special case of the quantal-response equilibrium defined in McKelvey and Palfrey (1995). It is in this sense that the logit equilibrium arises as a limit case of the noisy introspective process defined in (2).

[31] For an interesting alternative approach, see Capra (1999). In her model, beliefs are represented by degenerate distributions that put all probability mass at a single point. The location of the belief points is, of course, stochastic.

other player's decision probabilities in a mixed-strategy Nash equilibrium. The data from matching pennies experiments show strong "own-payoff" effects that are not predicted by the unique (mixed-strategy) Nash equilibrium. The Nash analysis seems to work only by coincidence, when the payoff structure is symmetric and deviation risks are balanced.

(3) Effort choices are strongly influenced by the cost of effort in coordination games, an intuitive result that is not explained by standard theory, since *any* common effort is a Nash equilibrium in such games. Moreover, as Kreps conjectured, it is possible to design coordination games where the majority of one player's decisions correspond to the only action that is not part of any Nash equilibrium.

(4) Subjects often do not trust others to be rational when irrationality is relatively costless. Moreover, "threats" that are *not* credible in a technical sense may nevertheless alter behavior in simple two-stage games when carrying out these threats is not costly.

(5) Deviations from Nash predictions in alternating-offer bargaining games and in private-value auctions are inversely related to the costs of such deviations. The effects of these biases can be quite large in the games considered.

(6) It is possible to set up a simple signaling game in which the decisions reveal the signaler's type (separation), even though the equilibrium involves pooling.

So what should be done? Reinhard Selten, one of the three game theorists to share the 1994 Nobel Prize, has said: "Game theory is for proving theorems, not for playing games."[28] Indeed, the internal elegance of traditional game theory is appealing, and it has been defended as being a normative theory about how perfectly rational people should play games with each other, rather than a positive theory that predicts actual behavior (Rubinstein, 1982). It is natural to separate normative and positive studies of individual decision-making, which allows one to compare actual and optimal decision-making. This normative-based defense is not convincing for games, however, since the best way for one to play a game depends on how others *actually* play, not on how some theory dictates that rational people *should* play. John Nash, one of the other Nobel recipients, saw no way around this dilemma, and when his experiments were not providing support to theory, he lost whatever confidence he had in the relevance of game theory and focused on more purely mathematical topics in his later research (Nasar, 1998).

Nash seems to have undersold the importance of his insight, and we will be the first to admit that we begin the analysis of a new strategic problem by considering the equilibria derived from standard game theory, before considering the effects of payoff and risk asymmetries on incentives to deviate. But in an interactive, strategic context, biases can have reinforcing effects that drive behavior well away from Nash predictions, and economists are starting to explain such deviations using computer simulations and theoretical analyses of learning and decision error processes. There has been relatively little theoretical analysis of one-shot games where learning is impossible. The models of iterated introspection discussed here offer some promise in explaining the qualitative features of deviations from Nash predictions enumerated above. Taken together, these new approaches to a stochastic game theory enhance the behavioral relevance of standard game theory. And looking at laboratory data is a lot less stressful than before.

[28] Selten reiterated this point of view in a personal communication to the authors.

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Answer to Problem 1 in Problem Set 5

(i) When Z = 40000, the following are the three Nash equilibria of the simultaneous-move "Claim Subgame" in the final stage:

(i) [***Y***: 60K; ***X***: 60K], generating payoff of 60K for each player

(ii) [***Y***: 40K; ***X***: 40K], generating payoff of 40K for each player

(iii) [***Y***: {40K, 60K; 3/4, 1/4}; *X*: {40K, 60K; 3/4, 1/4}], generating payoff of 37.5K for each player.

As a consequence, the following two results are true:

<u>Result 1</u> There are three *SPNE* of the whole game:

1. [***Y***: *B*, 60K if (*B*, *B*); ***X***: *B*, 60K if (*B*, *B*)]
2. [***Y***: *A*, 40K if (*B*, *B*); ***X***: *A*, 40K if (*B*, *B*)]
3. [***Y***: *A*, {40K, 60K; 3/4, 1/4} if (*B*, *B*); *X*: *A*, {40K, 60K; 3/4, 1/4} if (*B*, *B*)]

<u>Result 2</u> Of the three *SPNE* of the whole game, only the first *SPNE* survives the Forward Induction Refinement, while the other two fail the Forward Induction Refinement.

Reason: If ***Y*** starts the game by playing *B*, ***X*** must infer that ***Y*** will play 60K in the final Claim Subgame as that is the unique continuation ***SPNE*** in which ***Y*** will achieve a greater payoff than 45K. And when ***X*** infers that ***Y*** will play 60K in the final Claim Subgame, it will be in ***X***'s self-interest to make statement *B* and then play 60K in the final Claim Subgame.

(ii) When Z = 50000, the following are the three Nash equilibria of the simultaneous-move "Claim Subgame" in the final stage:

(i) [***Y***: 60K; ***X***: 60K], generating payoff of 60K for each player

(ii) [***Y***: 40K; ***X***: 40K], generating payoff of 50K for each player

(iii) [***Y***: {40K, 60K; 3/5, 2/5}; *X*: {40K, 60K; 3/5, 2/5}], generating payoff of 42K for each player.

As a consequence, the following two results are true:

<u>Result 1</u> There are three *SPNE* of the whole game:

1. [***Y***: *B*, 60K if (*B*, *B*); ***X***: *B*, 60K if (*B*, *B*)]
2. [***Y***: *B*, 40K if (*B*, *B*); ***X***: *B*, 40K if (*B*, *B*)]
3. [***Y***: *A*, {40K, 60K; 3/5, 2/5} if (*B*, *B*); *X*: *A*, {40K, 60K; 3/5, 2/5} if (*B*, *B*)]

<u>Result 2</u> All three *SPNE* of the whole game survive the Forward Induction Refinement.

Reason: If ***Y*** starts the game by playing *B*, ***X*** can only deduce that ***Y*** might either play 60K or 40K in the final Claim Subgame (as there is *no unique* continuation ***SPNE*** when ***Y*** starts the game by playing *B* in which ***Y*** will achieve a greater payoff than 45K). In that case, there exists no continuation strategy for ***X*** that will guarantee her a payoff greater than 44K (which she can ensure by making statement *A*). That is why, we are required to be "conservative" and to refrain from claiming that any of the three *SPNE* of the whole game fail the Forward Induction Refinement.

Viewing social / economic interactions
through the lens of Game Theory

… and learning about extensions
in Game Theory as we go along

Conflicting Coordination: *Battle of the Sexes*

Friday Night Out

Husband \ Wife	Spy Movie	Classical Concert
Spy Movie	1st, 2nd	3rd, 3rd
Classical Concert	4th, 4th	2nd, 1st

- Players have to coordinate on identical actions, but disagree about which one, multiple ***Pareto-incomparable*** coordinated pure-strategy Nash equilibria
- All strategies *rationalizable*; Spy / Concert *maximin* for H / W
- If players can stagger their moves, that coordinated outcome which *favours the 'first-mover'* will obtain as the unique *SPNE*
- Pre-play communication before a one-time interaction can lead to 'haggling' among players to select a specific coordination
- Similar issues arise when firms have to adopt a common 'standard', and every standard is *proprietary* to a specific firm (e.g., Sony's Blu-Ray and Toshiba's HD DVD technologies)

Games of Coordination, Anti-coordination, & Dis-coordination

- In many strategic interactions, the critical issue for players is to achieve ***specific kinds of coordination / anti-coordination / dis-coordination in strategy choices***
- Simplest such games are games of **pure non-conflicting coordination**, where there are "multiple equilibria", and *failing to coordinate* gives worse payoffs compared to *coordination*

Driving convention

Me \ You	Right hand	Left hand
Right hand	good, good	bad, bad
Left hand	bad, bad	good, good

- In most games of coordination, anti-coordination and dis-coordination, a unique pure-strategy Nash does not exist (while there might exist a unique mixed-strategy Nash equilibrium)

Anti-Coordination: Players need to coordinate on an asymmetric action pair, and disagree regarding the pair to be chosen

1. Minority Game

Us \ Others	El Farol bar	another bar
El Farol bar	3rd, 3rd	1st, 2nd
another bar	2nd, 1st	4th, 4th

2. Game of Chicken

Budweiser \ Miller	hold course	swerve
hold course	4th, 4th	1st, 2nd/3rd
swerve	2nd/3rd, 1st	2nd, 2nd

- Analogous problems arise when:

1. two firms compete for two "differentially profitable" markets, or
2. two firms compete in one small market with "monopoly profits > exit payoff > duopoly profits"

Ranked Non-conflicting Coordination

The Keyboard Puzzle

Typist A \ Typist B	Dvorak	QWERTY
Dvorak	best, best	bad, bad
QWERTY	bad, bad	avg, avg

- Multiple ***Pareto-ranked*** coordinated Nash equilibria
- All strategies are *rationalizable* and *maximin*
- If the game is between a few familiar players, 'pre-play communication' and / or 'any staggering of move order' will enable coordination
- But what happens when there are many players 'sitting' with the 'average choice', and individually considering transition to the 'better choice' without knowing others' transition costs?

  Cliometricians ask: Can this be the reason why historically, 'better technologies' have not always been adopted?

  We will revisit this game after formalizing it as a dynamic game of incomplete information

Achieving fair outcomes in conflicting- & anti-coordination games

- In conflicting-coordination and anti-coordination games, every pure Nash eqm gives dissimilar payoffs to *ex ante* symmetric players
- One way to achieve payoff-symmetry ("fairness") is to focus on "mixed strategy equilibria"
  - but it is not necessarily the best way as the payoffs may be quite small

Minority Game

	El Farol	Another
El Farol	1, 1	4, 2
Another	2, 4	0, 0

- In Minority Game
  - Mixed-strategy Nash eqm (4/5, 1/5) giving symmetric expected payoffs (1.6, 1.6)

Battle of Sexes

	Movie	Concert
Movie	4, 2	1, 1
Concert	0, 0	2, 4

- In Battle of Sexes:
  - Mixed Nash {(4/5, 1/5), (1/5, 4/5)} giving symmetric expected payoffs (1.6, 1.6)

Games of Dis-Coordination

1. Unilateral Dis-coordination: *One player wants dis-coordination while the other wants coordination (like in the matching-pennies game)*

- In tennis, if server aims at receiver's backhand then latter wants to prepare for backhand;
  - in which case server wants to serve to receiver's forehand;
  - in which case receiver wants to prepare for forehand; …
- **Attack-Defense games**: defender wants to *coordinate* with attacker, while attacker wants to *dis-coordinate* with defender

2. Bilateral Dis-coordination: ***Both players desire dis-coordination***

In "Rock-Scissors-Paper" – Rock breaks Scissors, Scissors cut Paper, Paper covers Rock

- The **Colonel Blotto Game** (look it up)

- ***Dis-coordination games*** have no pure-strategy Nash equilibria
- There is *second-mover advantage* in the sequential-move version of a dis-coordination game

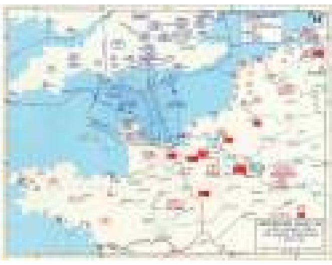
Best Response Functions and Mixed-Strategy Equilibrium

		Attacker: target C	Attacker: target N
Defender	defend C	90 , 10	20 , 80
	defend N	30 , 70	60 , 40

- y = probability that Defender defends C
- x = probability that Attacker attacks C

Attack-Defense games

- The greatest 'matching pennies game' ever played: ***Operation Overlord*** in June 1944

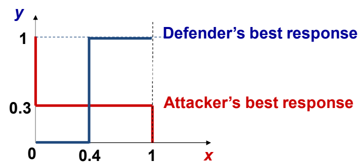


- Q: Do the Allies target Calais coast or Normandy coast? Do the Axis powers defend Calais or Normandy?
- Espionage and deception are valuable in such games

Best Response Functions and Mixed-Strategy Equilibrium

		Attacker: target C	Attacker: target N
Defender	defend C	90 , 10	20 , 80
	defend N	30 , 70	60 , 40

- y = probability that Defender defends C
- x = probability that Attacker attacks C



$\Rightarrow x^* = 40\%$, and $y^* = 30\%$,

Unique Nash eqm: Attacker {C, N; 0.4, 0.6}; Defender {C, N; 0.3, 0.7}

An Attack-Defense Game

- Two viable locations of conflict: C and N

		Attacker: target C	Attacker: target N
Defender	defend C	90 , 10	20 , 80
	defend N	30 , 70	60 , 40

Defender's payoff = probability that defender will be win × 100

Attacker's payoff = probability that attacker will win × 100

- It is a "constant sum game" [a strictly competitive game] with defender being *more vulnerable* at N [$60 < 90$ & $20 < 30$]
- In this game, we will engage in a "direct method" of determining mixed-strategy Nash equilibria, by examining "best response functions" expressed in terms of a player's probability of playing different mixed strategies

Minimax Theorem & 'Payoff Invariance' Result in constant-sum games

		Attacker: target C	Attacker: target N
Defender	defend C	90 , 10	20 , 80
	defend N	30 , 70	60 , 40

Attacker: {C, N; 0.4, 0.6}; Defender: {C, N; 0.3, 0.7}

- Of course, when Defender plays {C, N; 0.3, 0.7}, Attacker is indifferent between attacking at C and at N [her expected payoff is **52** in either case]
- And surprisingly, the following is also true!
- When the Attacker commits to playing {C, N; 0.4, 0.6}, she gets the same expected payoff **52** irrespective of where the Defender chooses to defend
- Of course, when Attacker plays {C, N; 0.4, 0.6}, Defender is indifferent between defending at C and at N [her expected payoff is **48** in either case]
- When the Defender commits to playing {C, N; 0.3, 0.7}, he gets the same expected payoff **48** irrespective of where the Attacker chooses to attack
- ***Minimax Theorem***: In a 2-player constant-sum (**S**) game, there exists a value **V** < **S** such that 'player one' can guarantee herself the expected payoff **V** irrespective of the strategy executed by 'player two', and 'player two' can guarantee himself the expected payoff [**S – V**] irrespective of the strategy executed by 'player one'

Testing the Payoff Invariance Result in Sports

- Walker and Wooders (*American Economic Review*, 2001) followed ten Wimbledon men's finals, coded serves as 'to backhand' and 'to forehand', and determined return outcome

  The data passed the statistical test that the probability of a serve being returned does not depend on its placement

- Palacios-Huerta (*Review of Economic Studies*, 2005) tracked forty-two soccer penalty kicks in World Cups between leading scorers and goalkeepers

  - coded kicks as 'to left', 'to centre', and 'to right' of the goalkeeper

  The data passed the statistical test that the probability of a goal being scored does not depend on shot placement

A constant-sum game with a pure-strategy Nash equilibrium

	left	centre	right
up	10, 10	20, 00	12, 08
middle	00, 20	10, 10	15, 05
down	08, 12	05, 15	10, 10

Unique (dominance-solved) Nash equilibrium: {**up**, **left**}

up is Row's maximin strategy & **left** is Column's maximin strategy

Result: If there exists a pure-strategy Nash equilibrium in a constant-sum game then it is necessarily unique, and in that unique pure-strategy Nash equilibrium each player's Nash strategy is necessarily her maximin strategy

The Great Keyboard Debate: QWERTY versus Dvorak

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Abstract. Many layouts have been proposed for keyboards since the invention of the typewriter in the 1800s. The two most popular formats, QWERTY and Dvorak, have been at the centre of a great debate for decades—both layouts are claimed to be optimal. Academic evidence is available supporting each layout, yet over 75 years since the debate began there is still no clear answer. This paper discusses the history of each format and presents new scientific experiments designed to give a final determination of the optimal layout.

1 Introduction

Since the beginning of the 20th Century a battle has quietly been fought over the optimal layout of alphanumeric keys on a keyboard. One might surmise the winner to be QWERTY because this is virtually the only keyboard layout available for purchase today. Although not universally accepted, the most commonly cited reason that QWERTY is so popular is because such an overwhelming market share was gained by QWERTY keyboards that it was unprofitable to continue making Dvorak ones [1]. The question of which keyboard format is optimal still rages on. Academic writing seems to point to QWERTY and Dvorak reaching a draw; equal numbers of papers exist in favour of each format [2–4].

This paper investigates which layout is the most efficient for text entry on a keyboard. The research conducted in this paper is part of a PhD research project on identifying a typist by the way they type. Whilst the cognitive difficulties associated with text entry are ignored in this paper because they are largely irrelevant with regards to keyboard layout, these difficulties can be used to identify a computer user. This paper is related to, but not part of, the main goal of the PhD: typist recognition. The PhD is mainly focused on implementation of machine learning techniques to process typing recordings into profiles of users. The work in this paper resulted from experimentation of building profiles based on finger movement instead of absolute times. The PhD project began in March 2006.

The next section in this paper describes the history of the two formats. Section 3 describes the design of the experiments and Section 4 gives the results of the experiments. Finally some conclusions are drawn in Section 5 and future work is discussed in Section 6.

2 The History of the Keyboard Layout

The typewriter was first invented in the early 1800s and the keyboard was laid out in alphabetical order with piano-like keys. The modern typewriter, invented in 1868 by Christopher Sholes, also had an alphabetical layout (see Figure 1) [5]. Because the key hammers were not spring-loaded and required gravity to return to their rest position, the hammers frequently jammed when two neighbouring keys were pressed in quick succession. Sholes continued to make improvements to his typewriting machine and eventually patented the QWERTY layout in 1878 (see Figure 2) [6]. The new layout was designed to place popular letter pairs onto opposite hands in order to reduce the occurrence of jamming. The introduction of QWERTY had the side effect of slowing typists down because the keys were no longer laid out as intuitively as before. Over time some minor improvements were made to the layout, until it finally became a standard layout (as seen in Figure 3).

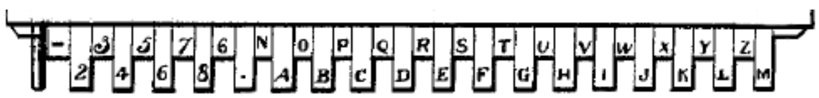

Fig. 1. Shole's alphabetic layout [5].

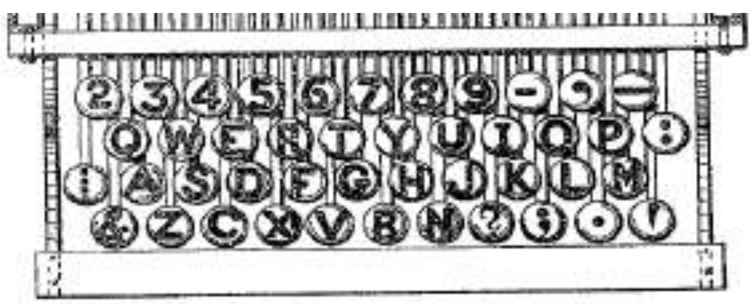

Fig. 2. The original QWERTY keyboard layout [6].

Fig. 3. The modern QWERTY keyboard layout.

To solve the problem of slow typists, regimented lessons were introduced. These lessons helped transition typists from a hunt-and-peck technique through to touch typing and by the early 1900's typists were typing faster than ever before.

In 1936 August Dvorak et al. published *Typing Behavior* [7], a book cataloging an 11 year study into keyboard layouts. Dvorak believed that the QWERTY format was not optimal because it did not place the popular letter pairs in easily accessible positions. He patented his own format, the Dvorak Simplified Keyboard (also known as DSK or Dvorak) layout, which placed popular letters like t, h and e directly under the typist's fingers. The Dvorak layout can be seen in Figure 4. Even Sholes was not happy with his original QWERTY layout, and in his final typewriter patent he included a new layout that has striking similarities to the Dvorak layout—it places all the vowels on one row and under one hand (see Figure 5) [8]. Sholes's final layout was never included in any manufactured typewriting machines.

Fig. 4. The Dvorak Simplified Keyboard layout.

Fig. 5. Sholes's final keyboard layout [8].

As proof that his format was superior, Dvorak trained up some typists at the Navy base where he worked and sent these typists around to various typing competitions. These typists were so successful that the Dvorak layout was eventually barred from all the competitions. Even today, the fastest known typing speed was recorded in the 1940's using the Dvorak Layout—212 words per minute (wpm) burst speed and 150 wpm sustained over 50 minutes [9]. In marketing the layout, Dr. Dvorak invested heavily in producing a Remington typewriter featuring his layout. Unfortunately this particular typewriter was “silent” (it did not make the typical click-clack noises whilst typing) and did not sell well because many typists disliked this aspect. The Dvorak layout failed to take off and QWERTY is the common format we see in practice today.

3 Experimental Design

Rather than retraining several typists to the Dvorak layout, we assume that there are no cognitive differences caused by laying out a keyboard in a different format: a movement (such as the index finger moving from the home to top row) takes a fixed amount of time, regardless of the keyboard layout. Of course, typing is affected by cognitive aspects—typists may pause between syllables or words as they prepare to type the next section. Assuming equality between touch typing lessons, the pauses should be caused by the typist’s language and keyboarding abilities, not the keyboard layout, and therefore occur in the same places regardless of the format being used.

There are 676 possible combinations of digraphs (also known as letter pairs) among the alphabetical characters on a QWERTY keyboard, and 1936 if the 9 punctuation keys operated by the right little finger are included. To simplify all the possible combinations, the following types of movements are defined for digraphs:

tap Both letters in the digraph are the same, meaning the finger can simply ‘tap’ the same key, as in `ee`, `tt`.
reach A digraph typed by the same finger on the same hand, moving over a distance of only one key, as in `ft`, `ju`, `ed`.
hurdle A digraph typed by the same finger on the same hand, hurdling over the home row, as in `ce`, `un`.
trill A digraph typed by adjacent fingers on the same hand, as in `fe`, `op`, `te`.
rock A digraph typed by remote fingers on the same hand, called so as the fingers often move in a rocking motion, as in `af`, `jp`, `on`.
opposite A digraph typed by different hands, as in `if`, `od`, `ma`.

It is possible to have up, down and side reaches, and up and down hurdles. Here these movements are only used to determine key distance so direction is irrelevant—regardless of the movement being upwards, downwards or to the side, a reach always moves the finger one key distance and a hurdle moves two. Taps and opposites are 0 key distance digraphs. A trill or rock could be any of a 0, 1 or 2 key distance depending on the row distance between the first and second keystrokes. For example, the trill `et` is typed all on the top row and the index finger is already over the position `t` when `e` is pressed, so it has a 0 key distance. Similarly, `ct` has a 2 key distance and `ef` has a 1 key distance. The actual key distances are simplified to the set $\{0,1,2\}$ to eliminate varying distances caused by typing style and hand, finger and keyboard sizes. The format that is the most distance efficient will have a lower key distance total across a given set of digraphs. Using key distances is an approximate way of utilising Fitts' law [10] for the keyboard where movements are in three dimensions rather than the required single dimension. Fitts' law is used to calculate efficiency of moving between two objects based on the distance between them and the size of the objects.

Instead of assuming that one key distance will always take a fixed amount of time, the keyboard layouts are also evaluated in terms of timing efficiency. Timing data from two separate research projects focusing on continuous typist recognition were used to obtain the times between two subsequent key presses for all digraphs that appeared in the datasets (for more details on the datasets see [11, 12]). The digraph time was recorded against the key position rather than the letters, meaning that the time on a Dvorak keyboard could be inferred by using times recorded on a QWERTY layout. For example, the digraph `ff` with a 200ms time on a QWERTY layout would be recorded as {`uu`, 200ms} on the Dvorak layout. Given a set of digraphs, any that did not have a recorded time under either layout were discarded. Of the 1029 unique digraphs found in the books, 474 were discarded. Although this seems like half the data was not used, the 474 unique digraphs make up only 13.4% of the total number of digraphs encountered in the corpus.

The two datasets used were not ideally suited to this task. One of the datasets is composed entirely of Italian text and is recorded with a coarse resolution of 10ms [12], while the other is in English and is recorded with an accurate resolution of 1ms [11]. The keyboard was laid out in the same format in each dataset (it was in QWERTY), but it is not known whether any of the typists were skilled touch typists or not. To combat the inaccuracies in this dataset only the average time for each digraph was used in evaluating the timing efficiency.

In order to have a measure of distance, a corpus of 21 free classic books (including *Moby Dick* and *The Last of the Mohicans*) was mined for digraph appearance. This particular corpus was chosen because it was freely available and has been used in at least one similar experiment (see [13]). Digraphs featuring one or more characters not in the bottom, home or top rows of either keyboard layout were discarded. The space character is always typed by the thumb in touch typing and because the thumb always rests on this key it can be discarded. Numerical keys (which feature above the top row and again on the number pad) and modifiers are discarded because of ambiguity—it is unclear which hand or finger may press each key. Incidently, these keys are also in the same position in both formats.

The counts of each digraph found in the corpus were multiplied by their key distance and were totaled to give an overall comparison of QWERTY versus Dvorak in terms of distance. In measuring the timing efficiency, key-specific timings were initially used, relating each recorded time on a QWERTY keyboard with the same keys on a Dvorak keyboard (which may have different letters on those keys). The average time for each digraph was then multiplied by the number of times it appeared in the corpus, summing up to give a total amount of time taken to type the corpus under each format. To prevent the data being skewed because of key placement changes between QWERTY and Dvorak, each layout was also measured using movement specific timings. That is, the average time for a particular type of motion on the keyboard (as defined previously) was recorded, then used to calculate the total time by multiplying the total number of the movements in each book by the average time for that type of movement. The results of the measurements are presented in the next section.

4 Experimental Results

The results from using the 21 books to determine the distance efficiency are presented in Table 1 below. The Dvorak layout clearly requires the fingers to move less over the keys, almost half as much as the QWERTY layout does. Most of the gains from the key distance can be seen in the number of hurdles—an 8 percentage point drop is seen between the QWERTY and Dvorak layouts for digraphs with distances that require the fingers to move across at least two keys. Dvorak is the more distance-efficient layout, and Dvorak users require less effort for text entry because the overall amount of movement around the keyboard is less.

Measures	QWERTY	Dvorak
Total Key Distance	3203413	1774265
Percentage of 0 Key Distance Digraphs	69.3	79.7
Percentage of 1 Key Distance Digraphs	19.7	17.5
Percentage of 2 Key Distance Digraphs	11.0	2.8

Table 1. Key distances over a corpus of 21 books.

Although Dvorak may require the fingers to move less over the keyboard, it does not mean that people will be able to type faster using this layout. As seen in Table 2, using the timing datasets to calculate the timing efficiency produces an entirely different view of the two layouts. After discarding counts of letter pairs which did not appear in both formats, it appears that QWERTY was faster to type on than Dvorak, even though the typist's fingers had to move further. However, the movement-specific times show that Dvorak is slightly faster, contradicting the key-specific measurements. Table 3 shows the average times for each movement (used to calculate the overall movement time), found by averaging all the times for a given movement from both datasets. The timing data used is biased towards QWERTY because it may be possible to achieve different average movement or average key times using the Dvorak layout.

Measures	QWERTY	Dvorak
Total Time - Key Specific (ms)	1.7×10^9	2.1×10^9
Total Time - Movement Specific (ms)	1.83×10^9	1.81×10^9

Table 2. Total time over a corpus of 21 books.

Movement type	Average Time (ms)
tap	170
opposite	238
trill	224
rock	245
hurdle	458
reach	244

Table 3. Average time taken for each movement.

The movement type times, as seen in Table 3, reinforce the ideas behind the Dvorak layout—it is faster to press the same key in succession or use different hands to type a key pair than it is for the fingers to reach or "hurdle" to a different row. This is expected considering that pressing the same key or using alternating hands allows for the finger to be moved into position over the key before it is pressed. Hurdles and reaches require additional time for the finger to leave the key before moving to the next one. However, it was surprising that although the Dvorak layout significantly reduces the incidence of moving over rows, it not quicker than QWERTY in producing the same text. On closer inspection it was discovered that Dvorak has a lower average time for each of 0, 1 and 2 key distances but the incidence of digraphs with longer times was greater. This is an example of bias favouring QWERTY because it is likely that frequently made movements will have lower times since they are more practiced. Often unpopular QWERTY movements are more common in the Dvorak format due to the different positions of letters on the keys.

5 Conclusion

The results in this paper are not conclusive enough to end the debate over which keyboard layout is the optimal one. The distance measure clearly shows that the Dvorak layout requires you to move your hands almost half as much as the QWERTY layout. The timing measures contradict each other and are biased towards QWERTY. We settle on the following conclusion: the Dvorak layout is the most efficient because it requires the least amount of effort to type some given text, even though it may take approximately the same amount of time as the QWERTY layout.

6 Future Work

Although no further work is intended in this particular area as part of this PhD research project, somewhat related to this is the intention of collecting a better dataset of keyboard recordings. As mentioned earlier, the two datasets are not suited for the task of evaluating keyboard layouts—one is not in English, contains coarse timings but has lots of data, the other is accurate and in English but contains very little data. The non-English dataset also does not contain key release events and although this is of no significance in the context of keyboard layout comparisons, it has a huge impact on being able to use this dataset for typist recognition (the subject of the PhD under which this research was conducted). Therefore the next logical step is to collect some more typist data that is in English and contains both key press and key release events. Unfortunately the easiest way to collect such data is by using a web-based approach, where timings have a coarse resolution of approximately 10ms. Work is currently underway to collect recordings of emails using a web-based client that has been altered to be able to record keyboard patterns. The data collection period will last 3 months, hopefully collecting at least one email per day for each participant during that time.

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Foreword to The Minority Game

(book of Damien Challet, Matteo Marsili, and Yi-Cheng Zhang)

By W. Brian Arthur

June, 2004

Once in a while a model problem appears that is simple to describe but offers a wealth of lessons. If the model problem is a classic — like the Ising model or the Prisoner's Dilemma — it both opens up our insight and gives us analytical pathways into an intriguing world. Challet, Marsili and Zhang's Minority Game problem is such a model. It is a classic. The game itself is easy to visualize. A number of agents wish to take some particular action, but they do not benefit if the majority of others take it too. The circumstances where this occurs may seem to be very specific, but in fact they arise *everywhere* in the economy when agents must commit in advance to something that shows diminishing returns. (Diminishing returns, that is, to the numbers committing). Financial trades must be committed to in advance, and the benefits of many strategies may be arbitraged away if others use them too. Hence the minority situation arises also in financial markets.

Damien Challet and Yi-Cheng Zhang conceived of the Minority Game in 1997. They, and their co-worker Matteo Marsili who joined them later, were intrigued not only by the problem itself, but by the possibility of applying statistical mechanics techniques to economics and game theory more generally. Over the previous two decades, physics had become adept at analyzing systems with many particles interacting heterogeneously, and markets clearly consisted of such particles — investors. But these economic "particles" reacted by attempting to forecast the outcome that would be caused by the forecasts of other particles, and to react to that in advance. They were strategic. This was a recognizable world for physicists but an intriguingly different one, and the minority game provided a pathway into it. The problem was quickly taken up by other physicists. Its formulation was simplified and progressively redefined to be closer to what physics was used to dealing with. Versions appeared, results appeared, and advances in understanding appeared. This book chronicles the progress of thought and the considerable insights gained. And it collects the papers that became the main milestones.

One distinctive feature of this work is its emphasis on phases in parameter space — different qualitative properties of the outcome that obtain under different parameter sets. Delineating phases is second nature to physicists but not to us economists, and we can learn much from this practice. In the minority game, agents align their strategies to the condition of the market: they want to choose the strategy that on average minimizes their chance of being in the majority. As a result, strategies co-organize themselves so as to minimize collective dissatisfaction. But this collective result has two phases. With few players, the recent history of the game contains some useful information that strategies can exploit. But once the number of players passes a critical value, all useful information has been used up. The properties of the outcome differ in these two regimes. Information, in fact, is central to the situation. And its role can be explicitly observed: players act as if to minimize a Hamiltonian that is itself a measure of available information. So the process by how information gets eaten up by players is made explicit, and this is another useful pathway into exploring financial markets.

Because some information is left unexploited if few players are in the market, and all information is used up as players increase past a critical number, Challet, Marsili and Zhang point out that players may enter the market until this critical number is reached. The market will therefore display self-ordered criticality. This is an important conjecture, and not just a theoretical one. It tells us that speculative investors — technical traders, at least — will seek out thin markets where possible and avoid deep ones. Markets may therefore hover on the edge of efficiency — a significant insight that can be made explicit with the techniques used here and one that is worth further investigation.

The Minority Game grew out of my El Farol bar problem, and to fill in some pre-history I should say a few words about that. Legend is indeed correct: in 1988 on Thursday nights Galway musician Gerry Carty played Irish music at El Farol, and the bar was not pleasant if crowded. Each week I mulled whether it was worth showing up, and I mulled that others also mulled. It occurred to me that situation was something like the forecasting equivalent of prisoner's dilemma. Few showing up was best; but it was in everyone's interest to be in that few. But my motivation was not to find a forecasting prisoner's dilemma. I was interested at the time in rational expectations (or forecasts) and their meaning for economics. In solving forecasting problems, economics had found it useful to imagine that everyone had the same forecasting machine and used it. You could then ask what forecasting machine would lead to agents' actions that would produce aggregate outcomes that on average would validate the machine's forecasts. This would be the rational expectations solution — the forecasting method that "rational" agents would choose. As an analytical strategy this worked, at least in the cases studied. But it bothered me. If someone were not smart enough to figure the proper forecast they would skew the outcome. Then forecasts should deviate; then I should deviate. Then others should too, and the situation would unravel. Rational expectations may have been good theory, but it was not good science. I realized that El Farol made this plain. If we postulated a "correct" bar-attendance forecasting machine that everyone coordinated on, their actions would collectively invalidate it. Therefore there could be no such correct machine. But if no deductive forecasting solution was possible, what should agents in the economy do? The problem was behaviorally ill-defined. And so were most situations involving the future in economics, I realized. This fascinated me.

In 1992 I stripped the El Farol situation to its essentials and programmed it. I wrote the problem up and presented it at the January 1994 American Economic Association meetings in a session on complexity in the economy chaired by Paul Krugman. The paper was received politely enough, but my discussant was irritated. He pointed out the problem had a solution in mixed Nash strategies: the bar-goers could toss coins to reach a satisfactory outcome. I had thought of that, but to me that missed the point. To me, El Farol was not a problem of how to arrive at a coordinated solution (although the Minority Game very much is). I saw it as a conundrum for economics: How do you proceed analytically when there is no deductive, rational solution? Defining the problem as a game lost this — all games have at least one Nash mixed-strategy equilibrium — so I resisted any game-theoretic formulation. My paper duly appeared, but economists didn't quite know what to make of it. My colleague at Santa Fe, Per Bak, did know however. He saw the manuscript and began to fax it to his physics friends. The physics community took it up, and in the hands of Challet, Marsili and Zhang, it inspired something different than I expected — the Minority Game. El Farol emphasized (for me) the difficulties of formulating economic behavior in ill-defined problems. The

Minority Game emphasizes something different: the efficiency of the solution. This is as should be. The investigation reveals explicitly how strategies co-adapt and how efficiency is related to information. This opens an important door to understanding financial markets.

As I write this, there are now over 140 papers on the Minority Game, and a growing community of econophysicists who have become deeply immersed in the dynamics of markets — especially financial markets. Economists wonder at times whether all this work in physics is not just a lengthy exercise in physicists learning what economists know. It is certainly not — and indeed this book shows it is not. Modern physics can offer much to economics. Not just different tools and different methods of analysis, but different concepts such as phase transitions, critical values, and power laws. And not just the analysis of pattern at stasis, but the analysis of patterns in formation. Economics needs this. In fact economics is changing currently from an emphasis on equilibrium and homogeneity to an emphasis on the formation of pattern and heterogeneity. And so, economics in due course would be forced to use the kind of tools that are brought to bear in this book. Luckily physics is there to supply them.

The papers here and the text are an important part of a movement — looking at the economy under heterogeneity. Challet, Marsili and Zhang and the others described here have done much to power this movement, and I congratulate them for this. It is indeed a benefit to physicists and economists to have this work collected in one place.

W. Brian Arthur,
Santa Fe, New Mexico.

Problem Set 6 (with answer)

Problem: Reconsider the Attack-Defense Game presented in class. Suppose that due to changes in weather conditions, it has become a bit easier to defend "location *N*". Specifically, if the defender defends "location *N*" and the attacker attacks "location *N*", then the probability that the defense will succeed rises from 60% to 65% (and consequently, the probability that the attack will succeed falls from 40% to 35%). There is no other change in the payoff matrix.

(i) Calculate the new mixed strategy Nash equilibrium of the Attack-Defense game.

(ii) In the new equilibrium, what is the probability that an attack will succeed?

Answer: **(i)** Attacker targets "C" with probability 42.86% (targets "N" with complementary probability); defender defends "C" with probability 33.33% (defends "N" with complementary probability). **(ii)** Probability that an attack will succeed is 50%.

Comment: In the "altered" Attack-Defense game, the "more fortified location N" is attacked with a lower probability and defended with a lower probability in equilibrium. This makes perfect sense. If I spend a lot of effort to improve my backhand return of serve, presumably you will serve to my backhand less often and (anticipating that) I will need to "prepare" for a backhand return less often. Does that mean that all my effort is wasted? Of course not, if the overall probability of my returning your serve increases. That's exactly what happens in the new mixed-strategy Nash equilibrium of the changed Attack-Defense game: probability that the defense is successful in equilibrium rises to 50% irrespective of the location of attack.

Information on the Mid-Term Exam

Exam Syllabus

The following will constitute the comprehensive study material for the Mid-term Exam:

(i) The posted Session Slides **1– 10**,

(ii) Chapters **1– 4,** and **6.1 – 6.3** in the Course Lecture Notes, and the posted notes on Expected Utility Theory, and

(iii) The Problem Sets **1–6**.

Exam Structure

The mid-term exam will be a **70**-minute closed-book exam. There will be **FIVE** questions that will have to be answered, for a total of **35** marks.

Students will be free to use calculators, but no other device.

Total marks for every question will be clearly indicated, and (after performing legible rough work in allotted spaces) answers have to be written in ink in the spaces provided in the exam paper (in writing final answers, if you scratch and/or over-write, then you will forfeit re-evaluation opportunities).

All rough work must be done on the exam paper itself, in the provided spaces.

I will look for evidence in the rough work that sensible calculations that led to the answer(s) were present. Absence of such calculations can/will lead to loss of marks.

Assumptions to be made for answering every question

1. Unless specified otherwise, assume that all players are risk-neutral expected utility maximizers, with Bernoulli utility function $u(x) = x$. Further, in every game payoff matrix specified in the exam questions, the Row Player's payoff is presented first and the Column Player's payoff is presented second.

2. Except when it is explicitly stated otherwise, assume that information about all aspects of a game and about players' rationality are *common knowledge*.

3. In some cases, players' costs and benefits might accrue at different points in time. Unless specified otherwise, assume that the players do not discount the future and there is no inflation.

Two Sample Mid-term Questions

QUESTION 1: (4 marks) In the following simultaneous-move game, the Row Player has three pure strategies: *a*, *b*, *c*; and the Column Player has three pure strategies: *x*, *y*, *z*.

	x	*y*	*z*
a	10, 15	8, 11	5, 10
b	12, 18	10, 15	12, 22
c	6, 20	5, 14	20, 12

The game has a unique Nash equilibrium.

State the Row player's strategy: ___________________________

State the Column player's strategy: _________________________

QUESTION 2: (6 marks) Two firms, Gamma and Delta, are engaged in an 'R&D race' to make a patentable innovation. There can only be one winner in this race, and the winner will get the patent whose market value is Rs.200 million (the loser gets nothing). It is a peculiar R&D race in that if one firm has a larger research lab than its rival it will win the patent for sure; and if both firms have labs of the same size, each will win the patent with equal probability. Currently, Gamma has a research lab of size X and Delta has a research lab of size $0.9X$. The only strategic decision facing each firm is whether to 'stay as it is' or to 'double its lab size'. For each firm, lab size can be doubled by spending Rs.70 million. The firms have to make their decisions simultaneously to maximize expected payoff (market value – lab size expenditure).

In the unique Nash equilibrium of the game,

State Gamma's strategy: ________________________________,

and Delta's strategy: ____________________________________.

State the Nash equilibrium expected payoff to Gamma: _______________________,
and to Delta:____________________________________.

Answers to the sample questions:

Question 1: In the unique Nash equilibrium, Row's strategy is {*b*, *c*; 2/3, 1/3}, and Column's strategy is {*x*, *z*; 4/7, 3/7}.

Question 2: In the unique Nash equilibrium, Gamma's strategy is {stay as is, double lab size; 7/20, 13/20}, and Delta's strategy is {stay as is, double lab size; 13/20, 7/20}. The Nash equilibrium expected payoff to Gamma is Rs. 130 million, and to Delta is zero.

Here, we equate the utility from call and don't call.

Randomized Strategies and Reality

1. When a game has both pure-strategy and mixed-strategy equilibria, considering individuals playing mixed strategies may be a fruitful / better way to understand behaviour in '*large social interactions*'

***Understanding Social Inaction*: The Volunteer's Dilemma**

- 38 people witnessed the brutal murder of Catherine Genovese over a period of half an hour in Queens, New York on the night of March 14, 1964
- No one responded to her screams; no one called the police
- Is it just the proverbial apathy of citizens of a large city?
- Experiments suggest that a lone witness of a crime is likely to intervene; but as the number of witnesses increases, there is a fall in the probability that any one person will intervene
- Social psychology hypotheses:
  (*a*) "diffusion of responsibility": larger the group, lower the psychological cost of not helping
  (*b*) "social inference": a person infers the appropriateness of helping from others' behavior
- Economic hypothesis: a strategic 'free-rider' problem

Formalizing the 'free rider problem': *N-player Chicken*

- N people witness an ongoing crime; each gets benefit B when ***someone*** calls the police, and 0 otherwise
- Each incurs cost of C in calling the police, with $B > C > 0$
- This makes it a 'game of chicken' among N players
- There are N asymmetric pure-strategy Nash equilibria, in each of which exactly one person calls the police
- There is a symmetric mixed-strategy Nash equilibrium where each person calls the police with probability p^*
- Here p^* satisfies: $B - C = [(1-p^*)^{N-1}]\times 0 + [1-(1-p^*)^{N-1}]\times B$
- So each person calls police with probability $p^*(N) = \{1-(C/B)^{1/(N-1)}\}$
- This probability $p^*(N)$ ***falls*** from $\{(B-C)/B\}$ to 0 as N increases from 2 to ∞
- Probability that ***not even one person*** calls the police:
  $(1-p^*)^N$ ***rises*** from $(C/B)^2$ to (C/B) as N increases from 2 to infinity

Obviously, C/B < 1

Randomized Strategies and Reality

2. *There can be strict preference for strategy randomization in distinct classes of complete-information game; for example, in Monitoring/Auditing Games*:

Monitoring Games

Consider "principal-agent games" with the following "order of moves":

1. Principal 'commits' to a monitoring/auditing scheme
2. Agents choose their actions
3. Principal's pre-committed scheme is 'implemented'

In a class of principal-agent games, pre-committing to a randomized monitoring/auditing scheme can be strictly preferred by the principal

- *i.e., principal might strictly prefer to implement a* ***stochastic scheme***

In a lot of real-world scenarios, randomization may work as a better alternative rather than committing to a strategy as it would be too expensive to audit every time and too risky not to audit at all.

The Teacher/Student Exam Game

Payoff Structure (*NOT Game Structure*)

		Student: *study*	Student: *shirk*
Teacher	*set exam*	15 , 30	0 , 0
	do not	35 , 30	10 , 60

Game Structure:
Teacher commits to prob. of exam; then students decide to study or not; then an external authority implements exam according to committed prob.

⇒ ***order of moves is neither "simultaneous" nor "sequential"***

- If teacher commits to exam with prob x, student will study iff $x > 30/60 = ½$
- Teacher's choice is between "setting $x = 0$ and getting a payoff 10" and "setting $x = 51\%$ and getting a payoff: **15(0.51) + 35(0.49)**"

⇒ Teacher's optimal strategy is to **commit** to the following *exam scheme*: "with some probability just over 50%, an exam will be held"

Nash equilibria and *pre-play self-enforcing recommendations*

❖ In *certain complete-information games*, a strategy vector is a Nash equilibrium if and only if it is a pre-play self-enforcing recommendation

- obviously holds for the case of dominance-solved Nash equilibria

❖ In *some other complete-information games*, there might exist additional self-enforcing recommendations that are not Nash equilibria

- in games with multiple Nash equilibria (for example, games of coordination and anti-coordination), when **correlated stochastic pre-play recommendations/agreements** are permitted, some non-Nash outcomes can be supported by self-enforcing recommendations

❖ In ***still other complete-information games***, a Nash equilibrium strategy vector might not be a pre-play self-enforcing recommendation

1. In games with more than two players (where **communication between player sub-groups** are possible), a Nash equilibrium might not be self-enforcing
2. In sequential-move games where **renegotiation is permitted**, some "subgame-perfect Nash equilibria" might not be self-enforcing

Examples and clarifications of each issue follow ...

Self-enforcing pre play recommendations is a pre-agreed situation which is in everyone's interests if others follow it.

Self-enforcing recommendations and "correlated equilibria"

A Traffic Coordination Game

		North-South: *drive thru*	North-South: *wait a minute*
East-West	*drive thru*	0 , 0	4 , 2
	wait a minute	2 , 4	2 , 2

Nash Equilibria: **[A]** *E-W* drives thru : payoffs (4, 2)
[B] *N-S* drives thru : payoffs (2, 4)
[C] Each drives thru with prob ½ : exp. payoffs (2, 2)

- Each of these equilibria is a pre-play self-enforcing recommendation
- But if an impartial strategy recommender is indeed present, (s)he can do better by using a ***public correlation device*** for example, a ***traffic light system*** that is green in "*E-W* direction" in every *odd minute* and green in "*N-S* direction" in every *even minute*
- Such traffic lights correspond to the following ***correlated strategy choices*** {With prob ½ *E-W* will drive thru while *N-S* will wait, & with prob ½ *N-S* will drive thru while *E-W* will wait} ⇒ expected payoffs (3, 3)

Self-enforcing recommendations and "correlated equilibria"

- A mixed-strategy Nash equilibrium is a pre-play self-enforcing recommendation in the same way that a pure-strategy Nash eqm is
- However, players' randomizations are necessarily ***statistically independent*** in any mixed-strategy Nash equilibrium
- If there is indeed a strategy consultant who makes pre-play strategy recommendations, he ***might*** be able to make "better self-enforcing recommendations" by using a ***public correlation device***
- A ***correlated equilibrium*** is a pre-agreement on a 'public correlation device' (e.g., publicly observed roll of a dice) whose different outcomes lead to recommendations of (mixed or pure) strategies to be played by each player, where the recommendations are ***mutual best responses***
- Every Nash equilibrium is (trivially) a correlated equilibrium, but not every correlated equilibrium is a Nash equilibrium
  - every correlated eqm is necessarily a ***self-enforcing recommendation***

Chicken

	Go straight	Swerve
Go straight	0, 0	4, 2
Swerve	2, 4	2, 2

- In Chicken:
  - Symmetric mixed Nash eqm (½, ½) giving expected payoffs (2, 2)
  - Additional "correlated equilibrium": {G, S w/ prob ½; S, G w/ prob ½} giving expected payoffs (3, 3)

Battle of Sexes

	Movie	Concert
Movie	4, 2	1, 1
Concert	0, 0	2, 4

- In Battle of Sexes:
  - Mixed Nash eqm {(4/5, 1/5), (1/5, 4/5)} giving expected payoffs (1.6, 1.6)
  - Additional "correlated equilibrium": {M, M w/ prob ½; C, C w/ prob ½} giving expected payoffs (3, 3)

- In many coordination games, players get symmetric eqm payoffs only in a mixed-strategy Nash eqm
  - but these symmetric payoffs can be "quite small"
  - that is precisely where "correlated equilibrium recommendations" can generate significantly larger symmetric payoffs to the players

Cooperation Dilemmas

- One of the central insights of Game Theory is that there are many strategic situations where players cannot help but compete against each other, while cooperation would benefit all players
- **Cooperation Dilemmas exist** in business, political, social interactions
- The weakest cooperation dilemma is the ***Assurance Game*** where it is a player's best response to cooperate if and only if others cooperate

Stag Hunt Game

	hunt stag	*hunt hare*
hunt stag	best, best	worst, OK
hunt hare	OK, worst	OK, OK

- The strongest cooperation dilemma is the ***Prisoners' Dilemma*** where non-cooperation is the strictly dominant strategy

	put high-beam on	*not*
put high-beam on	bad, bad	best, worst
not	worst, best	OK, OK

***Sub-coalition among players*: Choosing the Direction Game**

Possible strategy combinations and payoffs:
[A] All four arrows placed 'pointing north': each gets Rs. 2000
[B] All four arrows placed 'pointing south': each gets Rs. 1000
[C] Two arrows 'south' & two 'north': former get Rs.2500 each; latter 0
[D] Three arrows in one direction & one in another: Rs.750 to each

- There are exactly two pure-strategy Nash equilibria: [A] & [B]
- [A] is ***vulnerable to credible coalitional deviation*** by a ***2-player sub-group*** who can agree to deviate 'south' in credible coalition-proof way
- [B] is a ***coalition-proof pure-strategy Nash equilibrium***
  - immune to "credible coalitional deviations" by any sub-group
  [joint deviation by all four from 'south' to 'north' not credible]
- A coalition-proof Nash eqm is a Nash eqm that is immune to credible coalitional deviations by every sub-group of players; a credible coalitional deviation being one that constitutes a coalition-proof Nash equilibrium for the deviating coalition
  [***There may be no* coalition-proof Nash equilibrium *in a game***]

The last part of this definition implies that a coalition is credible only if once they deviate, no other group would want to deviate again.

A credible coalition is if a group wants to deviate and stick to their new agreement.

So, in case of B, if the entire group deviates to North, then 2 out of them would want to deviate to South. As a result, it is coalition proof as the subgroup does not want to switch.

This is practical as it would not be an agreement that would stand in the real world if everyone thinks that it is not gonna be held.

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- One of the central insights of Game Theory is that there are many strategic situations where players cannot help but compete against each other, while cooperation would benefit all players
- **Cooperation Dilemmas exist** in business, political, social interactions
- The weakest cooperation dilemma is the ***Assurance Game*** where it is a player's best response to cooperate if and only if others cooperate

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	hunt stag	*hunt hare*
hunt stag	best, best	worst, OK
hunt hare	OK, worst	OK, OK

- The strongest cooperation dilemma is the ***Prisoners' Dilemma*** where non-cooperation is the strictly dominant strategy

	put high-beam on	*not*
put high-beam on	bad, bad	best, worst
not	worst, best	OK, OK

Another 2-stage Cooperation Dilemma *with renegotiation possibilities*

- A & B play the following two-stage game & each wants to maximize sum of payoffs ($v_1 + v_2$):

Stage 1 PD game: (simultaneous-move)

	Coop	Defect
Coop	100, 100	0, 150
Defect	150, 0	50, 50

Stage 2 SH game: (simultaneous move)

	Stag	Hare
Stag	X > 50, X > 50	0, 50
Hare	50, 0	50, 50

Two pure SPNE outcomes: **both 'defect' and then 'coordinate'**

The Pareto-superior outcome: **both 'cooperate' and then 'stag'**

can be supported as an ***SPNE*** outcome if and only if $X - 50 > 150 - 100$

by a ***Carrot** and **stick strategy***: "I'll play ***coop*** in Stage 1. If both play ***coop*** then I will play ***stag*** in Stage 2; otherwise, I will play ***hare*** in Stage 2."

- There is one little problem: Suppose that after playing *Defect* in Stg 1, **Row** pleads **Column** to 'forgive & forget' and suggests that both play Stag in Stg 2
- ***The SPNE employing carrot-&-stick strategy is not renegotiation-proof***

Like supply volumes is an example of strategic substitutes, as the best response is to decrease volumes when competitor increases it. Advertising game is an example of strategic complements.

Nash equilibria and *pre-play self-enforcing recommendations*

❖ In *certain complete-information games*, a strategy vector is a Nash equilibrium if and only if it is a pre-play self-enforcing recommendation

- obviously holds for the case of dominance-solved Nash equilibria

❖ In *some other complete-information games*, there might exist additional self-enforcing recommendations that are not Nash equilibria

- in games with multiple Nash equilibria (for example, games of coordination and anti-coordination), when **correlated stochastic pre-play recommendations/agreements** are permitted, some non-Nash outcomes can be supported by self-enforcing recommendations

❖ In *still other complete-information games*, a Nash equilibrium strategy vector might not be a pre-play self-enforcing recommendation

1. In games with more than two players (where **communication between player sub-groups** are possible), a Nash equilibrium might not be self-enforcing
2. In sequential-move games where **renegotiation is permitted**, some "subgame-perfect Nash equilibria" might not be self-enforcing

Strategizing in Dominance-solved Cooperation Dilemmas

Competition in supply volumes

	30	*40*	*60*
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

Competition in advertising

	low	*medium*	*high*
low	50, 50	32, 60	25, 55
med	60, 32	41, 41	28, 42
high	55, 25	42, 28	30, 30

- Players try to solve/mitigate **cooperation dilemmas** by various means

1. by setting up institutions to enforce cooperation
2. *by making pre-emptive moves*
3. *by inducing other players to join the game*
4. ***by designing creative strategies (expanding/contracting strategy sets)***
5. ***by utilizing repetitiveness of strategic interactions***

Cooperation Dilemmas can be games of ***strategic substitutes*** (one player's 'best response' moves in the opposite direction to the other player's strategy), or ***strategic complements*** (one player's 'best response' moves in the same direction to the other player's strategy)

- The nature and extent of mitigation of cooperation dilemmas by the ***meta-strategies*** **2 – 4** depend on whether the underlying game is one of ***strategic substitutes*** or ***strategic complements***

Strategic complements are the more you do, the more I want to do something.
Strategic substitutes are the more you do, the less I want to do something.

A 2-stage Cooperation Dilemma

- A & B play the following two-stage game & each wants to maximize sum of payoffs ($v_1 + v_2$):

Stage 1 SH game: (simultaneous-move)

	Stag	Hare
Stag	X > 50, X > 50	0, 50
Hare	50, 0	50, 50

Stage 2 PD game: (simultaneous move)

	Coop	Defect
Coop	100, 100	0, 150
Defect	150, 0	50, 50

Two pure SPNE outcomes: **both 'coordinate' and then 'defect'**

The Pareto-superior outcome: **both 'coordinate on stag' and then 'defect' so that each gets payoff of (X + 50)**

Coop Dilemma: *The Cournot Game* (a game of strategic substitutes)

P = 140 - Q
TC = 20q

Firm A \ Firm B	*30*	*40*	*60*
30	1800, 1800	1500*, 2000	900, 1800
40	2000, 1500	1600*, 1600	800, 1200
60	1800*, 900	1200, 800	0, 0

- If A can decide first its volume of supply, and be able to confront B with that supply as a *fait accompli*, how much supply volume should A choose?
  A should choose *60* and thus ensure a profit of 1800
- **That meta-strategy of 'preemptive first move' does not lessen the 'cooperation dilemma', it simply twists it to the advantage of one player (the first mover)**
- The Cournot Game of 'strategic substitutes' has a ***first-mover advantage***
- But there will be a **commitment problem** for A : If B is convinced to supply *30*, then *ex post* A will want to cut back own supply to *40*
- Another is the **simultaneity problem**: If B also tried to change the game by committing to supply *60*, the two meta-strategies would hurt both the players

Strategic government subsidy in the Cournot game

- Suppose ***A*** and ***B*** are firms in countries *Alpha* and *Beta*, and sell all their bottled water in a third country *Gamma*
- ***B*** goes to its government and says: "If you publicly commit to subsidize me for my entire production costs,
  and then impose upon me a profits tax of 41%, then your 'budget will be more than balanced' and I will gain unambiguously."

TC = 20q

	30	40	60
30	1800, 1800 [2400]	1500, 2000 [2800]	900, 1800 [3000]
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

Resolving Cooperation Dilemmas

Cooperation Dilemmas are resolved most comprehensively when a strategic interaction is repeated unendingly over time for the same players, or when there are effective institutions to enforce cooperation

- When neither is the case, players can use some ***meta-strategies*** – that might involve ***pre-emptive commitment moves*** – to change strategy spaces / rules of the game
  (i) change order-of-moves, (ii) involve additional players,
  (iii) introduce new strategies
  - ***credibility*** of meta-strategies / commitment moves is often a concern
- In games of ***strategic substitutes***, such a meta-strategy employed by a player enables him to make own payoff gains ***at the cost of rival losses***
  - further, if the rival 'mimics' the meta-strategy, then everyone loses
- In contrast, in games of ***strategic complements***, a suitably-chosen meta-strategy employed by a player can generate payoff gains ***for all players***
  and if a rival 'mimics' the meta-strategy, then everyone might gain more

Government subsidy is like introducing more players into the game.

Strategic government subsidy in the Cournot game

- Suppose ***A*** and ***B*** are firms in countries *Alpha* and *Beta*, and sell all their bottled water in a third country *Gamma*
- ***B*** goes to its government and says: "If you publicly commit to subsidize me for my entire production costs,
  and then impose upon me a profits tax of 41%, then your 'budget will be more than balanced' and I will gain unambiguously."
- ***A voluminous literature on 'strategic trade policy' is based on this logic (think Boeing vs. Airbus)***

P = 140 - Q
TC = 20q

	30	40	60
30	1800, 2400	1500, 2800	900, 3000
40	2000, 2100	1600, 2400	800, 2400
60	1800, 1500	1200, 1600	0, 1200

Order-of-move Advantages

- Given a sequential-move game between two players, a player can ask: would I pay to move first or to move second?
- If answer is "first", there is ***first-mover advantage*** in the game; if answer is "second", there is ***second-mover advantage***
- In a symmetric ***Cooperation Dilemma game*** with scalar decision variables:
  - if the game is one of ***strategic substitutes***:
    Players have first-mover advantage: First-mover SPNE payoff > simultaneous-move Nash payoff > Second-mover SPNE payoff
  - if the game is one of ***strategic complements***:
    Players have second-mover advantage:
    Second-mover SPNE payoff > First-mover SPNE payoff > simultaneous-move Nash payoff

Coop Dilemma: *An Advertising Game* (a game of strategic complements)

		Firm B		
		low	medium	high
Firm A	low	50, 50	32*, 60	25, 55
	med	60, 32	41, 41	28*, 42
	high	55, 25	42, 28	30*, 30

- If A can decide first its 'ad level', and be able to confront B with that choice as a *fait accompli*, how much supply volume should A choose?
  A should choose *low* and thus ensure a profit of 32
- That meta-strategy of 'preemptive first move' ***does*** lessen the 'cooperation dilemma'
- The Advertising Game of 'strategic compliments' has a ***second-mover advantage***
- There will still be a **commitment problem** for A : If B is convinced to play *medium*, then *ex post* A will want to raise own advertising to *high*
- But **no simultaneity problem**: If B also tries to change the game by committing to *low* ad level, meta-strategies will help both the players greatly (if they can commit)

Commitment problems of First-moves

- While a first-mover in a cooperation dilemma can 'reduce' dilemma either for himself or for all players, he faces a commitment problem
  - once 'second-move' is made, first-mover will want to "revise" own move
- If this is foreseen, the efficacy of the "first move" is lost
- So to be effective, the "first-move" strategies should be so chosen to have "commitment value"
  - there must be little desire to change them *ex post*
- In Cournot games, "first moves" in terms of capacity choices have commitment value as capacity-building costs are largely irrecoverable
- In Bertrand pricing games, "first-moves" in terms of price-setting do not typically have significant commitment value
  - creative strategizing is necessary to "generate" commitment
  - a "most favoured customer" clause that states that any future price reductions will be passed on to current / past customers can generate such commitment

Innovative Pricing in 'Price Competition'

Example: ***A Bar Pricing Game***

- Price competition between two bars – **Alpha Bar** and **Beta Bar** – on a Greek island
- Each can charge a price of $2, $4, or $5 per bottle of beer
- Daily customer base consists of tourists and locals
  - – 6,000 tourists drink from the first bar they hit (randomly)
  - – 4,000 locals select the lowest price bar, and divide equally if prices are the same
- Each bar has a "fountain" in its backyard from which beer flows for free

The Bar Pricing Game

Beta \ Alpha	$2	$4	$5
$2	10 , 10	14 , 12	14 , 15
$4	12 , 14	20 , 20	28 , 15
$5	15 , 14	15 , 28	25 , 25

daily profits in thousands of dollars

- ***Dominance-solved*** Nash equilibrium: {$4, $4}
  - a cooperation dilemma that can't be resolved by price leadership
- Consider the case where **Alpha Bar** announces a ***price matching scheme***: "If we charge $5, and if a customer then sees a lower price offered at **Beta Bar**, we will refund the difference."
- Assume locals are aware of prices in both bars, and each tourist gets to see price at the other bar after drinking from one

Pricing-matching by Alpha Bar

	$2	$4	$5	$5 + match
$2	10 , 10	14 , 12	14 , 15	10 , 10
$4	12 , 14	20 , 20	28 , 15	20 , 20
$5	15 , 14	15 , 28	25 , 25	25 , 25

- Unique pure-strategy Nash equilibrium: {$4, $4}
- This is also a '*weakly dominance-solved*' equilibrium:
  - $2 is "strictly dominated" for both bars and $5+match is "weakly dominated" for Alpla Bar in the full game – delete them first
  - after pruning, $5 is "strictly dominated" for both bars
- Next, consider the case where after observing **Alpha**'s creativity, **Beta** decides to enlarge its strategy space in the same way

Pricing-matching by both bars

	$2	$4	$5	$5 + match
$2	10 , 10	14 , 12	14 , 15	10 , 10
$4	12 , 14	20 , 20	28 , 15	20 , 20
$5	15 , 14	15 , 28	25 , 25	25 , 25
$5 + match	10 , 10	20 , 20	25 , 25	25 , 25

- 2 pure-strategy Nash equilibria: {$4, $4} & {$5+match; $5+match}
- {$5 + match; $5 + match} is not only the Pareto-dominant Nash eqm, it is also the unique '*weakly dominance-solved*' Nash eqm:
  - $2 strictly dominated strategy for each bar in the full game – delete
  - after pruning $2, $5 *weakly dominated strategy* for each bar – delete
  - after pruning $5, $4 *weakly dominated strategy* for each bar – delete
- *Here, strategy mimicking "resolves" the cooperation dilemma*

Presidential Address
Pacific Division of the American Philosophical Association
March 2001

The Stag Hunt

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I: The Stag Hunt

The Stag Hunt is a story that became a game. The game is a prototype of the social contract. The story is briefly told by Rousseau, in *A Discourse on Inequality*:

> If it was a matter of hunting a deer, everyone well realized that he must remain faithful to his post; but if a hare happened to pass within reach of one of them, we cannot doubt that he would have gone off in pursuit of it without scruple..."

Rousseau's story of the hunt leaves many questions open. What are the values of a hare and of an individual's share of the deer given a successful hunt? What is the probability that the hunt will be successful if all participants remain faithful to the hunt? Might two deer hunters decide to chase the hare?

Let us suppose that the hunters each have just the choice of hunting hare or hunting deer. The chances of getting a hare are independent of what others do. There is no chance of bagging a deer by oneself, but the chances of a successful deer hunt go up sharply with the number of hunters. A deer is much more valuable than a hare. Then we have the kind of interaction that that is now generally known as the Stag Hunt.

Once you have formed this abstract representation of the Stag Hunt game, you can see Stag Hunts in many places. David Hume also has the Stag Hunt. His most famous illustration of a convention has the structure of a two-person Stag Hunt game:

> Two men who pull at the oars of a boat, do it by an agreement or convention, tho' they have never given promises to each other. ...

Both men can either row or not row. If both row, they get the outcome that is best for each - just as in Rousseau's example, when both hunt the stag. If one decides not to row then it makes no difference if the other does or not - they don't get anywhere. The worst outcome for you is if you row and the other doesn't, for then you lose your effort for nothing, just as the worst outcome for you in the Stag Hunt is if you hunt stag by yourself.

We meet the Stag Hunt again in the meadow-draining problem of Hume's *Treatise*:

> Two neighbors may agree to drain a meadow, which they possess in common; because 'tis easy for them to know each others mind, and each may perceive that the immediate consequence of failing in his part is the abandoning of the whole project. But 'tis difficult, and indeed impossible, that a thousand persons shou'd agree in any such action ...

where Hume observes that achieving cooperation in a many-person Stag Hunt is more difficult than achieving cooperation in a two-person stag hunt.[1]

The Stag Hunt does not have the same melodramatic quality as the Prisoner's Dilemma. It raises its own set of issues, which are at least as worthy of serious consideration. Let us focus, for the moment, on a two-person Stag Hunt for comparison to the familiar two-person Prisoner's Dilemma.

If two people cooperate in Prisoner's Dilemma, each is choosing less rather than more. In Prisoner's Dilemma, there is a conflict between individual rationality and mutual benefit.

In the Stag Hunt, what is rational for one player to choose depends on his beliefs about what the other will choose. Both stag hunting and hare hunting are *equilibria*. That is just to say that it is best to hunt stag if the other player hunts stag and it is best to hunt hare if the other player hunts hare. A player who chooses to hunt stag takes a risk that the other will choose not to cooperate in the Stag Hunt. A player who chooses to hunt hare runs no such risk, since his payoff does not depend on the choice of action of the other player, but he foregoes the potential payoff of a successful stag hunt. Here rational players are pulled in one direction by considerations of mutual benefit and in the other by considerations of personal risk.

Suppose that hunting hare has an expected payoff of 3, no matter what the other does. Hunting stag with another has an expected payoff of 4. Hunting Stag alone is doomed to failure and has a payoff of zero. It is clear that a pessimist, who always expects the worst, would hunt hare. But it is also true with these payoffs that a cautious player, who was so uncertain that he thought the other player was as likely to do one thing as another, would also hunt hare.[2] That is not to say that rational players could not coordinate on the stag hunt equilibrium that gives them both better payoff, but it is to say that they need a measure of trust to do so.

I told the story so that the payoff of hunting hare is absolutely independent of how others act. We could vary this slightly without affecting the underlying theme. Perhaps if you hunt hare, it is even better for you if the other hunts stag for you avoid competition for the hare. If the effect is small we still have an interaction that is much like the Stag Hunt. It displays the same tension between risk and mutual benefit. It raises the same question of trust. This small variation on the Stag Hunt is sometimes also called a Stag Hunt [3] and we will follow this more inclusive usage here.

Compared to the Prisoner’s Dilemma, the Stag Hunt has received relatively little discussion in contemporary social philosophy – although there are some notable exceptions.[4] But I think that the Stag Hunt should be a focal point for social contract theory.

The two mentioned games, Prisoner's Dilemma and the Stag Hunt, are not unrelated. Considerations raised by both Hobbes and Hume can show that a seeming Prisoner's Dilemma is really a Stag Hunt. Suppose that Prisoner’s Dilemma is repeated. Then your actions on one play may affect your partner's actions on other plays, and considerations of reputation may assume an importance that they cannot have if there is no repetition. Such considerations form the substance of Hobbes' reply to the Foole. Hobbes does not believe that the Foole has made a mistake concerning the nature of rational decision. Rather, he accuses the Foole of a shortsighted mis-specification of the relevant game:

> He, therefore, that breaketh his Covenant, and consequently declareth that he think that he may with reason do so, cannot be received into any society that unite themselves for Peace and Defense, but by the error of them that receive him.[5]

According to Hobbes, the Foole's mistake is to ignore the future.

David Hume invokes the same considerations in a more general setting:

> Hence I learn to do a service to another, without bearing him any real kindness; because I foresee, that he will return my service, in expectation of another of the same kind, and in order to maintain the same correspondence of good offices with me and with others. [6]

Hobbes and Hume are invoking the *shadow of the future*.

How can we analyze the shadow of the future? We can use the theory of indefinitely repeated games. Suppose that the probability that the Prisoner's Dilemma is will be repeated another time is constant. In the repeated game, the Foole has the strategy *Always Defect*. Hobbes argues that if someone defects, others will never cooperate with him. Those who initially cooperate, but who retaliate as Hobbes suggests against defectors, have a *Trigger* strategy.

If we suppose that *Always Defect* and *Trigger* are the only strategies available in the repeated game and that the probability of another trial is .6, then the Shadow of the Future transforms the two-person Prisoner's Dilemma:

	Cooperate	Defect
Cooperate	3	1
Defect	4	2

into the two-person Stag Hunt: [7]

	Trigger	All Defect
Trigger	7.5	4
All Defect	7	5

This is an exact version of the informal arguments of Hume and Hobbes.[8]

But for the argument to be effective against a fool, he must believe that the others with whom he interacts are not fools. Those who play it safe will choose *Always Defect*.

Rawls' maximin player is Hobbes' Foole.[9] The Shadow of the Future has not solved the problem of cooperation in the Prisoner's Dilemma; it has transformed it into the problem of cooperation in the Stag Hunt.

In a larger sense, the whole problem of adopting or modifying the social contract for mutual benefit can be seen as a Stag Hunt. For a social contract theory to make sense, the state of nature must be an equilibrium. Otherwise there would not be the problem of transcending it. And the state where the social contract has been adopted must also be an equilibrium. Otherwise, the social contract would not be viable. Suppose that you can either *devote energy to instituting the new social contract* or not. If everyone takes the first course the social contract equilibrium is achieved; if everyone takes the second course the state of nature equilibrium results. But the second course carries no risk, while the first does. This is all quite nicely illustrated in miniature by the meadow-draining problem of Hume.

The problem of reforming the social contract has the same structure. Here, following Binmore, we can then take the relevant "state of nature" to be the *status quo,* and the relevant social contract to be the projected reform. The problem of instituting, or improving, the social contract can be thought of as the problem of moving from riskless Hunt Hare equilibrium to the risky but rewarding Stag Hunt equilibrium.

II: Game Dynamics

How do we get from the Hunt Hare equilibrium to the Stag Hunt equilibrium? We could approach the problem in two different ways. We could follow Hobbes in asking the question in terms of rational self-interest. Or we could follow Hume by asking the question in a dynamic setting. We can ask these questions using modern tools – which are more that Hobbes and Hume had available, but still less than we need for fully adequate answers. The news from the frontiers of game theory is rather pessimistic about the transition from hare hunting to stag hunting.

The modern embodiment of Hobbes' approach is rational choice based game theory. It tells us that what a rational player will do in the Stag Hunt depends on what he thinks the other will do. It agrees with Hume's contention that a thousand person stag-hunt would be more difficult to achieve than a two-person stag hunt, because –assuming that everyone must cooperate for a successful outcome to the hunt – the problem of trust is multiplied. But if we ask how people can get from a Hare Hunt equilibrium to a Stag hunt equilibrium, it does not have much to offer. From the standpoint of rational choice, for the Hare Hunters to decide to be Stag Hunters, each must *change her beliefs* about what the other will do. But rational choice based game theory as usually conceived, has nothing to say about how or why such a change of mind might take place.

Let us turn to the tradition of Hume. Hume emphasized that social norms can evolve slowly:

> "Nor is the rule regarding the stability of possession the less derived from human conventions, that it arises gradually, and acquires force by a slow progression..."

We can reframe our problem in terms of the most thoroughly studied model of cultural evolution, the replicator dynamics. If we ask in this framework, how one can get from the Hunt Hare equilibrium to the Hunt Stag equilibrium, the answer is that you can't! In the vicinity of the state where all Hunt Hare, Hunting Hare has the greatest payoff. If you are close to it, the dynamics carries you back to it. This reasoning holds good over a large class of adaptive dynamics. The transition from non-cooperation to cooperation seems impossible.

Perhaps the restriction to deterministic dynamics is the problem. We may just need to add some chance variation. We could add some chance shocks to the replicator dynamics[10] or look at a finite population where people have some chance of doing the wrong thing, or just experimenting to see what will happen.[11] If we wait long enough, chance variation will bounce the population out of hare hunting and into stag hunting.

But in the same way, chance variation can bounce the population out of stag hunting into hare hunting. Can we say anything more than that the population bounces between these two states? Yes, we know how to analyze this system[12] but the news is not good. When the chance variation is small, the population spends almost all its time in a state where everyone hunts hare.[13] It seems that all we have achieved so far is to show how the social contract might degenerate spontaneously into the state of nature.

Social contracts do sometime spontaneously dissolve. But social contracts also form. And there is experimental evidence that people will hunt stag even when it is a risk to do so.[14]   This suggests the need for a richer theory.

III: Local Interaction

The foregoing discussion proceeded in terms of models designed for random encounters in large populations. But cooperation in the Stag Hunt may well have originated in small populations with non-random encounters. How should we think about this setting?

Perhaps, instead of interacting at random with other members of the population, we interact with our neighbors. Can local interaction make a difference? We know that it can, from investigations the dynamics of other games played with neighbors. Philosophers have contributed these developments and in this regard I would like to mention Jason Alexander, Peter Danielson, Patrick Grim, Bill Harms, Rainer Hegselmann, Gary Mar, and Elliott Sober. In some cases local interactions make a spectacular difference in the outcome of the evolutionary (or learning) process.

Does local interaction make a difference in the Stag Hunt? Can it explain the institution of the social contract? The news gets worse. Glenn Ellison (1993) investigates the dynamics of the Stag Hunt played with neighbors, where the players are arranged on a circle. He finds limiting behavior not much different than that in the large population with random encounters. With a small chance of error, the population spends most of its time hunting hare. The difference in the dynamics of the two cases is that given local interaction, the population approaches its long run behavior much more rapidly. The moral for us, if any, is that in small groups with local interaction the degeneration of the social contract into the state of nature can occur with great rapidity.[15]

There is, however, a small glimmer of light from the following fable.[16] There is a central figure in the group, who interacts (pairwise) with all others and with whom they interact - you might think of him as the Boatman in honor of Hume or the Huntsman in honor of Rousseau.

(figure here)

Every round, players revise their strategies by choosing the best response to their partners' prior play, with the exception that the Huntsman revises his strategy much less often. We add a small chance of spontaneously changing strategy for each player. If everyone hunts stag and the Huntsman spontaneously switches to hunting hare, in two rounds probably everyone will hunt hare. But conversely, if everyone is hunting hare and the Huntsman spontaneously switches to hunting stag, in two rounds probably everyone will hunt stag. In the long run the population spends approximately half its time hunting hare and half its time hunting stag. The story does not speak to all our concerns, but at least it shows that the social contract need not have negligible probability in the long run and that the *structure* of local interaction can make a difference[17].

IV: Dynamics of Interaction Structure

Still, it is hard not to feel that there must be something fundamental that is missing from our analysis. I would like to suggest that what is missing is an account of the evolution of the structure of interactions. Game theory takes the interaction structure as fixed. But in real life individuals adjust with whom they interact on the basis of past experience. This as a fundamental aspect of social behavior that is completely absent from the theory of games.

Let me show you how the analysis of the Stag Hunt is changed if individuals can learn with whom to interact. What follows is the result of joint work with Robin Pemantle.[18] Suppose that we have a small group of agents, some disposed to hunt stag and some disposed to hunt hare. They start out interacting at random, but when agents interact they are reinforced for interaction with the same agent by the payoff that they receive from the interaction. Reinforcement modifies the probability that one agent will choose to interact with another.

How do these interaction probabilities evolve? It can be shown, both by simulation and analytically, that stag hunters learn to interact with other stag hunters. This, perhaps, should come as no great surprise. It is not quite so obvious that hare hunters will end up interacting with other hare hunters. Hare hunters do not care with whom they interact. Nevertheless it is so.[19] Learning dynamics leads to a structure of interaction probabilities quite different from that of a random pairing model. In this environment, stag hunters prosper.

Now we can add the further consideration of players revising their strategies. Suppose that once in a while a player looks around the little group, sees who is doing best, and imitates that strategy. If interaction structure were fixed at random pairing, we would be back were we started and the most likely outcome would be that everyone would end up hunting hare. But if structure is fluid and the learning dynamics for structure is fast relative to the strategy revision dynamics, stag hunters will find each other and then imitation will slowly convert the hare hunters to stag hunters. This conclusion is robust to the addition of a little chance. Here, we finally have a model that can explain the institution of a modest social contract.

In between the extremes, the limiting outcome in a small group may be either all hare hunters or all stag hunters. Which outcome one gets depends somewhat on the vicissitudes of chance in the early stages of the evolution of the group. But it is also strongly influenced by the relative speeds of structure and strategy dynamics. Rapid structural adaptation favors the stag hunters. Our social contract may form in some circumstances but not in others.

We can consider different strategy revision dynamics. Suppose, as before, that structure dynamics is fast relative to strategy revision, but that the individuals never look around the group for successful models, but rather choose the best response to the strategy that they have encountered on the last play. Then the ultimate outcome will be that the group is divided into two stable classes, the stag hunters and the hare hunters, which never interact with one another. A different kind of strategy revision leads to a different social contract. There are other interesting possibilities to explore.

Conclusion

We should pay more attention to the Stag Hunt. There is a lot to think about. Real stag hunts are complex interactions between more than two people. So are the social contracts - large and small - that we have used the Stag Hunt to represent. In our analysis of two-person Stag Hunts we finally focused on the coevolution of strategy and interaction structure. I believe that this is a key to the larger question of the emergence of social structure.

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NOTES:

[1] For evidence that this is true in laboratory experiments, see Van Huyck, Battalio and Beil (1990).

[2] Hunting hare is said to be the *risk dominant* equilibrium.

[3] Sometimes it is called an Assurance Game, following Sen (1967).

[4] Ed Curley, Jean Hampton, Magnus Jiborn, and Peter Vanderschraaf.

[5] Hobbes, *Leviathan*, xv,5, 205.

[6] Hume, *Treatise*, 521.

[7] If the probability of repetition is less than .5, the repeated game is still a Prisoner's Dilemma. If the probability of repetition is high enough, the stag hunting equilibrium becomes risk dominant.

[8] And of Curley's remarks in his introduction to his edition of the Leviathan, p. xxviii.

[9] Rawls recommends that agents choose a social contract according to the maximin principle which would have each agent play it safe by maximizing her minimum gain. If agents followed this advice here, they would choose *Always Defect.*

[10] Foster and Young (1990).

[11] Kandori, Mailath and Rob (1993).

[12] Kandori, Mailath and Rob (1993), Young (1998).

[13] In the risk dominant equilibrium.

[14] See Van Huyck, Battalio and Beil (1990).

[15] Peyton Young (1998) finds the same story true much more generally for local interaction on structures different from the circle.

[16] adapted from an example of Jackson and Watts.

[17] If the huntsman were not patient, then the population would spend a smaller proportion of its time hunting stag, but that proportion would still not be negligible and it would still be true that structure makes a difference. That is the form of the example in Jackson and Watts.

[18] for details and analysis, see Skyrms and Pemantle (2000).

[19] Skyrms and Pemantle (2000).

Problem Set 7 (with added question)

Problem 1: Consider the following *Volunteer' Dilemma* game (which will be studied in class on 19.7.22): $N \geq 2$ persons witness an ongoing crime. The following benefits/costs hold for each person P: If P calls the police she gets a gross benefit of $[B + \Delta]$ and incurs a logistical cost of C. If someone else calls the police, P gets a gross benefit of B and incurs no cost [Δ is the "warm glow effect" of being a "good Samaritan"]. If no one calls the police, P receives no benefit and incurs no cost.

(i) Determine the symmetric Nash equilibrium of the game when $B > C$ and $\Delta < C$.

(ii) Determine the symmetric Nash equilibrium of the game when $B > C$ and $\Delta > C$.

Problem 2: *A Social Cooperation Dilemma*

Consider the conundrum of two urban Indian neighbours – Mr. ***A*** and Ms. ***B*** – deciding simultaneously on whether or not to install 'booster pumps' in their homes. The goal of such a pump is to divert a part of the public water supply from a neighbour's home to own home.

Specifically, the payoffs from installing booster pumps (as dependent on neighbor behavior) are given by the following matrix, where $\boldsymbol{V}$ = total volume of public water supply, $\boldsymbol{c}$ = utility cost of installing a booster pump, and $\boldsymbol{x}$ = volume of water diverted by a booster pump:

		Ms. B	
		install	***not***
Mr. A	***install***	$\sqrt{(0.5V)} - c,\ \sqrt{(0.5V)} - c$	$\sqrt{(0.5V + x)} - c,\ \sqrt{(0.5V - x)}$
	not	$\sqrt{(0.5V - x)},\ \sqrt{(0.5V + x)} - c$	$\sqrt{(0.5V)},\ \sqrt{(0.5V)}$

In what follows, we will set $\boldsymbol{c}$ = 25, and $\boldsymbol{x}$ = 10,000.

(***i***) Initially, the public water supply is adequate: $\boldsymbol{V}$ = 180,000. Determine the optimal decisions of the two neighbours regarding the installation of booster pumps.

(***ii***) Then, the public system worsens, and $\boldsymbol{V}$ falls to 80,000. How do the optimal installation decisions of the neighbours change?

(***iii***) Subsequently, the public system deteriorates even further, and $\boldsymbol{V}$ is reduced to 40,000. Now what are the optimal decisions of the neighbours regarding booster pumps?

(***iv***) In the worst public system, booster pumps are made illegal. However, it remains possible for a person to circumvent the law and install a booster pump by bribing government officials. If the 'additional utility cost of bribery' (which is a composite of monetary and ethical costs) is 5 for each neighbour, what will be the outcome of their decisions? Alternatively, if the 'additional utility cost of bribery' is 5 for Mr. ***A*** and is 20 for Ms. ***B***, what will be the outcome of their decisions?

Problem 3. Four players – Alice, Bob, Cathy, and Dave (who haven't met each other before) – are participate in a the following variant of the '*Choosing the Direction*' game.

I give each person an 'arrow', and lead him/her to a private cubicle. There, each player is required to place his/her arrow "pointing north" or "pointing south" on a table. Then, along with the players, I verify the placement of the four arrows in the four cubicles, and make payments to the players according to the following rules:

(*a*) if all four arrows are placed "pointing north", I pay each player Rs. 2000; (*b*) if all four arrows are placed "pointing south", I pay each player Rs. 1000; (*c*) if exactly two arrows are placed "pointing south", I play *those players* Rs. 1500 each while the others receive nothing; and (*d*) for any other placement of the arrows, I pay each player Rs.750.

We note that Alice, Bob, Cathy, and Dave can chat with each other (in groups of two and/or three and/or four) before they individually (and privately) make their 'direction choices'.

State all the "pure-strategy Nash equilibrium outcomes", and all the "pure-strategy ***coalition-proof*** Nash equilibrium outcomes" of the game.

Problem 4. Consider a "Cournot triopoly" with three identical firms A, B, and C, each with cost function: total cost = $20 \times \{\text{own output}\}$, selling an identical product. The market demand function for the product is: per unit price = 140 – total supply volume. Each firm can only choose to supply one of three volume levels: 20, or 30, or 34.

Find all the "pure-strategy symmetric (where all firms choose to supply the same volume) Nash equilibria" of the game. Identify those Nash equilibria that are "coalition-proof" (when subgroups of firms can *jointly deviate*).

Resolving Cooperation Dilemmas in Repeated Games

- Many interesting strategic interactions are intertemporal in nature
- Intertemporal interactions might allow players to achieve outcomes that they cannot achieve in 'one-shot interactions'
  - most importantly, 'repeated interactions' can **mitigate cooperation dilemmas**
- This is because use of history-dependent strategies in dynamic games make certain ***promises*** (carrots) & ***threats*** (sticks) effective & credible
  - ***promises*** are of the form: if one plays in a "desirable way" currently, she will be "rewarded" in the future
  - ***threats*** are of the form: if one plays in an "undesirable way" currently, she will be "punished" in the future in a particular way
- Sometimes promises and threats are "intertwined" in a strategy
- In many scenarios of intertemporal interactions, it makes sense to think that players' strategic behaviour arise from ***pre-play communication and/or a public recommendation from an impartial consultant***

For the finite repetition high beam game, we can just look at backward induction to solve for the SPNE. For an infinitely repeated game, backward induction cannot be used so we propose a strategy and check whether it is credible and effective.

Tacit Cooperation / Collusion in Repeated Games

Structure of Repeated Games

- In a *repeated game*, a 'stage game' is repeated more than once (possibly probabilistically), and players are forward-looking – they are motivated at every stage by their present discounted payoffs at that stage
- We need to specify each player's 'one-period discount factor' δ

  $\delta = d \times p$, where d = each player's 'one period patience factor' and p = probability that stage game will be repeated in period τ
- Player i's present value of expected future payoffs evaluated at the beginning of period t [assuming that p.v. payoff is zero if game ends]:

  $v^i_t + \delta\{v^i_{t+1} + \delta(v^i_{t+2} + \ldots)\}$
- Typically, in a finitely repeated game, we specify that $p = 1$ for $t < T$, and $p = 0$ for $t \geq T$ for a finite T

  and in an infinitely repeated game, we specify that p is a constant in (0, 1) for all t [and, of course, d is a constant in (0, 1)]

Discounted Repeated Cooperation Dilemmas

High-beam Game

Row \ Column	not	put high-beam on
not	7, 7	0, 10
put high-beam on	10, 0	4, 4

Preliminary results for cooperation dilemmas:

1. For finite repetition (T finite), per-period repetition of the unique stage-game Nash eqm strategy vector is the unique ***SPNE*** of the finitely repeated game
2. For infinite repetition (T infinite), per-period repetition of the unique stage-game Nash eqm strategy vector is one ***SPNE*** in the infinitely repeated game, and a continuation ***SPNE*** after every period

Two comments: **1.** For finite T, Result 1 is routinely violated in experiments.
2. For infinite T, there typically exist many other "mutually better" ***SPNE*** especially when "future matters sufficiently" to the players

The Infinitely-repeated High-beam Game

East-bound \ West-bound	low beam	high beam
low	7, 7	0, 10
high	10, 0	4, 4

Obvious cooperation strategy: each of us promises that (s)he will not be the first one to put on the high beam

Obvious punishment strategy: each of us threatens that if the other player breaks his/her promise then (s)he will 'put on the high beam'

The 'threats' are **credible** because the per-period repetition of the unique stage-game Nash eqm strategy vector is one ***SPNE*** in the continuation infinitely repeated game

The stated threats will be **effective** if and only if:

$$7/(1-\delta) \geq 10 + 4.\delta/(1-\delta)$$

Are other cooperation structures and punishment structures possibilities?

Credibility is because once the punishment starts, no player has an incentive to deviate. That is why, an SPNE works as a credible punishment.

Cooperation Dilemma: *Repeated Cournot Game* (strategic substitutes)

Firm A \ Firm B	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

A & B compete in repeated volume-setting for T pds given $\delta \in (0, 1)$; in each pd, each can set low vol = *30* or medium vol = *40* or high vol = *60*

- Unique ***SPNE*** for finite T: "Irrespective of history, each firm sets volume = *40* in every period" [.... This is ***one SPNE*** for infinite T]
- If repetition is infinite (probability of continuation always positive) and if δ is high enough (players are sufficiently patient & continuation probabilities are high), then firms can achieve better present-value payoffs
  - to do this players need to **cooperate by playing non-Nash strategies** and to sustain cooperation by **punishment threats following 'cheating'** where the **punishment threats have to be credible and effective**

Infinitely Repeated Cournot Game \w *grim strategies*

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

- Given $\delta \in (0, 1)$, consider the following 'two-phase' strategy vector: "**Each player starts with 30 and continues to play 30 so long as all have played 30 in the past; if any player plays a volume $\neq$ 30 in any period τ, each player plays 40 in all periods $T > \tau$.**"
- If players always stay in the **cooperation phase** then each will get **1800** in every period
- **Punishment phase** is one of "perpetual stage-game Nash reversion"
- If players ever enter **punishment phase** then each will get **1600** in every period and, more importantly, no one will want to "defect" from this punishment phase as it's a "continuation ***SPNE***"

  so punishment phase of 'perpetual stage-game Nash reversion' is ***credible***

Perpetual stage-game Nash reversion is that if anyone cheats, we will play the original one-shot Nash Equilibrium from there on. It is by definition credible, so we only need to check if it is effective.

Infinitely Repeated Cournot Game \w *grim strategies*

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

- Given $\delta \in (0, 1)$, consider the following 'two-phase' strategy vector: "**Each player starts with 30 and continues to play 30 so long as all have played 30 in the past; if any player plays a volume $\neq$ 30 in any period τ, each player plays 40 in all periods $T > \tau$.**"
- In order for punishment to be **effective**, threat of punishment must deter each player from cheating in any way (by deterring cheating in the optimal way) in the cooperation phase
- So punishment will be **effective** when following ***collusion constraint*** holds:

  $1800/(1-\delta) \geq 2000 + 1600.\delta/(1-\delta) \iff \delta \geq ½$

  where **1800** is per-pd cooperation-phase payoff, **2000** is one-pd payoff from 'optimal cheating', **1600** is per-pd punishment-phase payoff

Grim strategy is I will collaborate with you until you cheat once, after that I will never collaborate.

Infinitely Repeated Cournot Game \w *grim strategies*

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

- Given $\delta \in (0, 1)$, consider the following 'two-phase' strategy vector: "**Each player starts with 30 and continues to play 30 so long as all have played 30 in the past; if any player plays a volume $\neq$ 30 in any period τ, each player plays 40 in all periods $T > \tau$.**"
- **Cooperation phase** specifies the "cooperation target payoffs"
- **Punishment phase** specifies continuation strategies of the players if any player deviates from cooperation phase: here "*perpetual Nash reversion*"
- To be able to tacitly sustain cooperation at the specified target, *at the very least*, the punishment phase has to be "credible" and "effective"
- Credible ⇔ "continuation ***SPNE***" after relevant history
- Effective ⇔ deters cheating from cooperation ⇔ satisfies collusion constraint

Credibility is asking will you punish me whereas effectiveness is about asking even if I know that you are going to punish me, do I think that it is scary enough for me not to cheat.

Infinitely Repeated Cournot Game \w *grim strategies*

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

- Given $\delta \in (0, 1)$, consider the following 'two-phase' strategy vector: "**Each player starts with 30 and continues to play 30 so long as all have played 30 in the past; if any player plays a volume $\neq$ 30 in any period τ, each player plays 40 in all periods $T > \tau$.**"

Result: Cooperation on {*30*, *30*} will be **(credibly and effectively) *sustained*** by threat of perpetual Nash reversion if and only if $\delta \geq ½$

- Have to discuss the following for games of strategic substitutes and complements:
  - *possibility of renegotiation appeal, and how to deal with that*
  - *more forgiving punishment strategies (than grim strategies) for a given cooperation target and δ*
  - *possibility of coalitional cheating from collusion when > 2 players*
  - *idea of lowering collusion target if higher target cannot be sustained*

Alternative definitions of renegotiation-proofness in repeated games

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

- An ***SPNE*** in an infinitely repeated game is **strongly renegotiation-proof** if and only if, after a unilateral deviation by any player in any period ***t***, the "specified post-deviation continuation ***SPNE*** payoffs" of all non-deviating players are no less than what their "continuation payoffs" would be ***if the deviation did not occur in period t***.
- An ***SPNE*** in an infinitely repeated game is **weakly renegotiation-proof** if and only if, after a unilateral deviation by any player in any period ***t***, the "specified post-deviation continuation ***SPNE*** payoffs" of all non-deviating players are no less than what their "continuation payoffs" would be ***if the deviation occurred in every period after t***.

Strong renegotiation proofness is if the players who do not cheat are not worse off when punishing, whereas weak is when players are not worse off in punishing as compared to when A is cheating continuously.

Repeated Cournot Game: renegotiation-proofness of grim strategy

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

- "**Each player starts with 30 and continues to play 30 so long as all have played 30 in the past; if any player plays a volume $\neq$ 30 in any period τ, each player plays 40 in all future periods.**"

Result: Cooperation {*30*, *30*} will be ***credibly and effectively sustained in a weakly renegotiation-proof manner*** by the threat of perpetual Nash reversion if and only if $\delta \geq ½$

A forgiving* "Trigger strategy" in the *Repeated Cournot Game

P = 140 - Q
TC = 20q

Firm A \ Firm B	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

- Consider ***trigger strategy*** with punishment of ***T***-period Nash reversion: "**Each player starts with 30 and plays 30 so long as all have played 30 in the past; if any player 'deviates' in any period τ, each player plays 40 for next T periods, and then revert to cooperation.**"
- The ***punishment threat*** is ***credible*** and ***effective*** when δ is sufficiently high and ***T*** is sufficiently long
- *One can check whether a trigger strategy is effective by considering a one-shot deviation from cooperation; but when it is not effective the optimal way to cheat will involve multiple deviations*

SPNE, "one-stage deviation principle", and "collusion constraint"

- In "finite" sequential-move games, ***SPNE*** can be identified by backward induction algorithm
- In "infinite" sequential-move games, whether a proposed complete contingent strategy vector is ***SPNE*** or not can be checked by the following "***one stage deviation principle***" :

  A complete contingent strategy vector is an ***SPNE*** if an only if **no player can gain within any specific subgame by deviating from the continuation SPNE (specified for that subgame) at a single decision node/information set of that subgame**

  [*there is no need to check for gains from deviating at multiple successive decision nodes/information sets*]
- This logic motivates the checking of the "collusion constraint" as we have done

"Coalition-proofness" of Tacit Collusion

- In "repeated cooperation dilemmas" with more than two players, even if the posited "punishment strategy" is "credible and effective" (in our stated sense), that only prevents "cheating by a single player"
- That does not necessarily rule out "coordinated ***coalitional cheating*** by a sub-group of players" identical products & competing in volume
- While this can be a major issue in general cooperation dilemmas, it is typically not an issue in "*dominance-solved cooperation dilemmas involving either strategic substitutes or strategic complements*"
- That is basically because the "**stage-game Nash equilibrium** in these kind of games is a **coalition-proof Nash equilibrium**"

Advertising Competition in a year (a game in strategic complements)

		Firm B: low	Firm B: medium	Firm B: high
Firm A	low	40, 40	30, 46	15, 44
	med	46, 30	36, 36	16, 39
	high	44, 15	39, 16	19, 19

Repeated Advertising Game (looking for grim strategy to support collusion)

		Firm B: low	Firm B: medium	Firm B: high
Firm A	low	40, 40	30, 46	15, 44
	med	46, 30	36, 36	16, 39
	high	44, 15	39, 16	19, 19

- Given $\delta \in (0,1)$ & $T = \infty$, consider following ***grim strategy***:
  "Each player starts with *low* and continues to play *low* so long as all have played *low* in the past; if any player plays ad exp ≠ *low* in any period τ, each player plays *high* in all future periods."
- ***Collusion constraint*** : $40/(1-\delta) \geq 46 + 19.\delta/(1-\delta)$ ⇔ $\delta \geq 2/9$

Conclusion: The grim strategy ***credibly and effectively sustains*** the 'maximal collusion target' {*low*, *low*} **if** $\delta \geq 2/9$.

Further, maximal collusion {*low*, *low*} can be sustained in an ***SPNE*** *via* any feasible punishment strategy **only if** $\delta \geq 2/9$
because the grim strategy holds a 'cheater' to his ***maximin payoff***

Here, the grim strategy is neither strongly nor weakly renegotiation-proof

strategic substitutes

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

strategic complements

	low	medium	high
low	40, 40	30, 46	15, 44
medium	46, 30	36, 36	16, 39
high	44, 15	39, 16	19, 19

Two Important and Distinct Questions need to be addressed

1. For any given value of δ and a specific "collusion target", how do we determine whether there exists **any** set of credible punishment strategies that will effectively sustain the collusion target as an ***SPNE***?
2. How does the entire set of collusive outcomes that can be sustained by credible and effective punishments, depend upon different values of δ in (0, 1)?

Repeated Cournot Game **: use of a class of penal codes**

		Firm B: 30	Firm B: 40	Firm B: 60
Firm A	30	1800, 1800	1500, 2000	900, 1800
	40	2000, 1500	1600, 1600	800, 1200
	60	1800, 900	1200, 800	0, 0

- Define the following three 'codes' (following Dilip Abreu's research):
  1. Cooperation Code $\mathcal{C}$ = (*30*; *30*) in every period
  2. Penal Code $\mathcal{P}_A$ = (*30*; *60*) in every period [Stackelberg reversion]
  3. Penal Code $\mathcal{P}_B$ = (*60*; *30*) in every period
- Then consider the following 'three-phase' strategy profile:
  "Each firm starts by playing according to the cooperation code $\mathcal{C}$;
  a ***deviation*** by Firm A from any ongoing code, either $\mathcal{C}$ or $\mathcal{P}_A$ or $\mathcal{P}_B$, requires the firms to play according to $\mathcal{P}_A$ from then on forever;
  a ***deviation*** by Firm B from any ongoing code, either $\mathcal{C}$ or $\mathcal{P}_A$ or $\mathcal{P}_B$, requires the firms to play according to $\mathcal{P}_B$ from then on forever"

A maximin payoff is the maximum possible worst outcome in case of an action.

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

Consider strategy profile {C = (*30*; *30*); P_A = (*30*; *60*); P_B = (*60*; *30*)}

- This strategy profile is ***simple*** in that the penal code for firm ***i*** is the same irrespective of 'how and when it cheats'
- This strategy profile employs ***optimal (harshest) penal codes*** because the "punished player is held to his maximin payoff"

Question 1: After any history of the game that ends with **A** deviating from C, is the structure {P_A, P_B} 'credible', i.e., do the set of penal codes jointly constitute a continuation ***SPNE*** in which the first penal code path will be played forever after?

Answer: Yes, as long as $\delta \geq$ **2/11**.
Since $\delta \geq$ **2/11** implies: $1800/(1-\delta) \geq 2000 + 900.\delta/(1-\delta)$

Implication: When $\delta \geq$ **2/11**, punishment paths P_A, P_B will be "credible"

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

Consider strategy profile {C = (*30*; *30*); P_A = (*30*; *60*); P_B = (*60*; *30*)}

- This strategy profile is ***simple*** in that the penal code for firm ***i*** is the same irrespective of 'how and when it cheats'
- This strategy profile employs ***optimal (harshest) penal codes*** because the "punished player is held to his maximin payoff"

Question 2: Is a penal code P_i 'effective', i.e., does it deter player ***i*** from cheating when in the cooperation phase?

Answer: Yes, as long as $\delta \geq$ **2/11**.
Since $\delta \geq$ **2/11** implies: $1800/(1-\delta) \geq 2000 + 900.\delta/(1-\delta)$

Implication: When $\delta \geq$ **2/11**, punishment paths P_A, P_B will be "effective"

Setting Pa as A's maximin payoff ensures that A has no incentive to cheat from this punishment. So, for credibility, we only need to check B's pov.

Penal code is just a punishment. The optimal (or harshest) penal code holds each player to his/her own maximin payoff.

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

Consider strategy profile $\{\mathcal{C} = (30; 30); \mathcal{P}_A = (30; 60); \mathcal{P}_B = (60; 30)\}$

- This strategy profile is ***simple*** in that the penal code for firm ***i*** is the same irrespective of 'how and when it cheats'
- This strategy profile employs ***optimal (harshest) penal codes*** because the "punished player is held to his maximin payoff"

Question 3: Can cooperation on (30; 30) be sustained as an ***SPNE*** outcome when $\delta < 2/11$?

Answer: No, as there does not exist any penal code that can hold a punished player below his maximin payoff.

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

Consider strategy profile $\{\mathcal{C} = (30; 30); \mathcal{P}_A = (30; 60); \mathcal{P}_B = (60; 30)\}$

- This strategy profile is ***simple*** in that the penal code for firm ***i*** is the same irrespective of 'how and when it cheats'
- This strategy profile employs ***optimal (harshest) penal codes*** because the "punished player is held to his maximin payoff"

Conclusions: The proposed simple strategy profile credibly and effectively sustains ***tacit collusion*** on the desired cooperation target {30, 30} **if and only if** $\delta \geq 2/11$.

In this game, the proposed strategy profile $\{\mathcal{C} = (30; 30); \mathcal{P}_A = (30; 60)$; $\mathcal{P}_B = (60; 30)\}$ is ***strongly renegotiation-proof*** as p.v. payoff to firm ***I*** under the penal code $\mathcal{P}_J = 1800/(1-\delta)$

One can carry out such an exercise for a class of repeated cooperation dilemmas

This is strongly renegotiation proof as B is getting the same payoff of 1800 in Pa as compared to when there was no cheating.

Summary of Infinitely Repeated Cooperation Dilemmas

strategic substitutes

	30	40	60
30	1800, 1800	1500, 2000	900, 1800
40	2000, 1500	1600, 1600	800, 1200
60	1800, 900	1200, 800	0, 0

Cooperation target: (30; 30); penal codes: $\mathcal{P}_A = (30; 60)$, $\mathcal{P}_B = (60; 30)$
Result: Cooperation target (30; 30) is sustainable if and only if $\delta \geq 2/11$
Penal codes are ***most severe***: they give A and B their ***maximin*** payoffs
Penal codes are ***strongly*** renegotiation-proof given the cooperation target

strategic compliments

	low	medium	high
low	40, 40	30, 46	15, 44
medium	46, 30	36, 36	16, 39
high	44, 15	39, 16	19, 19

Cooperation target: (low; low); penal codes: $\mathcal{P}_A = \mathcal{P}_B$ = (high; high)
Result: Cooperation target (low; low) is sustainable if and only if $\delta \geq 2/9$
Penal codes are ***most severe***: they give A and B their ***maximin*** payoffs
Penal codes are ***not*** (may not be) renegotiation-proof given cooperation target

Consequences of high δ versus low δ (*Fudenberg and Tirole*)

- A big result in repeated cooperation dilemmas: when δ is close to 1, all 'collusion targets' can be achieved credibly and effectively
- However, that may not be the case when δ is low. Rather, only a 'less ambitious collusion target might be achievable, if at all.

strategic compliments

	low	medium	high
low	40, 40	30, 46	15, 44
medium	46, 30	36, 36	16, 39
high	44, 15	39, 16	19, 19

In this game, collusion on (low; low) achievable **iff** $\delta \geq 2/9$. ***What if*** $\delta = 0.2$**?**

Lower cooperation target to: (med; med)
Use penal codes: $\mathcal{P}_A = \mathcal{P}_B$ = (high; high)
Result: Cooperation target (med; med) is sustainable if and only if $\delta \geq 3/20$
Penal codes are ***most severe***: they give A and B their ***maximin*** payoffs
Penal codes ***are*** weakly renegotiation-proof given cooperation target

Over here, B would get 16 in case of A cheating but is getting 19 in punishment but less than 36 in case of no cheating so weakly renegotiation.

Possibility of *tacit collusion* in oligopoly (ignoring illegality)

- For any given **δ**, the larger the number of firms in an industry, the more difficult is it to sustain price collusion (as cooperative profit is smaller & 'gain from cheating' is larger)
- If firms have cost differences and/or different product portfolios, a 'collusion target' will be hard to agree on,

  and different firms will have differential incentives to cheat

  [generally, "weaker" firms will have greater incentives to cheat]
- If market environment is fluctuating, so are incentives to cheat (it's more profitable to cheat on price collusion during booms)
- **If 'cheating' changes the future game, it might be hard to punish such cheating, leading to break-down in collusion**

  Tacit collusion with respect to "product innovation", "capacity building", etc. have to be studied via ***dynamic games*** where current strategies affect future payoffs and feasible strategies

  Tacit collusion can be rationally sustainable in dynamic games, but the relationship with patience is not 'clear-cut'

'Climbing a Quality Ladder' Game

Per-period profit matrix:

	P1	P2
P1	10 , 10	0 , 30
P2	30 , 0	20 , 20

Climbing cost = 120

At $T = 0$, both firms at **P1**, and in every subsequent period ***t***, each can irreversibly climb up one position iff it is at **P1** at (***t***–1)

Q: In what positions will the firms end up for **δ=0.8** and for **δ=0.9**?

• The payoff matrix if this was a one-shot simultaneous move game:

One-shot game

	P1	P2
P1	10/(1-δ), 10/(1-δ)	0, 30/(1-δ) –120
P2	30/(1-δ) –120, 0	20/(1-δ) –120, 20/(1-δ) –120

'Climbing a Quality Ladder' Game (one-shot sim. move game)

One-shot δ=0.8

	P1	P2
P1	50, 50	0, 150 - 120
P2	150 - 120, 0	100 - 120, 100 - 120

One-shot δ=0.9

	P1	P2
P1	100, 100	0, 300 - 120
P2	300 - 120, 0	200 - 120, 200 - 120

In the one-shot simultaneous-move game:

- For **δ=0.8:** dominant strategy to stay at ***P1*** $\Rightarrow$ no cooperation dilemma
- For **δ=0.9:** dominant strategy to climb to ***P2*** $\Rightarrow$ cooperation dilemma

Climbing the Quality Ladder (dynamic game)

Per-period profit matrix:

	P1	P2
P1	10 , 10	0 , 30
P2	30 , 0	20 , 20

Climbing cost = 120

At T = 0, both firms at ***P*1**, and in every subsequent period t, each can irreversibly climb up one position iff it is at ***P*1** at (t–1)

Q: In what positions will the firms end up for **δ=0.8** and for **δ=0.9**?

- Unique ***SPNE*** strategy in the dynamic game when **δ = 0.8**: "I will stay at ***P*1** irrespective of rival behaviour"
- When **δ = 0.9**, there are two symmetric ***SPNE*** pure strategies:
  ***SPNE* 1**: "In every pd, I will climb if I am in ***P*1**"
  ***SPNE* 2**: "In every pd, I will climb **if** and **only if** I am in ***P*1** and rival is in ***P*2**".

So, when **δ = 0.9**, there is an SPNE that resolves the cooperation dilemma, but there is also another SPNE that does not

Thus, we had stated earlier that a higher delta can make cooperation easier but in this case, a higher delta is actually leading to a cooperation dilemma.

CAUGHT RIGGING: CARTELS UNCOVERED

September 2009 The building company Kier Regional Ltd is fined £17.89m, as part of £129.5m in fines issued to 103 construction firms for price-rigging.

September 2009 The recruitment specialist Hays is fined £30.36m for price-fixing with five other firms to drive out an emerging competitor. The six were fined a total of £39.27m.

December 2007 Sainsbury's is fined £26m after it admits being part of a dairy price-fixing cartel with rivals Morrison's, Ads and other dairy firms.

August 2007 British Airways is fined £121.5m for price-fixing with Virgin Atlantic on long-haul routes.

February 2003 Argos is fined £17.28m for fixing prices of Hasbro toys and games such as Action Man with Littlewoods – which was fined £5.4m.

RBS fined £28.6m for price collusion

Jack Jordan | The Independent, UK | 31 March 2010

The Office of Fair Trading (OFT) has issued a £28.59m fine to Royal Bank of Scotland – a bank 84 per cent owned by the taxpayer – for breaking competition law.

The fine was announced yesterday after RBS admitted revealing details of its loan pricing plans for professional services firms, such as lawyers and accountants, to its rival Barclays. The OFT reduced the fine from £33.6m to reflect RBS's admission and agreement to co-operate.

The regulator said that Barclays took the information into account to determine its own pricing. However, Barclays is unlikely to face any fine for its part in the matter because it blew the whistle on the affair, contacting the OFT in March 2008.

An RBS spokeswoman said "stringent" new competition-law training had been introduced to avoid similar breaches happening again. She said: "This is a deeply regrettable and isolated case from nearly two years ago, involving only two members of staff, one of whom has left the bank and one other who faces suspension and further investigation now the case has been settled."

According to the OFT, individuals in RBS's professional practices coverage unit disclosed information about their pricing plans to counterparts at Barclays on the fringes of social, client and industry events and in telephone conversations. Barclays and RBS are the main providers of loans to large professional services firms, such as solicitors, estate agents and accountants.

Ali Nikpay, the OFT's senior director of cartels and criminal enforcement, said: "Any company that discloses confidential future pricing information to its competitors risks a substantial penalty. It is important that companies operating in the UK understand the seriousness of such conduct and ensure effective competition compliance throughout their organisation."

Mr. Nikpay added: "This case underlines the OFT's commitment to protecting competition in the financial services sector. It also highlights the strong benefits of acting promptly to report anti-competitive conduct to the OFT and of co-operating with such investigations."

This fine for competition offences comes at a time when the banks already face public anger for their role in the recession, while investment bankers continue to collect bumper payouts.

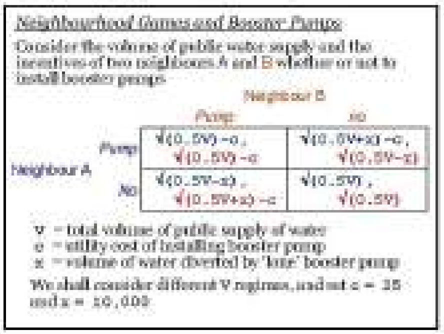
Under the Competition Act 1998, the OFT has the power to impose penalties on companies of up to 10 per cent of their worldwide turnover for breaches of competition law. However, the regulator can grant immunity from penalties or a significant reduction in fines if an organisation reports breaches of the Competition Act and then assists with any investigation.

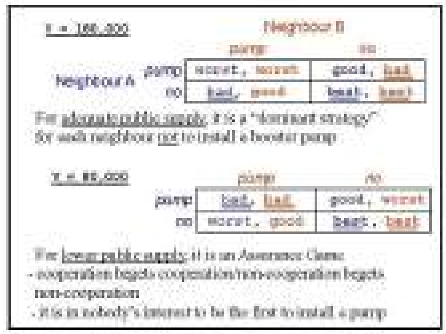
A statement from Barclays stressed it had voluntarily notified the OFT of the breach. It said: “Barclays has been co-operating throughout. The investigation operated solely within the confines of the professional services banking team in Barclays's commercial banking unit.”

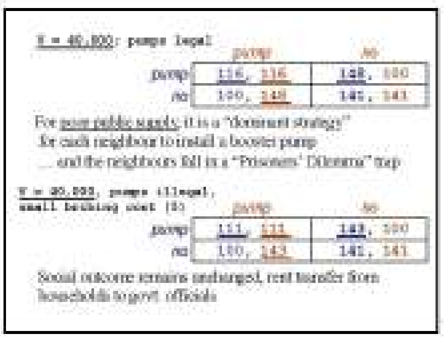
Answer Key 7

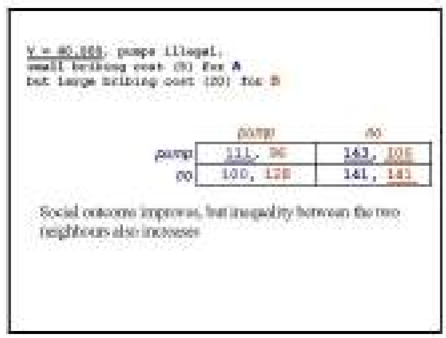
Problem 1: **(i)** Each witness calls the police with probability $\{1 - [(C - \Delta)/B]^{1/(N-1)}\}$.
(ii) Each witness has a dominant strategy to call the police.

Problem 2:









Problem 3. First argue that "all four pointing North" is a pure-strategy Nash equilibrium outcome [since no unilateral deviation can be profitable]. Next, argue that "all four pointing South" is another pure strategy Nash equilibrium outcome (for the same reason as before). Then, focusing on the first outcome, argue that it is a *coalition-proof* pure-strategy Nash equilibrium outcome because no (sub)group of players can jointly deviate and be better off. Finally, focusing on the second outcome, argue that it is **not** a *coalition-proof* pure-strategy Nash equilibrium outcome because all four players can jointly deviate and place their arrows pointing North rather than pointing South.

[This example clarifies that it can indeed be the case that the *Pareto-dominant* Nash equilibrium outcome can indeed be the *unique coalition-proof* Nash equilibrium outcome in some games.]

Problem 4. There are three candidate "pure-strategy symmetric Nash equilibria" of the game: {20, 20, 20}; {30, 30, 30}; {34, 34, 34}. Establish, by straightforward calculation, that (i) 20 is not a best response to {20, 20}; (ii) 30 ***is*** a best response to {30, 30}; and (iii) 34 is not a best response to {34, 34}. That proves that {30, 30, 30} is the ***unique*** "pure-strategy symmetric Nash equilibrium" of the game. In fact, it can be established, by more tedious calculations, that {30, 30, 30} is the ***unique*** Nash equilibrium of the game.

One now has to establish that this Nash equilibrium is ***coalition-proof***. This is to be done by establishing that if one firm supplies 30, then the two other firms cannot jointly profit by supplying either {20, 20}, or {34, 34}, or {20, 34}.

[What is the one-line proof that establishes that the two other firms cannot jointly profit by supplying either {20, 30}, or {34, 30}?]

Problem Set 8

Problem* 1: *Cheating against "trigger strategy"

Two firms *Row* and *Column* compete in "advertising budgets" from the beginning of the current year. At the beginning of every year, each firm can choose a "low" advertising budget, or a "medium" budget, or a "high" budget for the year. The yearly profits (in millions of rupees) for each strategy combination are given by the following payoff matrix:

	low	*medium*	*high*
low	50, 50	32, 60	25, 55
medium	60, 32	40, 40	26, 42
high	55, 25	42, 26	30, 30

The two firms play this game *repeatedly*. In every year, they know that there is **2/3** chance that the game will be played in the next year. Further, the annual 'discount factor' for each firm is (**3/4**) [i.e., one rupee a year from now is valued the same as (3/4) rupees today].

To avoid a fierce advertising battle and to sustain *self-enforcing collusion*, the firms consider playing the following 'trigger strategy':

"A firm will choose 'low advertising budget' on January 1 of the current year, and keep on choosing the low budget as long as both firms have chosen the low budget in previous years. If any firm chooses any other budget in any year, the two firms will choose high budget for the next two years, after which they will go back to choosing low budgets every year (with an identical two-year punishment threat for any subsequent deviation)." This is in effect a collusion strategy with a "two-period Nash reversion threat", rather than the "grim strategy" which specifies a threat of "perpetual Nash reversion".

Prove that the proposed trigger strategy rationally sustains collusion on 'low advertising'. Is the resultant ***SPNE*** renegotiation-proof?

Finally, consider a change in the probability of game continuation: Suppose that every year, the firms know that there is **4/9** chance that the game will be played in the next year. If *Column* expects *Row* to play the 'trigger strategy' described above, what will be *Column*'s "subgame-perfect (i.e., credible) best response"?

Problem* 2: *Tacitly sustaining 'non-stationary collusion'

Two firms *Micro* and *Macro* compete in "marketing outlays" from the current year onwards. At the beginning of every year, each firm chooses a "low" marketing outlay, or a "medium" outlay, or a "high" outlay for the year. Yearly profits (in thousand rupees, net of marketing outlays) for each outlay combination are given by the following payoff matrix:

Macro

Micro		low	medium	high
	low	18 , 18	14 , 30	10 , 23
	medium	30 , 14	15 , 15	8 , 13
	high	23, 10	13 , 8	5, 5

The firms play this game *repeatedly* every year, and have a common annual discount factor **δ.**

(i) In pre-play communication on January 1 of the current year, the firms attempt to sustain the 'cooperative outcome' of {*low*, *low*} every year by using the threat of 'perpetual Nash reversion'. For what values the common **δ** ∈ (0, 1) will they succeed?

(ii) Next, suppose that **δ = 0.9**, and that in pre-play communication on January 1 of the current year, the firms try to devise history-dependent strategies that will sustain cooperation under which each firm will get at least 210 thousand rupees in present value profits. As a strategy-consultant to the two firms, suggest a *renegotiation-proof SPNE* strategy profile that will enable the firms to attain this objective, and that has the following structure: each simple strategy prescribes a "cooperation code", and an "optimal penal code for each player" that specifies that cheating in the cooperation path or in any punishment path by any player ***i*** will initiate the penal code for player ***i*** (which will continue till a further instance of cheating).

Problem* 3: *Dilip Abreu's Example

Consider two symmetric firms playing a Cournot super-game in supply volumes, with the restriction that the only permissible choices are '*high*' volume, '*medium*' volume, and '*low*' volume. This is Dilip Abreu's example presented in the posted Notes on Repeated Games:

Firm 2

Firm 1		low	medium	high
	low	10 , 10	3 , 15	0 , 7
	medium	15 , 3	7 , 7	–4 , 5
	high	7, 0	5 , –4	–15, –15

Abreu shows that when the firms' common discount factor is δ = 4/7, the following symmetric (but non-stationary) penal code rationally sustains the cooperative outcome (*low*, *low*) in every period of the infinite super-game:

"Deviant firm is required to play *medium* in the first round after deviation, and *low* thereafter and ever after; the 'non-deviant' firm is required to play *high* in the first round after deviation, and *medium* thereafter and ever after."

Verify that the above penal code is "optimal" and "strongly renegotiation-proof" for each of the firms, and sustains collusion on (*low*, *low*) forever.

Dynamic Games of Timing

- In many business scenarios, firms' important strategic decisions are to determine *when* to implement an (almost) irreversible decision [e.g., adopt a new technology, introduce a new product, exit a business, ..]
- ***Games of timing*** are dynamic games that study such issues
  - in some games, players try to **achieve self-enforcing cooperation**
  - in some other games, players attempt to **resolve various coordination problems in real time** (like Chicken, Battle-of-Sexes, QWERTY, etc.)

Issue 1: ***Costly Preemption*** – Can desire to be first lead to value destruction?

Issue 2: ***Strategic Delay*** – Can firms delay adopting new innovations that would be beneficial for consumers?

Issue 3: ***Stake-out & Shake-out*** – In a declining industry, how do firms "shake out"/ "stake out" rivals?

shake out: thrust larger losses on rival

stake out: demonstrate greater ability to sustain losses than rival

Technology Adoption Game

Gamma / Lambda can adopt at every 6-mnth interval from 1-14 / 7-14

- Fixed cost ***F*** of adoption falls over time:

1-14	7-14	1-15	7-15	1-16	7-16	1-17	7-17	1-18	7-18 →
60	57	54	51	48	45	42	40	38	38 →

- Six-monthly profits (in millions of Rs) from market competition:
- Six-monthly discount factor: **δ** = 0.8

Gamma \ Lambda	adopt	not
adopt	15, 15	20, 8
not	8 , 20	10, 10

- If this was a one-shot simultaneous-move adoption game with ***F*** = 45 then it would be a Game of Chicken

Gamma \ Lambda	adopt	not
adopt	75 – 45, 75 – 45	100 – 45, 40
not	40 , 100 – 45	50, 50

Technology Adoption Game

Gamma / Lambda can adopt at every 6-mnth interval from 1-14 / 7-14

- Fixed cost *F* of adoption falls over time:

1-14	7-14	1-15	7-15	1-16	7-16	1-17	7-17	1-18	7-18 →
60	57	54	51	48	45	42	40	38	38 →

- Six-monthly profits (in millions of Rs) from market competition:
- Six-monthly discount factor: **δ** = 0.8

Gamma \ Lambda	adopt	not
adopt	15, 15	20, 8
not	8 , 20	10, 10

- Three features of the game:

[1] **Adoption is imitation-deterring**: if one firm adopts at T, then rival's best-response: never adopt thereafter [$40 > 75 - F(t)$ for all t]

[2] **Proprietary adoption is eventually profitable**: If tech is proprietary, a firm will adopt when F falls a lot: Gamma at **1-17** / Lambda at **7-17**

[proprietary adoption at earliest t for which $100 - F(t) > (1+\delta)10 + \delta^2(100 - F(t+2))$]

[3] **After** 1-14 **, preemptive adoption is better that being deterred**: adopting *before* rival better than *being deterred*: $100 - F(t) > 10 + \delta.40$

Proprietary adoption means adopting the technology while understanding that the rival will never adopt.

Technology Adoption Game (continued)

- Fixed cost *F* of adoption falls over time:

1-14	7-14	1-15	7-15	1-16	7-16	1-17	7-17	1-18	7-18 →
60	57	54	51	48	45	42	40	38	38 →

- Six-monthly profits (in millions of Rs) from market competition:
- Six-monthly discount factor: **δ** = 0.8

Gamma \ Lambda	adopt	not
adopt	15, 15	20, 8
not	8 , 20	10, 10

- ***Best response to rival adopting at t***: If J conjectures rival will adopt at t, J should preempt at $(t-1)$ if & only if $100 - F(t-1) > 10 + 0.8(40) = 42$
- ***Preemption in technology adoption***:
  - IF no adoption till 1-17, Gamma will at 1-17
  - given that, Lambda will preempt and adopt at 7-16
  - given that, Gamma will preempt and adopt at 1-16
  - …..
  - given that, Lambda will preempt and adopt at 7-14

Technology Adoption Game (continued)

- Fixed cost *F* of adoption falls over time:

1-14	7-14*	1-15	7-15	1-16	7-16	1-17	7-17	1-18	7-18 →
60	57	54	51	48	45	42	40	38	38 →

- Six-monthly profits (in millions of Rs) from market competition:
- Six-monthly discount factor: **δ** = 0.8

Gamma \ Lambda	adopt	not
adopt	15, 15	20, 8
not	8 , 20	10, 10

- Unique ***SPNE*** outcome: Lambda adopts at 7-14, and Gamma never adopts
- Lambda's adoption date is such that if Lambda waited any longer, it would be preempted by Gamma
- Victory achieved at high price: Lambda adopts "too early" paying a large adoption cost: ***rent dissipation*** from competitive preemption
- If adoption costs fell continually over time, the 'follower' will also eventually adopt, but only after a significant delay

Alternative payoffs & "Strategic Delay" in Adoption

- Fixed cost ***F*** of adoption falls over time:

1-14	7-14	1-15	7-15	1-16	7-16	1-17	7-17	1-18	7-18 →
60	57	54	51	48	45	42	40	38	38 →

- Six-monthly profits (in millions of Rs) from market competition:
- Six-monthly discount factor: **δ** = 0.8

Gamma \ Lambda	adopt	not
adopt	15, 15	17, 3
not	3 , 17	13, 13

- If this was a one-shot adoption game with ***F*** = 45 it would be a "non-conflicting ranked coordination game" with two Pareto-ranked pure Nash equilibria**:** (*not*, *not*) >> (*adopt*, *adopt*)

	adopt	not
adopt	75 - 45, 75 - 45	85 - 45, 15
not	15 , 85 - 45	65, 65

Alternative payoffs & "Strategic Delay" in Adoption

- Fixed cost F of adoption falls over time:

1-14	7-14	1-15	7-15	1-16	7-16	1-17	7-17	1-18	7-18 →
60	57	54	51	48	45	42	40	38	38 →

- Six-monthly profits (in millions of Rs) from market competition:

	adopt	not
adopt	15, 15	17, 3
not	3, 17	13, 13

- Six-monthly discount factor: $\delta = 0.8$
- Three features of this dynamic adoption game:

[1] **Adoption is imitation-inducing**: if one firm adopts at T, rival best-response: adopt immediately thereafter [$15 < 75 - F(t)$]

[2] **Proprietary adoption is never profitable**: If rival "will never adopt", a firm will never find it profitable to be the single adopter [$65 > 85 - F(t)$]

[3] **Preemptive adoption is always worse that imitation**: adopting *before* rival is worse than adopting *after* rival: [$17 + \delta.75 - F(t) < 13 + \delta.3 + \delta^2(75 - F(t{+}2))$]

Alternative payoffs & "Strategic Delay" in Adoption

- Fixed cost F of adoption falls over time:

1-14	7-14	1-15	7-15	1-16	7-16	1-17	7-17	1-18	7-18 →
60	57	54	51	48	45	42	40	38	38 →

- Six-monthly profits (in millions of Rs) from market competition:

	adopt	not
adopt	15, 15	17, 3
not	3, 17	13, 13

- Six-monthly discount factor: $\delta = 0.8$

- Unique ***SPNE*** outcome: neither firm adopts
  [supported by strategy: "I will adopt if and only if rival adopts"]
- If adoption costs fell continually over time, one firm will adopt after a significant delay, and rival will follow suit

There is real-world evidence that imitation-deterring innovations are adopted faster than imitation-inducing innovations

e.g., installing MRI machines by hospitals vs. ATM machines by banks

An Exit Game

In some declining chemical/metallurgical industries in the US (synthetic soda ash, steel casting, etc.), firms with larger capacities exited earlier than those with smaller capacities

Stake-out versus Shake-out

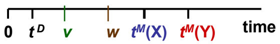
shake out: thrust larger losses on rival

stake out: demonstrate greater ability to sustain losses than rival

Stake-out: Exit from a declining industry [Ghemawat-Nalebuff]

- Consider chemical industries where firms ***have*** to sell up to capacity
- 2 firms X & Y have weekly capacities *40* & *30* & costs $c_X = c_Y = 20$
- Demand is shrinking; demand in week t : $P = (100 - t) - Q$
- Each firm has to decide on ***when to exit*** the industry
- Week t duopoly profits for J : $[\{(100 - t) - (40 + 30)\} - 20] \times K_J$
- Duopoly profits of each firm reach 0 on: $t^D = 100 - 70 - 20 = 10$
- Should any of the firms stay on beyond $t^D = 10$ weeks? Which one?
- Week t monopoly profits for J : $[\{(100 - t) - K_J\} - 20] \times K_J$
- Monopoly profits of firm J reach 0 on: $t^M(J) = 80 - K_J$
- Thus, $t^M(\text{X}) = 40$ and $t^M(\text{Y}) = 50$
- The fact that smaller Y can out-last larger X as a monopolist will allow the former to ***stake-out*** the latter <u>at a much earlier date</u>

Given $t^D < t^M(\text{X}) < t^M(\text{Y})$:



- At any $t > t^M(J)$, it is dominant strategy for firm J to "exit" and at any $t < t^D$, it is dominant strategy for each firm to "stay"
- Let week $\boldsymbol{w}$ be such that p.v. of duopoly losses for Y during $[\boldsymbol{w}, t^M(\text{X})]$ equals p.v. monopoly profits during $[t^M(\text{X}), t^M(\text{Y})]$
- At any $t > \boldsymbol{w}$, it is dominant strategy for Y to "stay" if it has survived till then, as rival will definitely exit by $t^M(\text{X})$
- Then, dominant strategy for X to exit at every $t \geq \boldsymbol{w}$
- Let week $\boldsymbol{v}$ be such that p.v. of duopoly losses for Y during $[\boldsymbol{v}, \boldsymbol{w}]$ equals p.v. monopoly profits during $[\boldsymbol{w}, t^M(\text{Y})]$
- Then, dominant strategy for X to exit at any $t \geq \boldsymbol{v}$
- So it goes all the way up to t^D [iterated deletion of ***S-D-S***s]
- Unique ***SPNE*** outcome: firm X exits at t^D, firm Y exits at $t^M(\text{Y})$
- *What are the unique SPNE strategies of the two firms?*

The Logic of Stake-out *vs.* Shake-out

- For duopolists X & Y facing continually declining demand, four dates are crucial for 'exit behaviour': $t^D(\text{X})$, $t^D(\text{Y})$, $t^M(\text{X})$, $t^M(\text{Y})$

<u>Case 1</u>: $t^D(\text{X}) \leq t^D(\text{Y}) < t^M(\text{X}) < t^M(\text{Y}) < \infty$ [when $K_X > K_Y$, $c_X \geq c_Y$]

- firm X will exit at $t^D(\text{X})$, and firm Y will exit at $t^M(\text{Y})$ {***stake out***}

<u>Case 2</u>: $t^D(\text{Y}) < t^D(\text{X}) < t^M(\text{X}) < t^M(\text{Y}) < \infty$ [when $K_X > K_Y$, and $c_X < c_Y$ by a small amount]

- if cumulative duopoly losses for firm Y over $[t^D(\text{Y}), t^D(\text{X})]$ is "covered" by cumulative monopoly profits over $[t^D(\text{X}), t^M(\text{Y})]$, firm X will exit at $t^D(\text{X})$, and firm Y will exit at $t^M(\text{Y})$ {***stake out***}
- if not firm Y will exit at $t^D(\text{Y})$, firm X will exit at $t^M(\text{X})$ {***shake out***}

Problem Set 9

***Problem* 1:** Consider the "timing of technology adoption game" as presented in the posted Session 14 Slides, but with the following altered six-monthly profit matrix:

		Lambda	
		new tech	*old tech*
Gamma	*new tech*	15, 15	20, 2
	old tech	2, 20	10, 10

State TWO pure-strategy Subgame-perfect Nash equilibria of this dynamic technology adoption game, and their associated outcomes.

***Problem* 2:** Consider the Ghemawat-Nalebuff "Exit" game as presented in the posted Session Slides. State the unique *SPNE* strategies of firms X and Y in that game.

***Problem* 3:** A particular chemical processing industry is in a decline, with demand shrinking over time. The daily market demand function is $P(t) = [1000 - t] - Q$, where t denotes the time period and Q the total industry output. The present is denoted by $t = 0$, and $t = n > 0$ denotes the "n'th day" in the future.

At present, there are three firms in this industry: firms A, B, and C, each having one plant with a fixed daily production capacity. Firm A's daily capacity is 150 units, firm B's daily capacity is 300 units, and firm C's daily capacity is 100 units. Each firm is constrained to produce and sell *all of its capacity* every day.

Further, the unit production cost of each firm is 100 (so a firm with capacity K incurs total production cost of $100K$ every day that it is in business). Every day, the equilibrium market price is that price where demand equals supply.

The only strategic decision that each firm has to make is when to "exit" from the industry, i.e., when to irreversibly shut down its plant. Determine the subgame-perfect Nash equilibrium exit dates of the three firms.

Games of Incomplete (Asymmetric) Information

- In many strategic situations, it is realistic to assume that one player does not know the "true" motivations/payoffs of other players
- A simple way to model this is to say that from a player's perspective, her rivals have different payoff functions with different probabilities

Example: Game of Chicken

		Col: Go	Col: Swerve
Row	Go	2, 2	6, 3
	Swerve	3, 6	4, 4

- We will study incomplete-information games in which players have different payoff structures with different probabilities, and that this "payoff distribution structure" **is common knowledge among players**

G-o-C w/ info asymmetry

	Go	Swerve
Go	2, 2	6, {3, 1; 0.3, 0.7}
Swerve	3, 6	4, 4

- It'll be helpful to view this as a 3-player game in which Row plays against Col(3) with prob 0.3 and against Col(1) with prob 0.7

G-o-C w/ info asymmetry

		Col: Go	Col: Swerve
Row	Go	2, 2	6, {3, 1; 0.3, 0.7}
	Swerve	3, 6	4, 4

***Best-responses of* Col(1), Col(3), *& Row given their conjectures, and given that it is common knowledge that Col is 3 or 1 with probability 0.3 & 0.7*:**

Col(1) has a dominant strategy to play *Go*
If Col(3) conjectures that Row will play *Go* then she will play *Swerve*,
if Col(3) conjectures that Row will play *Swerve* she will play *Go*
If Row conjectures that Col(3) will play *Go* (given Col(1)'s *Go*), Row will have to respond to the pure strategy *Go* and so will play *Swerve*

If Row conjectures that Col(3) will *Swerve* (given Col(1)'s *Go*), Row will have to respond to ***mixed strategy*** {*Swerve*, *Go*; **0.3**, **0.7**} & play *Swerve*

Row Conjecture	Row Best Response
Col(1) and Col(3) will play *Go*	*Swerve* because 3 > 2
Col(1) will play *Go* & Col(3) will play *Swerve*	*Swerve* because 0.3×4 + 0.7×3 > 0.3×6 + 0.7×2

G-o-C w/ info asymmetry

		Col: Go	Col: Swerve
Row	Go	2, 2	6, {3, 1; 0.3, 0.7}
	Swerve	3, 6	4, 4

***Best-responses of* Col(1), Col(3), *& Row given their conjectures, and given that it is common knowledge that Col is 3 or 1 with probability 0.3 & 0.7*:**

Col(1) has a dominant strategy to play *Go*
If Col(3) conjectures that Row will play *Go* then she will play *Swerve*,
if Col(3) conjectures that Row will play *Swerve* she will play *Go*
If Row conjectures that Col(3) will play *Go* (given Col(1)'s *Go*), Row will have to respond to the pure strategy *Go* and so will play *Swerve*

If Row conjectures that Col(3) will *Swerve* (given Col(1)'s *Go*), Row will have to respond to ***mixed strategy*** {*Swerve*, *Go*; **0.3**, **0.7**} & play *Swerve*

- There exists exactly one intersection of these three best-response functions:

Unique Eqm: Row will play *Swerve* and Col(3) & Col(1) will play *Go*

Here, Col *will successfully leverage his reputation of being* 'macho' *for gain*

G-o-C w/ info asymmetry

		Col: Go	Col: Swerve
Row	Go	2, 2	6, {3, 1; 0.3, 0.7}
	Swerve	3, 6	4, 4

There exists exactly one intersection of these three best-response functions:
Unique Eqm: Row will play *Swerve* and Col(3) & Col(1) will play *Go*

- In games of incomplete info, a player might have to best-respond to a "**mixed strategy**" even when every player of every type is playing ***pure***
- Our ability of predicting a specific equilibrium in this game depends crucially on the players' "payoff distribution structures" and on the fact that they are **common knowledge among the players**
- And when it is the case that players' "payoff distribution structures" will eventually (over time) become common knowledge among them, there is value to acquiring specific **"reputations"**

Cooperation Dilemmas under Asymmetric Information

- Nuclear arms races are some of the worst cooperation dilemmas. How do we explain them? Do we believe that some countries have dominant strategies to build nuclear arsenal? Or, are arms races consequences of information asymmetries among nations?

Question: Baliga & Sjostrom (2004) ask whether the following situation can arise:

No party has an *ex ante* dominant strategy 'not to cooperate', but all are *sucked into* non-cooperation simply because each believes that there's a *small chance* that rivals have a dominant strategy to 'not to cooperate'?

The real issue: Does information asymmetry make it harder to achieve cooperative outcomes?

An Arms Race Game

		Country B: *escalate*	Country B: *desist*
Country A	*escalate*	1 , 1	*vA* , 0
	desist	0 , *vB*	4 , 4

- If it is common knowledge that either $v_A > 4$ or $v_B > 4$ (or both), then {*escalate*, *escalate*} will be the dominance-solved (or dominant strategy) equilibrium in this Cooperation Dilemma game
- Alternatively, if it is common knowledge that $v_A < 4$ and $v_B < 4$, then {*desist*, *desist*} & {*escalate*, *escalate*} will be the two Nash equilibria of an Assurance Game, with the former being Pareto-superior
- Now suppose each country I does not know precise rival 'type' v_J and believes that v_J takes on some value in the interval [2, 5]
- How much "belief pessimism about rival" is needed to induce a country I to "escalate for sure" even if it has $v_I < 4$?
  - belief pessimism ≡ probability assessment that rival value $v_J > 4$

Arms Race w/ Information Asymmetry

		Country B: escalate	Country B: desist
Country A	escalate	1 , 1	vA , 0
	desist	0 , vB	4 , 4

The "discrete types" case : Each country I is one of two "types": either a **Hawk** with $vI = 5$, or a **Dove** with $vI = 2$

- Each country knows its own type, and holds 'commonly known prior belief' that rival is a **Dove** with prob p and a **Hawk** with prob $1 - p$
- We will view this as a 4-player game in which:
  each of **Hawk** A and **Dove** A plays against **Hawk** B with prob $1 - p$ and **Dove** B with prob p;
  while each of **Hawk** B and **Dove** B plays against **Hawk** A with prob $1 - p$ and **Dove** A with prob p

		Country B: escalate	Country B: desist
Country A	escalate	1 , 1	vA , 0
	desist	0 , vB	4 , 4

Best-responses of two 'types' of each player, given each player's prior commonly known belief that rival is **Dove** (vJ=2) ***with*** prob. $p = 0.3$:

- A **Hawk** [A(5) / B(5)] has a dominant strategy to *escalate*
- If a **Dove** [A(2) / B(2)] conjectures that "**rival Dove**" will *escalate*, it has to respond to the ***pure strategy*** *escalate,* and so it will *escalate*
- If a **Dove** [A(2) / B(2)] conjectures that "**rival Dove**" will *desist*, it has to respond to the ***mixed strategy*** {*escalate*, *desist* ; **0.7**, **0.3**}
  - if it escalates, its expected utility is: $(0.7)\times1 + 0.3\times2 = 1.3$
  - if it desists, its expected utility is: $(0.7)\times0 + 0.3\times4 = 1.2$
- So a **Dove** will *escalate* irrespective of its conjecture about rival
- For $p = 0.3$, unique ***dominance-solved Bayes-Nash equilibrium***:
  all "*four*" players {**Hawk** A, **Dove** A, **Hawk** B, **Dove** B} *escalate*
  - ***Here 'escalate' is the only rationalizable strategy***

		Country B: escalate	Country B: desist
Country A	escalate	1 , 1	vA , 0
	desist	0 , vB	4 , 4

Here, p is each country's prior belief that rival is a **Dove** ($v = 2$)

- A **Hawk** [A(5) / B(5)] has a dominant strategy to *escalate*
- If a **Dove** [A(2) / B(2)] conjectures that "**rival Dove**" will *escalate*, it has to respond to the ***pure strategy*** *escalate,* and so it will *escalate*
- If a **Dove** [A(2) / B(2)] conjectures that "**rival Dove**" will *desist*, it has to respond to the ***mixed strategy*** {*escalate*, *desist* ; $1 - p$, p}
  - if it escalates, its expected utility is: $(1-p)\times1 + p\times2 = 1 + p$
  - if it desists, its expected utility is: $(1-p)\times0 + p\times4 = 4p$

Result: Let p be each country's prior belief that rival is a **Dove**
- If $p < 1/3$, both countries will escalate irrespective of their 'types'
- If $p > 1/3$, there'll be 2 ***equilibria***: in one eqm., each country will escalate for sure; in the other eqm., each country will escalate if and only if it is a **Hawk**

Equilibrium Notion in Sim-move Games with Incomplete Information

- Game description includes specification of different possible "***types***" of a player (capturing her private information), and each player's common knowledge prior beliefs about rival type
- A player's type (information) certainly affects his own payoffs; it can also directly affect rivals' payoffs ['independent-values games' *vs.* 'interdependent-values games']
- In a ***simultaneous-move game of incomplete information***, a ***Bayes-Nash equilibrium*** [***BNE***] is a vector of strategies, one for **each player of each type** such that **each player of each type best-responds** to the vector of ***expected*** ***rival strategies***
- ***Expected rival strategy*** is the ***mixed strategy*** generated by the strategies chosen deterministically by the ***different types*** of a rival
- In specific simultaneous-move games with incomplete information, there can be a unique ***BNE*** that is "dominance-solved"

The Four Dominant Equilibrium Concepts in Game Theory

	Simultaneous-move	*Sequential-move*
Complete Info	**Nash Eqm**	**Subgame-perfect Nash Eqm**
Incomplete Info	**Bayes-Nash Eqm**	**Perfect Bayesian Nash Eqm**

Arms Race with diffuse priors

		Country B: escalate	Country B: desist
Country A	escalate	1 , 1	vA , 0
	desist	0 , vB	4 , 4

- For each country I, vI is any number between [2, 5]
- each country I knows vI and (following Laplace's logic of insufficient reason) holds 'uniform prior' on rival vJ & this is common knowledge

- ***Iterated deletion of strictly dominated strategies*** under asymmetric info:
  for own $v > 4$, each country's dominant strategy: *escalate*
  $\Rightarrow$ for own $v > 3.5$, each country's dominance-solved strategy: *escalate*
  $\Rightarrow$ for own $v > 3$, each country's dominance-solved strategy: *escalate*
  $\Rightarrow$ for own $v > 2$, each country's dominance-solved strategy: *escalate*
- ***Dominance-solved Bayes-Nash equilibrium***: both countries will *escalate* irrespective of its 'type'
  [***a negative "strategy contagion effect"***]

The strategy contagion argument in details…

- If country $\boldsymbol{i}$ is a Hawk ($\boldsymbol{v_i > 4}$), dominant strategy to escalate
- When it is a Dove ($\boldsymbol{v_i \leq 4}$), if it holds optimistic view that rival Dove will desist, it has to respond to **{*escalate*, *desist*; 1/3, 2/3}** and so will desist if and only if $\boldsymbol{v_i < 3.5}$ [if escalates expected utility is $\boldsymbol{\{(1/3)\times 1 + (2/3)\times v_i\}}$, and if desists expected utility is $\boldsymbol{\{(2/3)\times 4\}}$]
- If country $\boldsymbol{i}$ has 'first-order knowledge' about rival rationality, it will realize that rival country $\boldsymbol{j}$ will escalate whenever $\boldsymbol{v_j > 3.5}$
- So the most optimistic view that country $\boldsymbol{i}$ can hold is that rival $\boldsymbol{j}$ will desist if and only if $\boldsymbol{v_j \leq 3.5}$; given that, i has to respond to **{*escalate*, *desist*; 1/2, 1/2}**, and so will desist if and only if $\boldsymbol{v_i < 3}$
- 'Second-order knowledge' about rationality implies that most optimistic view that country $\boldsymbol{i}$ can hold is that rival $\boldsymbol{j}$ will desist iff $\boldsymbol{v_j \leq 3}$; given that, i has to respond to **{*escalate*, *desist*; 2/3, 1/3}**, and so will strictly prefer to desist if and only if $\boldsymbol{v_i < 2}$

Lessons from the Baliga-Sjostrom study…

- Even in an incomplete information games, the unique Bayes Nash equilibrium might be dominance-solved, and thus be uniquely rationalizable
- Such an equilibrium might represent a cooperation dilemma among the players, whose 'depth' will depend on the players' initial beliefs about rival 'types'
- One or more players might exhibit "strategy contagion" in that all (or many) player types might play an equilibrium strategy which is the *ex ante* dominant strategy only for a small set of player types
- Strategy-contagion might or might not be symmetric across symmetric players, and might be partial or complete for a player

Answer Key to Problem Set 9

***Problem* 1:** Subgame-perfect Nash equilibrium I

Each firm plays the strategy: "I will adopt the new technology in period ***t*** if and only if rival has adopted in period ***t*–1**".

Eqm outcome: Neither will ever adopt new technology (Pareto-optimal outcome for the firms).

Subgame-perfect Nash equilibrium II

Lambda's strategy: "Before 7-2016, I will adopt in period ***t*** if and only if rival has adopted in period ***t*–1**. From 7-2016 onwards, I will adopt in every period that I have a chance to adopt if I have not adopted before".

Gamma's strategy: "Before 1-2017, I will adopt in period ***t*** if and only if rival has adopted in period ***t*–1**. From 1-2017 onwards, I will adopt in every period that I have a chance to adopt if I have not adopted before".

Equilibrium outcome: Lambda will adopt new technology in 7-2016, and Gamma will follow suit in 1-2017.

Explanation: In the above game, technology adoption is "imitation inducing". So, if one player is assured that the other will adopt if and only if the former adopts, there exists a Pareto-superior equilibrium in which each player plays "after you". However, the following statements are also true: When the cost of adoption is relatively high (before 7-2016) while one will want to imitate rival adoption, one would not want to preempt it. But when the cost of adoption is relatively low (after 7-2016) one will want to preempt rival adoption if possible.

[Verify the following: Suppose that today is 1/16, and I know that rival will adopt in 7/16. Then if I "imitate" him in 1/17, my p.v.profit (calculated today) is: $10 + \delta 2 + \delta^2.(75 - 42) = 32.72$. But if I "preempt" him in 1/16, my p.v.profit (calculated today) is: $20 - 48 + \delta.75 = 32$. So I will not preempt my rival, but imitate him after he adopts at 7/16. And he will adopt at 7/15 because he thinks I will adopt at 1/16.]

***Problem* 2**:

Complete contingent *SPNE* strategy of larger Firm *X*: (i) in every week $t < t^{D}$, I will stay in the industry irrespective of whether firm *Y* is in the industry or has exited, (ii) in every week *t* such that $t^{D} \leq t < t^{M}(X)$, I will exit the industry if and only if firm *Y* is still in the industry, and (iii) in every week $t \geq t^{M}(X)$, I will exit the industry irrespective of whether firm *Y* is in the industry or has exited.

Complete contingent *SPNE* strategy of smaller Firm *Y*: (i) in every week $t < t^{M}(Y)$, I will stay in the industry irrespective of whether firm *X* is in the industry or has exited, and (ii) in every week $t \geq t^{M}(Y)$, I will exit the industry irrespective of whether firm *X* is in the industry or has exited.

[*Convince yourself that the above are indeed the unique* "*complete contingent SPNE strategies*" *of the two firms*; *and think about how firm Y can rationalize 'not exiting' in a week t between t^D and $t^M(X)$ if firm X is still in the industry at that t*.]

***Problem* 3:** The "largest" firm will exit first when the "triopoly" profits go to zero. The "middle" firm will exit next when the duopoly profits (in a duopoly between the "middle" and the "smallest" firms) go to zero. And the "smallest" firm will exit last when its monopoly profits go to zero.

Arms Race with diffuse priors

		Country B: escalate	Country B: desist
Country A	escalate	1 , 1	vA , 0
	desist	0 , vB	4 , 4

- For each country I, vI is any number between [2, 5]
- each country I knows vI and (following Laplace's logic of insufficient reason) holds 'uniform prior' on rival vJ & this is common knowledge
- ***Iterated deletion of strictly dominated strategies*** under asymmetric info:

for own $v > 4$, each country's dominant strategy: *escalate*

$\Rightarrow$ for own $v > 3.5$, each country's dominance-solved strategy: *escalate*

$\Rightarrow$ for own $v > 3$, each country's dominance-solved strategy: *escalate*

$\Rightarrow$ for own $v > 2$, each country's dominance-solved strategy: *escalate*

- ***Dominance-solved Bayes-Nash equilibrium***: both countries will *escalate* irrespective of its 'type'
[***a negative "strategy contagion effect"***]

Logic of "Strategy Contagion" in Sim-move Games of Incomplete Information

- A game between two players **A** and **B**
- The 'type' of each is continuously distributed on line-interval [m, n]
- Each player has two pure strategies **E** and **D** to chose from
- **E** is a player's dominant strategy if his type is above a critical value $v^* \in (m, n)$
- The probability that a player's type is above v is a number $p^* \in (0, 1)$
- *Necessary condition for **mutual strategy contagion** to occur*:
Each player's desire to play **E** rises in the probability that the other will play **E**
- *Sufficient condition for **mutual strategy contagion** to occur*:
If a player **I** conjectures that rival player will play **E** with at least probability p^*, then player **I**'s best response will be **E** as long as his realized type is no less than $v^* - \varepsilon$ for some $\varepsilon > 0$.
- "Extent" of ***mutual strategy contagion*** depends upon all parameters of the game

The Hawk Dove Game

		Beta: act hawkish = aggressive	Beta: dovish = passive
Alpha	act hawkish	$10 - 0.5C\alpha$, $10 - 0.5C\beta$	20 , 0
	act dovish	0 , 20	12 , 12

- Each player i ($i = \alpha$ or β) knows his/her "cost of fighting" Ci, which is either **30** (for '**Weak**') or **10** (for '**Tough**').
- Each player believes that rival is **Weak** (with $Cj = 30$) with probability p, and **Tough** (with $Cj = 10$) with probability $(1 - p)$
- If the animals' costs of fighting was common knowledge, then the pure-strategy Nash eqm outcomes in complete-info game will be:
  - In a game between two Toughies, both would be *hawkish* (dominant strategy)
  - In a game between a Toughie and a Weakling, the Toughie would be *hawkish* and the Weakling would be *dovish*.
  - In a game between two Weaklings, one would be *hawkish* and other *dovish*.

The Hawk Dove Game

		Beta: act hawkish = aggressive	Beta: dovish = passive
Alpha	act hawkish	$10 - 0.5C\alpha$, $10 - 0.5C\beta$	20 , 0
	act dovish	0 , 20	12 , 12

- Each player i ($i = \alpha$ or β) knows his/her "cost of fighting" Ci, which is either **30** (for '**Weak**') or **10** (for '**Tough**').
- Each player believes that rival is **Weak** (with $Cj = 30$) with probability p, and **Tough** (with $Cj = 10$) with probability $(1 - p)$

In the simultaneous-move incomplete-info game:

- **Tough** has a dominant strategy to be ***hawkish (aggressive)***
- If a **Weak** conjectures that 'rival **Weak**' will act ***hawkish***, (s)he has to respond to the pure strategy ***hawkish***, and so will act ***dovish (passive)***
- If a **Weak** conjectures that rival **Weak** will act ***dovish***, (s)he will have to best-respond to the ***mixed strategy*** {***dovish***, ***hawkish***; p, $1-p$}

and so, she will strictly prefer to be ***dovish*** if and only if $p < 5/13$

The Hawk Dove Game

		Beta: act hawkish = aggressive	Beta: dovish = passive
Alpha	act hawkish	$10 - 0.5C\alpha$, $10 - 0.5C\beta$	20 , 0
	act dovish	0 , 20	12 , 12

- Each player i ($i = \alpha$ or β) knows his/her "cost of fighting" Ci, which is either **30** (for '**Weak**') or **10** (for '**Tough**').
- Each player believes that rival is **Weak** (with $Cj = 30$) with probability p, and **Tough** (with $Cj = 10$) with probability $(1 - p)$

Bayes-Nash equilibria:

A. If $p < 5/13$, then in the unique *dominance-solved* ***BNE*** : each animal is ***hawkish*** if (s)he is a Toughie and ***dovish*** if (s)he is a Weakling.

B. If $p > 5/13$, then there are 2 pure-strategy ***BNE*** which are *asymmetric*:
In each, one animal is ***hawkish*** irrespective of his/her 'type', and the other animal is ***hawkish*** if (s)he is **Tough** and ***dovish*** if (s)he is **Weak**

** If each animal's cost of fighting is equally likely to be any no. betwn 10 & 30, the only pure-strategy **BNE** will be B. $\Rightarrow$ ***no 'mutual' strategy contagion***

Note that in the complete-info game between two Weaklings, the "fair" outcome would be one where both the animals are *dovish*. But that cannot be a pure-strategy Nash equilibrium outcome under complete information.

But under asymmetric info, when each animal is "sufficiently skeptical" that its rival is a Toughie, then each animal – if it happens to be a Weakling – will be *dovish*; and consequently, when two Weaklings face each other the "fair" outcome mentioned above will indeed arise as the equilibrium outcome.

In the asymmetric-info game, the necessary condition for "mutual strategy contagion" is not satisfied.

The Four Dominant Equilibrium Concepts in Game Theory

	Simultaneous-move	*Sequential-move*
Complete Info	Nash Eqm	Subgame-perfect Nash Eqm
Incomplete Info	Bayes-Nash Eqm	Perfect Bayesian Nash Eqm

Dynamic Games of Incomplete Information

We will study four dynamic scenarios where players are asymmetrically informed about each others' preferences/payoffs

1. Arms Race with two countries taking turns in nuclear stockpiling
2. Costly migration from a worse to a better technology
3. Reputation Effects in a Centipede Game
4. Costly Signalling of Human Capital

In these games, relevant eqm concept is ***Perfect Bayesian Nash Eqm***

Strategy vectors and beliefs must satisfy -

(1) ***Sequential optimality***: at every subgame, continuation strategies constitute Bayes-Nash equilibria given appropriately updated beliefs

(2) ***Belief consistency***: Updated beliefs about rival 'types' derived using Bayes' rule (where ever possible), given players' past strategies

1 & 2 are independent-values games; **3 & 4** are interdependent-values games

In the former, a player's "updated" beliefs about her rival's 'type' does not influence her optimal action choice; in the latter, a player's "updated" beliefs about her rival's 'type' ***can*** influence her optimal action choice

Arms Race w/ sequential moves

Country A \ Country B	*escalate*	*desist*
escalate	1 , 1	*vA* , 0
desist	0 , *vB*	4 , 4

- First *A* chooses its strategy irreversibly, then *B* chooses its strategy irreversibly after observing *A*'s move
- This is a *multi-stage game of incomplete information*, and the relevant eqm concept is ***Perfect Bayesian Nash Equilibrium***

Arms Race w/ sequential moves

Country A \ Country B	*escalate*	*desist*
escalate	1 , 1	*vA* , 0
desist	0 , *vB*	4 , 4

- ***vA*** and ***vB*** are distributed over [2, 5] acc. to some distribution function
- First *A* chooses 'escalate' or 'desist'; then *B* chooses 'escalate' or 'desist'
- ***B*** will play *desist* if and only if "***vB*** < 4 and *A* has played *desist*"
- Suppose ***A***'s *ex ante* belief that ***vB*** < 4 is π
- Given belief, if ***A*** *escalates* exp. utility is 1, if it *desists* exp. utility is 4 π and so ***A*** will *desist* iff π > 1/4 (irrespective of ***A***'s own type)
- Unique ***PBNE*** outcome for π < ¼: ***A*** *escalates*, then ***B*** *escalates*
  Unique ***PBNE*** outcome for π > ¼: ***A*** *desists*, then ***B*** *escalates* if ***vB*** > 4 and ***B*** *desists* if ***vB*** < 4
- Sequential move likely to lessen intensity of cooperation dilemma, but belief pessimism is still key to cooperation failure

Ranked Non-conflicting Coordination

The Keyboard Puzzle

Typist A \ Typist B	Dvorak	QWERTY
Dvorak	best, best	bad, bad
QWERTY	bad, bad	avg, avg

- Multiple ***Pareto-ranked*** coordinated Nash equilibria
- What happens when there are many players 'sitting' with the 'average choice' and individually considering transition to the 'better choice' without precisely knowing others' transition costs?

  [*Cliometricians ask*: Can this be the reason why historically, 'better technologies' have not always been adopted?]

The "Migrating Across Technology Standards" Game

Migration from 'old' technology to 'better' one, where both have 'network effects', and migration costs are private information:

Firm A \ Firm B	*migrate*	*stay*
migrate	$19 - c_A$, $19 - c_B$	$13 - c_A$, 6
stay	6, $13 - c_B$	7.5, 7.5

c = cost of migration

- If *c* was commonly known to be '10' for both firms, it would be a ***ranked coordination game*** with two Pareto-ranked Nash equilibria
- Suppose for each firm *I*, c_I can be any number between [4, 16]
- For $c_I < 5.5$: strictly dominant strategy to *migrate*,
  for $c_I > 13$: strictly dominant strategy to *stay*
  for $c_I = 10$: best response to *migrate* is *migrate* and to *stay* is *stay*
- Each firm *I* knows c_I and vide Laplace's logic of insufficient reason, holds ***uniform prior*** for rival c_J (and this is common knowledge)

"Migrating Across Technology Standards" Game

		Firm B: *migrate*	Firm B: *stay*
Firm A	***migrate***	$19 - c_A$, $19 - c_B$	$13 - c_A$, 6
	stay	6, $13 - c_B$	7.5, 7.5

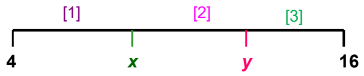
- *Question*: If the two firms can irreversibly migrate in any time period $t = 1, \ldots, T$ what will each firm do?
- In this *dynamic game of incomplete information*, each firm has 'essentially' three strategies to choose from:
  [1] *Me first*: Migrate in period $t = 1$
  [2] *After you*: Migrate in t if and only if rival migrated in $t - 1$
  [3] *Never*: Do not migrate in any time period $t = 1, \ldots, T$

  [An *ex ante* strategy "Migrate after τ periods if no migration by either" is dominated by [1]]

"Migrating Across Technology Standards" Game

	migrate	*stay*
migrate	$19 - c_A$, $19 - c_B$	$13 - c_A$, 6
stay	6, $13 - c_B$	7.5, 7.5

- Consider range of costs for which a firm will play each strategy:
  [1] *Me first*: Migrate in period $t = 1$
  [2] *After you*: Migrate in t if and only if rival migrated in $t - 1$
  [3] *Never*: Do not migrate in any time period $t = 1, 2, \ldots, T$



- y given by indifference between 'after you' & 'never' $\Rightarrow y = 13$
- x given by indifference between 'me first' & 'after you'

  $$[(13-4)/12]\times(19-x) + [(16-13)/12]\times(13-x) = [(x-4)/12]\times(19-x) + [(16-x)/12]\times(7.5) \Rightarrow x \approx 9$$

Unique Perfect Bayesian Nash Equilibrium

	migrate	*stay*
migrate	$19 - c_A$, $19 - c_B$	$13 - c_A$, 6
stay	6, $13 - c_B$	7.5, 7.5

Strategy of each firm I: if $c_I \leq 9$ play "me first"; if $9 < c_I \leq 13$ play "after you", if $c_I > 13$ play "never"

Updated beliefs of each firm I: if Firm J has played "me first" then believe $c_J \in [4, 9]$; if Firm J has played "after you" then believe $c_J \in [9, 13]$; and if Firm J has played "never then believe $c_J \in [13, 16]$

Unique Perfect Bayesian Nash Equilibrium outcome:
With probability $(5/12)^2$ both firms migrate in pd 1; with probability 2(5/12)(4/12) one firm migrates in pd 1 and the other in pd 2; with probability 2(5/12)(3/12) one firm migrates in pd 1 and the other does not migrate; with probability $(7/12)^2$ neither firm migrates.

Bandwagon Strategies: *Excess Inertia & Excess Momentum*

	migrate	*stay*
migrate	$19 - c_A$, $19 - c_B$	$13 - c_A$, 6
stay	6, $13 - c_B$	7.5, 7.5

- ***In the unique Perfect Bayesian Nash Equilibrium***:
  Each firm I plays "me first" if $c_I \leq 9$;
  plays "after you" if $9 < c_I \leq 13$; and plays "never" if $c_I > 13$
- ***Equilibrium Bandwagon Strategy***: A "low migration cost" firm is willing to start a migration bandwagon by preemptively migrating; an "intermediate migration cost" firm is willing to ***join*** a migration bandwagon, but is not willing to ***start*** one
- ***Excess Inertia property***: When both firms have intermediate migration costs in (9, 11.5), both wait for a bandwagon to start, and that is 'Pareto-inferior coordination' for both
- ***Excess Momentum property***: When a firm has migration costs in (11.5, 13), it may be "dragged into migration" (with resultant payoff < 7.5) by a "low migration cost" firm

Problem Set 10

***Problem* 1:** Reconsider the *Arms Race Game with Diffuse Priors* presented in Chapter 9.3 of the Lecture Notes (page 86). Suppose that each country $\boldsymbol{i}$ knows the value of its own $\boldsymbol{v_i}$ in the interval [1, 5] and assesses that rival $\boldsymbol{v_j}$ is drawn from a uniform distribution over the interval [1, 5]. Determine that pure-strategy symmetric Bayes-Nash equilibrium of this game in which the probability that each country *escalates* is the least.

***Problem* 2:** Study the *Hawk-Dove Game under Asymmetric Information* presented in Chapter 9.4 of the Lecture Notes (pages 87-88). Prove the statements made in the last paragraph of the section (i.e., the last paragraph in page 88).

Player 1 \ Player 2	I	N
I	−1, −1	1, 0
N	0, 1	0, 0

Figure 11.9
A game of "grab the dollar."

struction, let us consider the one-period version of the "grab the dollar" game represented in figure 11.9. Each player has two possible actions: investment and no investment. In the complete-information version of the game, a firm gains 1 (i.e., wins) if it is the only one to make the investment, loses 1 if both firms invest, and breaks even if it does not invest. (We can view this game as an extremely crude representation of a natural monopoly market.) The only symmetric equilibrium involves mixed strategies: Each firm invests with probability $\frac{1}{2}$. This clearly is an equilibrium: Each firm makes 0 if it does not invest and makes $\frac{1}{2}(1) + \frac{1}{2}(-1) = 0$ if it does invest. Now consider the same game with the following incomplete information: Each firm has the same payoff structure, except that when it wins it gets $(1 + t)$, where t is uniformly distributed on $[-\varepsilon, +\varepsilon]$. Each firm knows its type t, but not that of the other firm. Now, it is easily seen that the symmetric *pure* strategies, "$a(t < 0)$ = do not invest, $a(t \geq 0)$ = invest," form a Bayesian equilibrium. From the point of view of each firm, the other firm invests with probability $\frac{1}{2}$. Thus, the firm should invest if and only if $\frac{1}{2}(1 + t) + \frac{1}{2}(-1) \geq 0$, i.e., $t \geq 0$. Because for a given type a player has a unique best action, there is no problem with the player resisting the prescription to randomize with a given probability ($\frac{1}{2}$ before). When ε converges to zero, the pure-strategy Bayesian equilibrium converges to the mixed-strategy Nash equilibrium of the complete-information game.

Because games of complete information are an idealization (in practice, everyone has at least a slight amount of incomplete information about the other players' objectives), Harsanyi's argument shows that it is hard to make a strong case against mixed-strategy equilibria on the ground that they require a randomizing device.[19]

11.5 Perfect Bayesian Equilibrium

We now study dynamic games of incomplete (or imperfect) information. The tricky feature in these games is that a player who reacts to another player's move can extract information from that move. It is natural to suppose that the inference process takes the form of Bayesian updating from the latter player's supposed equilibrium strategy and his observed action. The equilibrium notion is a combination of the (subgame) perfect-equilibrium concept for dynamic games and the Bayesian-equilibrium concept for games of incomplete information. This section introduces the simplest such notion: perfect Bayesian equilibrium (PBE)[20] and gives four simple applications of the concept.

To motivate the notion of equilibrium, we begin with the simple game of imperfect information depicted in figure 11.10. This game has three players and takes place over three "periods." In period 1, player 1 can choose from among three actions: "Left" (L_1), "Middle" (M_1), and "Right" (R_1). If player 1 chooses one of the latter two, player 2 gets to choose between "Left" (L_2) and "Right" (R_2), although he is not informed of player 1's exact choice (he knows only that player 1 did not choose L_1). The imperfect information of player 2 is represented by an information set $\{M_1, R_1\}$, characterized by an oval around the two corresponding nodes (n_2 and n_3). Given his state of information, player 2 is faced with the same choices at nodes n_2 and n_3. Finally, for move $\{M_1, R_2\}$ or $\{R_1, L_2\}$ player 3 must choose between "Left" (L_3) and "Right" (R_3) in the third period without knowing which of the nodes (n_4 or n_5) the game has attained. The values of the objective functions are written at the bottom of the tree. For example, for moves (M_1, L_2, R_3) player 1 receives 3, player 2 receives 2, and player 3 receives 0.

The following parable should prove helpful in understanding the equilibrium concept. Suppose that an econo-

19. See Milgrom and Weber 1986 for sufficient conditions on the objective functions and information structure so that the limit of Bayesian equilibrium strategies when the uncertainty becomes "negligible" forms a Nash equilibrium of the complete-information limit game.

20. The refinement of this notion is discussed in section 11.6.

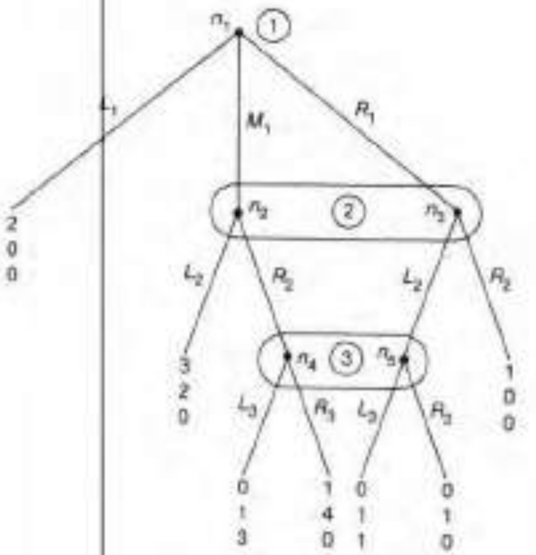


Figure 11.10
Game 5.

mist has to solve game 5 and, not knowing what to do, he turns to two people who have very specific talents.

The first consultant is a game theorist. He is well versed in the method of solving perfect equilibria of dynamic games, as described in section 11.3. On the other hand, he is sufficiently familiar with decision theory to understand the concept of expected utility (payoff). However, he is not aware of Bayes' law for calculating posterior probability distributions. Glancing at the problem described in figure 11.10, he first tries to solve player 3's decision problem. Unfortunately, he realizes that he can formulate this problem as a classical decision problem only so long as player 3 attributes some subjective probability (say, μ_3) to the game being at node n_4 (so the conditional probability of n_5 is $1 - \mu_3$). Similarly, player 2's decision problem depends on the probability (μ_2) that he attributes to the game being at node n_2. Then the game theorist behaves in a scientific manner. Admitting his ignorance of the probabilities attached to each informational set (here, $\mu = (\mu_2, \mu_3)$), he makes the following remark: "If I am given subjective probabilities μ, the game is identical to a perfect-information game, which I can solve." In fact, given μ_3, player 3 chooses the action that maximizes his expected payoff. Player 2 knows the optimal strategy of player 3, and he also maximizes his expected utility given the subjective probability μ_2. Finally, given the other players' optimal strategies, player 1 chooses according to his own self-interest. Therefore, the game theorist reports to the economist the correspondence between the subjective probabilities μ and the optimal strategies obtained by backward induction:

$$a^*(\mu) = \{a_1^*(\mu), a_2^*(\mu), a_3^*(\mu)\},$$

where a_i^* can be a mixed strategy.

The second consultant is a Bayesian statistician. Calculating posterior probability distributions is second nature to him; however, he is familiar with neither game theory nor decision theory. Possessing a scientific mind, he makes the following remark: "If I am given the set of strategies $a = \{a_1, a_2, a_3\}$, I can calculate the probabilities the players should attribute to the various nodes." For example, if player 1 plays each of his actions with probability $\frac{1}{3}$, player 2 must assign a probability of $\mu_2 = \frac{1}{2}$ to node n_2 if the game attains his informational set. Moreover, if $a_2 = R_2$, player 3 assigns probability $\mu_3 = 1$ to node n_4. What happens if player 1 plays $a_1 = L_1$ (with probability 1)? Then player 2 can attribute any probability μ_2 to node n_2, since every probability is compatible with Bayes' law (because the event "the game attains player 2's informational set" has zero probability).[21] The statistician reports for each set of strategies the set of beliefs that are compatible with these strategies in a Bayesian sense: $\mu^{\text{Bay}}(a)$.

The conclusion to this parable is easy to guess. The economist calls in his two consultants. The statistician proposes that the strategies he uses to calculate the probability be the strategies actually chosen by the players, which the game theorist should give him. In turn, the game theorist suggests that the players' subjective probabilities be based on a study of the other players' behaviors, and calls on the statistician to provide him with these probabilities. Then both agree that the notion of equilibrium should make explicit these two types of compatibility. Consequently, they define a *perfect Bayesian equilibrium* as a set of strategies a that satisfy

21. However, we can, following Kreps and Wilson 1982, require that μ_3 be consistent with μ_2. For example, if $\mu_2 = \frac{1}{2}$ and player 2 plays L_2 with probability $\frac{1}{3}$, then $\mu_3 = \frac{2}{3}$ (see below).

$$a \in a^*(\mu^{Bay}(a)), \tag{11.1}$$

and they associate with these strategies a system of beliefs that sustain the equilibrium by satisfying

$$\mu \in \mu^{Bay}(a^*(\mu)). \tag{11.2}$$

Thus, the optimal strategies (and the associated beliefs) satisfy a fixed-point condition:

(P) Strategies are optimal given beliefs.
(B) Beliefs are obtained from strategies and observed actions using Bayes' rule.

The concept of perfect Bayesian equilibrium was introduced into the formal game-theory literature by Selten (1975) as "perfect equilibrium" and by Kreps and Wilson (1982) as "sequential equilibrium."[22] Although Selten pioneered this concept, Kreps and Wilson must be credited for putting more emphasis on beliefs,[23] making the concept more applicable, and paving the way for refinements based on restrictions on beliefs for zero-probability events (see section 11.6 for an example).

Remark We have been very informal about what Bayesian updating means. At the very least, the information sets that are reached with positive probability on the equilibrium path should be assigned beliefs consistent with Bayes' rule. This yields a very weak definition of PBE, which coincides with sequential equilibrium only in some simple games (e.g., the signaling game of section 11.6). In more complex games, further restrictions on the consistency of beliefs off the equilibrium path must be added to produce a stronger and more reasonable version of PBE. The consistency requirement of Kreps and Wilson (see subsection 11.6.1) is one such restriction.

How do we calculate the perfect Bayesian equilibrium (or equilibria) of a game? The characterization of a PBE as a fixed point of a correspondence suggests a method of calculation. However, calculating a fixed point is extremely tedious. Quite often, intuition allows us to solve the problem directly if the definition is well understood. There exists no general method; however, a few systematic tricks are of use in solving these games. The game depicted in figure 11.10 is actually trivial. Player 3 has a dominant strategy that consists of playing L_3; therefore, regardless of μ_3, $a_3 = L_3$. Hence, we can convert the game to the two-period, two-player game depicted in figure 11.11. (Since player 3 does not play in this game, we omit the values of his objective function.)

We observe that player 2 now has a dominant strategy: L_2. Therefore, player 1's optimal action is M_1. Consequently, the unique PBE is given by $a = (M_1, L_2, L_3)$, and it has associated to it a set of beliefs $(\mu_2 = 1, \mu_3 \in [0, 1])$.

This game is indeed trivial. Because of the existence of dominating strategies, the game theorist finds the equilibrium path without the help of the statistician.

We now solve four simple but nontrivial games that are relevant to industrial organization. In these games (in contrast with the previous game) there is interaction between backward induction and forward Bayesian inference. The first two are games of incomplete information in which an informed party may reveal information when choosing between two actions. In the third game, a player's private information can be revealed through a complex (two-dimensional) action. There, simple economic reasoning indicates what the economic path must be. The fourth game gives an example of a game of imperfect information.

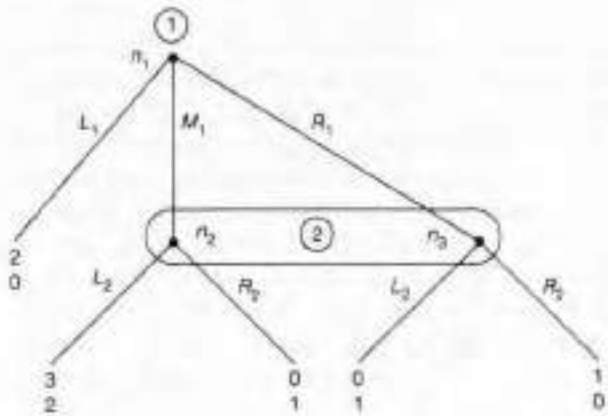


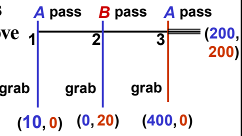
Figure 11.11
Game 6.

22. Kreps and Wilson have shown that perfect and sequential equilibria coincide for almost all games.

23. Selten defined his concept of "trembling hand" perfect equilibrium on the normal form. Beliefs, although easy to compute, are implicit rather than explicit.

The "Grab or Pass" Game under incomplete information

- **B** 'normal': payoff = number of coins received
- **A** 'normal' with 89% probability: payoff = number of coins received
- **A** 'saintly' with 11% probability: *always passes*
- **It's an *interdependent-values game* since B cares directly about A's type to predict A's future move**

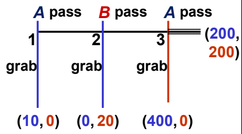


- At Stage 3: A(N) will grab & A(S) will pass
- At Stage 2, **B**'s decision must depend on his ***updated belief*** about **A**'s type given past play
- At Stage 2: **If B**'s updated belief is that **A** is saintly with probability **less than 10%** / **greater than 10%** / **equal to 10%** then **B** will **grab** / **pass** / **be indifferent**
- **B**'s updated belief must obey ***belief consistency*** - which requires that his belief-updating must follow **Bayes Rule given A's stage 1 strategy**

If **B** believes that normal **A** passes with prob $\pi \in [0, 1]$, then **B**'s updated belief that **A** is **saintly** after observing 'pass' in Stage 1 should be: **[0.11] / [0.11 + 0.89π]**

The "Grab or Pass" Game

- In Stage 1: If **A** plays deterministically, she can do one of two things
  - she can "***separate***": play *grab* if normal, and *pass* if saintly
  - or she can "***pool***": play *pass* irrespective of own type



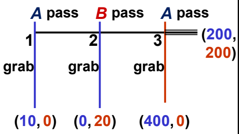
Definition: In an incomplete-information game, a player is said to play
- a "***pooling***" strategy at a decision node if her strategy is "the same independent of" her private information
- a "***separating***" strategy at a decision node if her strategy is "different depending on" her private information

- In a ***PBNE*** of the *Grab-or-Pass Game*, **B**'s Stage 2 updated belief must be Bayes-consistent with **A**'s Stage 1 strategy, and **B**'s Stage 2 strategy must be sequentially optimal given **B**'s Stage 2 updated belief and **A**'s backward-inducted Stage 3 strategy
- To check for the above, let's appeal to "self-enforcing pre-play reco"

The "Grab or Pass" Game

Q1: Given the previous conclusions, can an impartial consultant give a public self-enforcing recommendation in which she asks **A** "to separate" in Stage 1 (i.e., asks **A**(Normal) "to grab" in Stage 1)?

Ans: NO. If she does so, she must also ask **B** to pass in Stage 2, because **B**'s updated belief if Stage 2 is reached will be that A *saintly* with prob 1. But if **B** passes in Stage 2, then **A**(Normal) will strictly prefer to pass in Stage 1.

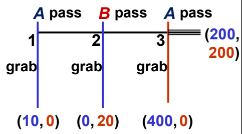


Q2: Can the consultant ask **A** "to pool" in Stage 1 (i.e., ask **A**(Normal) "to pass" in Stage 1)?

Ans: YES. She will also ask **B** to pass in Stg 2. **B** will pass since **B**'s updated belief if Stag 2 is reached will be that **A** is saintly with prob 11%. Finally, she will ask **A**(Normal) to grab in Stg 3, which she will do.

Unique (pure-strategy) Perfect Bayesian Nash Equilibrium given B's prior belief that A is saintly with 11% probability:

Strategies:
- Normal **A** will pass in Stage 1, and grab in Stage 3
- Saintly **A** will always pass
- **B** will pass in Stage 2



B's beliefs:
- If **B** sees **A** pass in stage 1, he will believe she's saintly with prob 0.11
- **If B sees A grab in Stage 1, he will believe she's saintly with prob *p* for any $p \in [0, 1]$**
  {here all belief are Bayes-consistent & don't matter as grabbing ends game}

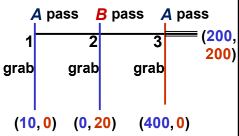
- In the **unique *PBNE***, **A** "maintains" her saintly reputation till Stage 3

Note: According to eqm strategies, **A** grabbing at 1 is a zero probability event, and so Bayes' Rule imposes no belief restriction at Node 2; but rational thinking should indicate that a 'grabbing' **A** cannot be saintly, and so "inferring" ***p*** = 0 is "intuitive"

Experiment Four : *"Grab or Pass" Game*

Beta 'normal': payoff = number of coins received
Alpha 'normal' with 89% probability: payoff = no. of coins received
Alpha 'saintly' with 11% probability: *always passes*

Unique PBNE strategies:
- **Normal A will pass at stg 1, grab at stg 3**
- **B will pass at stage 2**



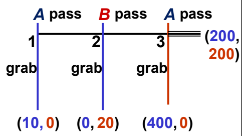
Normal Alpha (8)	"grab in 1"	"pass in 1, grab in 3"
	4	18
	35%	65%

BETA (7)	"grab in 2"	"pass in 2"
	1	17
	5%	95%

The "Grab or Pass" game with a worse initial reputation for A

Consider the case where prior probability that **A** is 'saintly' is 5%

- In this case, if and only if **A** can ***enhance*** reputation at node 2 to the critical level **10%** will **B** pass
- There cannot exist a ***PBNE*** in which **A** either deterministically 'pools' or deterministically 'separates' in Stage 1



Unique mixed-strategy ***PBNE*** is as described below:
- normal **A** grabs with prob 10/19 and passes with prob 9/19 in Stage 1
- so **B**'s inference re: saint is (1/20)/[(1/20) + (19/20)(9/19)] = 10%
- indifferent **B** grabs with prob. 39/40 and passes with prob. 1/40
- normal **A** grabs in Stage 3
- this ensures that so that **A**'s expected payoff in Stg 1 is (1/40).400 =10

- In this ***fragile*** eqm, **A** enhances reputation at node 2 to the critical level (if node 2 is reached), but gains nothing by doing so

Signalling Games

- The Spence Game is the first example of a ***signalling game*** (a "sender-receiver game") in the game theory literature (1972)

  - where a privately informed 'sender' aims to transmit "favourable private information" by sending a "costly signal"

  - in equilibrium, such a signal will be ***meaningful*** (as opposed to being ***cheap talk***) if and only if the sender would not have found it profitable to signal if the information was not 'favourable'

  - the signal itself **can** be "valueless" (a ***dissipative*** signal)

Michael Spence's Recruitment Game : Model 1 $c_{EX} = 5$

- There are *N* people who are qualified to work for Sleaze Company. Half of them are "**exceptional**", other half are "**average**" – exceptional will generate p.v. profits of Rs. **10** million, average will generate Rs. **4** million.
- Each job-applicant knows own type, HR Head of Sleaze does not.
- Each job-applicant can get MBA from Crazy Institute of Management (CIM).
- CIM charges Rs. **1** million, adds nothing to a person's ability, but admits students through a *Complex & Amazingly Tough Entrance Exam* (CATEE)
- In order to crack the CATEE: each exceptional applicant needs to spend **4** million on tutorials, each average applicant needs to spend **6** million.
- HR Head ***has to*** hire every applicant, and offer each p.v. compensation of either Rs. **4** million, or Rs. **7** million, or Rs. **10** million.
- If an applicant is **exceptional** [or **average**] & if (s)he is hired at ***X*** million, HR Head is paid Rs. $\mathbf{10{,}000 - 100.(10 - X)^2}$ [or Rs. $\mathbf{10{,}000 - 100.(4 - X)^2}$] once true ability of the applicant is discovered after (s)he starts working.
- Payment rule for HR Head implies that if, at time of hiring, (s)he believes that an applicant is exceptional with probability close to **one** / **half** / **zero**, (s)he should offer p.v. compensation of **10** / **7** / **4** million to the applicant.

A "common sense" approach to the Spence Game when $c_{EX} = 5$

- It is a strictly-dominated-strategy for an average applicant to get education certificate, as **4 > 10 – 7** $[\pi_{AV} > \pi_{EX} - c_{AV}]$
- So on seeing 'certificate', makes sense for Recruiter to believe that applicant is exceptional and offer $\{\pi_E\} = \mathbf{10}$
- Question is: what should Recruiter offer on seeing 'no certificate'?
  - Depends on fraction ***f*** of 'exceptionals' he expects to get certs
  - If ***f*** close to 1, Recruiter should offer $\{\pi_{AV}\} = \mathbf{4}$ to the 'cert-less'
  - If ***f*** close to 0, he should offer $\{½(\pi_{EX}+\pi_{AV})\} = \mathbf{7}$ to 'cert-less'
- Recognizing this, if an 'exceptional' expects ***f*** to be close to 1, she should get 'cert' as **10 – 5 > 4** $[\pi_{EX} - c_{EX} > \pi_{AV}]$ (taking ***f*** closer to 1)
  - and if 'exceptional' expects ***f*** to be close to 0, she should not get 'cert', as **7 > 10 – 5** $[½(\pi_{EX}+\pi_{AV}) > \pi_{EX} - c_{EX}]$ (taking ***f*** closer to 0)
- Common sense suggests two "self-sustaining" configurations:

1. exceptionals expected to get cert ⇒ cert-less get **4** ⇒ exceptionals get cert
2. nobody expected to get cert ⇒ cert-less get **7** ⇒ nobody gets cert

Michael Spence's Recruitment Game : Model 2 $c_{EX} = 2$

- There are *N* people who are qualified to work for Sleaze Company. Half of them are "**exceptional**", other half are "**average**" – exceptional will generate p.v. profits of Rs. **10** million, average will generate Rs. **4** million.
- Each job-applicant knows own type, HR Head of Sleaze does not.
- Each job-applicant can get MBA from Crazy Institute of Management (CIM).
- CIM charges Rs. **1** million, adds nothing to a person's ability, but admits students through a *Complex & Amazingly Tough Entrance Exam* (CATEE)
- In order to crack the CATEE: each exceptional applicant needs to spend **1** million on tutorials, each average applicant needs to spend **6** million.
- HR Head ***has to*** hire every applicant, and offer each p.v. compensation of either Rs. **4** million, or Rs. **7** million, or Rs. **10** million.
- If an applicant is **exceptional** [or **average**] & if (s)he is hired at ***X*** million, HR Head is paid Rs. $\mathbf{10{,}000 - 100.(10 - X)^2}$ [or Rs. $\mathbf{10{,}000 - 100.(4 - X)^2}$] once true ability of the applicant is discovered after (s)he starts working.
- Payment rule for HR Head implies that if, at time of hiring, (s)he believes that an applicant is exceptional with probability close to **one** / **half** / **zero**, (s)he should offer p.v. compensation of **10** / **7** / **4** million to the applicant.

A "common sense" approach to the Spence Game when $c_{EX} = 2$

- It is a strictly-dominated-strategy for an average applicant to get education certificate, as **4 > 10 – 7** $[\pi_{AV} > \pi_{EX} - c_{AV}]$
- So on seeing 'certificate', makes sense for Recruiter to believe that applicant is exceptional and offer $\{\pi_E\} = \mathbf{10}$
- Question is: what should Recruiter offer on seeing 'no certificate'?
  - The answer need not depend on fraction ***f*** of 'exceptionals' he expects to get certificates, for the following reason
  - Suppose the Recruiter expects ***f*** to be 0, and has thus decided to pay $\{½(\pi_{EX}+\pi_{AV})\} = \mathbf{7}$ to the 'cert-less'

    he should realize that if he unexpectedly sees an applicant with certificate and decides to optimistically view applicant as exceptional and offer 10

    the applicant will gain from the Recruiter's optimism **if and only if (s)he is indeed exceptional** [**10 – 2 > 7 > > 4 > 10 – 7**]

    so 'optimism is justified' when seeing a 'cert' while 'pessimism is not'
- Common sense implies a ***stable*** self-sustaining configuration in this case: Only exceptionals get cert and get paid **10**, while the cert-less get paid **4**

PBNE in the Spence Game [Model 1: $c_{EX} = 5$]

- Recruiter has to hire each applicant at p.v. compensation of {"inferred" profit-creation}

Applicant decisions: In deterministic play she can do one of two things

- she can "separate": 'get cert' if exceptional, and 'not' if average
- or she can "pool": 'not get cert' irrespective of own type

Q. 1: Can there be a self-enforcing recommendation for applicants to **separate**?

Answer: YES. If applicants separate & reveal their types by using certification signal, Recruiter will be "Bayes-correct" to offer **10** to 'certs' and **4** to 'cert-less' sustaining only exceptional's desire to signal since **10 – 5 > 4** $[\pi_{EX} - c_{EX} > \pi_{AV}]$ … while **4 > 10 – 7** $[\pi_{AV} > \pi_{EX} - c_{AV}]$

Unique Separating Equilibrium

Strategies: Each applicant will 'get cert' if exceptional, 'not' if avg
Recruiter offers 'certs' **10** (π_{EX}), and 'cert-less' **4** (π_{AV})

Recruiter belief ***p***: If sees 'cert' then $\boldsymbol{p}=\mathbf{1}$ (ex), if sees 'no cert' then $\boldsymbol{p}=\mathbf{0}$ (av)

PBNE in the Spence Game [Model 1: $c_{EX} = 5$]

- Recruiter has to hire each applicant at p.v. compensation of {"inferred" profit-creation}

Applicant decisions: In deterministic play she can do one of two things
- she can "separate": 'get cert' if exceptional, and 'not' if average
- or she can "pool": 'not get cert' irrespective of own type

Q. 2: Can there be a self-enforcing recommendation for applicants to **pool**?
Answer: YES. If applicants 'pool' and nobody gets a 'cert', Recruiter will be "Bayes-correct" to offer **7** to every applicant

Pooling Equilibrium I
Strategies: Each applicant will 'not get cert'
Recruiter offers 'cert-less' **7** (π_{AV}) and 'certs' **10** (π_{EX}),
*Recruiter belief **p***: If sees 'no cert' then $p = ½$ & if sees 'cert' then $p = 1$

Pooling Equilibrium II
Strategies: Each applicant will 'not get cert'
Recruiter offers **7** ($½\pi_{EX} + ½\pi_{AV}$) to 'cert-less' & 'certs'
Recruiter belief p: If sees 'no cert' then $p = ½$, & if sees 'cert' then $p = ½$

All *PBNE in* Model 1: $½\pi_{EX} + ½\pi_{AV} > \pi_{EX} - c_{EX} > \pi_{AV} > \pi_{EX} - c_{AV}$

Unique Separating Equilibrium
Strategies: Each applicant will 'get cert' if exceptional, 'not' if avg
Recruiter offers 'certs' **10**, and 'cert-less' **4**
Recruiter beliefs: If sees 'cert' then $p = 1$, if sees 'no cert' then $p = 0$

Pooling Equilibrium I
Strategies: Each applicant will 'not get cert'
Recruiter offers 'cert-less' **7** and 'certs' **10**
Recruiter beliefs: If sees 'no cert' then $p = ½$, & if sees 'cert' then $p = 1$

Pooling Equilibrium II
Strategies: Each applicant will 'not get cert'
Recruiter offers **7** to 'cert-less' & 'certs'
Recruiter beliefs: If sees 'no cert' then $p = ½$, & if sees 'cert' then $p = ½$

- *Both pooling equilibria generate same outcome, and are Pareto-preferred by applicants* in Model 1; **Pooling Eqm I** is more ***intuitive***

EC 246: Decisions & Games

Answer Key to Problem Set 10

Problem 1: Consider the case where there exists some number $x \in (1, 5)$ such that each country i will *escalate* if and only if $v_i > x$ and will *desist* if and only if $v_i < x$, with $v_i = x$ being indifferent. Then country 1 will expect to best respond to $\{escalate, \text{desist}; (5–x)/4, (x–1)/4\}$. So if country 1's valuation v_1 equals x, it must be indifferent between playing *escalate* and getting $\{[(5–x)/4] \times 1 + [(x–1)/4] \times x\}$ and playing desist and getting $\{[(5–x)/4] \times 0 + [(x–1)/4] \times 4\}$. The two expected utilities will be equal if and only if $x = 3$. Thus, the specific pure-strategy symmetric Bayes-Nash equilibrium of this game in which the probability that each country escalates the least is the one where each country i escalates if and only if $v_i > 3$. Of course, there is also another pure-strategy symmetric Bayes-Nash equilibrium of this game in which each country escalates with probability 1 irrespective of the value of its $v_i \in [1, 5]$.

[Note that there is still a contagion effect in the above game, because for all values of $v_i \in [3, 4]$, it is not a dominant strategy for country i to escalate, but it will escalate in every Bayes-Nash equilibrium.]

Problem 2: In the case where each player's prior beliefs are that rival's cost of fighting is "equally likely to be any number between 10 and 30", there will be exactly two pure-strategy Bayes-Nash equilibria which are asymmetric: in each equilibrium, one player is hawkish irrespective of his/her realized cost of fighting between 10 and 30, and the other player is hawkish if her realized cost of fighting is less than 20 and dovish if her realized cost of fighting is greater than 20. To prove the above result, note that the stated strategies of the two animals are best responses to each other.

[Note that in contrast to the Arms Race game, there will be no contagion effect in the above Hawk-Dove game.]

A "common sense" approach to the Spence Game when $c_{EX} = 5$

- It is a strictly-dominated-strategy for an average applicant to get education certificate, as $4 > 10 - 7$ [$\pi_{AV} > \pi_{EX} - c_{AV}$]
- So on seeing 'certificate', makes sense for Recruiter to believe that applicant is exceptional and offer $\{\pi_E\} = 10$
- Question is: what should Recruiter offer on seeing 'no certificate'?
  - Depends on fraction f of 'exceptionals' he expects to get certs
  - If f close to 1, Recruiter should offer $\{\pi_{AV}\} = 4$ to the 'cert-less'
  - If f close to 0, he should offer $\{½(\pi_{EX}+\pi_{AV})\} = 7$ to 'cert-less'
- Recognizing this, if an 'exceptional' expects f to be close to 1, she should get 'cert' as $10 - 5 > 4$ [$\pi_{EX} - c_{EX} > \pi_{AV}$] (taking f closer to 1)
  - and if 'exceptional' expects f to be close to 0, she should not get 'cert', as $7 > 10 - 5$ [$½(\pi_{EX}+\pi_{AV}) > \pi_{EX} - c_{EX}$] (taking f closer to 0)
- Common sense suggests two "self-sustaining" configurations:

1. exceptionals expected to get cert ⇒ cert-less get 4 ⇒ exceptionals get cert
2. nobody expected to get cert ⇒ cert-less get 7 ⇒ nobody gets cert

All *PBNE in* Model 1: $½\pi_{EX} + ½\pi_{AV} > \pi_{EX} - c_{EX} > \pi_{AV} > \pi_{EX} - c_{AV}$

Unique Separating Equilibrium

Strategies: Each applicant will 'get cert' if exceptional, 'not' if avg
Recruiter offers 'certs' **10**, and 'cert-less' **4**

Recruiter beliefs: If sees 'cert' then $p = 1$, if sees 'no cert' then $p = 0$

Pooling Equilibrium I

Strategies: Each applicant will 'not get cert'
Recruiter offers 'cert-less' **7** and 'certs' **10**

Recruiter beliefs: If sees 'no cert' then $p = ½$, & if sees 'cert' then $p = 1$

Pooling Equilibrium II

Strategies: Each applicant will 'not get cert'
Recruiter offers **7** to 'cert-less' & 'certs'

Recruiter beliefs: If sees 'no cert' then $p = ½$, & if sees 'cert' then $p = ½$

- *The **pooling equilibria** generate same **outcome,** which is Pareto-preferred by applicants to the **separating eqm outcome***; **Pooling Eqm I** is more ***intuitive***

A "common sense" approach to the Spence Game when $c_{EX} = 2$

- It is a strictly-dominated-strategy for an average applicant to get education certificate, as $4 > 10 - 7$ [$\pi_{AV} > \pi_{EX} - c_{AV}$]
- So on seeing 'certificate', makes sense for Recruiter to believe that applicant is exceptional and offer $\{\pi_E\} = 10$
- Question is: what should Recruiter offer on seeing 'no certificate'?
  - The answer need not depend on fraction f of 'exceptionals' he expects to get certificates, for the following reason
  - Suppose the Recruiter expects f to be 0, and has thus decided to pay $\{½(\pi_{EX}+\pi_{AV})\} = 7$ to the 'cert-less'

  he should realize that if he unexpectedly sees an applicant with certificate and decides to optimistically view applicant as exceptional and offer 10 the applicant will gain from the Recruiter's optimism **if and only if (s)he is indeed exceptional** [$10 - 2 > 7 >> 4 > 10 - 7$]

  so "optimism when seeing a 'cert' is justified" while "pessimism is not"
- Common sense implies a ***unique*** self-sustaining configuration in this case: Only exceptionals get cert and get paid **10**, while the cert-less get paid **4**

PBNE in the Spence Game [Model 2: $c_{EX} = 2$]

- Recruiter has to hire each applicant at p.v. compensation of {"inferred" profit-creation}

Applicant decisions: In deterministic play she can do one of two things
- she can "separate": 'get cert' if exceptional, and 'not' if average
- or she can "pool": 'not get cert' irrespective of own type

Q. 1: Can there be a self-enforcing recommendation for applicants to **separate**?

Answer: YES. If applicants separate & reveal their types by using certification signal, Recruiter will be "Bayes-correct" to offer **10** to 'certs' and **4** to 'cert-less' sustaining only exceptional's desire to signal

Unique Separating Equilibrium

Strategies: Each applicant will 'get cert' if exceptional, 'not' if avg
Recruiter offers 'certs' 10 (π_{EX}), and 'cert-less' 4 (π_{AV})

Recruiter beliefs: If sees 'cert' then $p = 1$, if sees 'no cert' then $p = 0$

PBNE in the Spence Game [Model 2: $c_{EX} = 2$]

- Recruiter has to hire each applicant at p.v. compensation of {"inferred" profit-creation}

Applicant decisions: In deterministic play she can do one of two things
- she can "separate": 'get cert' if exceptional, and 'not' if average
- or she can "pool": 'not get cert' irrespective of own type

Q. 2: Can there be a self-enforcing recommendation for applicants to **pool**?

Answer: YES. If applicants 'pool' and nobody gets a 'cert', Recruiter will be "Bayes-correct" to offer **7** to every applicant

A Pooling Equilibrium

Strategies: Each applicant will 'not get cert'
Recruiter offers 7 ($\{½\pi_{EX} + ½\pi_{AV}\}$) to 'cert-less' & 'certs'

Recruiter beliefs: If sees either 'cert' or 'no cert': $p = ½$

Recall the '**unintuitive Pooling Equilibrium II**' in **Model 1**

Strategies: Each applicant will 'not get cert'
Recruiter offers **7** to 'cert-less' & 'certs'

Recruiter beliefs: If sees 'no cert' then $p = ½$, & if sees 'cert' then $p = ½$

Set of *PBNE in* Model 2: $\pi_{EX} - c_{EX} > ½\pi_{EX} + ½\pi_{AV} > \pi_{AV} > \pi_{EX} - c_{AV}$

Unique Separating Equilibrium

Strategies: Each applicant will 'get cert' if exceptional, 'not' if avg
Recruiter offers 'certs' 10 (π_{EX}), and 'cert-less' 4 (π_{AV})

Recruiter beliefs: If sees 'cert' then $p = 1$, if sees 'no cert' then $p = 0$

A Pooling Equilibrium

Strategies: Each applicant will 'not get cert'
Recruiter offers 7 ($\{½\pi_{EX} + ½\pi_{AV}\}$) to 'cert-less' & 'certs'

Recruiter beliefs: If sees either 'cert' or 'no cert': $p = ½$

Comment: This "pooling equilibrium" needs a belief specification on seeing a "cert" that is Bayes-correct ***but NOT intuitive***

Problem with the Pooling Eqm in Model 2: $c_{EX} = 2$

A Pooling Equilibrium

Strategies: Each applicant will 'not get cert'

Recruiter offers 7 ($\{½\pi_{EX} + ½\pi_{AV}\}$) to 'cert-less' & 'certs'

Recruiter beliefs: If sees either 'cert' or 'no cert': $p = ½$

Comment: This "pooling equilibrium" needs a belief specification on seeing a "cert" that ***is NOT intuitive*** for the following reason …..

- If the players are expected to play the pooling eqm, an 'exceptional' might deviate and 'get cert', and make following speech to Recruiter: "As I 'have cert', you must believe I am exceptional & offer 10 because (*a*) **If** you offer 10, ***as exceptional*** I'll get more than my specified eqm payoff & (*b*) **even if** you offer 10, ***were I average*** I would get less than my specified eqm payoff."
- This 'forward induction refinement' is called ***the intuitive criterion***
  - it 'refines' the ***PBNE*** concept: it rules out **pooling eqm** in Model 2
  - note that it does not rule out **Pooling Equilibrium II** in Model 1

***The Spence Signalling Game Experiment* when $c_{EX} = 2$:**
Model 2:

Applicant (each of two incarnations)

Strategy (22 players)	Frequency
MBA if exceptional, non if average	19 [71%]
Non if exceptional, non if average	1 [9%]
MBA if exceptional, MBA if average	2 [14%]
Non if exceptional, MBA if average	0 [6%]

HR Head

Strategy (18 players)	Frequency
10 for MBA, 4 for non	14 [74%]
10 for MBA, 7 for non	0 [4%]
7 for MBA, 4 for non	0 [2%]
7 for MBA, 7 for non	4 [18%]
7 for MBA, 10 for non	0 [1%]

Forward Induction under Complete Information

- ***Off-eqm play taken as a signal about what a player will do in the future***

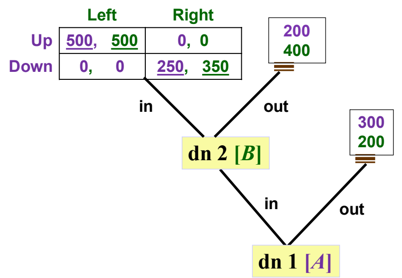
Two pure ***SPNE***:

[1] *A*: out at dn1, down at dn3][*B*: out at dn2, right at is4]

[2] *A*: in at dn1, up at dn3][*B*: in at dn2, left at is4]

- We can rule out [1] by the ***forward induction refinement***

Game I:



***Intuitive Criterion*: Forward Induction under Incomplete Info**

- ***Off-eqm play taken as a signal about what a player's type is***

- In a Signalling Game, if Sender sends an "unexpected" signal, Bayes' Rule does not restrict the inference Receiver can draw from the signal
- In some cases, this allows the Receiver to "infer/believe whatever (s)he wants to, and then play according to his inference/belief"
- The ***Intuitive Criterion*** is a "forward induction refinement" that constrains the Receiver from making arbitrary inferences following unexpected play, and thus rules out certain "unintuitive" ***PBNE***
- A formal statement of the ***Intuitive Criterion***:
  Consider a ***PBNE*** in which a signal s^* is not used in equilibrium.

If (*i*) for all 'types' of Sender except for some type t^*, s^* is unprofitable (relative to their eqm payoffs) under "**all**" Receiver responses

& (*ii*) for Sender 'type' t^*, s^* is profitable (relative to her eqm payoff) under a "**plausible**" Receiver response against type-t^* Sender

then upon observing s^*, Receiver must infer that Sender is of type t^*.

Another Signaling Game: *The Wooing Game*

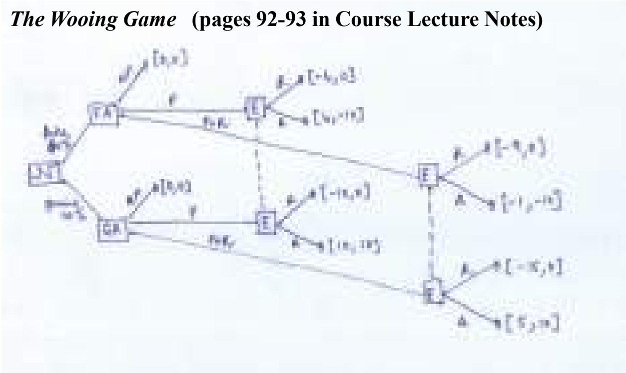
Adam, who resides and works in Fun-city, is wooing Eve, who resides and works in Joy-city. Eve is not sure whether Adam is a 'genuine' suitor (i.e., whether he truly loves her) or a 'fake' (who will soon break up); she believes that Adam is genuine only with a 40% probability.

In a proposal-response game, Adam moves first (obviously knowing his own 'type'). His choices are: (*N*) not to propose marriage; (*P*) to propose marriage while continuing to live in Fun-city; (*P+R*) to propose marriage and simultaneously relocate to Joy-city (incurring a cost in the process). If Adam plays (*N*), the game is over and each person gets a long-term payoff of zero. If Adam does propose, the game moves on to a second-stage where Eve has to accept or reject the proposal.

If Eve rejects Adam's proposal, her long-term payoff is zero. If she accepts the proposal, her payoff depends on Adam's 'true type': if Adam is 'Genuine' her payoff is 10 while if he is 'Fake' her payoff is –10.

If Adam plays (*P*), his acceptance and rejection payoffs are 10 and –10 respectively *if* he is Genuine, and are 4 and –4 respectively if he is Fake. If Adam plays (*P+R*), the corresponding payoffs for him are all reduced by 5, which is his utility cost of relocation (irrespective of his type).

The Wooing Game **(pages 92-93 in Course Lecture Notes)**



In a pooling equilibrium, the two types of Adam must make "no proposal" due to his adverse initial reputation

In a separating equilibrium, 'fake Adam' must make "no proposal" and 'genuine Adam' must "propose and relocate" (and this is robust to 'all interior initial reputations')

General Comments about Signalling Games

- In the Spence game, the features $[\pi_{EX} > \pi_{AV}]$ and $[c_{EX} < c_{AV}]$ open up the possibility of signalling "ability" by "getting certification"
- Given that, the parameter conditions $\pi_{EX} - c_{EX} > \pi_{AV} > \pi_{EX} - c_{AV}$ are sufficient for existence of at least one *PBNE* where education is used as an effective signal by the exceptionals
  - If $\pi_{AV} > \pi_{EX} - c_{EX} > \pi_{EX} - c_{AV}$, then no applicant would 'get cert' (signal is too costly)
  - If $\pi_{EX} - c_{EX} > \pi_{EX} - c_{AV} > \pi_{AV}$, then any attempt by an exceptional applicant to use 'certification' as a signal would be "mimicked" by an average applicant (signal is too cheap)
- When "signal intensities" can be <u>chosen</u> by the Sender, it is possible that a signal of "lower intensity" will be "mimicked" but one of a "higher intensity" will not be mimicked

  In such a case, we will have "separating equilibria" where a Sender with favourable information will invest in "excessive (socially wasteful) signalling" to effectively transmit information

Problem Set 11

***Problem* 1:** Consider the following game of technology adoption with network effects: Two firms, Alpha and Beta, are currently using an 'old' technology, and have the opportunity to migrate to a 'new and better' technology. Both technologies exhibit 'positive network effects' in that the individual return to using a particular technology rises when other firms also use the same technology (think of this as compatibility and/or standardization benefits). The payoff matrix for the two firms – which gives the net present value payoffs as a function of migration decisions – is as follows:

Beta

Alpha		*migrate to new*	*stay with old*
	migrate to new	$20 - c_A$, $20 - c_B$	$16 - c_A$, 8
	stay with old	8, $16 - c_B$	10, 10

For each firm i, its cost of 'migrating to new technology' c_i is a number between 5 and 15. Each firm i knows its own c_i, and (following Laplace's 'logic of insufficient reason') believes that the other firm's cost c_j is uniformly distributed on the interval [5, 15].

Each firm can decide to migrate in any one of N periods: $T = 1, 2, 3, \ldots, N$, and can observe the rival's migration decision during that time. [The payoffs described in the above matrix obtain after these N periods are over.] This structure of the game allows each firm to either "lead the other" or "follow the other" in migrating to the new technology. Determine the unique symmetric Perfect Bayesian Nash Equilibrium strategies of the two firms in this game.

***Problem* 2:** Reconsider the "Wooing Game" presented in pages 92–93 of the Course Lecture Notes. Specify an alternative (but reasonable) set of integer payoffs for 'fake' Adam when he plays (P) such that the 'candidate' pooling equilibrium outcome described in the Notes becomes a *PBNE* that survives the forward induction refinement 'intuitive criterion'.

Information about the End Term Exam

End-term Exam Syllabus

- The material covered in the post-midterm Class Sessions, and the post-midterm Problem Sets.
- The relevant chapters in the Course Lecture Notes are : Ch. 5 (ignoring problems where choice sets are "continuous intervals"); Ch. 6.4; Ch. 7; Problems I and II in Ch. 8; Ch. 9; Ch. 10.
- Even though there will be no direct exam questions on the pre-midterm material, you cannot afford to "forget" that material.

Exam Structure

It will be a **100**-minute closed-book exam, worth 50 marks. There will be **SIX** problems, three on "complete information games" and three on "incomplete information games". The exam format will be similar to the midterm exam. The "exam instructions" will be identical to those in the mid-term exam.

Answer Key 11

***Problem* 1:** Unique PBNE outcome when each firm can decide to migrate in any one of N periods $T = 1, 2, 3, \ldots, N$: A firm with $\boldsymbol{c} \in (12, 15]$ will never migrate, a firm with $\boldsymbol{c} \in (8.225, 12)$ will migrate in period $\boldsymbol{t}$ if and only if the other firm has migrated in period $(\boldsymbol{t}–1)$, and a firm with $\boldsymbol{c} \in [5, 8.225)$ will migrate in period 1. A firm with $\boldsymbol{c} = 12$ will be indifferent between never migrating and migrating in period $\boldsymbol{t}$ if the other firm has migrated in period $(\boldsymbol{t}–1)$, and a firm with $\boldsymbol{c} = 8.225$ will be indifferent between migrating in period 1 and migrating in period $\boldsymbol{t}$ if the other firm has migrated in period $(\boldsymbol{t}–1)$.

***Problem* 2:** In the "Wooing Game", let us specify that when 'fake' Adam plays (P), he gets $x > 0$ if Eve accepts his proposal, while), he gets $[–x]$ if Eve rejects his proposal. No other payoff specifications are changed. Convince yourself that for any integer x between (and including) the values 6 and 9, the 'candidate' pooling equilibrium outcome described in the Notes becomes a *PBNE* that survives the forward induction refinement 'intuitive criterion'.